Pharmacology & Pathology

Comprehensive Lecture Notes for 400 Level MBBS

Pharmacology · Histopathology · Haematology · Chemical Pathology · Medical Microbiology (Virology & Parasitology)

Rearranged from the departmental course outline · Explained notes · Exam pearls · MCQs · Short-answer questions · Viva practice · Appendices · References

Compiled November 2026 as a study companion. These notes summarise standard textbooks and international guidelines (WHO, BNF, Robbins, Rang & Dale, Katzung, Hoffbrand, Marshall, Jawetz, Cheesbrough). Always confirm doses and local protocols with your lecturers, the current Nigerian National Treatment Guidelines and an up-to-date formulary before clinical use.

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Pathology (Histopathology)
Haematology
Chemical Pathology
Microbiology, Virology and Parasitology
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How to Use These Notes

These notes take the topics in your departmental course outline and put them in a **logical teaching order**. Each topic builds on the one before it, in the same way that standard international textbooks (Robbins, Rang & Dale, Katzung, Hoffbrand, Marshall & Bangert, Jawetz, Cheesbrough) organise the subjects.

Every chapter follows the same pattern:

1. **Definition and overview.** What the topic is and why it matters clinically.
2. **Core science.** The physiology, mechanism or pathogenesis, explained step by step.
3. **Classification tables.** Use these to revise quickly.
4. **Clinical application.** Uses, adverse effects, diagnosis and treatment.
5. **Exam pearls** (green boxes) and **viva corner** (blue boxes). These hold the points examiners ask about most.

At the end of each book there is a **self-assessment section**: single best answer questions (MCQs) with answers and explanations, short-answer/essay questions with model answer outlines, and a **viva voce bank** of the questions examiners usually ask, with model answers.

> **⚠ Caution**
>
> Doses are given as typical **adult** doses for orientation only. Always check a current formulary (BNF, Nigerian Essential Medicines List) and the current Nigerian National Treatment Guidelines before prescribing.

How the Outline Was Rearranged

The original outline listed topics **by lecturer**. Here they are grouped **by system and concept** so that the background is covered before the applications. For example, pharmacokinetics comes before antihypertensives, and diuretics come before heart failure. Topics that overlapped have been merged: Rhabdoviridae with rabies virus, and coccidia with the apicomplexans. The table shows where each original topic now sits.

New book / part	Topics (original lecturer in brackets)
Book 1 Pharmacology. Part A: General principles	Introduction, pharmacokinetics, pharmacodynamics, pharmacogenetics, drug interactions, new drug development (Dr Amalizu / Dr Agbasi)
Part B: Autonomic pharmacology	ANS pharmacology (Dr Amalizu / Dr Agbasi)
Part C: Cardiovascular, renal and blood	Diuretics (Dr Okolo); antihypertensives, anti-angina, heart failure, antiarrhythmics (Dr Nwogu); hyperlipidaemia, blood coagulation (Dr Okolo)

New book / part	Topics (original lecturer in brackets)
Part D: Gastrointestinal	Anti-ulcer (Dr Okolo); antiemetics, drugs affecting motility, antidiarrhoeals and laxatives, haemorrhoid preparations (Dr Nwogu)
Part E: Pain, inflammation, anaesthesia	NSAIDs, opioids and non-opioids, anaesthetic agents (Dr Nwogu); gout (Dr Okolo)
Part F: Endocrine	Calcium metabolism (Dr Nwogu)
Part G: Chemotherapy	Anti-TB, anti-leprosy, antifungals, antimalarials, anticancer (Dr Nwogu); anthelminthics (Dr Okolo)
Part H: Immunopharmacology	Immune pharmacology, vaccines and adjuvants (Dr Nwogu)
Book 2 Histopathology	Cell injury, necrosis, apoptosis, cell cycle, neoplasia, haemodynamics, environmental and nutritional pathology (Dr Mmadi); inflammation and mediators (Dr Nweke)
Book 3 Haematology	Haemopoiesis, HIM, blood cell development, erythropoiesis, myelopoiesis, megakaryocytopoiesis, lymphocyte ontogeny, iron, IDA, B12, folate (Dr Ifeanacho)
Book 4 Chemical Pathology	Laboratory medicine principles (requests, specimens, errors, POCT, reference intervals, decision limits, physician's role) (Drs Emedoh, Onuegbu, Otti); K^+, acid–base, renal function, calcium/phosphate and bone, diabetes, lipoproteins (introduction only) (Drs Okoana, Otti, Emedoh, Onuegbu). Not included in this edition: metabolic syndrome, thyroid axis, haem/bilirubin, porphyrias
Book 5 Microbiology	Virology (Dr Chigbo); protozoology (Dr Nworeji); helminthology (Dr Obiajuru)
Appendices and references	Reference values, antidotes, treatment regimens, immunisation schedule, mnemonics, abbreviations, bibliography

Note on names and page numbers. The lecturer names were read from a handwritten list, so please check the spellings. The haematology page numbers in your outline (pg 1, 11, 42, 50, 96, 141) correspond closely to the chapter order of *Hoffbrand's Essential Haematology*, which is the core haematology reference used here.

Book 1 — Pharmacology

Part A: General Principles · Part B: Autonomic Pharmacology · Part C: Cardiovascular, Renal & Blood · Part D: Gastrointestinal · Part E: Pain, Inflammation & Anaesthesia · Part F: Endocrine · Part G: Chemotherapy · Part H: Immunopharmacology

Chapter 1 · Introduction to Pharmacology

Pharmacology is the study of how chemicals (drugs) interact with living systems. A **drug** is any substance that changes physiological function and is used to diagnose, prevent or treat disease (WHO).

Branches of pharmacology

Branch	Concern
Pharmacokinetics (PK)	What the body does to the drug: absorption, distribution, metabolism, excretion (ADME)
Pharmacodynamics (PD)	What the drug does to the body: mechanism, receptor action, dose–response
Pharmacotherapeutics	Using drugs to prevent or treat disease
Toxicology	Harmful effects of chemicals, poisoning and its management
Pharmacogenetics / pharmacogenomics	How genes affect drug response
Pharmacovigilance	Detecting, assessing and preventing adverse drug reactions (ADRs) after marketing
Clinical pharmacology	The scientific study of drugs in humans
Chemotherapy	Drugs used against microbes, parasites and cancer cells

Sources and nomenclature of drugs

- **Sources:** plants (morphine, digoxin, quinine, artemisinin), animals (insulin and heparin, historically), minerals (iron, lithium), microbes (penicillin, streptomycin), synthetic chemistry (most modern drugs), and biotechnology (recombinant insulin, monoclonal antibodies).
- **Names:** the *chemical name* describes the structure; the *non-proprietary* (*generic*, INN) name is the one to use when prescribing, e.g. paracetamol; the *proprietary* (*brand*) name is the manufacturer's, e.g. Panadol®.

Routes of administration

Route	Advantages	Disadvantages
Oral	Safe, convenient, cheap	First-pass metabolism, slow, needs cooperation, GI irritation

Route	Advantages	Disadvantages
Sublingual / buccal	Rapid; avoids first-pass (glyceryl trinitrate)	Only small, lipid-soluble doses
Intravenous	100% bioavailability, immediate, can be titrated	Cannot be reversed; risk of infection, embolism, anaphylaxis
Intramuscular / subcutaneous	Depot preparations possible	Painful; absorption depends on blood flow
Rectal	Useful when vomiting or unconscious; ~50% bypasses the portal circulation	Erratic absorption
Inhalational	Rapid; local action in the lung (salbutamol); gaseous anaesthetics	Needs technique and devices
Transdermal / topical	Steady levels (fentanyl, nitrate patches)	Skin irritation; only potent lipophilic drugs
Intrathecal / epidural	Bypasses the blood–brain barrier	Specialist technique

★ Exam pearls

- Generic names are the international standard for prescribing.
- The first-pass effect matters most for propranolol, morphine, lidocaine (ineffective orally), GTN and verapamil.
- “Pharmacokinetics = body → drug; pharmacodynamics = drug → body.”

Chapter 2 · Pharmacokinetics

Absorption

Most drugs cross membranes by **passive diffusion**, which depends on lipid solubility, the concentration gradient and how much of the drug is ionised.

Henderson–Hasselbalch: for a weak acid, $pH - pKa = \log([A^-]/[HA])$; for a weak base, $pKa - pH = \log([BH^+]/[B])$.

- Weak **acids** (aspirin, barbiturates) are unionised in acid, so they are absorbed better from the stomach in theory. In practice the huge surface area of the small intestine means most absorption happens there.
- **Ion trapping** is used in poisoning: alkalinising the urine (sodium bicarbonate) ionises aspirin and phenobarbital and so increases their excretion.

Other transport mechanisms: facilitated diffusion, active transport (levodopa, 5-FU), endocytosis (vitamin B_{12} with intrinsic factor). **P-glycoprotein** efflux limits the absorption of digoxin and ciclosporin.

Bioavailability (F): the fraction of an administered dose that reaches the systemic circulation unchanged. F = AUC(oral) ÷ AUC(IV). It is reduced by first-pass metabolism, poor solubility and efflux.

Bioequivalence: two formulations are bioequivalent when their AUC and Cmax 90% confidence intervals fall within 80–125%.

Distribution

- **Volume of distribution (Vd)** = amount in body ÷ plasma concentration. It is an *apparent* volume.
  - Low Vd (about 3–5 L): drug stays mostly in plasma, e.g. warfarin and heparin (highly protein-bound or large molecules).
  - Very high Vd (hundreds of litres): drug is concentrated in tissues, e.g. chloroquine (~13,000 L), digoxin (~500 L), amiodarone.
- **Plasma protein binding:** acidic drugs bind albumin (warfarin, phenytoin, sulfonamides); basic drugs bind α_1-acid glycoprotein (lidocaine, propranolol). Only the *free* drug is active. **Displacement** matters only for highly bound drugs with a low Vd and narrow therapeutic index. For example, sulfonamides displace bilirubin and can cause kernicterus in neonates.
- **Barriers:** the blood–brain barrier (tight junctions plus P-glycoprotein) and the placenta, which most lipophilic drugs cross.
- **Redistribution** ends the action of a single IV bolus of thiopental or propofol.

Metabolism (biotransformation)

Its purpose is to make drugs more water-soluble so they can be excreted. The liver is the main site.

Phase	Reactions	Examples
Phase I (functionalisation)	Oxidation (CYP450), reduction, hydrolysis	Codeine → morphine (CYP2D6); diazepam → nordiazepam
Phase II (conjugation)	Glucuronidation, sulfation, acetylation, methylation, glutathione conjugation	Paracetamol glucuronide; isoniazid acetylation (NAT2)

- **Prodrugs** are activated by metabolism: enalapril, levodopa, clopidogrel, codeine, cyclophosphamide, proguanil, azathioprine.
- **CYP3A4** metabolises about 50% of drugs.

CYP inducers ("GPRS Cell Phone")	CYP inhibitors ("SICKFACES.COM + grapefruit")
Griseofulvin, **P**henytoin, **R**ifampicin, **S**t John's wort, **C**arbamazepine, **Phe**nobarbital, chronic alcohol, smoking (CYP1A2)	**S**odium valproate, **I**soniazid, **C**imetidine, **K**etoconazole, **F**luconazole, **A**cute alcohol, **C**hloramphenicol, **E**rythromycin/clarithromycin, **S**ulfonamides, **Cip**rofloxacin, **O**meprazole, **M**etronidazole, ritonavir, grapefruit juice

Excretion

- **Renal excretion** = glomerular filtration (free drug only) + active tubular secretion (organic anion transporters handle penicillin and methotrexate, which probenecid blocks) − passive reabsorption (lipophilic, unionised drug is reabsorbed).
- **Biliary excretion and enterohepatic circulation** prolong the action of oral contraceptives, digitoxin and morphine. Broad-spectrum antibiotics interrupt this circulation.
- Other routes: lungs (anaesthetic gases), breast milk, sweat and saliva.

Kinetic equations

Parameter	Formula / meaning
Clearance (CL)	Volume of plasma cleared of drug per unit time; CL = rate of elimination ÷ Cp
Half-life (t½)	t½ = 0.693 × Vd ÷ CL
Steady state	Reached after about **4–5 half-lives**, whatever the dose
Loading dose	LD = Vd × target Cp ÷ F
Maintenance dose	MD = CL × target Cp × dosing interval ÷ F

First-order kinetics: a constant *fraction* of drug is eliminated per unit time, so t½ is constant. Most drugs behave this way.

Zero-order (saturation) kinetics: a constant *amount* is eliminated per unit time, because the enzyme is saturated. Examples are **p**henytoin, **e**thanol and **a**spirin at high doses ("PEA"). A small increase in dose

can cause a large rise in plasma level and toxicity.

Therapeutic drug monitoring (TDM) is needed for drugs with a narrow therapeutic index: digoxin, lithium, phenytoin, aminoglycosides, vancomycin, theophylline, ciclosporin and tacrolimus.

★ Exam pearls

- Steady state takes 4–5 half-lives; a loading dose gets there sooner (amiodarone, digoxin, phenytoin, chloroquine).
- Neonates have immature glucuronidation, which explains **grey baby syndrome** with chloramphenicol.
- Old age reduces GFR, so reduce the doses of renally excreted drugs. Estimate GFR with Cockcroft–Gault or eGFR.

Chapter 3 · Pharmacodynamics

Drug targets

Drugs act on four main kinds of target: **receptors**, **ion channels**, **enzymes** and **carrier/transport molecules**. A few act in other ways: chemically (antacids, chelators), physically (osmotic laxatives), or on DNA and microbes.

Receptor superfamilies

Type	Time scale	Example
1. Ligand-gated ion channels (ionotropic)	Milliseconds	Nicotinic ACh, GABA-A, 5-HT_3, NMDA
2. G-protein-coupled receptors (GPCR, metabotropic)	Seconds	Muscarinic, adrenergic, opioid, histamine. Gs ↑cAMP; Gi ↓cAMP; Gq → PLC → IP_3/DAG → ↑Ca^{2+}
3. Kinase-linked receptors	Minutes	Insulin (tyrosine kinase), cytokines (JAK–STAT), growth factors
4. Nuclear (intracellular) receptors	Hours	Steroids, thyroid hormone, vitamin D, PPAR (fibrates, glitazones)

Key definitions

- **Affinity:** how strongly a drug binds its receptor (K_D).
- **Efficacy (intrinsic activity):** the ability of a bound drug to activate the receptor.
- **Potency:** the dose needed for a given effect (EC_{50}). A more potent drug has a curve shifted further *left*.
- **Full agonist:** efficacy = 1. **Partial agonist:** efficacy between 0 and 1, e.g. buprenorphine and pindolol. A partial agonist can act as an antagonist when a full agonist is present.
- **Inverse agonist:** reduces constitutive receptor activity (some antihistamines, β-carbolines).
- **Competitive (surmountable) antagonist:** shifts the dose–response curve to the right in parallel; Emax is unchanged (atropine, propranolol, naloxone).
- **Non-competitive / irreversible antagonist:** lowers Emax (phenoxybenzamine, aspirin on COX).
- **Physiological antagonism:** two drugs act on different receptors with opposite effects, e.g. adrenaline against histamine in anaphylaxis.
- **Chemical antagonism:** the drugs combine directly, e.g. protamine with heparin, chelators with metals.
- **Therapeutic index (TI)** = TD_{50} ÷ ED_{50}. A narrow TI means toxicity is close to the therapeutic dose (warfarin, digoxin, lithium).

- **Tolerance:** a reduced response after repeated doses (nitrates, opioids). **Tachyphylaxis:** tolerance that develops rapidly (ephedrine, which depletes stores).
- **Desensitisation / down-regulation:** after chronic agonist exposure. **Up-regulation (supersensitivity):** after chronic antagonist exposure, which is why β-blockers cause rebound when stopped abruptly.

Dose–response relationships

- **Graded curve:** response in one individual. Plotted against log dose it gives a sigmoid curve.
- **Quantal curve:** the percentage of a population that responds. It is used to derive ED_{50}, TD_{50} and LD_{50}.

Adverse drug reactions (ADRs): the Rawlins–Thompson classification

Type	Description	Example
A Augmented	Dose-related; predictable from pharmacology	Hypoglycaemia with insulin; bleeding with warfarin
B Bizarre	Not dose-related, idiosyncratic or immunological	Anaphylaxis to penicillin; malignant hyperthermia
C Chronic	Time- and dose-related	Cushingoid features with steroids
D Delayed	Appears late	Teratogenicity (thalidomide); carcinogenesis
E End of use	Withdrawal	Rebound hypertension after clonidine; adrenal crisis after steroids
F Failure	Unexpected loss of efficacy	Contraceptive failure with enzyme inducers

🎤 **Viva corner**

Q: What is the difference between potency and efficacy? **A:** Potency is how much drug is needed (EC_{50}). Efficacy is the maximum effect the drug can produce (Emax). Clinically, efficacy matters more. Furosemide is more efficacious than thiazides, but bumetanide is more potent than furosemide.

Chapter 4 · Pharmacogenetics

Pharmacogenetics studies how inherited variation in single genes changes drug response. **Pharmacogenomics** studies the whole genome. The aim is *personalised medicine*: the right drug, at the right dose, for the right patient.

Polymorphism	Effect	Clinical consequence
N-acetyltransferase 2 (NAT2) slow/fast acetylators	Slow acetylators: ~50% of Caucasians and Africans; fewer East Asians	Slow: isoniazid peripheral neuropathy (give pyridoxine), hydralazine/procainamide drug-induced lupus, sulfasalazine toxicity. Fast: isoniazid hepatitis (debated) and lower efficacy
Plasma cholinesterase (butyrylcholinesterase) atypical	Cannot hydrolyse suxamethonium	Prolonged apnoea ("scoline apnoea") lasting hours; ventilate until it wears off
G6PD deficiency (X-linked; common in Nigeria, ~20–25% of males in some areas)	Little NADPH, so RBCs cannot withstand oxidant stress	Haemolysis with primaquine, tafenoquine, dapsone, sulfonamides, nitrofurantoin, rasburicase, quinolones, fava beans
CYP2D6 poor / ultra-rapid metabolisers	Altered activation of codeine and tramadol; altered clearance of TCAs and metoprolol	Ultra-rapid: morphine toxicity from codeine (child deaths after tonsillectomy). Poor: no analgesia
CYP2C9 + VKORC1	Warfarin sensitivity	Lower warfarin dose needed
CYP2C19 loss of function	Clopidogrel cannot be activated	Stent thrombosis; use prasugrel or ticagrelor
TPMT / NUDT15 deficiency	Azathioprine and 6-mercaptopurine accumulate	Severe myelosuppression; test before starting
DPYD deficiency	5-FU / capecitabine cannot be broken down	Fatal toxicity
HLA-B*57:01	Abacavir hypersensitivity	Screen before prescribing
HLA-B*15:02 (Han Chinese, SE Asian)	Carbamazepine SJS/TEN	Screen before prescribing
HLA-B*58:01	Allopurinol SJS/TEN	Screen in high-risk groups
Malignant hyperthermia (RYR1 mutation)	Volatile anaesthetics + suxamethonium cause uncontrolled muscle Ca^{2+} release	Hyperthermia, rigidity, ↑CO_2. Treat with **dantrolene**
Acute porphyrias	Drugs that induce ALA synthase	Barbiturates, sulfonamides and griseofulvin precipitate attacks

Polymorphism	Effect	Clinical consequence
UGT1A1*28 (Gilbert)	Reduced glucuronidation	Irinotecan toxicity

★ Exam pearls

G6PD deficiency and primaquine is the classic exam pairing in a Nigerian setting. Always test G6PD before giving primaquine for radical cure of *P. vivax/ovale*, or use the single low dose (0.25 mg/kg) that WHO recommends for *P. falciparum* transmission blocking.

Chapter 5 · Drug Interactions

A **drug interaction** happens when the effect of one drug is changed by another drug, food, herb or disease.

Pharmaceutical (in vitro) interactions

These happen outside the body, before the drugs are given. Examples: phenytoin precipitates in dextrose; gentamicin is inactivated by penicillins in the same infusion; diazepam binds to PVC infusion bags.

Pharmacokinetic interactions

Stage	Mechanism	Example
Absorption	Chelation	Tetracyclines or quinolones + antacids/iron/Ca^{2+} (take them 2 h apart)
	Altered gastric pH	PPIs ↓ ketoconazole and atazanavir absorption
	Altered motility	Metoclopramide speeds paracetamol absorption
Distribution	Protein-binding displacement	Sulfonamides displace bilirubin in neonates; aspirin displaces methotrexate
Metabolism	Enzyme induction	Rifampicin ↓ oral contraceptives, warfarin, ARVs (nevirapine) and steroids, causing failure
	Enzyme inhibition	Erythromycin + simvastatin → myopathy; fluconazole + warfarin → bleeding; cimetidine + theophylline
	Monoamine oxidase inhibition	MAOI + tyramine (cheese) → hypertensive crisis (“cheese reaction”)
	Xanthine oxidase inhibition	Allopurinol + azathioprine/6-MP → marrow toxicity (reduce dose to 25%)
Excretion	Competition for tubular secretion	Probenecid ↑ penicillin levels (useful); NSAIDs ↑ methotrexate (dangerous)

Stage	Mechanism	Example
	Altered urine pH / Na^+	Thiazides and NSAIDs ↑ lithium levels

Pharmacodynamic interactions

- **Synergism / addition:** alcohol + benzodiazepines → respiratory depression; sulfamethoxazole + trimethoprim (sequential blockade); ACE inhibitor + spironolactone → hyperkalaemia; QT-prolonging combinations (macrolides + antiarrhythmics + antipsychotics).
- **Antagonism:** naloxone against opioids; flumazenil against benzodiazepines; NSAIDs blunt diuretics and antihypertensives.
- **Indirect:** hypokalaemia from diuretics increases digoxin toxicity.

> **⚠ Caution**
>
> High-risk "object" drugs that are easily affected by interactions: **warfarin, digoxin, lithium, theophylline, phenytoin, ciclosporin, methotrexate, oral contraceptives and antiretrovirals**. All have a narrow therapeutic index or are critical to treatment.

Chapter 6 · New Drug Development

Developing a new drug takes about **10–15 years** and costs over **US$1–2 billion**. Only about 1 in 10,000 compounds screened reaches the market.

Stage	Key features
Discovery	Target identification, lead compound (natural products, high-throughput screening, rational/computer-aided design), lead optimisation
Preclinical	In vitro and animal studies: pharmacology, PK, acute/subacute/chronic toxicity, reproductive toxicity, mutagenicity (Ames test), carcinogenicity. Must follow **Good Laboratory Practice (GLP)**. Then an IND application is filed with the regulator (FDA; NAFDAC in Nigeria)
Phase I	20–100 healthy volunteers (patients for cytotoxics). Tests **safety, tolerability, PK** and maximum tolerated dose. First-in-human dosing uses microdosing / phase 0
Phase II	100–500 patients. Tests **efficacy, dose-finding** and short-term side effects (IIa proof of concept, IIb dose-ranging)
Phase III	1,000–5,000+ patients in **randomised controlled, usually double-blind** multicentre trials against placebo or standard treatment. Confirms efficacy and safety, then a New Drug Application (NDA) is filed for marketing approval
Phase IV	**Post-marketing surveillance / pharmacovigilance**: detects rare ADRs (about 1 in 10,000), long-term effects and new indications. Uses spontaneous reporting (Nigeria: NAFDAC pharmacovigilance yellow form; UK: Yellow Card)

Ethics: trials must follow the Declaration of Helsinki and **Good Clinical Practice (ICH-GCP)**, with informed consent and approval from an ethics committee or IRB (in Nigeria, NHREC).

Generic drugs only need to show bioequivalence. **Orphan drugs** are for rare diseases. **Biosimilars** are copies of biologics.

★ Exam pearls

- The **thalidomide** disaster (1957–61; phocomelia) led to modern regulation and reproductive toxicity testing.
- Phase I is about safety in healthy volunteers, phase II about efficacy, phase III is the confirmatory RCT, and phase IV is pharmacovigilance.

Chapter 7 · Autonomic Nervous System Pharmacology

Physiology review

Feature	Sympathetic	Parasympathetic
Outflow	Thoracolumbar (T1–L2)	Craniosacral (CN III, VII, IX, X; S2–S4)
Ganglia	Paravertebral, close to the cord (short preganglionic fibres)	Near or in the target organ (long preganglionic fibres)
Ganglionic transmitter	ACh → nicotinic N_N	ACh → nicotinic N_N
Postganglionic transmitter	Noradrenaline (α, β); ACh at sweat glands (muscarinic)	ACh → muscarinic
Function	"Fight or flight"	"Rest and digest"

The adrenal medulla is a modified sympathetic ganglion that releases adrenaline (~80%) and noradrenaline.

Cholinergic transmission

Choline + acetyl-CoA → (choline acetyltransferase) → ACh. ACh is stored in vesicles by VAChT, which **vesamicol** blocks. Ca^{2+}-dependent release is blocked by **botulinum toxin**. ACh is then hydrolysed by **acetylcholinesterase**. Choline reuptake is blocked by **hemicholinium**.

Muscarinic receptors: M1 (neural, gastric; Gq), M2 (heart; Gi, causes bradycardia), M3 (smooth muscle, glands, endothelial NO; Gq).

Class	Drugs	Uses	Adverse effects / notes
Direct muscarinic agonists	Bethanechol, pilocarpine, carbachol	Urinary retention; glaucoma and dry mouth (pilocarpine)	**DUMBBELSS**: Diarrhoea, Urination, Miosis, Bradycardia, Bronchospasm, Emesis, Lacrimation, Salivation, Sweating
Reversible anticholinesterases	Neostigmine, pyridostigmine, physostigmine (crosses the BBB), edrophonium, donepezil, rivastigmine	Myasthenia gravis, reversal of NMB, Alzheimer's disease, anticholinergic poisoning (physostigmine)	Cholinergic crisis

Class	Drugs	Uses	Adverse effects / notes
Irreversible anticholinesterases	Organophosphates (malathion, parathion, nerve agents), ecothiopate	Insecticides	**Organophosphate poisoning**: treat with **atropine** (muscarinic effects) + **pralidoxime** (reactivates the enzyme before "ageing"), plus diazepam for seizures
Muscarinic antagonists	Atropine, hyoscine (scopolamine), ipratropium, tiotropium, tropicamide, oxybutynin, glycopyrrolate, hyoscine butylbromide	Bradycardia, organophosphate poisoning, premedication, mydriasis, COPD, overactive bladder, motion sickness, colic	"Dry as a bone, red as a beet, blind as a bat, hot as a hare, mad as a hatter". Can precipitate angle-closure glaucoma and urinary retention
Ganglion blockers	Hexamethonium, trimetaphan	Historical	Block both sympathetic and parasympathetic outflow
Neuromuscular blockers	See Chapter 23		

Adrenergic transmission

Tyrosine → (tyrosine hydroxylase, the rate-limiting step; **metyrosine** inhibits it) → L-DOPA → dopamine → (dopamine β-hydroxylase) → NA → (PNMT in the adrenal medulla) → adrenaline.

NA action is ended mainly by **reuptake-1 (NET)**, which **cocaine and TCAs** block, and is metabolised by MAO and COMT. **Reserpine** blocks VMAT.

Receptor	G-protein	Main effects
α_1	Gq	Vasoconstriction, mydriasis (radial muscle), bladder sphincter contraction
α_2	Gi	↓ NA release (presynaptic), ↓ insulin, central ↓ sympathetic outflow
β_1	Gs	↑ HR and contractility, ↑ renin release
β_2	Gs	Bronchodilation, vasodilation (skeletal muscle), uterine relaxation, glycogenolysis, K^+ moves into cells
β_3	Gs	Lipolysis, detrusor relaxation (mirabegron)
D_1	Gs	Renal and mesenteric vasodilation

Drug	Receptors	Key use
Adrenaline	α, β_1, β_2	**Anaphylaxis: 0.5 mg (0.5 mL of 1:1000) IM**; cardiac arrest (1 mg IV); with local anaesthetics
Noradrenaline	$\alpha_1 > \beta_1$	Septic shock (causes reflex bradycardia)
Dopamine	D_1 (low dose), β_1 (medium), α (high)	Shock
Dobutamine	β_1	Cardiogenic shock, stress echocardiography
Isoprenaline	β_1, β_2	Heart block and bradycardia (historical)
Salbutamol, terbutaline (short-acting); salmeterol, formoterol (long-acting)	β_2	Asthma; tocolysis; hyperkalaemia (salbutamol)
Phenylephrine	α_1	Nasal decongestion, mydriasis, hypotension under anaesthesia
Clonidine, methyldopa	α_2 (central)	Hypertension; methyldopa in pregnancy
Ephedrine, amphetamine, tyramine	Indirect (release NA)	Hypotension during spinal anaesthesia (ephedrine); ADHD (amphetamines)

Adrenergic antagonists

- **α-blockers:** prazosin, doxazosin, terazosin (α_1: hypertension, BPH; cause *first-dose hypotension*); tamsulosin (α_1A: BPH); phenoxybenzamine (irreversible, non-selective: **phaeochromocytoma**, given *before* β-blockade); phentolamine (reversible: hypertensive crisis, extravasation of NA).
- **β-blockers:** see Chapter 9.

> 🎤 **Viva corner**
>
> **Q:** Why must an α-blocker be started before a β-blocker in phaeochromocytoma? **A:** If the β_2 receptors (vasodilator) are blocked alone, adrenaline acts unopposed on α_1 receptors. This causes severe vasoconstriction and a hypertensive crisis.
>
> **Q:** Explain "adrenaline reversal". **A:** After an α-blocker, adrenaline causes a *fall* in BP because only its β_2 vasodilator action remains.

Chapter 8 · Diuretics

Diuretics increase urine output by increasing the excretion of Na^+ (natriuresis) and water. Their potency depends on how much Na^+ is reabsorbed at their site of action: PCT ~65%, thick ascending limb ~25%, DCT ~5–8%, collecting duct ~2–3%.

Class (site)	Mechanism	Examples	Uses	Adverse effects
Carbonic anhydrase inhibitors (PCT)	Block CA, so ↓ HCO_3^- reabsorption	Acetazolamide	Glaucoma, altitude sickness, idiopathic intracranial hypertension, metabolic alkalosis	Hyperchloraemic metabolic acidosis, hypokalaemia, renal stones, sulfa allergy
Osmotic (PCT, descending limb)	Freely filtered, not reabsorbed; holds water in the tubule	Mannitol	Raised ICP, acute glaucoma	Pulmonary oedema (expands ECF), hyponatraemia then hypernatraemia. Contraindicated in anuria and heart failure
Loop (TAL)	Inhibit **$Na^+/K^+/2Cl^-$ (NKCC2)** cotransporter. "High-ceiling" diuretics	Furosemide, bumetanide, torasemide, ethacrynic acid	Acute pulmonary oedema, CCF, oedema in nephrotic syndrome and cirrhosis, hypercalcaemia, hyperkalaemia, renal failure	**OH DANG**: Ototoxicity, Hypokalaemia, Dehydration, Allergy (sulfa), Nephritis (interstitial), Gout; also hypomagnesaemia, hypocalcaemia, metabolic alkalosis
Thiazides and thiazide-like (DCT)	Inhibit **Na^+/Cl^- (NCC)** cotransporter	Hydrochlorothiazide, bendroflumethiazide, chlortalidone, indapamide, metolazone	Hypertension (first line in Africans), mild HF, **nephrogenic DI**, hypercalciuria (stones)	**Hyper-GLUC**: hyperGlycaemia, hyperLipidaemia, hyperUricaemia, hyperCalcaemia; hypokalaemia, hyponatraemia, metabolic alkalosis, impotence

Class (site)	Mechanism	Examples	Uses	Adverse effects
K^+-sparing: aldosterone antagonists (collecting duct)	Block mineralocorticoid receptors	Spironolactone, eplerenone	**Heart failure (↓ mortality)**, primary hyperaldosteronism (Conn's), ascites in cirrhosis, resistant hypertension, hirsutism	Hyperkalaemia, **gynaecomastia** (spironolactone, which has antiandrogen action; less with eplerenone)
K^+-sparing: ENaC blockers (collecting duct)	Block epithelial Na^+ channels	Amiloride, triamterene	With thiazides/loops to prevent hypokalaemia; Liddle syndrome; lithium-induced DI (amiloride)	Hyperkalaemia
ADH (V_2) antagonists (vaptans)	Aquaretics	Tolvaptan, conivaptan	SIADH (hyponatraemia), ADPKD	Too-rapid Na^+ correction (osmotic demyelination), hepatotoxicity

Why do loop diuretics cause hypocalcaemia while thiazides cause hypercalcaemia? In the TAL, the positive lumen potential (from K^+ recycling) drives paracellular reabsorption of Ca^{2+} and Mg^{2+}; loops abolish it. In the DCT, a fall in intracellular Na^+ increases basolateral Na^+/Ca^{2+} exchange, so more Ca^{2+} is reabsorbed.

Why do loop and thiazide diuretics cause hypokalaemia? More Na^+ is delivered to the collecting duct, so more is exchanged for K^+ and H^+. Volume contraction also stimulates aldosterone.

★ Exam pearls

- In acute pulmonary oedema, IV furosemide also causes *venodilation* that relieves congestion before the diuresis starts.
- **Thiazides paradoxically reduce urine volume in nephrogenic DI.**
- The RALES trial showed spironolactone **reduces mortality** in HFrEF.
- Diuretic resistance: combine a loop diuretic with metolazone (sequential nephron blockade).

Chapter 9 · Antihypertensive Drugs

Background

Hypertension is defined as BP ≥ **140/90 mmHg** in clinic (ISH 2020, NICE, ESC). ACC/AHA 2017 uses ≥130/80. It is the leading risk factor for stroke, IHD, heart failure and CKD. Black Africans tend to have **low-renin, salt-sensitive hypertension**.

BP = cardiac output × total peripheral resistance. BP is controlled by four sites: arterioles (resistance), venules (capacitance), the heart (pump) and the kidneys (volume). These are regulated by the baroreflex, the sympathetic nervous system and the **renin–angiotensin–aldosterone system (RAAS)**.

RAAS: angiotensinogen → (renin) → Ang I → (ACE, which also breaks down bradykinin) → Ang II → AT_1 receptors cause vasoconstriction, aldosterone release, Na^+ retention, thirst, ADH release, and cardiac and vascular remodelling.

Classification

Class	Examples	Mechanism	Key adverse effects / contraindications
Thiazide / thiazide-like diuretics	HCTZ, chlortalidone, indapamide	↓ volume early; ↓ PVR later	See Chapter 8; gout
ACE inhibitors	Captopril, enalapril, lisinopril, ramipril	↓ Ang II, ↑ bradykinin	**Dry cough (~10–15%; bradykinin)**, **angio-oedema** (more common in Black patients), hyperkalaemia, first-dose hypotension, acute kidney injury in **bilateral renal artery stenosis**. **Teratogenic** (fetal renal dysgenesis): contraindicated in pregnancy
ARBs	Losartan, valsartan, candesartan, telmisartan	Block AT_1	Like ACEIs but **no cough** (rare angio-oedema). Losartan is uricosuric
Direct renin inhibitor	Aliskiren	Inhibits renin	Avoid combining with ACEI/ARB
ARNI	Sacubitril/valsartan	Neprilysin inhibition ↑ natriuretic peptides	Used in heart failure

Class	Examples	Mechanism	Key adverse effects / contraindications
Calcium channel blockers (CCBs): dihydropyridines	Amlodipine, nifedipine, felodipine	Block L-type channels in vascular smooth muscle → arteriolar dilation	**Ankle oedema**, flushing, headache, reflex tachycardia, gum hypertrophy. Short-acting nifedipine is no longer used routinely
CCBs: non-dihydropyridines	Verapamil (cardiac > vascular), diltiazem	↓ HR, AV conduction, contractility	Bradycardia, heart block, **constipation (verapamil)**. **Never give IV verapamil with a β-blocker** (asystole)
β-blockers	Cardioselective (β_1): atenolol, metoprolol, bisoprolol, nebivolol (also releases NO). Non-selective: propranolol, timolol. With α-blockade: labetalol, carvedilol. With ISA: pindolol, acebutolol	↓ CO, ↓ renin, central effects	Bronchospasm (contraindicated in asthma), bradycardia, fatigue, cold extremities, mask hypoglycaemia, dyslipidaemia, worsened psoriasis, **rebound on abrupt withdrawal**
α_1-blockers	Prazosin, doxazosin	↓ PVR	First-dose syncope, postural hypotension. Useful in BPH
Central α_2 agonists	**Methyldopa** (pregnancy), clonidine, moxonidine	↓ sympathetic outflow	Methyldopa: sedation, positive Coombs test (haemolytic anaemia), hepatitis. Clonidine: rebound hypertension
Direct vasodilators	Hydralazine, minoxidil	Arteriolar dilation (minoxidil opens K_ATP channels)	Reflex tachycardia, fluid retention (give with β-blocker + diuretic). Hydralazine: **drug-induced lupus** (slow acetylators). Minoxidil: hirsutism
Parenteral agents (emergencies)	Sodium nitroprusside, labetalol IV, hydralazine IV, nicardipine, GTN, esmolol		Nitroprusside → cyanide/thiocyanate toxicity (treat with hydroxocobalamin and thiosulfate)
Adrenergic neurone blockers (historical)	Reserpine, guanethidine	Deplete NA	Depression (reserpine)

Choosing a drug (NICE 2019/2023, ISH 2020, Nigerian practice)

- **Black African or African-Caribbean patients, or age ≥55:** start with a **CCB** (or a thiazide-like diuretic). If a second drug is needed, add an **ARB in preference to an ACEI** (less angio-oedema) or a thiazide-like diuretic.
- **Age <55 and non-Black, or diabetes:** start with an ACEI or ARB.
- **Step 3:** A + C + D. **Step 4 (resistant hypertension):** add spironolactone if $K^+ \leq 4.5$ mmol/L, otherwise an α- or β-blocker.

Compelling indication	Preferred drug
Diabetes with proteinuria / CKD	ACEI or ARB
Heart failure	ACEI/ARB/ARNI + β-blocker (bisoprolol, carvedilol, metoprolol succinate) + MRA + SGLT2 inhibitor
Post-MI	β-blocker + ACEI
Angina	β-blocker or CCB
Pregnancy	**Methyldopa, labetalol, nifedipine (modified release)**, hydralazine
BPH	α-blocker
Isolated systolic hypertension in the elderly	CCB or thiazide-like diuretic

Hypertensive emergencies

Hypertensive emergency means BP ≥180/120 with acute target-organ damage (encephalopathy, LVF, aortic dissection, eclampsia). Lower BP by **no more than 25% in the first hour**, using IV labetalol, nitroprusside, GTN or nicardipine. Aortic dissection is the exception: target SBP <120 within 20 minutes using esmolol or labetalol.

In **pre-eclampsia with severe features**, give IV labetalol, IV hydralazine or oral nifedipine, plus **magnesium sulfate** (Pritchard or Zuspan regimen) to prevent eclamptic fits.

🎤 Viva corner

Q: Why does an ACE inhibitor cause cough but an ARB does not? **A:** ACE (kininase II) also breaks down bradykinin and substance P. Inhibiting it lets these build up in the airway and trigger cough. ARBs block the AT_1 receptor directly and leave bradykinin metabolism alone.

Q: Name four antihypertensives that are safe in pregnancy. **A:** Methyldopa, labetalol, nifedipine and hydralazine.

Chapter 10 · Anti-anginal Drugs

Angina pectoris is chest pain caused by myocardial ischaemia, an imbalance between myocardial O_2 supply and demand. **Stable angina** comes on with exertion (fixed atheroma). **Unstable angina** occurs at rest or is worsening (plaque rupture with thrombus; part of acute coronary syndrome). **Variant (Prinzmetal) angina** is caused by coronary vasospasm.

O_2 demand depends on **heart rate, contractility and wall tension** (preload and afterload). Anti-anginal drugs work by ↓ demand, ↑ supply, or both.

Class	Mechanism	Examples	Notes
Organic nitrates	Release **NO** → ↑ guanylyl cyclase → ↑ cGMP → MLC dephosphorylation → **venodilation** (↓ preload) more than arteriolar and coronary dilation	**GTN (sublingual for acute attacks)**, isosorbide dinitrate, **isosorbide mononitrate** (prophylaxis)	Headache, flushing, postural hypotension, reflex tachycardia. **Tolerance**: needs a nitrate-free interval of 8–12 h. **Contraindicated with PDE-5 inhibitors (sildenafil)** because of profound hypotension
β-blockers	↓ HR and contractility, so ↓ demand; longer diastole improves perfusion	Atenolol, metoprolol, bisoprolol	First line for stable angina; reduce mortality after MI. **Avoid in variant angina** (unopposed α spasm)
CCBs	Vasodilate (↓ afterload), ↓ HR (non-DHP), relieve spasm	Amlodipine, diltiazem, verapamil	**Drug of choice for Prinzmetal angina**
K^+-channel opener	Opens K_ATP + nitrate-like action	Nicorandil	Mouth, GI and anal ulceration
Late Na^+-current inhibitor	↓ Na^+ then Ca^{2+} overload; metabolic	Ranolazine	Prolongs QT
I_f ("funny" current) inhibitor	↓ SA node rate only	Ivabradine	Visual disturbances (phosphenes). Used when β-blockers cannot be taken
Metabolic modulator	Shifts myocardium from fatty acid to glucose oxidation	Trimetazidine	
Antiplatelets and statins (secondary prevention)	Prevent thrombosis; stabilise plaque	Aspirin 75 mg, clopidogrel, atorvastatin	Reduce MI and death

Management of stable angina: sublingual GTN for attacks, plus a β-blocker *or* CCB (both if needed), plus aspirin, a statin and an ACEI if diabetic or with LV dysfunction. Revascularisation (PCI or CABG) if symptoms persist.

ACS (MONA-BASH, modern version): O_2 only if SpO_2 <90%; aspirin 300 mg + a $P2Y_{12}$ inhibitor (ticagrelor or clopidogrel); nitrates; morphine if needed; anticoagulation (fondaparinux or heparin); β-blocker; statin; ACEI. STEMI: primary PCI or thrombolysis (streptokinase, alteplase, tenecteplase) within 12 h.

★ Exam pearls

Nitrates mainly reduce **preload** (they dilate veins). In the cell, NO is released by mitochondrial aldehyde dehydrogenase. Tolerance comes from depletion of sulfhydryl groups and oxidative stress, so include a nitrate-free period every day.

Chapter 11 · Drugs for Heart Failure

Heart failure (HF) is a clinical syndrome in which the heart cannot pump enough blood to meet the body's metabolic needs, or can do so only at raised filling pressures. **HFrEF:** EF ≤40%. **HFmrEF:** 41–49%. **HFpEF:** ≥50%.

Pathophysiology and drug targets

Low cardiac output triggers compensatory mechanisms that are helpful at first but harmful over time:

- Sympathetic activation (tachycardia, vasoconstriction, myocyte toxicity)
- RAAS activation (Na^+ and water retention, fibrosis)
- ADH release
- Ventricular remodelling

Modern treatment blocks these neurohormonal pathways. These are the drugs that improve survival.

The "four pillars" of HFrEF (ESC 2021/2023, AHA 2022)

Pillar	Drugs	Mortality benefit
1. **ACEI / ARB, or ARNI** (sacubitril/valsartan, preferred)	Enalapril, ramipril, sacubitril/valsartan	Yes (CONSENSUS, SOLVD, PARADIGM-HF)
2. **β-blockers** (start low, go slow)	Bisoprolol, carvedilol, metoprolol succinate, nebivolol	Yes (CIBIS-II, MERIT-HF, COPERNICUS)
3. **Mineralocorticoid receptor antagonists (MRA)**	Spironolactone, eplerenone	Yes (RALES, EMPHASIS)
4. **SGLT2 inhibitors**	Dapagliflozin, empagliflozin	Yes (DAPA-HF, EMPEROR-Reduced), even in non-diabetics

Symptom relief and other drugs

Drug	Mechanism	Notes
Loop diuretics	Relieve congestion	Improve symptoms; **no mortality benefit**
Digoxin	Inhibits **Na^+/K^+-ATPase** → ↑ intracellular Na^+ → less Ca^{2+} leaves via NCX → ↑ **Ca^{2+}** → **positive inotropy**. Vagal stimulation slows AV conduction	Reduces hospitalisation but **not mortality** (DIG trial). Useful in **HF with atrial fibrillation**. Narrow TI (0.5–0.9 ng/mL)

Drug	Mechanism	Notes
Hydralazine + isosorbide dinitrate	↓ afterload + preload	Improves survival **in Black patients** (A-HeFT) or when ACEI/ARB cannot be used
Ivabradine	↓ HR	If sinus rhythm with HR ≥70 despite a β-blocker
Inotropes (acute only)	Dobutamine (β_1), milrinone (PDE-3 inhibitor), levosimendan (Ca^{2+} sensitiser)	Cardiogenic shock; ↑ **mortality with long-term use**
Vericiguat	sGC stimulator	Worsening HF
IV iron		Corrects iron deficiency

Digoxin toxicity

- **Precipitants: hypokalaemia** (K^+ and digoxin compete for the Na^+/K^+-ATPase), hypomagnesaemia, hypercalcaemia, renal impairment, elderly patients, and interacting drugs (**amiodarone, verapamil, quinidine, macrolides**, which raise digoxin levels).
- **Features:** anorexia, nausea and vomiting (early), diarrhoea, **xanthopsia (yellow vision)**, confusion. Arrhythmias: ventricular ectopics, bigeminy, **paroxysmal atrial tachycardia with block**, AV block, VT. Hyperkalaemia in acute overdose.
- **ECG at therapeutic levels:** "reverse tick" (sagging ST), short QT, inverted T waves. This is *not* toxicity.
- **Treatment:** stop digoxin, correct K^+ and Mg^{2+}, atropine for bradycardia, lidocaine or phenytoin for ventricular arrhythmias, and **digoxin-specific antibody fragments (Fab, DigiFab)** for severe toxicity. Avoid DC cardioversion if possible.

Acute pulmonary oedema (LVF) management

Sit the patient up. Give O_2, **IV furosemide**, and **IV/SL nitrates** if SBP >90–100. Use CPAP if hypoxic, and inotropes if hypotensive. Morphine is used with caution (it increases mortality in registries).

> 🎤 **Viva corner**
>
> **Q:** Which drugs improve survival in heart failure, and which only improve symptoms? **A:** ACEI/ARB/ARNI, β-blockers, MRAs, SGLT2 inhibitors, and hydralazine + nitrate in Black patients improve survival. Diuretics and digoxin improve symptoms and hospitalisation only.

Chapter 12 · Antiarrhythmic Drugs

Cardiac action potential (ventricular myocyte)

Phase	Event
0	Rapid depolarisation: **Na^+ influx**
1	Early repolarisation: transient K^+ efflux (I_to)
2	Plateau: **Ca^{2+} influx (L-type)** balanced by K^+ efflux
3	Repolarisation: **K^+ efflux** (I_Kr, I_Ks)
4	Resting potential. In SA/AV nodal cells phase 4 shows *spontaneous depolarisation* (I_f "funny" current) and phase 0 is **Ca^{2+}-dependent**

Mechanisms of arrhythmia: abnormal impulse formation (enhanced automaticity, early or delayed **afterdepolarisations / triggered activity**) or abnormal conduction (**re-entry**, heart block).

Vaughan Williams classification (with modern additions)

Class	Action	Drugs	Key features / adverse effects
Ia	Na^+ block (intermediate) + K^+ block → ↑ APD, ↑ QT	**Quinidine, procainamide, disopyramide** ("**Double Quarter Pounder**")	Quinidine: **cinchonism**, torsades, diarrhoea, ↑ digoxin. Procainamide: **SLE-like** syndrome. Disopyramide: anticholinergic effects
Ib	Na^+ block (fast on/off), ↓ APD; acts on ischaemic and depolarised tissue	**Lidocaine, mexiletine, phenytoin**	Ventricular arrhythmias after MI. Lidocaine has high first-pass metabolism (IV only); CNS toxicity (seizures)
Ic	Na^+ block (slow), marked ↓ conduction; no APD change	**Flecainide, propafenone** ("**Fries Please**")	Paroxysmal AF in a *structurally normal* heart ("pill-in-the-pocket"). ↑ **mortality after MI (CAST trial)**: contraindicated in structural or ischaemic heart disease

Class	Action	Drugs	Key features / adverse effects
II	**β-blockers**: ↓ phase 4 slope, ↓ AV conduction	Propranolol, metoprolol, **esmolol** (ultra-short, IV), atenolol	Rate control in AF and SVT; ↓ sudden death after MI
III	**K^+ block → ↑ APD and refractory period, ↑ QT**	**Amiodarone, sotalol** (also β-blocker), dronedarone, dofetilide, ibutilide ("**AIDS**": Amiodarone, Ibutilide, Dofetilide, Sotalol)	Risk of torsades (less with amiodarone)
IV	**CCBs (non-DHP)**: ↓ AV nodal conduction	**Verapamil, diltiazem**	SVT, AF rate control. **Contraindicated in WPW with AF** and with IV β-blockers
Others		**Adenosine** (A_1 receptor → ↑ K^+ efflux → transient AV block; t½ <10 s)	**Drug of choice to terminate SVT**: 6 mg rapid IV push, then 12 mg, then 12 mg. Flushing, chest tightness, sense of doom, bronchospasm (avoid in asthma). Effect blocked by theophylline/caffeine; increased by dipyridamole
		Digoxin	AF rate control
		Magnesium sulfate	**Torsades de pointes** (2 g IV); digoxin-induced arrhythmias
		Atropine	Symptomatic bradycardia (0.5 mg IV, up to 3 mg)

Amiodarone adverse effects (check thyroid, liver and lung function)

The mnemonic is **"PFT, LFT, TFT"**:

- **Pulmonary fibrosis**
- **Hepatotoxicity**
- **Hypo- or hyperthyroidism** (it contains iodine and resembles T_4)
- Corneal microdeposits
- **Blue-grey skin** discolouration and photosensitivity
- Peripheral neuropathy and optic neuritis
- Bradycardia, QT prolongation
- Interactions: ↑ warfarin and digoxin

It has a very long t½ (~50 days) and a huge Vd.

Treatment summary

Arrhythmia	Treatment
SVT (AVNRT/AVRT)	Vagal manoeuvres (modified Valsalva), then **adenosine**, then verapamil/β-blocker. DC cardioversion if unstable
AF	Rate control (β-blocker, diltiazem/verapamil, digoxin) or rhythm control (amiodarone, flecainide, DC cardioversion). **Anticoagulate by CHA_2DS_2-VASc** (DOAC or warfarin)
VT with pulse, stable	Amiodarone 300 mg IV over 20–60 min. Unstable: synchronised DC shock
VF / pulseless VT	Defibrillation; adrenaline 1 mg every 3–5 min after the 3rd shock; amiodarone 300 mg after the 3rd shock
Torsades de pointes	**IV magnesium sulfate**; stop QT-prolonging drugs; overdrive pacing or isoprenaline
Sinus bradycardia / heart block	Atropine, isoprenaline/adrenaline infusion, pacing

⚠ Caution

QT-prolonging drugs (risk of torsades): class Ia and III antiarrhythmics, macrolides, fluoroquinolones, antipsychotics (haloperidol), TCAs, methadone, ondansetron, **quinine, halofantrine, chloroquine**, antifungal azoles, and hypokalaemia/hypomagnesaemia.

Chapter 13 · Drugs for Hyperlipidaemia

Lipoprotein metabolism is introduced in Book 4, Chapter 55. The main drug targets are LDL-C (the major atherogenic particle) and triglycerides.

Class	Mechanism	Effect on lipids	Adverse effects / notes
Statins (atorvastatin, rosuvastatin, simvastatin, pravastatin)	Inhibit **HMG-CoA reductase** (rate-limiting step of cholesterol synthesis) → ↑ hepatic **LDL-receptor** expression	**LDL ↓↓↓ 20–60%**; TG ↓; HDL ↑ slightly. Pleiotropic: plaque stabilisation, anti-inflammatory	**Myopathy / rhabdomyolysis** (↑ with fibrates (gemfibrozil), macrolides, azoles, ciclosporin, grapefruit), ↑ transaminases, new-onset diabetes (small). **Contraindicated in pregnancy**. Give simvastatin at night
Ezetimibe	Inhibits **NPC1L1** intestinal cholesterol absorption	LDL ↓ ~20%	Added to statins (IMPROVE-IT)
PCSK9 inhibitors (evolocumab, alirocumab, s.c. monoclonal antibodies; inclisiran, an siRNA)	Prevent LDL-receptor degradation	LDL ↓ ~60%	Familial hypercholesterolaemia; very high risk. Expensive
Bile acid sequestrants (cholestyramine, colestipol, colesevelam)	Bind bile acids → liver uses cholesterol to make more → ↑ LDL-R	LDL ↓; **TG may** ↑	GI upset, ↓ absorption of fat-soluble vitamins, warfarin, digoxin and thyroxine. Also used for pruritus in cholestasis
Fibrates (fenofibrate, gemfibrozil, bezafibrate)	**PPAR-α agonists** → ↑ lipoprotein lipase, ↓ VLDL	**TG ↓↓↓**, HDL ↑	Myopathy (especially gemfibrozil + statin), gallstones, ↑ warfarin effect
Niacin (nicotinic acid)	↓ lipolysis in adipose tissue → ↓ VLDL	TG ↓, LDL ↓, **HDL ↑↑ (most)**, ↓ Lp(a)	**Flushing (prostaglandin; give aspirin first)**, hyperuricaemia, hyperglycaemia, hepatotoxicity. Little outcome benefit
Omega-3 fatty acids (icosapent ethyl)	↓ VLDL synthesis	TG ↓	REDUCE-IT: CV benefit
Bempedoic acid	ATP-citrate lyase inhibitor	LDL ↓	For statin-intolerant patients

★ Exam pearls

- Statins are the **only class with strong mortality benefit** in primary and secondary prevention (4S, HPS, WOSCOPS, JUPITER).
- Myopathy: check CK if there are muscle symptoms; stop the statin if CK >10× the upper limit.
- Severe hypertriglyceridaemia (>10 mmol/L) carries a risk of **acute pancreatitis**: use a fibrate.

Chapter 14 · Drugs Affecting Blood Coagulation

Haemostasis review

1. **Vascular phase:** vasoconstriction.
2. **Primary haemostasis:** platelet adhesion (vWF binding to GPIb), activation (release of ADP and TXA_2), and aggregation (fibrinogen bridging **GPIIb/IIIa**).
3. **Secondary haemostasis:** the coagulation cascade. The modern **cell-based model** has three steps:
   - *Initiation:* tissue factor + VIIa → small amounts of Xa and thrombin.
   - *Amplification:* thrombin activates V, VIII, XI and platelets.
   - *Propagation:* the tenase (IXa–VIIIa) and prothrombinase (Xa–Va) complexes on platelet surfaces produce a thrombin burst.
   - Thrombin turns fibrinogen into fibrin, which is cross-linked by XIIIa.
4. **Natural anticoagulants:** antithrombin (inhibits IIa, Xa, IXa), protein C/S (inactivate Va and VIIIa), TFPI.
5. **Fibrinolysis:** plasminogen → (tPA, uPA) → plasmin → fibrin breakdown → **D-dimers**.

Vitamin K-dependent factors: II, VII, IX, X ("1972") and **proteins C and S**. They need γ-carboxylation of glutamate residues.

Anticoagulants

Drug	Mechanism	Monitoring	Reversal	Notes
Unfractionated heparin (UFH)	Binds **antithrombin**, increasing its activity ~1000×; inhibits **IIa and Xa equally**	**aPTT** (1.5–2.5× control)	**Protamine sulfate**	IV/SC; immediate action; safe in pregnancy (does not cross the placenta). Adverse effects: bleeding, **heparin-induced thrombocytopenia (HIT type II**, IgG against PF4–heparin, causes *thrombosis*), osteoporosis, hyperkalaemia (↓ aldosterone)
Low-molecular-weight heparin (LMWH) (enoxaparin, dalteparin)	Antithrombin-mediated; **anti-Xa > anti-IIa**	Usually none; anti-Xa levels in renal failure, pregnancy and obesity	Protamine (partial)	SC once or twice daily; predictable; less HIT. Avoid if CrCl <30

Drug	Mechanism	Monitoring	Reversal	Notes
Fondaparinux	Synthetic pentasaccharide; selective indirect Xa inhibitor		None specific	No HIT
Warfarin	**Vitamin K epoxide reductase (VKORC1) inhibitor** → no γ-carboxylation of II, VII, IX, X, C, S	**PT/INR** (target 2–3; 2.5–3.5 for mechanical mitral valves)	**Vitamin K** (slow, 6–24 h); **prothrombin complex concentrate** or FFP (immediate)	Oral; delayed onset (3–5 days; needs heparin overlap). Initially *prothrombotic* because protein C falls first (t½ 8 h) → **warfarin skin necrosis**. **Teratogenic** (nasal hypoplasia, stippled epiphyses: fetal warfarin syndrome). Many interactions (CYP2C9)
Direct thrombin inhibitors	Dabigatran (oral), argatroban, bivalirudin (IV)	Not routine	**Idarucizumab** (dabigatran)	Argatroban used in HIT
Direct Xa inhibitors	Rivaroxaban, apixaban, edoxaban	Not routine	**Andexanet alfa**, PCC	Fixed dose; fewer interactions; avoid with mechanical valves and in antiphospholipid syndrome

Antiplatelet drugs

Drug	Mechanism	Notes
Aspirin	**Irreversible COX-1 acetylation** → ↓ TXA_2 for the platelet's whole life (7–10 days)	75–150 mg daily. GI bleeding
$P2Y_{12}$ (ADP receptor) blockers	Clopidogrel, prasugrel (irreversible prodrugs), ticagrelor (reversible), ticlopidine	Clopidogrel needs CYP2C19 activation (interaction with omeprazole). Ticlopidine: neutropenia, TTP. Ticagrelor: dyspnoea
GPIIb/IIIa antagonists	Abciximab, eptifibatide, tirofiban (IV)	During PCI. Thrombocytopenia

Drug	Mechanism	Notes
PDE inhibitors	Dipyridamole (with aspirin for stroke), cilostazol (claudication)	Headache
PAR-1 antagonist	Vorapaxar	

Thrombolytics (fibrinolytics)

Streptokinase (antigenic, causes allergy; *cannot be repeated* within 1 year), **alteplase (rtPA)**, **tenecteplase**, reteplase. They turn plasminogen into plasmin. Used in STEMI (<12 h), ischaemic stroke (<4.5 h, alteplase or tenecteplase) and massive PE.

Contraindications: previous haemorrhagic stroke, active bleeding, recent major surgery or trauma, aortic dissection, severe uncontrolled hypertension.

Antifibrinolytics and procoagulants

Tranexamic acid (blocks lysine binding sites on plasminogen) is used for menorrhagia, trauma (CRASH-2: within 3 h) and **postpartum haemorrhage (WOMAN trial)**. Others: vitamin K_1 (phytomenadione; give to newborns to prevent haemorrhagic disease), desmopressin (↑ vWF and VIII: haemophilia A, von Willebrand disease), factor concentrates, and emicizumab.

> 🎤 **Viva corner**
>
> **Q:** Why do we overlap heparin with warfarin for at least 5 days? **A:** Warfarin's anticoagulant effect is delayed until the existing clotting factors are cleared (prothrombin t½ ≈ 60 h). Its early fall in protein C (t½ ≈ 8 h) can also make the blood *hypercoagulable*. Heparin covers this period until the INR has been in range for 24 h.

Chapter 15 · Anti-ulcer Drugs (Peptic Ulcer and GORD)

Physiology of acid secretion

The **parietal cell** secretes H^+ through the **H^+/K^+-ATPase (proton pump)**, the final common pathway. Three stimuli drive it:

- **Histamine** from ECL cells, acting on H_2 receptors (Gs → cAMP)
- **Gastrin** from G cells, acting on CCK_2 receptors (Gq)
- **ACh** from the vagus, acting on M_3 receptors (Gq)

Prostaglandins (PGE_2, PGI_2) reduce acid and increase mucus and HCO_3^- secretion. This is why NSAIDs cause ulcers. **Somatostatin** also inhibits secretion.

Peptic ulcer disease is an imbalance between *aggressive factors* (acid, pepsin, ***H. pylori***, NSAIDs, smoking) and *defensive factors* (mucus, HCO_3^-, PGs, blood flow). *H. pylori* is a Gram-negative, urease-positive, microaerophilic curved rod. It causes ~90% of duodenal and ~70% of gastric ulcers, and is linked to gastric adenocarcinoma and MALT lymphoma.

Drug classes

Class	Examples	Mechanism	Notes / adverse effects
Proton pump inhibitors (PPIs)	Omeprazole, esomeprazole, lansoprazole, pantoprazole, rabeprazole	**Irreversibly inhibit H^+/K^+-ATPase**. They are *prodrugs* activated in the acid canaliculus, so give them 30–60 min before breakfast	Most effective acid suppressors. Long-term use: **hypomagnesaemia**, B_{12} deficiency, osteoporotic fractures, *C. difficile* infection, pneumonia, rebound acid hypersecretion. Omeprazole inhibits CYP2C19 (↓ clopidogrel activation)
H_2-receptor antagonists	Cimetidine, ranitidine (withdrawn 2020 for NDMA contamination), famotidine, nizatidine	Competitive H_2 blockade; especially ↓ nocturnal acid	**Cimetidine**: CYP inhibitor, antiandrogenic (gynaecomastia, impotence), confusion in the elderly

Class	Examples	Mechanism	Notes / adverse effects
Antacids	$Mg(OH)_2$ (causes diarrhoea), $Al(OH)_3$ (causes constipation), $CaCO_3$, $NaHCO_3$	Neutralise acid	Symptom relief. Chelate tetracyclines and quinolones. Al: hypophosphataemia. $NaHCO_3$: alkalosis, Na^+ load. $CaCO_3$: milk-alkali syndrome
Mucosal protectants	**Sucralfate**	Polymerises in acid and coats the ulcer base	Constipation, Al accumulation in renal failure
	Bismuth chelate (bismuth subcitrate/subsalicylate)	Coats the ulcer; **anti-*H. pylori***	Black tongue and stools
PG analogue	**Misoprostol** (PGE_1)	↓ acid, ↑ mucus	Prevents NSAID ulcers. Diarrhoea. **Abortifacient: contraindicated in pregnancy** (also used for PPH and induction of labour)
Antimuscarinic	Pirenzepine (M_1)	↓ acid	Rarely used
Alginates	Gaviscon	Form a raft on top of stomach contents	GORD
Potassium-competitive acid blocker	Vonoprazan	Reversible H^+/K^+-ATPase block	Newer

H. pylori eradication regimens (Maastricht VI / ACG 2024)

- **Standard triple therapy for 14 days:** PPI twice daily + **amoxicillin 1 g twice daily** + **clarithromycin 500 mg twice daily**. If penicillin-allergic, use metronidazole instead of amoxicillin.
- **Bismuth quadruple therapy for 14 days** (preferred where clarithromycin resistance is >15%, which is common in Nigeria): PPI + bismuth + tetracycline + metronidazole.
- **Confirm eradication** with a urea breath test or stool antigen test ≥4 weeks after antibiotics and ≥2 weeks after stopping the PPI.

★ Exam pearls

- PPIs are prodrugs. They need acid to activate, so give them before meals and don't combine them routinely with H_2RAs.
- **Zollinger–Ellison syndrome** (gastrinoma) needs high-dose PPI.
- **Stress ulcer prophylaxis** in ICU uses a PPI or H_2RA.

Chapter 16 · Antiemetics

Physiology of vomiting

The **vomiting centre** (nucleus tractus solitarius / central pattern generator in the medulla) receives input from four sources:

1. **Chemoreceptor trigger zone (CTZ)** in the area postrema, *outside* the blood–brain barrier. It has D_2, 5-HT_3, NK_1 and opioid receptors and detects drugs, toxins and uraemia.
2. **Vestibular system** (H_1, M_1 receptors). This drives motion sickness.
3. **GI tract** via vagal afferents (5-HT_3 receptors, activated by serotonin released from enterochromaffin cells by cytotoxics and radiation).
4. **Higher centres** (cortex): anticipation, smells, raised ICP.

Class	Drugs	Best for	Adverse effects
5-HT_3 antagonists	**Ondansetron**, granisetron, palonosetron	**Chemotherapy-induced** nausea and vomiting, postoperative, radiotherapy	Constipation, headache, **QT prolongation**
D_2 antagonists	**Metoclopramide** (also prokinetic: 5-HT_4 agonist), **domperidone** (does not cross the BBB), prochlorperazine, haloperidol, droperidol	Gastroenteritis, migraine, postoperative, drug-induced, palliative	**Extrapyramidal effects / acute dystonia** (young women; oculogyric crisis; treat with procyclidine or benzatropine), hyperprolactinaemia, tardive dyskinesia. Domperidone: QT prolongation. Metoclopramide is **contraindicated in GI obstruction**
H_1 antihistamines	Cyclizine, promethazine, cinnarizine, meclizine, doxylamine (+ pyridoxine)	**Motion sickness**, vestibular causes, **pregnancy** (promethazine, cyclizine)	Sedation, antimuscarinic effects
Antimuscarinics	**Hyoscine hydrobromide** (scopolamine patch)	**Motion sickness (most effective; take before travel)**	Dry mouth, drowsiness, blurred vision
NK_1 antagonists	**Aprepitant**, fosaprepitant	**Delayed** chemotherapy-induced nausea and vomiting (cisplatin)	CYP3A4 interactions

Class	Drugs	Best for	Adverse effects
Corticosteroids	Dexamethasone	Chemotherapy-induced (mechanism unclear), raised ICP	
Cannabinoids	Nabilone, dronabinol	Refractory chemotherapy-induced vomiting	Psychotropic effects
Benzodiazepines	Lorazepam	**Anticipatory** vomiting	Sedation
Others	Olanzapine (CINV), ginger, pyridoxine		

★ Exam pearls

- **Pregnancy (hyperemesis):** first line is antihistamines (cyclizine, promethazine) or pyridoxine–doxylamine; then metoclopramide or ondansetron.
- **Highly emetogenic chemotherapy** (cisplatin): 5-HT_3 antagonist + NK_1 antagonist + dexamethasone ± olanzapine.
- **Parkinson's disease:** use **domperidone** (does not enter the CNS). *Avoid metoclopramide and prochlorperazine.*

Chapter 17 · Drugs Affecting Intestinal Motility

Category	Drugs	Mechanism	Uses
Prokinetics	Metoclopramide	D_2 antagonist + 5-HT_4 agonist → ↑ ACh release; ↑ lower oesophageal sphincter tone and gastric emptying	Gastroparesis, GORD, before emergency anaesthesia, migraine (speeds analgesic absorption)
	Domperidone	Peripheral D_2 antagonist	Gastroparesis, nausea
	Erythromycin	**Motilin receptor agonist**	Diabetic gastroparesis; before endoscopy for upper GI bleeding
	Prucalopride	Selective 5-HT_4 agonist	Chronic constipation
	Neostigmine	Anticholinesterase	Acute colonic pseudo-obstruction (Ogilvie syndrome)
	(Cisapride)	5-HT_4	**Withdrawn**: QT/torsades
Antispasmodics (↓ motility)	**Hyoscine butylbromide (Buscopan)**, dicycloverine, propantheline	Antimuscarinic	Abdominal colic, IBS, during radiological procedures
	Mebeverine, alverine, peppermint oil	Direct smooth muscle relaxants	**IBS**
Antimotility	Loperamide, diphenoxylate (+ atropine: Lomotil), codeine	μ-opioid agonists in the gut	Diarrhoea (Chapter 18)
Drugs for IBS	Linaclotide (guanylate cyclase-C agonist), lubiprostone (ClC-2 activator), alosetron (5-HT_3; IBS-D), low-dose TCAs		

Chapter 18 · Antidiarrhoeal Drugs and Laxatives

A. Diarrhoea

Diarrhoea is three or more loose or watery stools in 24 h (WHO). It is **acute** if <14 days and **persistent** if ≥14 days. It is a leading cause of death in under-fives in Nigeria. **Dehydration, not the infection itself, is the killer.**

Principles of management:

1. **Rehydration (the most important step).**
   - **Low-osmolarity ORS** (WHO 2002; 245 mOsm/L: Na^+ 75, glucose 75 mmol/L). It works by **Na^+–glucose cotransport (SGLT1)**, which stays intact even in cholera.
   - Plan A: no dehydration, treat at home. Plan B: some dehydration, give ORS 75 mL/kg over 4 h. Plan C: severe dehydration, give IV **Ringer's lactate** 100 mL/kg.
2. **Zinc** (WHO): 20 mg/day for 10–14 days (10 mg if <6 months). It shortens the episode and reduces recurrence.
3. **Continued feeding**, including breastfeeding.
4. **Antimicrobials only when indicated:**

Condition	Antimicrobial
Dysentery (*Shigella*)	Ciprofloxacin or azithromycin
Cholera	Doxycycline single dose or azithromycin
Amoebiasis, giardiasis	Metronidazole/tinidazole
C. *difficile*	Oral vancomycin or fidaxomicin
Typhoid	Ceftriaxone or azithromycin

5. **Antimotility drugs** (**loperamide**: μ agonist that does not cross the BBB because of P-gp efflux) are for adults with non-bloody traveller's diarrhoea only.

> **⚠ Caution**
>
> **Avoid antimotility agents** in children, in bloody diarrhoea or dysentery, and in C. *difficile* infection. They risk toxic megacolon and prolonged infection.

Other antidiarrhoeals:

- Adsorbents: kaolin, activated charcoal (little evidence)
- **Racecadotril** (enkephalinase inhibitor; antisecretory)
- Bismuth subsalicylate

- Octreotide (carcinoid, VIPoma)
- Probiotics

B. Laxatives (purgatives)

Class	Examples	Mechanism	Onset	Notes
Bulk-forming	Ispaghula husk (psyllium), methylcellulose, sterculia, bran	Retain water and increase stool mass, stimulating peristalsis	2–4 days	Take with plenty of fluid; risk of obstruction. Good for IBS and haemorrhoids
Osmotic	**Lactulose**, macrogol (PEG), Mg salts (Epsom salt), sodium phosphate	Hold water in the lumen. Lactulose → colonic bacteria → acids → ↓ pH → traps NH_3 as NH_4^+	1–3 days (salts: 1–3 h)	**Lactulose: hepatic encephalopathy.** PEG: bowel preparation, faecal impaction
Stimulant	**Bisacodyl**, **senna** (anthraquinones), sodium picosulfate, castor oil, glycerol suppository	Stimulate the myenteric plexus and secretion	6–12 h (suppository 15–60 min)	Cramps; chronic abuse → cathartic colon, **melanosis coli**, hypokalaemia
Stool softeners	Docusate sodium, liquid paraffin, arachis oil enema	Surfactant / lubricant	1–3 days	Liquid paraffin: lipoid pneumonia, ↓ fat-soluble vitamins (avoid)
Secretagogues	Lubiprostone, linaclotide	Increase intestinal fluid		Chronic idiopathic constipation, IBS-C
Peripherally acting μ-opioid antagonists (PAMORAs)	Methylnaltrexone, naloxegol	Block gut μ receptors		**Opioid-induced constipation**
5-HT_4 agonist	Prucalopride			Chronic constipation

★ Exam pearls

- **Contraindications to laxatives:** intestinal obstruction, undiagnosed abdominal pain, and inflammatory bowel disease flares (stimulants).
- **Opioid constipation:** give a stimulant + softener (senna + docusate), or a PAMORA. Do not use bulk laxatives.

Chapter 19 · Preparations for Haemorrhoids

Haemorrhoids (piles) are abnormally enlarged or displaced anal vascular cushions. **Internal** haemorrhoids lie above the dentate line, are covered by mucosa and are *painless*. They are graded I–IV by prolapse: I none; II prolapse and reduce spontaneously; III need manual reduction; IV irreducible. **External** haemorrhoids lie below the dentate line, are covered by somatically innervated anoderm, and are *painful*, especially when thrombosed.

Pharmacological management (grades I–II, and as an adjunct)

1. **Dietary and bowel measures (first line).** High-fibre diet, fluids, **bulk laxatives**, stool softeners, avoid straining. Sitz baths.
2. **Topical preparations.** These relieve symptoms only and don't cure; use them for short courses (≤7 days for steroid-containing products, to avoid skin atrophy).
   - **Local anaesthetics:** lidocaine, benzocaine, cinchocaine. Relieve pain and itch; may sensitise the skin.
   - **Corticosteroids:** hydrocortisone, prednisolone. ↓ inflammation and oedema.
   - **Astringents:** zinc oxide, bismuth subgallate, hamamelis (witch hazel), aluminium acetate. Precipitate surface proteins and protect the skin.
   - **Soothing protectants / emollients:** zinc oxide, Peru balsam.
   - **Vasoconstrictors:** phenylephrine.
   - **Combination products,** e.g. Anusol-HC®, Proctosedyl® (cinchocaine + hydrocortisone), Scheriproct®.
   - **GTN 0.4% ointment or diltiazem 2% cream** relax the internal sphincter. Used for anal fissure and thrombosed haemorrhoids.
3. **Oral phlebotonics (venoactive drugs): micronised purified flavonoid fraction (diosmin + hesperidin, Daflon®)**, calcium dobesilate, rutosides. They improve venous tone and capillary resistance and reduce bleeding in acute episodes. They are widely used in Africa, Europe and Asia.
4. **Analgesics:** paracetamol. Avoid codeine because it constipates.

Non-pharmacological / procedural

- **Sclerotherapy:** injection of **5% phenol in almond (arachis) oil** or **sodium tetradecyl sulfate / polidocanol** above the dentate line. Used for grades I–II.
- Rubber band ligation (grades II–III; most effective outpatient treatment), infrared coagulation, Doppler-guided haemorrhoidal artery ligation, haemorrhoidectomy (grades III–IV), stapled haemorrhoidopexy.

★ Exam pearls

The dentate line is the key anatomical landmark. Injections and bands go **above** it, where there are no somatic pain fibres. Topical anaesthetics act **below** it.

Chapter 20 · NSAIDs and Non-opioid Analgesics

Mechanism

Arachidonic acid (released from membrane phospholipids by **phospholipase A_2**, which glucocorticoids inhibit indirectly via annexin-1/lipocortin) is converted by two kinds of enzyme:

- **Cyclooxygenase (COX)** → prostaglandins (PGE_2, PGI_2), thromboxane A_2.
  - **COX-1** is constitutive ("housekeeping"): gastric protection, platelet TXA_2, renal blood flow.
  - **COX-2** is inducible at sites of inflammation; it is constitutive in the kidney, brain and endothelium (PGI_2).
- **Lipoxygenase** → leukotrienes.

NSAIDs inhibit COX, which gives them three actions:

1. **Analgesic:** ↓ PG sensitisation of nociceptors.
2. **Antipyretic:** ↓ PGE_2 in the hypothalamus, which lowers the raised set point.
3. **Anti-inflammatory:** ↓ vasodilation and oedema.

Classification

Group	Examples
Salicylates	**Aspirin** (irreversible COX acetylation), diflunisal
Propionic acids	**Ibuprofen**, naproxen, ketoprofen, flurbiprofen
Acetic acids	**Diclofenac**, **indomethacin**, ketorolac, sulindac, etodolac
Enolic acids (oxicams)	Piroxicam, meloxicam (relatively COX-2 preferential), tenoxicam
Fenamates	**Mefenamic acid** (dysmenorrhoea)
Pyrazolones	Metamizole (dipyrone; agranulocytosis), phenylbutazone
Selective COX-2 inhibitors (coxibs)	Celecoxib, etoricoxib, parecoxib (IV). Rofecoxib was **withdrawn in 2004 (↑ MI and stroke, VIGOR/APPROVe)**
Para-aminophenol (non-NSAID analgesic)	**Paracetamol (acetaminophen)**: central COX inhibition, weak peripheral anti-inflammatory action

Adverse effects of NSAIDs

System	Effect	Mechanism
GIT	Dyspepsia, **peptic ulcer, GI bleeding**	↓ protective PGs (COX-1). Use a PPI or misoprostol, or a coxib, in high-risk patients
Kidney	**AKI** (especially with ACEI + diuretic: the "**triple whammy**"), Na^+/water retention, hyperkalaemia, interstitial nephritis, papillary necrosis (analgesic nephropathy)	↓ renal PGs → afferent arteriolar constriction
CVS	↑ BP, worsened heart failure, ↑ **MI and stroke risk** (coxibs, diclofenac; least with naproxen)	↓ PGI_2 with unopposed TXA_2
Respiratory	**Aspirin-exacerbated respiratory disease** (Samter's triad: asthma, nasal polyps, aspirin sensitivity)	Shunting to leukotrienes
Blood	Bleeding (aspirin)	↓ TXA_2
Pregnancy	**Premature closure of the ductus arteriosus** (3rd trimester), oligohydramnios, delayed labour	↓ PGE_2
Others	Hypersensitivity, hepatotoxicity (diclofenac), Reye syndrome (aspirin)	

Aspirin: special points

- Dose-dependent actions: 75–150 mg/day is antiplatelet; 300–600 mg is analgesic/antipyretic; 4–6 g/day is anti-inflammatory (now obsolete).
- **Reye syndrome:** aspirin in children <16 years with a viral infection (influenza, varicella) → hepatic failure + encephalopathy (mitochondrial damage). **Avoid aspirin in children** except in Kawasaki disease.
- **Salicylate poisoning:**
  - Features: tinnitus, hyperventilation (**respiratory alkalosis first**, then **high anion gap metabolic acidosis**), hyperthermia, hypoglycaemia, non-cardiogenic pulmonary oedema.
  - Treatment: activated charcoal, **urinary alkalinisation with $NaHCO_3$**, and **haemodialysis** if level >700 mg/L, with severe acidosis, CNS features or renal failure.

Paracetamol and paracetamol poisoning

- Paracetamol is analgesic and antipyretic with **no significant anti-inflammatory action**. It does not irritate the GI tract and is safe in pregnancy and children. Usual dose is 1 g every 4–6 h, max 4 g/day.
- **Overdose:**
  1. Normally paracetamol is glucuronidated and sulfated.

2. In overdose these pathways saturate, so more goes through CYP2E1 → **NAPQI** (toxic metabolite).
3. NAPQI is normally detoxified by **glutathione**. Once glutathione is depleted, NAPQI binds hepatocyte proteins → **centrilobular (zone 3) hepatic necrosis**.

- **Antidote: N-acetylcysteine (NAC)**, which replenishes glutathione. Give it according to the **Rumack–Matthew nomogram** (4-hour level), or empirically if the patient presents >8 h after ingestion or the timing is unknown. NAC is most effective within 8 h.
- **Risk factors:** enzyme inducers, chronic alcohol use, malnutrition and anorexia (low glutathione).

> 🎤 **Viva corner**
>
> **Q:** Why does low-dose aspirin inhibit platelets but not endothelial PGI_2 for long? **A:** Platelets have no nucleus, so they cannot make new COX-1. The irreversible inhibition lasts the platelet's lifespan (7–10 days). Endothelial cells resynthesise COX within hours. Low dose also acts mainly on platelets in the portal circulation before reaching the systemic endothelium.

Chapter 21 · Drugs for Gout

Gout is an inflammatory arthritis caused by deposition of **monosodium urate (MSU) crystals** when serum urate is persistently above its saturation point (~6.8 mg/dL or 405 µmol/L). Crystals are **needle-shaped and strongly negatively birefringent** under polarised light. **Pseudogout** (CPPD) crystals are rhomboid and *positively* birefringent.

Pathogenesis: purines → hypoxanthine → (xanthine oxidase) → xanthine → (xanthine oxidase) → **uric acid**.

- **Overproduction (~10%):** Lesch–Nyhan syndrome (HGPRT deficiency), tumour lysis, myeloproliferative disorders, psoriasis.
- **Underexcretion (~90%):** CKD, **thiazide and loop diuretics**, low-dose aspirin, **pyrazinamide, ethambutol**, ciclosporin, alcohol (lactate competes for excretion), lead.

MSU crystals → phagocytosis → **NLRP3 inflammasome** → IL-1β → neutrophil influx → acute arthritis. This classically affects the **first MTP joint (podagra)**.

Drugs for the acute attack (anti-inflammatory; do not lower urate)

Drug	Mechanism	Notes
NSAIDs (naproxen, indomethacin, diclofenac, etoricoxib)	COX inhibition	First line if there are no contraindications. **Avoid aspirin** (low doses retain urate)
Colchicine	Binds **tubulin**, preventing microtubule polymerisation → ↓ neutrophil migration, phagocytosis and inflammasome activation	Low dose (EULAR/ACR): 1 mg, then 0.5 mg 1 h later. **Diarrhoea**, vomiting, myopathy, bone marrow suppression. Toxicity rises with CYP3A4/P-gp inhibitors (clarithromycin) and renal failure
Corticosteroids	Oral prednisolone, intra-articular	When NSAIDs and colchicine are contraindicated (renal failure)
IL-1 blockers	Anakinra, canakinumab	Refractory cases

Urate-lowering therapy (ULT; chronic, prophylactic)

Class	Drugs	Mechanism	Notes
Xanthine oxidase inhibitors	**Allopurinol** (prodrug → oxipurinol), febuxostat	↓ uric acid production	**First line.** Start low (100 mg, less in CKD) and titrate to **target urate <6 mg/dL (<360 µmol/L)**; <5 mg/dL if tophi. **Do not start during an acute attack** in classical teaching (mobilising urate can flare it), though recent guidelines allow it under NSAID or colchicine cover. Adverse effects: rash, **SJS/TEN, DRESS (HLA-B*58:01)**. **Interaction: azathioprine / 6-MP** (reduce their dose to 25%). Febuxostat: CV mortality warning
Uricosurics	**Probenecid**, sulfinpyrazone, benzbromarone, lesinurad	Inhibit URAT1 → ↓ tubular reabsorption of urate	For underexcreters with normal renal function. Increase fluid intake (stones). Ineffective in CKD. Probenecid ↑ penicillin levels
Uricase (urate oxidase)	**Rasburicase** (recombinant), pegloticase	Converts urate to soluble allantoin	**Tumour lysis syndrome prevention**. Contraindicated in **G6PD deficiency** (H_2O_2 → haemolysis, methaemoglobinaemia)

Flare prophylaxis when starting ULT: low-dose colchicine (0.5 mg once or twice daily) or an NSAID for 3–6 months.

★ Exam pearls

Treat-to-target ULT is indicated for: ≥2 attacks per year, tophi, gouty arthropathy on X-ray, urate stones, or CKD stage ≥3.

Chapter 22 · Opioid and Non-opioid Analgesics

Pain physiology

Nociceptors carry pain along **Aδ fibres** (fast, sharp) and **C fibres** (slow, burning) to the dorsal horn (substance P, glutamate). From there the spinothalamic tract takes it to the thalamus and cortex. The **gate control theory** and **descending inhibitory pathways** (periaqueductal grey → rostral ventromedial medulla → dorsal horn, using enkephalins, 5-HT and NA) modulate it.

WHO analgesic ladder (1986; now also applied "bidirectionally"):

1. Non-opioid (paracetamol/NSAID) ± adjuvant
2. Weak opioid (codeine, tramadol) ± non-opioid ± adjuvant
3. Strong opioid (morphine) ± non-opioid ± adjuvant

Adjuvants: TCAs (amitriptyline), anticonvulsants (gabapentin, pregabalin, carbamazepine for trigeminal neuralgia), steroids, bisphosphonates and radiotherapy for bone pain.

Opioid receptors (all Gi/o GPCRs)

They open K^+ channels (hyperpolarisation), close presynaptic Ca^{2+} channels (↓ transmitter release) and reduce cAMP.

Receptor	Effects
μ (MOP)	Supraspinal and spinal analgesia, **respiratory depression**, euphoria, miosis, constipation, dependence, bradycardia
κ (KOP)	Spinal analgesia, dysphoria, diuresis, sedation, miosis
δ (DOP)	Analgesia, seizures (animals)
NOP (nociceptin)	Anti-opioid effects

Classification

Category	Drugs	Notes
Strong agonists	**Morphine**, diamorphine (heroin), **pethidine (meperidine)**, **fentanyl** (patch, IV; 100× morphine), remifentanil (ester hydrolysis; ultra-short), alfentanil, sufentanil, methadone (long t½; NMDA antagonist; opioid maintenance; QT), oxycodone, hydromorphone	Morphine: active metabolite **M6G** (more potent; accumulates in renal failure). Pethidine: **norpethidine → seizures**; serotonin syndrome with MAOIs; less miosis

Category	Drugs	Notes
Weak / moderate agonists	**Codeine** (CYP2D6 → morphine), dihydrocodeine, **tramadol** (weak μ + SNRI → seizures, serotonin syndrome)	Codeine is antitussive and constipating. Contraindicated <12 yr and after tonsillectomy (ultra-rapid metabolisers)
Partial agonists / mixed agonist–antagonists	**Buprenorphine** (partial μ, κ antagonist; high affinity; patch; opioid substitution), pentazocine (κ agonist, weak μ antagonist; dysphoria; can precipitate withdrawal), butorphanol, nalbuphine	**Ceiling effect** on respiratory depression; naloxone reverses buprenorphine poorly
Antagonists	**Naloxone** (IV/IM/IN; short t½ 30–80 min, so **repeat dose or infusion needed**), naltrexone (oral; alcohol and opioid dependence), methylnaltrexone, naloxegol (peripheral)	Can precipitate acute withdrawal
Antitussives / antidiarrhoeals	Pholcodine, dextromethorphan (NMDA); loperamide, diphenoxylate	

Actions and adverse effects of morphine

CNS	Peripheral
Analgesia (especially dull, continuous pain), euphoria, sedation	**Constipation** (no tolerance develops)
Respiratory depression (↓ brainstem sensitivity to CO_2; the commonest cause of death)	↑ biliary pressure (sphincter of Oddi spasm)
Cough suppression	Urinary retention
Miosis (pin-point pupils; Edinger–Westphal nucleus; no tolerance)	**Histamine release** → itch, urticaria, bronchospasm, hypotension
Nausea and vomiting (CTZ)	Bradycardia
Truncal rigidity (fentanyl)	↓ GI secretions
Hormonal: ↑ ADH, ↑ prolactin, ↓ LH	

Opioid overdose triad: coma + **pin-point pupils + respiratory depression**. Treatment: **naloxone 0.4–2 mg IV**, repeated as needed, plus ventilatory support.

Withdrawal syndrome: yawning, lacrimation, rhinorrhoea, **mydriasis**, piloerection ("cold turkey"), diarrhoea, cramps, restlessness. It is unpleasant but rarely fatal in adults. Manage with methadone or buprenorphine tapering, or **clonidine/lofexidine** for symptoms.

Contraindications / cautions: head injury (↑ ICP from CO_2 retention; masks pupils), respiratory disease (asthma, COPD), hypothyroidism, Addison's disease, liver failure, biliary colic (pethidine was historically preferred, a debatable point), the elderly, MAOIs (pethidine), and pregnancy near term (neonatal respiratory depression).

🎤 Viva corner

Q: Why is pethidine avoided for chronic pain and in renal failure? **A:** Its metabolite norpethidine accumulates and causes CNS excitation and seizures. Pethidine is also short-acting and interacts dangerously with MAOIs (serotonin syndrome).

Chapter 23 · Anaesthetic Agents

A. General anaesthesia

General anaesthesia is a reversible, drug-induced loss of consciousness. Its components are the **triad of anaesthesia**: hypnosis (unconsciousness), analgesia and muscle relaxation. Amnesia and reflex suppression are often added.

Guedel's stages (described with ether): I analgesia → II excitement/delirium (dangerous) → III surgical anaesthesia (4 planes) → IV medullary paralysis (respiratory arrest). Modern IV induction passes through stage II rapidly.

Mechanism: most agents **potentiate GABA-A** receptors (propofol, etomidate, barbiturates, volatiles). Some **antagonise NMDA** receptors (ketamine, N_2O, xenon). Others act on two-pore K^+ channels (volatiles).

Inhalational agents

- **Minimum alveolar concentration (MAC):** the alveolar concentration that prevents movement in 50% of patients in response to a surgical incision. It measures **potency**: a low MAC means a potent agent.
- **Blood:gas partition coefficient** determines **speed of induction and recovery**. A *low* coefficient (low blood solubility) gives *fast* onset and offset (desflurane, N_2O).
- The **oil:gas** coefficient correlates with potency (Meyer–Overton rule).

Agent	Blood:gas	MAC (%)	Features
Nitrous oxide	0.47	104	Good analgesic, weak anaesthetic. **Second gas effect, diffusion hypoxia** (give 100% O_2 at the end). Expands air-filled spaces (pneumothorax, bowel). Prolonged use **inactivates vitamin B_{12}** (megaloblastic anaemia, subacute combined degeneration). Entonox = 50:50 with O_2

Agent	Blood:gas	MAC (%)	Features
Halothane	2.4	0.75	Smooth, non-irritant, cheap (still used in some Nigerian centres). **Halothane hepatitis** (repeated exposure; immune), **sensitises the myocardium to catecholamines** (arrhythmias), bronchodilator, uterine relaxation
Isoflurane	1.4	1.15	Pungent; "coronary steal" (theoretical); widely used
Sevoflurane	0.65	2.0	**Non-pungent: inhalational induction in children**; fast. Compound A with soda lime; fluoride
Desflurane	0.42	6.0	Fastest recovery; very pungent (airway irritation); needs a heated vaporiser
Ether (historical)	12	1.9	Flammable; slow induction; used in rural settings in the past

All volatile agents can trigger **malignant hyperthermia** (as can suxamethonium). They cause dose-dependent hypotension and respiratory depression, and relax the uterus.

Intravenous agents

Agent	Mechanism	Features
Propofol	GABA-A	**Most common induction agent.** Rapid, clear-headed recovery; **antiemetic**; used for TIVA. Pain on injection, **hypotension**, apnoea. *Propofol infusion syndrome* (prolonged high dose; metabolic acidosis)
Thiopental (thiopentone)	GABA-A (barbiturate)	Rapid onset; recovery depends on **redistribution**. Anticonvulsant, ↓ ICP. Accumulates with repeated doses. Severe tissue damage on extravasation or **intra-arterial injection**. **Contraindicated in acute porphyria**

Agent	Mechanism	Features
Ketamine	**NMDA antagonist** → "dissociative anaesthesia"	**Analgesia, preserved airway reflexes and respiration,** ↑ **BP/HR (sympathomimetic), bronchodilator.** Ideal for **shocked patients, asthmatics, field and low-resource settings, children**. **Emergence delirium / hallucinations** (reduce with a benzodiazepine), ↑ salivation, ↑ ICP (debated)
Etomidate	GABA-A	**Cardiovascularly stable. Adrenal suppression** (11β-hydroxylase inhibition), myoclonus, PONV
Midazolam	GABA-A (benzodiazepine)	Sedation, premedication, amnesia. Reverse with **flumazenil**
Opioids (fentanyl, remifentanil)	μ	Analgesia component
Dexmedetomidine	α_2 agonist	Sedation without respiratory depression

B. Neuromuscular blocking drugs (NMBDs)

Type	Drugs	Mechanism	Notes
Depolarising	**Suxamethonium (succinylcholine)**	Nicotinic agonist → persistent depolarisation → **fasciculations** then flaccid paralysis (phase I). Prolonged exposure → phase II (desensitisation)	**Fastest onset (~60 s), shortest duration (~5 min)**. Used for **rapid sequence induction**. Metabolised by plasma cholinesterase. Adverse effects: **hyperkalaemia** (burns, crush injury, denervation: avoid after 48 h), malignant hyperthermia, **scoline apnoea**, myalgia, bradycardia, ↑ intraocular and intragastric pressure. **Anticholinesterases potentiate it (do not reverse)**

Type	Drugs	Mechanism	Notes
Non-depolarising (competitive)	Aminosteroids: **rocuronium** (fast onset; RSI alternative), vecuronium, pancuronium. Benzylisoquinoliniums: **atracurium** (**Hofmann elimination**: safe in hepatic and renal failure; histamine release), cisatracurium, mivacurium (plasma cholinesterase); tubocurarine (historical)	Competitive antagonists at nicotinic N_M receptors	**Reversal: neostigmine + glycopyrrolate/atropine** (blocks muscarinic effects). **Sugammadex** encapsulates rocuronium and vecuronium. Potentiated by aminoglycosides, volatiles, Mg^{2+}, hypothermia, myasthenia gravis

C. Local anaesthetics (LAs)

Mechanism: LAs are weak bases (pKa 7.6–9). The **unionised form crosses the membrane**; inside the cell the **ionised form blocks voltage-gated Na^+ channels from the inside** (use-dependent block). Small, myelinated fibres are blocked first, so the order of loss is: pain → temperature → touch → pressure → motor. *Infected (acidic) tissue reduces their efficacy*, because more drug is ionised and cannot enter.

Group	Drugs	Notes
Esters (one "i")	Cocaine, procaine, tetracaine (amethocaine), benzocaine, chloroprocaine	Hydrolysed by plasma cholinesterase → **PABA** → **allergy** more common. Cocaine: vasoconstrictor (blocks NA reuptake), topical for ENT
Amides (two "i"s)	**Lidocaine (lignocaine)**, **bupivacaine** (long-acting; **most cardiotoxic**; "heavy" bupivacaine for spinal), levobupivacaine, ropivacaine (less cardiotoxic), prilocaine (**methaemoglobinaemia**; IV regional anaesthesia, Bier's block), articaine	Hepatic metabolism

Maximum doses:

Drug	Plain	With adrenaline
Lidocaine	3 mg/kg	7 mg/kg
Bupivacaine	2 mg/kg	2 mg/kg (adrenaline doesn't raise the max)

Adrenaline (1:200,000) prolongs action and reduces bleeding and systemic toxicity. **Avoid it at end-arteries**: fingers, toes, penis, nose, ear lobes.

Local anaesthetic systemic toxicity (LAST):

- **CNS:** circumoral and tongue numbness, metallic taste, tinnitus, lightheadedness, twitching → **seizures** → coma.
- **CVS:** arrhythmias, cardiovascular collapse (bupivacaine).
- **Treatment:** ABC, benzodiazepines for seizures, **20% lipid emulsion (Intralipid) 1.5 mL/kg bolus**, prolonged CPR.

Techniques: topical, infiltration, nerve or plexus block, IV regional (Bier's), **spinal (subarachnoid)** and **epidural**. Complications of spinal anaesthesia: hypotension (sympathetic block; treat with fluids and ephedrine/phenylephrine), post-dural puncture headache, high or total spinal block, urinary retention.

★ Exam pearls

- **Ketamine** is the anaesthetic of choice for the haemodynamically unstable patient and in low-resource settings.
- **Thiopental** is contraindicated in porphyria.
- **Suxamethonium** is contraindicated in hyperkalaemia, burns >48 h, spinal cord injury and myopathies.
- **Atracurium** (Hofmann elimination) is the NMBD of choice in hepatic or renal failure.

Chapter 24 · Calcium Metabolism and Drugs Affecting Bone

(The biochemistry and laboratory assessment are in Book 4, Chapter 53.)

Regulation of calcium

Plasma Ca^{2+} is 2.2–2.6 mmol/L total: ~45% ionised (active), ~45% bound to albumin, ~10% complexed. It is controlled by three hormones:

Hormone	Source / trigger	Bone	Kidney	Gut	Net effect
PTH	Chief cells of the parathyroid, in response to ↓ ionised Ca^{2+} (calcium-sensing receptor)	↑ resorption (via osteoblast RANKL → osteoclasts)	↑ Ca^{2+} reabsorption (DCT), ↓ **PO_4^{3-} reabsorption**, ↑ 1α-hydroxylase	Indirect (via vitamin D)	↑ **Ca^{2+}**, ↓ **PO_4^{3-}**
Calcitriol (1,25-$(OH)_2D_3$)	Skin (UVB) → D_3 → liver 25-hydroxylation → kidney 1α-hydroxylation	Mineralisation (with high doses, resorption)	↑ Ca^{2+} and PO_4^{3-} reabsorption	↑ **Ca^{2+} and PO_4^{3-} absorption**	↑ **Ca^{2+}**, ↑ **PO_4^{3-}**
Calcitonin	Thyroid C cells, in response to ↑ Ca^{2+}	↓ osteoclast activity	↑ Ca^{2+} excretion	—	↓ Ca^{2+} (minor role in humans)
FGF-23	Osteocytes	—	↓ **PO_4^{3-} reabsorption**, ↓ 1α-hydroxylase	—	↓ PO_4^{3-}

Drugs

Class	Drugs	Mechanism	Uses	Adverse effects
Calcium salts	Calcium carbonate, gluconate (10% IV), chloride	Replace Ca^{2+}	**Acute hypocalcaemia (10–20 mL 10% Ca gluconate IV slowly)**, **hyperkalaemia (cardioprotection)**, osteoporosis, CCB overdose	Constipation; extravasation necrosis (chloride)

Class	Drugs	Mechanism	Uses	Adverse effects
Vitamin D preparations	Ergocalciferol (D_2), colecalciferol (D_3), **alfacalcidol (1α-OH-D_3)** and **calcitriol** (do *not* need renal activation)	↑ Ca^{2+} absorption	Rickets, osteomalacia, **renal osteodystrophy (use alfacalcidol or calcitriol)**, hypoparathyroidism	**Hypercalcaemia**
Bisphosphonates	Alendronate, risedronate (oral weekly); **zoledronic acid**, pamidronate, ibandronate (IV)	Pyrophosphate analogues; bind hydroxyapatite; taken up by osteoclasts → **apoptosis** (nitrogen-containing: inhibit farnesyl pyrophosphate synthase)	**Osteoporosis**, **Paget's disease**, **hypercalcaemia of malignancy**, bone metastases, myeloma	**Oesophagitis**: take on an empty stomach with a full glass of water and **stay upright for 30 min**. **Osteonecrosis of the jaw**, **atypical femoral fractures**, acute phase reaction (IV), hypocalcaemia
RANKL inhibitor	**Denosumab** (s.c. 6-monthly)	Monoclonal antibody against RANKL → ↓ osteoclast formation	Osteoporosis, bone metastases	Hypocalcaemia, ONJ, **rebound vertebral fractures on stopping**
SERMs	Raloxifene	Oestrogen agonist on bone, antagonist on breast and uterus	Postmenopausal osteoporosis	VTE, hot flushes
Anabolic (PTH analogues)	**Teriparatide** (PTH 1-34), abaloparatide	**Intermittent** PTH → ↑ osteoblast activity (continuous PTH → resorption)	Severe osteoporosis	Hypercalcaemia; osteosarcoma (rats; limit to 2 years)
Sclerostin inhibitor	Romosozumab	↑ formation, ↓ resorption	Severe osteoporosis	CV events
Calcitonin	Salmon calcitonin	↓ resorption	Hypercalcaemia (rapid but short tachyphylaxis), Paget's	Nausea; malignancy risk with long use
Calcimimetics	**Cinacalcet**, etelcalcetide	↑ sensitivity of the calcium-sensing receptor → ↓ PTH	**Secondary hyperparathyroidism in CKD**, parathyroid carcinoma	Hypocalcaemia

Class	Drugs	Mechanism	Uses	Adverse effects
Others	HRT, strontium ranelate, **phosphate binders** (sevelamer, calcium acetate, lanthanum), gallium nitrate, glucocorticoids (↓ Ca absorption: myeloma, sarcoid, vitamin D toxicity)			

Treatment of hypercalcaemia (Ca^{2+} > 3.0 mmol/L symptomatic, or > 3.5)

"Stones, bones, abdominal groans, psychic moans": dehydration, polyuria, constipation, confusion, short QT.

1. **IV 0.9% saline rehydration** (3–4 L/24 h). This is the most important first step.
2. **IV bisphosphonate** (zoledronic acid 4 mg or pamidronate). Effect peaks at 2–4 days.
3. Calcitonin for rapid short-term effect; glucocorticoids (vitamin D-mediated: granulomas, lymphoma); denosumab if refractory; haemodialysis.
4. Loop diuretics only after rehydration.
5. Treat the cause: primary hyperparathyroidism or malignancy (~90% of cases together).

★ Exam pearls

Intermittent PTH builds bone (anabolic); continuous PTH destroys bone (catabolic). Examiners like this paradox.

Chapter 25 · Antituberculosis Drugs

Background

Mycobacterium tuberculosis is an acid-fast bacillus (Ziehl–Neelsen stain), an obligate aerobe and slow-growing (doubling time 15–20 h). Its cell wall is rich in **mycolic acid**. Nigeria is among the **top 10 high-burden countries** for TB and TB/HIV (WHO Global TB Report).

Why combination therapy?

1. **Prevents resistance.** Spontaneous mutants: about 1 in 10^6 for INH, 1 in 10^8 for rifampicin. A cavity holds about 10^8–10^9 bacilli.
2. **Kills different bacterial populations:**
   - Rapidly dividing extracellular bacilli in cavities: **INH** (most bactericidal).
   - Slow, intermittently dividing semi-dormant bacilli: **rifampicin** (sterilising).
   - Intracellular bacilli in an acidic environment (macrophages, caseous foci): **pyrazinamide** (sterilising).
3. **Shortens treatment.**

First-line drugs ("RIPE": Rifampicin, Isoniazid, Pyrazinamide, Ethambutol)

Drug	Mechanism	Key adverse effects	Notes
Isoniazid (H)	Prodrug activated by mycobacterial **catalase-peroxidase (KatG)** → inhibits **mycolic acid synthesis (InhA)**	**Peripheral neuropathy** (pyridoxine deficiency; prevent with **pyridoxine 10–25 mg/day**), **hepatitis** (↑ with age and alcohol), drug-induced lupus, seizures, optic neuritis, sideroblastic anaemia	Bactericidal. **NAT2 acetylator status.** Enzyme inhibitor (↑ phenytoin). Used for TB preventive therapy (IPT)
Rifampicin (R)	Inhibits **DNA-dependent RNA polymerase (rpoB)**	**Orange-red discoloration of urine, tears, sweat** (stains contact lenses), **hepatitis**, flu-like syndrome (intermittent dosing), thrombocytopenia, renal failure	Bactericidal and sterilising. **Potent CYP450 inducer** → contraceptive failure, ↓ warfarin, steroids, ARVs (nevirapine, PIs), methadone. Also used in leprosy and meningococcal prophylaxis

Drug	Mechanism	Key adverse effects	Notes
Pyrazinamide (Z)	Converted to pyrazinoic acid (pyrazinamidase); disrupts membrane energetics and trans-translation; active at **acidic pH**	**Hepatotoxicity (most hepatotoxic of the three)**, **hyperuricaemia / gout**, arthralgia	Sterilising; used only in the first 2 months
Ethambutol (E)	Inhibits **arabinosyl transferase** → ↓ arabinogalactan synthesis	**Retrobulbar (optic) neuritis**: ↓ visual acuity and **red–green colour blindness**; check vision before and during treatment. Avoid in young children who cannot report vision	Bacteriostatic; prevents resistance. Dose-adjust in renal impairment
Streptomycin (S)	Aminoglycoside (30S)	Ototoxicity (vestibular), nephrotoxicity	No longer in standard regimens

Hepatotoxicity ranking: pyrazinamide > isoniazid > rifampicin. Ethambutol is not hepatotoxic.

Regimens (WHO 2022; Nigerian National TB & Leprosy Control Programme)

Category	Regimen
Drug-susceptible pulmonary TB (new and previously treated)	**2HRZE / 4HR** (2 months intensive phase with HRZE as fixed-dose combinations, then 4 months continuation with HR), given daily under DOT
Shortened regimen, age ≥12 (WHO 2022)	4 months **HPMZ**: isoniazid + rifapentine + moxifloxacin + pyrazinamide
Non-severe TB in children 3 months–16 years	**4 months (2HRZ(E)/2HR)**
TB meningitis, bone and joint TB	**2HRZE / 10HR** (12 months). **Add a corticosteroid** (dexamethasone/prednisolone) in meningitis and pericarditis
Rifampicin-resistant / MDR-TB (resistant to at least H + R)	**BPaLM** for 6 months: **Bedaquiline + Pretomanid + Linezolid + Moxifloxacin**. Or longer individualised regimens
XDR-TB	Resistant to a fluoroquinolone + bedaquiline or linezolid as well; individualised
TB preventive therapy (TPT)	6H (isoniazid for 6 months), **3HP** (weekly isoniazid + rifapentine for 12 weeks), 3HR, 1HP

Second-line drugs and their key toxicities:

- Fluoroquinolones (levofloxacin, moxifloxacin)
- **Bedaquiline** (ATP synthase inhibitor; QT prolongation)

- **Linezolid** (myelosuppression, neuropathy)
- Clofazimine
- Cycloserine (psychosis)
- Delamanid, pretomanid
- Aminoglycosides (amikacin), ethionamide
- PAS

TB/HIV:

- Start **ART within 2 weeks** of starting TB treatment (or within 8 weeks if CD4 is high). In **TB meningitis, delay ART for 4–8 weeks**.
- Use **dolutegravir 50 mg twice daily** with rifampicin.
- Watch for **IRIS** (immune reconstitution inflammatory syndrome).
- Give **cotrimoxazole prophylaxis**.

Drug-induced liver injury (DILI)

If ALT >3× ULN with symptoms, or >5× ULN, **stop all hepatotoxic drugs**. If treatment must continue, give a non-hepatotoxic regimen (ethambutol + a fluoroquinolone + streptomycin/amikacin). Once the liver tests normalise, reintroduce the drugs **one at a time**: rifampicin, then isoniazid, then consider pyrazinamide.

★ Exam pearls

INH side effects: "INH = Injures Neurons and Hepatocytes". Ethambutol → Eyes. Rifampicin → Red secretions and enzyme **induction. Pyrazinamide → Painful gout (urate). Streptomycin → Sound (ears).**

Chapter 26 · Antileprosy Drugs

Background

Mycobacterium leprae is acid-fast, but weakly so compared with *M. tuberculosis* (decolourise with 5% H_2SO_4). It is an **obligate intracellular** organism that **cannot be cultured on artificial media**; it grows in mouse footpads and nine-banded armadillos. It prefers cooler temperatures (skin, peripheral nerves) and divides very slowly (~12–14 days). Spread is by nasal droplets. Nigeria still reports new cases, and the disease is targeted for elimination.

Classification:

- **Ridley–Jopling spectrum** (depends on the cell-mediated immune response):
  - **TT** (tuberculoid: strong CMI, few lesions, bacilli scanty)
  - BT, BB, BL
  - **LL** (lepromatous: poor CMI, Th2, numerous bacilli, leonine facies)
- **WHO operational classification:**
  - **Paucibacillary (PB):** 1–5 skin lesions, slit-skin smear negative.
  - **Multibacillary (MB):** >5 lesions, or nerve involvement, or smear positive.

Drugs

Drug	Mechanism	Adverse effects
Dapsone (diaminodiphenyl sulfone)	Folate synthesis inhibitor (PABA antagonist, like sulfonamides); bacteriostatic	**Haemolysis (G6PD deficiency)**, **methaemoglobinaemia**, agranulocytosis, **dapsone hypersensitivity syndrome** (fever, rash, hepatitis), peripheral neuropathy
Rifampicin	RNA polymerase inhibitor; **most bactericidal against *M. leprae*** (one monthly dose kills 99.99%)	As in Chapter 25; harmless red urine
Clofazimine	Phenazine dye; binds DNA; also **anti-inflammatory** (useful in ENL)	**Red-brown to black skin discolouration** (especially in lesions), ichthyosis, GI upset (crystal deposition causing enteropathy)
Alternatives	Ofloxacin, minocycline, clarithromycin	

WHO multidrug therapy (MDT), 2018 update: uniform 3-drug regimen

Type	Regimen (adult)	Duration
Paucibacillary	**Rifampicin 600 mg + clofazimine 300 mg once monthly (supervised); dapsone 100 mg + clofazimine 50 mg daily (self-administered)**	**6 months**
Multibacillary	Same drugs	**12 months**

(Before 2018, PB was treated with rifampicin + dapsone only.)

Single-dose rifampicin (SDR-PEP) is given to eligible contacts as post-exposure prophylaxis.

Lepra reactions (acute immunological episodes)

Feature	Type 1 (reversal) reaction	Type 2 reaction: erythema nodosum leprosum (ENL)
Mechanism	**Type IV** (delayed) hypersensitivity; ↑ CMI	**Type III** (immune complex)
Occurs in	Borderline leprosy (BT, BB, BL)	**LL and BL**
Features	Existing lesions become red and swollen; **acute neuritis** (risk of nerve damage)	Crops of tender red nodules, fever, arthritis, iritis, orchitis, neuritis, glomerulonephritis
Treatment	**Prednisolone** (e.g. 40 mg/day tapered over 20 weeks); continue MDT	Prednisolone, **thalidomide** (drug of choice; **teratogenic**), high-dose clofazimine, NSAIDs

★ Exam pearls

- **Never stop MDT during a reaction.**
- **Disability prevention:** nerve function assessment, self-care, protective footwear.
- **Lucio phenomenon** (necrotising vasculitis) occurs in diffuse LL.

Chapter 27 · Antifungal Drugs

Fungal targets

The fungal cell membrane contains **ergosterol** (instead of cholesterol). The cell wall contains **β-(1,3)-D-glucan** and chitin.

Class	Drugs	Mechanism	Spectrum / uses	Adverse effects
Polyenes	**Amphotericin B** (deoxycholate, liposomal), **nystatin** (topical/oral only, too toxic systemically)	Bind **ergosterol** → form **pores** → leakage of K^+ and Mg^{2+} (fungicidal)	Amphotericin: **broadest spectrum; drug of choice for severe systemic mycoses** (cryptococcal meningitis + flucytosine, mucormycosis, histoplasmosis, invasive candidiasis/aspergillosis), **visceral leishmaniasis**. Nystatin: oral and vaginal candidiasis	**"Amphoterrible"**: infusion reactions (fever, chills, rigors: "shake and bake"), **nephrotoxicity** (↓ with saline loading and liposomal forms), **hypokalaemia, hypomagnesaemia**, normocytic anaemia (↓ EPO), thrombophlebitis
Azoles: imidazoles	Ketoconazole, **clotrimazole, miconazole, econazole** (topical)	Inhibit fungal **CYP450 14α-demethylase** → ↓ ergosterol synthesis (fungistatic)	Topical: candidiasis, dermatophytoses, pityriasis versicolor	Ketoconazole (oral): **hepatotoxicity**, ↓ steroid synthesis → **gynaecomastia**, adrenal insufficiency (used for Cushing's). Oral use restricted
Azoles: triazoles	**Fluconazole**, itraconazole, **voriconazole**, posaconazole, isavuconazole	As above, more selective	Fluconazole: **candidiasis** (oral, vaginal single 150 mg dose, oesophageal), **cryptococcal meningitis (maintenance; CSF penetration)**, prophylaxis. Itraconazole: dermatophytes, onychomycosis, histoplasmosis, sporotrichosis. **Voriconazole: drug of choice for invasive aspergillosis.** Posaconazole/isavuconazole: mucormycosis	**CYP3A4 inhibition** → interactions (warfarin, statins, ciclosporin, QT drugs), hepatotoxicity, QT prolongation. Voriconazole: **visual disturbances**, photosensitivity, skin cancer. Itraconazole: negative inotrope (CHF). Teratogenic

Class	Drugs	Mechanism	Spectrum / uses	Adverse effects
Echinocandins	**Caspofungin, micafungin, anidulafungin**, rezafungin	Inhibit **β-(1,3)-D-glucan synthase** → cell wall damage	**Invasive candidiasis / candidaemia (first line)**, febrile neutropenia, salvage for aspergillosis. **Not active against *Cryptococcus***	Well tolerated; histamine-like infusion reactions. IV only
Antimetabolite	**Flucytosine (5-FC)**	Converted in the fungus (cytosine deaminase) → 5-FU → inhibits DNA/RNA synthesis	**Cryptococcal meningitis (with amphotericin B; synergy)**	**Bone marrow suppression**, hepatotoxicity. Resistance develops quickly, so never use alone
Allylamines	**Terbinafine** (oral and topical), naftifine	Inhibit **squalene epoxidase** → toxic squalene accumulates (fungicidal)	**Dermatophytes: onychomycosis (drug of choice, 6–12 weeks)**, tinea	Hepatotoxicity, taste disturbance, rash
Griseofulvin		Binds **microtubules** → disrupts the mitotic spindle; deposits in keratin	**Dermatophytes, especially tinea capitis** in children	Enzyme inducer (↓ OCP, warfarin), headache, photosensitivity, **teratogenic**, precipitates porphyria
Others	Whitfield's ointment (benzoic + salicylic acid), **selenium sulfide** and ketoconazole shampoo (pityriasis versicolor), tolnaftate, ciclopirox, amorolfine (nail lacquer), potassium iodide (sporotrichosis)			

WHO cryptococcal meningitis induction (HIV) 2022:

- **Single high-dose liposomal amphotericin B (10 mg/kg) + 14 days flucytosine + fluconazole.**
- Then consolidation with fluconazole 800 mg, then maintenance 200 mg.
- Repeated **therapeutic LPs** for raised pressure.
- Defer ART for 4–6 weeks.

★ Exam pearls

- Know the targets: **Polyenes poke holes in ergosterol. Azoles stop ergosterol synthesis. Echinocandins attack the wall. Flucytosine** is an antimetabolite. **Terbinafine** blocks squalene epoxidase. **Griseofulvin** blocks microtubules.
- Echinocandins don't cover *Cryptococcus*. Fluconazole doesn't cover *Aspergillus* or *C. krusei*.

Chapter 28 · Antimalarial Drugs

The parasite's life cycle and pathogenesis are covered in Book 5, Chapter 73.

Classification by the stage they act on

Stage	Drug type	Drugs
Blood schizonticides (treat the acute attack)	Fast-acting	**Artemisinins**, quinine, chloroquine, mefloquine, lumefantrine, amodiaquine, piperaquine, pyronaridine, atovaquone
	Slow-acting	Sulfadoxine–pyrimethamine (SP), proguanil, doxycycline, clindamycin
Tissue (hepatic) schizonticides	Causal prophylaxis (primary liver stage)	Primaquine, **proguanil**, atovaquone
Hypnozoitocides (radical cure: *P. vivax/ovale* relapse)		**Primaquine, tafenoquine** (only these two)
Gametocytocides (block transmission)		**Primaquine** (*P. falciparum*), artemisinins (partial). Chloroquine and quinine act on *vivax* gametocytes
Sporonticides		Proguanil, pyrimethamine (prevent sporogony in the mosquito)

Individual drugs

Drug	Mechanism	Key adverse effects / notes
Artemisinins (artesunate, artemether, dihydroartemisinin)	Endoperoxide bridge activated by **haem-Fe^{2+} → free radicals** damage parasite proteins. Fastest parasite clearance; kills young ring forms	Generally safe. **Delayed post-artesunate haemolysis**. Short t½, so always combined with a partner drug (ACT). Emerging partial resistance (**kelch13** mutations) in SE Asia and now East Africa. Avoid in the 1st trimester (WHO 2022 now allows AL in the 1st trimester)

Drug	Mechanism	Key adverse effects / notes
Chloroquine	Accumulates in the **acidic food vacuole**; **inhibits haem polymerase**, so toxic free haem builds up	*P. falciparum* resistance (**PfCRT** mutation) is widespread, so chloroquine is no longer used for falciparum in Nigeria (since 2005). Still used for *vivax* elsewhere. Pruritus (common in Africans), **retinopathy** (long-term "bull's-eye"), QT prolongation, toxicity in overdose (children). Also for lupus and RA (hydroxychloroquine)
Amodiaquine	Like chloroquine	Agranulocytosis and hepatitis (with prophylactic use). Partner in AS-AQ and **SMC (SPAQ)**
Quinine (cinchona alkaloid)	Like chloroquine	**Cinchonism** (tinnitus, deafness, headache, nausea, visual disturbance), **hypoglycaemia** (stimulates insulin release, especially in pregnancy), QT prolongation, blackwater fever. IV quinine is second line in severe malaria. Safe in pregnancy
Mefloquine	Like quinine	**Neuropsychiatric effects** (vivid dreams, depression, psychosis), bradycardia. Prophylaxis
Lumefantrine	Haem detoxification inhibitor	Take with **fatty food** (milk) for absorption. Partner in **artemether–lumefantrine (AL, Coartem®)**
Sulfadoxine–pyrimethamine (SP)	**Sequential folate blockade**: sulfadoxine inhibits dihydropteroate synthase; pyrimethamine inhibits **DHFR**	**SJS/TEN**, megaloblastic anaemia. **IPTp in pregnancy** (from 13 weeks, monthly, ≥3 doses). Avoid with cotrimoxazole
Proguanil	DHFR inhibitor (cycloguanil)	Mouth ulcers. Combined with atovaquone (**Malarone®**) for prophylaxis and treatment
Atovaquone	Mitochondrial **cytochrome bc_1** inhibitor	
Primaquine	8-aminoquinoline; oxidative damage to parasite mitochondria	**Haemolysis in G6PD deficiency**, methaemoglobinaemia. Contraindicated in pregnancy and breastfeeding (if infant's G6PD status is unknown)
Tafenoquine	Long-acting 8-aminoquinoline	Single-dose radical cure; G6PD testing required

Drug	Mechanism	Key adverse effects / notes
Doxycycline	Apicoplast protein synthesis	Prophylaxis; adjunct with quinine. Not for children <8 or in pregnancy
Pyronaridine	Haem inhibition	With artesunate (Pyramax®)

Nigerian National Malaria Treatment Policy (aligned with WHO Guidelines 2023)

Uncomplicated *P. falciparum* malaria: ACTs (3 days)

- **Artemether–lumefantrine (AL)**: first line. Twice daily for 3 days (doses at 0, 8, 24, 36, 48, 60 h) with fatty food.
- **Artesunate–amodiaquine (AS-AQ)**: first line.
- Others: dihydroartemisinin–piperaquine (DHA-PPQ), artesunate–mefloquine, artesunate–pyronaridine.
- WHO: confirm with **RDT or microscopy before treating** ("test, treat, track").

Severe malaria: IV (or IM) artesunate (drug of choice)

- Dose: **2.4 mg/kg at 0, 12 and 24 h, then daily** (3 mg/kg for children <20 kg) until the patient can take oral drugs. Then complete a full 3-day ACT course.
- **AQUAMAT and SEAQUAMAT trials:** artesunate reduces mortality compared with quinine (by ~22% in children, ~35% in adults).
- Alternatives: IM artemether, then IV quinine (20 mg/kg loading dose, then 10 mg/kg every 8 h in 5% dextrose; monitor glucose).
- **Pre-referral:** rectal artesunate (10 mg/kg) for children <6 years where injections are not available.
- **Supportive care:** glucose, anticonvulsants, transfusion (Hb <5 g/dL in children, <7 g/dL in adults), fluids with care.

Malaria in pregnancy:

- 1st trimester: WHO (2022) allows **AL**; quinine + clindamycin is the alternative.
- 2nd and 3rd trimesters: ACTs.
- Severe malaria in any trimester: **IV artesunate**.

Chemoprevention:

- **IPTp-SP** in pregnancy.
- **Seasonal malaria chemoprevention (SMC)** with **SP + amodiaquine (SPAQ)** monthly during the rainy season for children 3–59 months in the Sahel (northern Nigeria).
- **Perennial malaria chemoprevention (PMC / IPTi)** with SP.

Malaria vaccines (WHO-recommended): **RTS,S/AS01 (Mosquirix®)** and **R21/Matrix-M**. Nigeria introduced R21 in 2024.

🎤 Viva corner

Q: Why are artemisinins always given as combinations? **A:** Their very short half-life (~1 h) leaves residual parasites that cause recrudescence. Monotherapy also encourages resistance. A longer-acting partner drug clears the remaining parasites and protects each drug against resistance.

Q: What is the drug of choice for severe malaria? **A:** IV artesunate. It clears parasites faster and lowers mortality compared with quinine, and it does not cause hypoglycaemia.

Chapter 29 · Anthelminthic Drugs

The biology of the parasites is covered in Book 5, Chapters 75–79.

Drug	Mechanism	Spectrum / drug of choice for	Adverse effects / notes
Albendazole, mebendazole (benzimidazoles)	Bind **β-tubulin** → block microtubule polymerisation → ↓ glucose uptake	**Broad-spectrum intestinal nematodes**: *Ascaris*, hookworm, *Trichuris*, *Enterobius* (single dose; repeat after 2 weeks for pinworm). Albendazole: **neurocysticercosis**, **hydatid disease**, *Strongyloides* (2nd line), cutaneous larva migrans, filariasis (with ivermectin/DEC)	GI upset; hepatotoxicity and marrow suppression with prolonged high doses. **Avoid in the 1st trimester.** Mass deworming of school children
Ivermectin	Activates **glutamate-gated Cl^- channels** → hyperpolarisation → **paralysis** of the parasite	**Onchocerciasis (drug of choice; Mectizan® Donation Programme / APOC)**, **strongyloidiasis (drug of choice)**, lymphatic filariasis (with albendazole), scabies, head lice	**Mazzotti reaction** (fever, itch, rash, lymphadenopathy from dying microfilariae). **Severe encephalopathy in heavy *Loa loa* infection**. Does not cross the human BBB (P-gp). Kills microfilariae but not adult worms
Diethylcarbamazine (DEC)	Sensitises microfilariae to phagocytosis; alters arachidonic acid metabolism	**Lymphatic filariasis** (*Wuchereria*, *Brugia*), **loiasis**	**Contraindicated in onchocerciasis** (severe Mazzotti reaction, blindness)
Praziquantel	**↑ Ca^{2+} permeability** → spastic paralysis and tegument damage	**All trematodes (schistosomiasis: drug of choice**, single 40 mg/kg dose; *Clonorchis*, *Opisthorchis*, *Paragonimus*) **except *Fasciola***. Cestodes: *Taenia*, cysticercosis, *Hymenolepis*, *Diphyllobothrium*	Dizziness, abdominal pain. In neurocysticercosis give steroids (seizures from inflammation)
Triclabendazole	Benzimidazole	***Fasciola hepatica* (drug of choice)**	

Drug	Mechanism	Spectrum / drug of choice for	Adverse effects / notes
Niclosamide	Uncouples oxidative phosphorylation in the scolex	Intestinal tapeworms (not cysticercosis)	Not absorbed
Pyrantel pamoate	**Depolarising neuromuscular blocker** (nicotinic agonist) → spastic paralysis	Ascaris, hookworm, *Enterobius*. Safe in pregnancy (used in some settings)	**Antagonised by piperazine**
Piperazine	**GABA agonist** → flaccid paralysis	Ascaris, *Enterobius* (historical)	Neurotoxicity
Levamisole	Nicotinic agonist; also an immunomodulator	Ascaris	Agranulocytosis
Moxidectin	Like ivermectin, longer acting	Onchocerciasis (approved 2018)	
Doxycycline	Kills *Wolbachia* endosymbionts → sterilises and kills **adult** filariae	Onchocerciasis, lymphatic filariasis (6 weeks)	
Oxamniquine, metrifonate		Historical schistosomicides	

Dracunculiasis: there is **no specific drug**. Treat by slowly winding out the worm on a stick, plus wound care. Control depends on safe water (filtering, Abate® larvicide). Nigeria was certified free of transmission in 2013.

★ Exam pearls

Paralysis types:

Drug	Type of paralysis	Mechanism
Pyrantel / levamisole	**Spastic**	Nicotinic agonists
Piperazine	**Flaccid**	GABA agonist
Ivermectin	**Flaccid**	Glutamate-gated Cl^- channel
Praziquantel	**Spastic**	Ca^{2+} influx

Pyrantel and piperazine antagonise each other, so never combine them.

Chapter 30 · Anticancer Chemotherapy

Principles

- **The cell cycle:** G_0 (resting) → G_1 → S (DNA synthesis) → G_2 → M (mitosis).
  - **Cell cycle-specific (CCS)** drugs act in one phase (antimetabolites in S; vinca alkaloids and taxanes in M; bleomycin in G_2). They work best on rapidly dividing tumours and are given by **schedule** (infusions, repeated doses).
  - **Cell cycle-non-specific (CCNS)** drugs act in all phases, including G_0 (alkylating agents, anthracyclines, platinum compounds). They work against slow-growing tumours, and their effect is **dose-dependent**.
- **Log-kill hypothesis (Skipper):** each dose kills a constant *fraction* of cells, not a constant number. Repeated cycles are therefore needed to reduce the tumour burden (10^9 cells is about 1 g, the clinically detectable limit).
- **Combination chemotherapy:** use drugs with different mechanisms and non-overlapping toxicities. This gives greater kill and less resistance. Examples: ABVD, CHOP, BEP, FOLFOX.
- **Settings:**
  - *Adjuvant*: after surgery, to treat micrometastases.
  - *Neoadjuvant*: before surgery, to shrink the tumour.
  - *Palliative*.
  - *Curative*: e.g. Hodgkin lymphoma, testicular cancer, choriocarcinoma, childhood ALL, **Burkitt lymphoma** (Africa).
- **Resistance:**
  - **P-glycoprotein (MDR1)** efflux
  - ↓ drug uptake or activation
  - ↑ DNA repair
  - Target mutation or amplification (DHFR amplification with methotrexate)
  - Defective apoptosis (p53)

Classification

Class	Drugs	Mechanism	Key toxicity	Main uses
Alkylating agents (CCNS)	**Cyclophosphamide**, ifosfamide	Prodrugs (CYP activation) → cross-link DNA (N7 guanine)	**Haemorrhagic cystitis** (acrolein; prevent with **mesna** + hydration), SIADH, infertility, secondary leukaemia	Lymphomas, breast cancer, Burkitt, immunosuppression (lupus nephritis, vasculitis)

Class	Drugs	Mechanism	Key toxicity	Main uses
	Chlorambucil, melphalan, busulfan, bendamustine		Busulfan: **pulmonary fibrosis**, hyperpigmentation	CLL, myeloma, CML/transplant conditioning
	Nitrosoureas (carmustine, lomustine)	Lipophilic, so cross the BBB	Delayed myelosuppression	**Brain tumours**
	Dacarbazine, temozolomide, procarbazine	Methylating agents		Hodgkin (ABVD), melanoma, glioblastoma
Platinum compounds (CCNS)	**Cisplatin**, carboplatin, oxaliplatin	DNA cross-links	Cisplatin: **nephrotoxicity** (prevent with hydration and amifostine), **ototoxicity**, **severe emesis**, neuropathy, ↓ Mg^{2+}. Carboplatin: **myelosuppression**. Oxaliplatin: **cold-induced neuropathy**	Testicular, ovarian, lung, head and neck, cervical, bladder, colorectal (oxaliplatin)
Antimetabolites (S-phase)	**Methotrexate** (folate analogue)	Inhibits **DHFR** → ↓ THF → ↓ thymidylate and purines	Myelosuppression, **mucositis**, hepatotoxicity (fibrosis), **pneumonitis**, nephrotoxicity, teratogenic. **Rescue: folinic acid (leucovorin)**. NSAIDs, PPIs and penicillins ↓ its excretion	ALL, choriocarcinoma, osteosarcoma (high dose), breast cancer; **RA, psoriasis, ectopic pregnancy**
	5-Fluorouracil (pyrimidine analogue), capecitabine (oral prodrug)	→ 5-FdUMP inhibits **thymidylate synthase**. Leucovorin *enhances* it	Myelosuppression, mucositis, diarrhoea, **hand–foot syndrome**, coronary vasospasm. DPD deficiency → severe toxicity	**Colorectal**, breast, GI cancers; topical for actinic keratoses
	6-Mercaptopurine, 6-thioguanine, azathioprine (prodrug)	Purine analogues; activated by HGPRT	Myelosuppression; **TPMT** deficiency; **allopurinol interaction**	ALL maintenance; immunosuppression

Class	Drugs	Mechanism	Key toxicity	Main uses
	Cytarabine (Ara-C), gemcitabine	Inhibit DNA polymerase / chain termination	Myelosuppression; cytarabine: **cerebellar toxicity**, conjunctivitis	**AML**; pancreatic cancer (gemcitabine)
	Hydroxyurea (hydroxycarbamide)	**Ribonucleotide reductase** inhibitor	Myelosuppression	CML, polycythaemia vera, **sickle cell disease (↑ HbF)**
	Fludarabine, cladribine, pemetrexed			CLL, hairy cell leukaemia, mesothelioma
Cytotoxic antibiotics	**Doxorubicin**, daunorubicin, epirubicin (anthracyclines)	**Topoisomerase II inhibition**, DNA intercalation, **free radicals**	**Dose-dependent cardiotoxicity (dilated cardiomyopathy)**: cumulative limit ~450–550 mg/m²; prevent with **dexrazoxane**. Red urine, alopecia, extravasation necrosis	Breast cancer, lymphomas, sarcomas, leukaemias
	Bleomycin (G_2)	Free radicals → DNA strand breaks	**Pulmonary fibrosis**, skin pigmentation, fever. **Minimal myelosuppression**	Testicular cancer (BEP), Hodgkin (ABVD)
	Dactinomycin (actinomycin D)	Intercalation		**Wilms tumour**, Ewing sarcoma, rhabdomyosarcoma, choriocarcinoma
	Mitomycin C	Alkylation		Anal and bladder cancer
Plant derivatives / microtubule inhibitors (M-phase)	**Vincristine, vinblastine** (vinca alkaloids)	Bind tubulin → **prevent microtubule assembly**	Vincristine: **peripheral neuropathy** ("vincristine for nerves"), paralytic ileus, SIADH; **fatal if given intrathecally**; little myelosuppression. Vinblastine: **myelosuppression** ("vinblastine blasts the bone marrow")	ALL, lymphomas (CHOP/ABVD), Wilms, **Burkitt**

Class	Drugs	Mechanism	Key toxicity	Main uses
	Paclitaxel, docetaxel (taxanes)	**Stabilise microtubules** (prevent disassembly)	Myelosuppression, neuropathy, hypersensitivity (paclitaxel's vehicle; premedicate)	Breast, ovarian, lung, prostate
	Etoposide, teniposide (podophyllotoxins)	**Topoisomerase II** inhibitors	Myelosuppression, secondary AML	Testicular (BEP), small cell lung cancer
	Irinotecan, topotecan (camptothecins)	**Topoisomerase I** inhibitors	**Diarrhoea** (irinotecan; UGT1A1), myelosuppression	Colorectal, ovarian
Hormonal therapy	**Tamoxifen** (SERM), anastrozole/letrozole (aromatase inhibitors), fulvestrant	Block oestrogen action or synthesis	Tamoxifen: hot flushes, **endometrial cancer, VTE**. AIs: osteoporosis	**ER-positive breast cancer**
	GnRH agonists (goserelin, leuprolide; initial "flare"), antiandrogens (bicalutamide, enzalutamide), abiraterone (CYP17 inhibitor)			**Prostate cancer**
	Glucocorticoids (prednisolone)	Lymphocytolysis		ALL, lymphoma, myeloma
Targeted therapies	**Imatinib** (BCR-ABL tyrosine kinase inhibitor), dasatinib, nilotinib		Fluid retention	**CML**, GIST (c-KIT)
	Erlotinib, gefitinib, osimertinib (EGFR TKIs)		Acneiform rash, diarrhoea	NSCLC with EGFR mutation
	Vemurafenib (BRAF), sunitinib, sorafenib (multi-kinase/VEGFR), bortezomib (proteasome inhibitor)			Melanoma, RCC, HCC, myeloma
	ATRA (all-trans retinoic acid), arsenic trioxide	Differentiation of promyelocytes	**Differentiation syndrome**	**APL (AML-M3, t(15;17))**

Class	Drugs	Mechanism	Key toxicity	Main uses
Monoclonal antibodies ("-mab")	**Rituximab** (anti-CD20), **trastuzumab** (anti-HER2; **cardiotoxicity**), bevacizumab (anti-VEGF; bleeding, GI perforation, hypertension), cetuximab (anti-EGFR; RAS wild-type colorectal)		Infusion reactions	NHL, CLL; HER2+ breast cancer; colorectal cancer
Immune checkpoint inhibitors	Pembrolizumab, nivolumab (anti-PD-1), atezolizumab (anti-PD-L1), ipilimumab (anti-CTLA-4)	Release T-cell "brakes"	**Immune-related adverse events** (colitis, hepatitis, thyroiditis, pneumonitis, hypophysitis)	Melanoma, lung, many cancers

Common toxicities and their management

Toxicity	Management
Myelosuppression (nadir 7–14 days)	Check FBC before each cycle. **Febrile neutropenia** (ANC $<0.5 \times 10^9$/L + fever ≥38 °C) is an emergency: give broad-spectrum antibiotics (piperacillin–tazobactam) within 1 h. G-CSF (filgrastim)
Nausea and vomiting	5-HT_3 antagonist + NK_1 antagonist + dexamethasone
Mucositis, alopecia, infertility	Sperm or oocyte banking
Tumour lysis syndrome (↑ K^+, ↑ PO_4^{3-}, ↑ urate, ↓ Ca^{2+} → AKI) in Burkitt lymphoma and leukaemias	Hydration, **allopurinol or rasburicase**
Extravasation (vesicants: anthracyclines, vinca alkaloids)	Stop infusion; dexrazoxane for anthracyclines; hyaluronidase + warm compress for vinca

Specific antidotes ("rescue agents"):

Toxic drug	Rescue agent
Methotrexate	**Leucovorin (folinic acid)**
Cyclophosphamide / ifosfamide	**Mesna**
Doxorubicin	**Dexrazoxane**
Cisplatin	**Amifostine**
5-FU (DPD deficiency)	**Uridine triacetate**

★ Exam pearls

Organ-specific toxicities:

Drug	Toxicity
Bleomycin, busulfan, methotrexate	Lung
Doxorubicin, trastuzumab	Heart
Cisplatin	Kidney, ear
Cyclophosphamide	Bladder
Vincristine	Nerves

Chapter 31 · Immunopharmacology

Immunosuppressants

Class	Drugs	Mechanism	Uses	Toxicity
Glucocorticoids	Prednisolone, methylprednisolone, dexamethasone, hydrocortisone	Cytoplasmic receptor → ↓ NF-κB and AP-1 → ↓ cytokines (IL-1, IL-2, TNF), ↓ PLA_2 (annexin-1), ↓ COX-2; lymphocyte apoptosis; ↓ leucocyte migration	Asthma, autoimmune disease, transplant rejection, allergy, cancer, adrenal insufficiency	**Cushing's syndrome**, diabetes, osteoporosis, myopathy, peptic ulcer, cataract, glaucoma, infection, psychosis, growth retardation, **HPA suppression** (taper if >3 weeks)
Calcineurin inhibitors	**Ciclosporin** (binds cyclophilin), **tacrolimus** (binds FKBP-12)	Inhibit **calcineurin** → NFAT stays phosphorylated → ↓ **IL-2 transcription**	**Transplant rejection** (cornerstone), psoriasis, nephrotic syndrome, atopic dermatitis (topical tacrolimus)	**Nephrotoxicity**, hypertension, hyperkalaemia, neurotoxicity, diabetes (tacrolimus). Ciclosporin: **gum hyperplasia, hirsutism**. CYP3A4 interactions. **TDM required**
mTOR inhibitors	Sirolimus (rapamycin), everolimus	Bind FKBP-12 → inhibit **mTOR** → block the T-cell response to IL-2 ($G_1 \rightarrow S$)	Transplantation, drug-eluting stents, RCC	Hyperlipidaemia, poor wound healing, pneumonitis, thrombocytopenia
Antiproliferatives	**Azathioprine** (→ 6-MP), **mycophenolate mofetil** (inhibits **IMP dehydrogenase** → ↓ guanine synthesis in lymphocytes), methotrexate, cyclophosphamide, leflunomide (DHODH inhibitor)	↓ lymphocyte proliferation	Transplantation, lupus nephritis, autoimmune hepatitis, IBD	Myelosuppression. Azathioprine: TPMT, allopurinol interaction. MMF: GI upset, **teratogenic**

Class	Drugs	Mechanism	Uses	Toxicity
Antibodies / biologics	Anti-thymocyte globulin (ATG), basiliximab (anti-IL-2R / CD25), **anti-TNF** (infliximab, adalimumab, etanercept), rituximab (anti-CD20), tocilizumab (anti-IL-6R), abatacept (CTLA-4-Ig; ↓ costimulation), belatacept, ustekinumab (IL-12/23), secukinumab (IL-17), omalizumab (**anti-IgE**; asthma), mepolizumab (anti-IL-5), natalizumab (α4-integrin; **PML**), eculizumab (anti-C5; **meningococcal infection risk**)	Specific immune targets	RA, IBD, psoriasis, ankylosing spondylitis, transplantation, severe asthma, MS, PNH	**Anti-TNF → reactivation of latent TB** (screen with IGRA/Mantoux and CXR before starting), infections, lymphoma
JAK inhibitors	Tofacitinib, baricitinib, upadacitinib	Block JAK–STAT signalling	RA, UC, alopecia areata	Infection, VTE, herpes zoster

DMARDs for rheumatoid arthritis: methotrexate (anchor drug), sulfasalazine, hydroxychloroquine, leflunomide. Then biologics or JAK inhibitors.

Immunostimulants

- Interferons: IFN-α (hepatitis B/C, hairy cell leukaemia), IFN-β (MS), IFN-γ (chronic granulomatous disease).
- Interleukin-2 (aldesleukin): RCC, melanoma.
- G-CSF / GM-CSF.
- BCG (intravesical for bladder cancer).
- Levamisole.
- Imiquimod (topical; TLR-7 agonist: warts, basal cell carcinoma).
- Thalidomide, lenalidomide (myeloma).
- CAR-T cell therapy.

Antihistamines and mast-cell stabilisers (allergy)

- **H_1 blockers:**

  - First generation: chlorphenamine, promethazine, diphenhydramine, hydroxyzine. Sedating and anticholinergic.
  - Second generation: loratadine, cetirizine, fexofenadine. Non-sedating.

- **Mast cell stabilisers:** sodium cromoglicate, nedocromil, ketotifen.
- **Leukotriene antagonists:** montelukast, zafirlukast. Zileuton is a 5-LOX inhibitor.

Anaphylaxis (Resuscitation Council UK 2021 / WAO):

- **IM adrenaline 1:1000:** adult 0.5 mg; child 6–12 years 0.3 mg; 6 months–6 years 0.15 mg. Repeat after 5 min.
- Airway, high-flow O_2, IV fluids.
- Antihistamines and hydrocortisone are **second line and no longer routine**.

Hypersensitivity reactions (Gell and Coombs)

Type	Mediator	Examples (drug-related)
I (immediate)	IgE, mast cells	Anaphylaxis to penicillin, urticaria
II (cytotoxic)	IgG/IgM against cell surface	Methyldopa haemolytic anaemia, heparin-induced thrombocytopenia
III (immune complex)	Ag–Ab complexes	Serum sickness, ENL, drug-induced vasculitis
IV (delayed)	T cells	Contact dermatitis, SJS/TEN, DRESS, Mantoux test

Chapter 32 · Vaccines and Adjuvants

A **vaccine** is a biological preparation containing antigen(s) that induces **active, specific, protective immunity and immunological memory** without causing disease. **Passive immunisation** gives preformed antibodies (immunoglobulins, antitoxins, monoclonals) for immediate but short-lived protection, e.g. rabies immunoglobulin, HBIG, tetanus immunoglobulin, antivenom, nirsevimab (RSV).

Types of vaccine

Type	Description	Examples	Advantages / disadvantages
Live attenuated	Weakened organisms that replicate	**BCG, OPV (Sabin), measles, mumps, rubella (MMR), yellow fever, varicella, rotavirus, oral typhoid (Ty21a), LAIV**, dengue	Strong, long-lasting humoral + cellular (and mucosal) immunity, often after one dose. **Contraindicated in immunosuppression and pregnancy**; can revert (vaccine-derived poliovirus, cVDPV2); need a cold chain
Inactivated (killed) whole organism	Heat or formalin killed	**IPV (Salk)**, hepatitis A, rabies (cell culture), whole-cell pertussis, inactivated influenza, cholera (oral)	Safe in immunocompromised patients; need boosters and adjuvants; mainly humoral
Toxoids	Inactivated toxins (formaldehyde)	**Tetanus, diphtheria**	Antitoxin immunity; boosters needed (Td every 10 years)
Subunit / recombinant protein	Purified antigens	**Hepatitis B (HBsAg, recombinant yeast)**, **HPV (L1 VLP)**, acellular pertussis, recombinant zoster (gE + AS01B), R21 and RTS,S malaria	Safe; need adjuvants
Polysaccharide	Capsular polysaccharides	Pneumococcal PPSV23, Vi typhoid, meningococcal (old)	**T-independent: poor response in children <2 yr, no memory**

Type	Description	Examples	Advantages / disadvantages
Conjugate	Polysaccharide **linked to a protein carrier** (tetanus toxoid, CRM197)	**Hib, PCV10/13/15/20, MenACWY, MenAfriVac (MenA)**, typhoid conjugate (TCV)	**T-dependent: works in infants, gives memory and herd immunity**
Viral vector	Recombinant non-replicating virus carrying a gene	ChAdOx1 (Oxford/AstraZeneca) and Ad26 COVID-19 vaccines, rVSV-ZEBOV (**Ervebo®: Ebola**)	Strong cellular immunity
Nucleic acid (mRNA, DNA)	mRNA in lipid nanoparticles	Pfizer-BioNTech and Moderna COVID-19 vaccines; RSV mRNA	Rapid design; ultra-cold chain

Adjuvants

> An **adjuvant** (Latin *adjuvare*, to help) is a substance added to a vaccine to **enhance, accelerate and prolong the immune response** to an antigen. It allows less antigen (dose-sparing) or fewer doses, and improves responses in the elderly and infants.

Mechanisms:

- **Depot effect:** slow release of antigen.
- Recruitment and activation of **antigen-presenting cells** via **pattern-recognition receptors** (TLRs), and the inflammasome (alum activates NLRP3).
- Better antigen uptake and presentation.
- Skewing the Th1/Th2 balance.

Adjuvant	Composition / mechanism	Used in
Aluminium salts (alum): $Al(OH)_3$, $AlPO_4$	Depot + NLRP3 inflammasome; **Th2 (antibody)** bias. The oldest (1926, Glenny) and most widely used	DTP, HepB, HepA, Hib, PCV, HPV (some)
MF59	Oil-in-water squalene emulsion	Fluad® (elderly influenza)
AS03	Squalene + α-tocopherol emulsion	Pandemic H1N1 influenza
AS04	Alum + **MPL** (monophosphoryl lipid A, a **TLR4 agonist**)	Cervarix® (HPV), Fendrix® (HepB)
AS01	Liposome + MPL + **QS-21** (saponin from *Quillaja saponaria*)	**RTS,S malaria vaccine**, Shingrix®
Matrix-M	Saponin nanoparticles	**R21 malaria**, Novavax COVID-19
CpG 1018	TLR9 agonist (unmethylated CpG DNA)	Heplisav-B®

Adjuvant	Composition / mechanism	Used in
Freund's adjuvant	Mineral oil + killed mycobacteria (complete)	**Experimental animals only**: too reactogenic for humans
Virosomes, ISCOMs, cytokines (IL-12, GM-CSF)		Experimental/various

Nigerian National Programme on Immunisation (NPI) schedule (NPHCDA)

Age	Vaccines
Birth	**BCG, OPV0, HepB0** (within 24 h)
6 weeks	**Pentavalent 1** (DTwP-HepB-Hib), **OPV1, PCV1, Rota1**, (IPV1 in some schedules)
10 weeks	Penta 2, OPV2, PCV2, Rota2
14 weeks	Penta 3, OPV3, PCV3, **IPV**, (Rota3 depending on product)
6 months	Vitamin A (first dose; then every 6 months to 59 months); **malaria vaccine R21** dose 1 in eligible states (5 months+, 4 doses)
9 months	**Measles 1 (MCV1), yellow fever, meningococcal A conjugate (MenAfriVac)**
15 months	**Measles 2 (MCV2)**
9–14 years (girls)	**HPV vaccine** (single dose; introduced October 2023)
Women of childbearing age / pregnancy	**Td** (tetanus–diphtheria) 5-dose schedule

Cold chain, contraindications and AEFI

- **Cold chain:** most vaccines are stored at **+2 to +8 °C**. **OPV can be frozen** (−15 to −25 °C). **Never freeze** DTP, HepB, Hib or PCV (alum-adsorbed); check with the "**shake test**". **Vaccine vial monitors (VVM)** show cumulative heat exposure.
- **True contraindications:** anaphylaxis to a previous dose or component; live vaccines in severe immunosuppression and pregnancy; encephalopathy within 7 days of a pertussis vaccine.
- **Not contraindications:** mild illness, low-grade fever, malnutrition, prematurity (vaccinate at chronological age), breastfeeding, antibiotics. **HIV:** give BCG to asymptomatic infants per national policy (WHO advises withholding it if HIV infection is confirmed).
- **AEFI** (adverse events following immunisation) must be reported. Categories: vaccine product-related, quality defect, **immunisation error**, anxiety-related, coincidental.

🎤 Viva corner

Q: Why is a conjugate vaccine better than a plain polysaccharide vaccine in infants? **A:** Polysaccharides are T-independent antigens. Infants under 2 respond poorly to them, and they produce no memory or class switching. Linking the polysaccharide to a protein carrier recruits T-helper cells, which gives a T-dependent response with memory B cells, boostability, IgG class switching and mucosal (herd) protection.

Pharmacology Self-Assessment

Section A: Single best answer MCQs

1. A 30-year-old man on rifampicin-containing anti-TB therapy reports that his partner became pregnant despite taking the combined oral contraceptive pill. The mechanism is: A. Displacement from protein binding · B. Inhibition of CYP3A4 · C. **Induction of CYP3A4** · D. Reduced renal clearance · E. Pharmaceutical incompatibility

C. Rifampicin is a potent inducer of hepatic enzymes. It speeds up the metabolism of ethinylestradiol and progestogens, so contraceptive failure follows. Advise an IUD or depot medroxyprogesterone.

2. Which drug follows zero-order kinetics at therapeutic doses? A. Gentamicin · B. **Phenytoin** · C. Paracetamol · D. Amoxicillin · E. Atenolol

B. Phenytoin's metabolism saturates within the therapeutic range. Small increases in dose can therefore cause large rises in plasma level and toxicity (nystagmus, ataxia).

3. The drug most likely to cause a dry cough is: A. Losartan · B. Amlodipine · C. **Lisinopril** · D. Hydrochlorothiazide · E. Bisoprolol

C. ACE inhibitors prevent the breakdown of bradykinin, which builds up and causes cough. Switch the patient to an ARB.

4. The drug of choice for terminating paroxysmal SVT in a haemodynamically stable patient is: A. Digoxin · B. Amiodarone · C. **Adenosine** · D. Lidocaine · E. Flecainide

C. Adenosine acts on A_1 receptors to block the AV node transiently. Its half-life is under 10 seconds, so give it as a rapid IV push followed by a flush.

5. A patient with HFrEF is on furosemide and digoxin. Which addition will reduce mortality? A. Increase digoxin · B. **Spironolactone** · C. Nifedipine · D. Verapamil · E. Increase furosemide

B. In the RALES trial, mineralocorticoid antagonists reduced mortality. Non-DHP CCBs (verapamil) and short-acting nifedipine are harmful in HFrEF.

6. The antidote to heparin overdose is: A. Vitamin K · B. **Protamine sulfate** · C. Fresh frozen plasma · D. Tranexamic acid · E. Idarucizumab

B. Protamine is strongly basic and neutralises the acidic heparin (chemical antagonism). Vitamin K and PCC reverse warfarin. Idarucizumab reverses dabigatran.

7. A G6PD-deficient patient develops haemolysis after radical cure of *P. vivax*. The likely culprit is: A. Chloroquine · B. Artesunate · C. **Primaquine** · D. Lumefantrine · E. Doxycycline

C. Primaquine is an 8-aminoquinoline oxidant. Test G6PD status before giving it.

8. Red–green colour blindness during TB therapy is caused by: A. Isoniazid · B. Rifampicin · C. Pyrazinamide · D. **Ethambutol** · E. Streptomycin

D. Ethambutol causes retrobulbar optic neuritis. Test visual acuity and colour vision at baseline and monthly.

9. Mesna is given to prevent which toxicity of cyclophosphamide? A. Cardiomyopathy · B. Pulmonary fibrosis · C. **Haemorrhagic cystitis** · D. Neuropathy · E. Ototoxicity

C. Mesna binds acrolein in the bladder and neutralises it.

10. The local anaesthetic most likely to cause cardiovascular collapse on accidental IV injection is: A. Lidocaine · B. Prilocaine · C. **Bupivacaine** · D. Procaine · E. Benzocaine

C. Bupivacaine binds cardiac Na^+ channels strongly and dissociates slowly. Treat LAST with 20% lipid emulsion.

11. The induction agent best suited to a hypotensive trauma patient in a rural hospital is: A. Thiopental · B. Propofol · C. **Ketamine** · D. Halothane · E. Midazolam

C. Ketamine maintains BP through sympathetic stimulation and preserves the airway and breathing.

12. Which antiemetic is preferred in a patient with Parkinson's disease? A. Metoclopramide · B. Prochlorperazine · C. **Domperidone** · D. Haloperidol · E. Droperidol

C. Domperidone does not cross the blood–brain barrier, so it does not worsen parkinsonism.

13. The anthelminthic that causes flaccid paralysis by acting as a GABA agonist is: A. Pyrantel · B. **Piperazine** · C. Praziquantel · D. Levamisole · E. Albendazole

B. Piperazine causes flaccid paralysis. Pyrantel causes spastic paralysis, and the two antagonise each other.

14. A conjugate vaccine differs from a polysaccharide vaccine because it: A. Contains live organisms · B. **Induces T-cell-dependent memory in infants** · C. Needs no cold chain · D. Gives passive immunity · E. Contains an adjuvant only

B. See Chapter 32.

15. Allopurinol increases the toxicity of: A. Methotrexate · B. **Azathioprine** · C. Cyclophosphamide · D. Ciclosporin · E. Prednisolone

B. Xanthine oxidase breaks down 6-MP, the active metabolite of azathioprine. Allopurinol blocks this, so reduce the azathioprine dose to 25%.

16. The adjuvant in the R21 malaria vaccine is: A. Alum · B. MF59 · C. AS01 · D. **Matrix-M** · E. CpG 1018

D. RTS,S uses AS01; R21 uses Matrix-M.

17. Which statement about nitrates is FALSE? A. They release NO · B. They mainly reduce preload · C. Tolerance develops · D. **They are safe with sildenafil** · E. They cause headache

D. Nitrates with a PDE-5 inhibitor cause profound hypotension because cGMP builds up from both sides.

18. Long-term use of which drug causes hypomagnesaemia? A. Ranitidine · B. Sucralfate · C. **Omeprazole** · D. Misoprostol · E. Bismuth

C. PPIs reduce intestinal absorption of Mg^{2+} (via TRPM6/7).

19. Which drug is used for drug-induced dystonia from metoclopramide? A. Haloperidol · B. **Procyclidine** · C. Neostigmine · D. Ondansetron · E. Levodopa

B. Procyclidine and benzatropine are anticholinergics, given IM or IV.

20. Which drug inhibits ergosterol synthesis by blocking squalene epoxidase? A. Fluconazole · B. Amphotericin B · C. **Terbinafine** · D. Caspofungin · E. Griseofulvin

C. Terbinafine blocks squalene epoxidase. Fluconazole blocks 14α-demethylase. Amphotericin B binds ergosterol. Caspofungin blocks glucan synthase. Griseofulvin binds microtubules.

Section B: Short-answer and essay questions (with answer outlines)

1. Classify antihypertensive drugs. Discuss the pharmacology of ACE inhibitors. (20 marks)

Outline:

1. **Classification:** diuretics; RAAS inhibitors (ACEI, ARB, renin inhibitor, MRA); CCBs; β- and α-blockers; central sympatholytics; direct vasodilators; adrenergic neurone blockers; drugs for emergencies.
2. **ACEI mechanism:** RAAS diagram; ↓ Ang II → vasodilation, ↓ aldosterone, ↓ remodelling; ↑ bradykinin.
3. **Pharmacokinetics:** prodrugs (enalapril → enalaprilat); captopril has an –SH group and short t½; lisinopril is not a prodrug and is renally excreted.
4. **Uses:** hypertension; HF; post-MI; diabetic nephropathy and proteinuric CKD; scleroderma renal crisis.
5. **Adverse effects:** CAPTOPRIL mnemonic: **C**ough, **A**ngio-oedema, **P**roteinuria, **T**aste change, **O**rthostatic (first-dose) hypotension, **P**regnancy contraindication, **R**enal failure (bilateral renal artery stenosis), **I**ncreased K^+, **L**eucopenia.
6. **Interactions:** NSAIDs, K^+-sparing diuretics, lithium.

7. **Monitoring**: U&E 1–2 weeks after starting. A creatinine rise of up to 30% is acceptable.

2. Write short notes on: (a) organophosphate poisoning; (b) digoxin toxicity; (c) paracetamol poisoning.

(a) Organophosphate poisoning - Features: muscarinic (DUMBBELSS), nicotinic (fasciculation, weakness, paralysis) and CNS (seizures, coma) effects. - Treatment: decontamination; atropine titrated until secretions dry (chest clear, HR >80, SBP >80); pralidoxime early before "ageing"; benzodiazepines for seizures; ventilation. - Watch for intermediate syndrome at 24–96 h (proximal weakness, respiratory failure).

(b) Digoxin toxicity: see Chapter 11.

(c) Paracetamol poisoning: see Chapter 20. Cover the NAPQI mechanism, the four clinical stages, the nomogram, the NAC regimen, and the King's College criteria for liver transplant (pH <7.3, or INR >6.5 + creatinine >300 + grade III–IV encephalopathy).

3. Discuss the treatment of severe malaria in a 4-year-old Nigerian child.

1. Definition (WHO criteria): impaired consciousness, prostration, convulsions, acidosis, hypoglycaemia, severe anaemia, renal impairment, jaundice, pulmonary oedema, bleeding, shock, hyperparasitaemia >10%.
2. Resuscitation (ABC).
3. **IV artesunate 3 mg/kg** (child <20 kg) at 0, 12 and 24 h, then daily. Complete with oral ACT.
4. Correct hypoglycaemia with 10% dextrose 5 mL/kg. Treat seizures with diazepam or midazolam.
5. Transfuse if Hb <5 g/dL, or <7 g/dL with complications. Manage fluids carefully. Give antibiotics if sepsis is suspected.
6. Monitor glucose, Hb, urine output and level of consciousness (Blantyre Coma Scale).
7. Pre-referral: rectal artesunate.
8. Prevention: ITNs, SMC, R21 vaccine.

4. Classify anticancer drugs. Describe the mechanism and toxicity of methotrexate.

5. Write an essay on the pharmacology of general anaesthetics.

6. Describe the management of drug-susceptible pulmonary TB in an HIV-positive adult.

Section C: Viva voce bank (Pharmacology)

#	Examiner's question	Model answer
1	Define bioavailability.	The fraction of an unchanged drug that reaches the systemic circulation. IV is 100%. It is reduced by first-pass metabolism and poor absorption.

#	Examiner's question	Model answer
2	What is a prodrug? Give three examples.	An inactive drug that is converted in the body into its active form, e.g. enalapril, levodopa, codeine, cyclophosphamide, clopidogrel, azathioprine, proguanil.
3	Name three drugs requiring TDM.	Digoxin, lithium, aminoglycosides, phenytoin, vancomycin, ciclosporin.
4	What is the antidote for warfarin?	Vitamin K (slow); prothrombin complex concentrate or FFP (fast).
5	Antidote for benzodiazepines? Opioids? Organophosphates? Iron? Methanol?	Flumazenil; naloxone; atropine + pralidoxime; desferrioxamine; fomepizole or ethanol.
6	What is the "triple whammy"?	ACEI/ARB + diuretic + NSAID. Together they precipitate AKI.
7	Why is propranolol contraindicated in asthma?	β_2 blockade causes bronchoconstriction.
8	Which antihypertensive is preferred in pregnancy?	Methyldopa, labetalol or nifedipine. ACEIs and ARBs are contraindicated.
9	Which drug for torsades de pointes?	IV magnesium sulfate.
10	What are the four pillars of HFrEF therapy?	ARNI/ACEI/ARB, β-blocker, MRA, SGLT2 inhibitor.
11	Give the first-line regimen for uncomplicated malaria in Nigeria.	Artemether–lumefantrine or artesunate–amodiaquine for 3 days.
12	What is cinchonism?	Quinine/quinidine toxicity: tinnitus, deafness, headache, vertigo, blurred vision, nausea.
13	Why give pyridoxine with isoniazid?	INH binds pyridoxine and increases its excretion, which causes peripheral neuropathy.
14	Mechanism of PPIs?	Irreversible inhibition of the H^+/K^+-ATPase in parietal cells (prodrug activated in acid).
15	What is ORS composition and why glucose?	Low-osmolarity ORS: Na^+ 75, Cl^- 65, glucose 75, K^+ 20, citrate 10 mmol/L; total 245 mOsm/L. Glucose drives Na^+ (and water) absorption through SGLT1.
16	What is MAC?	Minimum alveolar concentration that prevents movement to a skin incision in 50% of patients. An inverse measure of potency.

#	Examiner's question	Model answer
17	Why is suxamethonium dangerous in burns?	Extrajunctional receptor proliferation → massive K^+ efflux → hyperkalaemic cardiac arrest.
18	Drug of choice for onchocerciasis? Schistosomiasis? Fasciola?	Ivermectin; praziquantel; triclabendazole.
19	What is the Mazzotti reaction?	An inflammatory reaction to dying microfilariae (itch, fever, rash, hypotension) after DEC or ivermectin in onchocerciasis.
20	What does an adjuvant do?	Enhances the immune response through a depot effect and activation of innate immunity (TLR, inflammasome), so less antigen or fewer doses are needed.
21	Which vaccines should not be frozen?	Alum-adsorbed vaccines: DTP/pentavalent, HepB, PCV, Td. Do the shake test.
22	Name drugs that cause gynaecomastia.	Spironolactone, cimetidine, ketoconazole, digoxin, oestrogens, GnRH analogues, antiandrogens, cannabis, alcohol (DISCO mnemonic).
23	Which antidiabetic and antihypertensive pair masks hypoglycaemia?	β-blockers with insulin or sulfonylureas: they mask the tremor and tachycardia.
24	Name drugs that are teratogens.	Warfarin, ACEIs, valproate, phenytoin, isotretinoin, thalidomide, methotrexate, tetracyclines, lithium, misoprostol.
25	Why is atracurium suitable in renal failure?	It is eliminated by Hofmann degradation (spontaneous breakdown at body pH and temperature) and ester hydrolysis, independent of the liver and kidney.

Book 2 — Histopathology (General Pathology)

Cell injury → cell death (necrosis, apoptosis) → cell cycle → neoplasia → inflammation and mediators → haemodynamic disorders → environmental and nutritional pathology

Chapter 33 · Cellular Adaptation and Cell Injury

Pathology is the study of disease: its **aetiology** (cause), **pathogenesis** (mechanism), **morphological changes** (gross and microscopic) and **clinical significance**. A cell under stress can **adapt**. If the stress is beyond its capacity, it suffers **reversible injury**. If the stress persists or is severe, it suffers **irreversible injury and cell death**.

Cellular adaptations

Adaptation	Definition	Physiological example	Pathological example
Hypertrophy	↑ cell **size** → ↑ organ size (no new cells). Occurs in non-dividing cells	Uterus in pregnancy, skeletal muscle in body-builders	**LVH in hypertension** and aortic stenosis
Hyperplasia	↑ cell **number** (only in cells that can divide)	Breast at puberty; liver regeneration after partial hepatectomy; endometrium in the proliferative phase	**Endometrial hyperplasia** (unopposed oestrogen → risk of carcinoma), BPH, warts (HPV)
Atrophy	↓ cell size and number; autophagy, ubiquitin–proteasome	Thymus after puberty, post-menopausal uterus	Disuse, denervation, ischaemia, malnutrition (marasmus), pressure, loss of endocrine stimulation, ageing (senile brain atrophy)
Metaplasia	**Reversible replacement of one differentiated cell type by another**. Reprogramming of stem cells	—	**Squamous metaplasia** of bronchial epithelium (smoking), cervix (transformation zone), bladder (*Schistosoma haematobium* → **squamous cell carcinoma**), vitamin A deficiency. **Columnar (intestinal) metaplasia** of the oesophagus (**Barrett's**, from GORD → adenocarcinoma). Osseous metaplasia (myositis ossificans)

Dysplasia is *disordered growth*. It is not a true adaptation. Features: loss of uniformity and architectural orientation, **pleomorphism, hyperchromatism, ↑ N:C ratio, ↑ mitoses**. It is a **pre-neoplastic** change (e.g. CIN). **Carcinoma in situ** means full-thickness dysplasia with an intact basement membrane.

Causes of cell injury

1. **Hypoxia** (the most common cause): ischaemia, hypoxaemia, anaemia, CO poisoning.
2. Physical agents: trauma, heat and cold, radiation, electricity.
3. Chemicals and drugs: CCl_4, paracetamol, alcohol.
4. Infectious agents.
5. Immunological reactions: autoimmunity, hypersensitivity.
6. Genetic defects.
7. Nutritional imbalance.
8. Ageing.

Mechanisms of cell injury

Mechanism	Details
ATP depletion	↓ oxidative phosphorylation, then: **Na^+/K^+-ATPase failure** → Na^+ and water enter → **cell swelling**, ER dilation. **Anaerobic glycolysis** → lactic acid → ↓ pH → **chromatin clumping**, ↓ glycogen. Ca^{2+} pump failure. Ribosomes detach → ↓ protein synthesis
Mitochondrial damage	Mitochondrial permeability transition pore opens → loss of membrane potential; cytochrome c leaks → **apoptosis**
Influx of Ca^{2+}	Activates phospholipases (membrane damage), proteases, endonucleases (nuclear damage) and ATPases (more ATP loss)
Free radicals / reactive oxygen species ($O_2\bullet^-$, H_2O_2, •OH via the **Fenton reaction** $Fe^{2+} + H_2O_2 \rightarrow$ •OH)	**Lipid peroxidation** of membranes, protein oxidation, DNA damage. Sources: radiation, inflammation (respiratory burst), reperfusion, drugs (paracetamol, CCl_4), transition metals. **Defences:** SOD, catalase, glutathione peroxidase, vitamins E, C and A, ferritin and caeruloplasmin
Membrane damage	Lysosomal enzymes leak → digestion. Plasma membrane damage → **leakage of enzymes, which is the basis of diagnostic markers** (troponin, CK-MB, ALT, AST)
DNA and protein damage	Misfolded proteins → **ER stress (unfolded protein response)** → apoptosis

Ischaemia–reperfusion injury: restoring blood flow can make injury worse through ROS, Ca^{2+} overload, neutrophil influx and complement activation.

Morphology of reversible vs irreversible injury

Reversible	Irreversible (point of no return)
Cellular swelling (hydropic / vacuolar change): the earliest change, seen in all injury	**Severe mitochondrial damage** (amorphous densities)
Fatty change (steatosis): liver, heart	**Plasma membrane rupture**
Membrane blebbing, loss of microvilli, myelin figures	**Lysosomal rupture**
Mitochondrial swelling, ER dilation, ribosome detachment	**Nuclear changes**: pyknosis → karyorrhexis → karyolysis
Chromatin clumping	

The two hallmarks of irreversibility are **inability to restore mitochondrial function** and **profound membrane damage**.

Intracellular accumulations and pigments

- **Steatosis (fatty liver):** alcohol, obesity and metabolic syndrome (NAFLD/MASLD), malnutrition (**kwashiorkor**: ↓ apoprotein), toxins, diabetes.
- **Cholesterol:** atherosclerosis (foam cells), xanthomas.
- **Proteins:** Russell bodies (plasma cells), Mallory–Denk hyaline (alcohol).
- **Glycogen:** diabetes, glycogen storage diseases.
- **Pigments:**
  - Exogenous: **anthracosis** (carbon), tattoo.
  - Endogenous: **lipofuscin** ("wear-and-tear" pigment; *brown atrophy*), **melanin**, **haemosiderin** (Prussian blue positive; haemorrhage, **haemochromatosis**), bilirubin.

Pathological calcification

	Dystrophic	Metastatic
Serum Ca^{2+}	**Normal**	**Raised (hypercalcaemia)**
Site	**Dead or degenerated tissue**	**Normal tissues** (interstitial tissue of lungs, kidneys (nephrocalcinosis), blood vessels, gastric mucosa)
Causes	Atheroma, damaged heart valves, TB caseation, fat necrosis, **psammoma bodies** (papillary thyroid carcinoma, serous ovarian cystadenocarcinoma, meningioma, mesothelioma)	Hyperparathyroidism, bone destruction (metastases, myeloma), vitamin D intoxication, sarcoidosis, **renal failure** (secondary hyperparathyroidism), milk-alkali syndrome

★ Exam pearls

- Metaplasia is reversible. Dysplasia may regress but is pre-malignant.
- The **earliest change** in reversible injury is **cell swelling**.
- The **best marker of irreversibility** is mitochondrial amorphous densities (on electron microscopy).

Chapter 34 · Necrosis

Necrosis is cell death after irreversible injury, marked by **loss of membrane integrity, enzymatic digestion of the cell (autolysis and heterolysis) and leakage of contents, which provokes inflammation**. It is always **pathological**.

Microscopy:

- **Cytoplasm:** ↑ **eosinophilia** (denatured proteins, loss of RNA), glassy appearance, vacuolation.
- **Nuclear changes:**
  - **Pyknosis:** shrinkage and ↑ basophilia.
  - **Karyorrhexis:** fragmentation.
  - **Karyolysis:** fading due to DNase.

Patterns of necrosis

Pattern	Mechanism / description	Examples
Coagulative	Protein **denaturation** predominates. **Tissue architecture is preserved** for days ("ghost cells")	**Ischaemic infarcts of all solid organs except the brain** (heart, kidney, spleen)
Liquefactive	**Enzymatic digestion** predominates. The tissue becomes a liquid, viscous mass	**Brain infarcts**; **abscesses** (bacterial/fungal infections; pus)
Caseous	"Cheese-like", friable, white. **Amorphous granular eosinophilic debris surrounded by a granuloma**, with architecture lost	**Tuberculosis**, histoplasmosis
Fat necrosis	Released **lipases** digest fat → fatty acids + Ca^{2+} → **saponification** (chalky white spots)	**Acute pancreatitis**. Traumatic fat necrosis of the **breast** (mimics carcinoma)
Fibrinoid	Immune complexes + fibrin deposited in vessel walls (bright pink, "smudgy")	**Vasculitis** (PAN), **malignant hypertension**, rheumatic fever (Aschoff bodies), pre-eclampsia (placenta)
Gangrenous (a clinical term)	Usually coagulative necrosis of a **limb** or bowel. *Dry gangrene*: coagulative, arterial occlusion. *Wet gangrene*: + bacterial infection → liquefactive. *Gas gangrene*: ***Clostridium perfringens*** (α-toxin, lecithinase)	Diabetic foot, peripheral arterial disease, strangulated bowel, frostbite

Feature	Dry gangrene	Wet gangrene
Cause	Arterial occlusion	Venous + arterial occlusion; infection
Site	Limbs	Bowel, lung, limbs (diabetes)
Line of demarcation	**Clear**	Poor
Spread	Slow	Rapid; septicaemia
Prognosis	Better	Poor

Fate of necrotic tissue:

- Phagocytosis and removal.
- Repair by fibrosis.
- **Dystrophic calcification**.
- Liquefaction and cyst formation (brain).

> **★ Exam pearls**
>
> The brain undergoes **liquefactive** necrosis after infarction because it is rich in lipids and enzymes and has little connective tissue. All other solid organs show coagulative necrosis.

Chapter 35 · Apoptosis

Apoptosis (Greek: "falling off") is **programmed cell death**. It is regulated, energy-dependent suicide in which the cell **activates caspases**, degrades its own DNA and proteins, and breaks up into **membrane-bound apoptotic bodies**. Phagocytes clear these **without inflammation**.

Causes

Physiological	Pathological
Embryogenesis (interdigital webbing, organogenesis)	DNA damage (radiation, cytotoxics), via p53
Hormone-dependent involution (endometrium at menstruation, breast after weaning, prostate after castration)	Misfolded proteins (ER stress): neurodegenerative diseases
Cell turnover (intestinal crypts)	**Viral infections**: **Councilman bodies** in viral hepatitis and yellow fever
Negative selection of self-reactive lymphocytes in the thymus	Killing by cytotoxic T cells (tumours, transplant rejection)
Death of neutrophils after inflammation	Atrophy after duct obstruction (pancreas, parotid)

Mechanisms

1. **Intrinsic (mitochondrial) pathway** (most common):
   - Cell stress (DNA damage, growth factor withdrawal) activates **BH3-only sensors** (Bim, Bid, Bad, **PUMA, NOXA**; p53 induces these).
   - These activate **Bax/Bak** and neutralise anti-apoptotic **Bcl-2/Bcl-xL/Mcl-1**.
   - Mitochondrial outer membrane permeabilisation → **cytochrome c** is released.
   - Cytochrome c + **Apaf-1** form the **apoptosome** → **caspase-9** (initiator).
   - Smac/DIABLO neutralises IAPs.
2. **Extrinsic (death receptor) pathway:**
   - **Fas ligand binds Fas (CD95)**, or TNF binds TNFR1.
   - This recruits **FADD** (death domain), forming the **DISC**.
   - **Caspase-8** is activated (initiator). It can also cleave Bid and so recruit the intrinsic pathway.
   - FLIP inhibits caspase-8.
3. **Cytotoxic T-cell pathway: perforin + granzyme B** directly activate caspases.
4. **Execution phase:**
   - **Executioner caspases 3, 6, 7** cleave nuclear proteins and the cytoskeleton.

- They activate **CAD (caspase-activated DNase) → internucleosomal DNA cleavage into 180–200 bp fragments** → "**DNA ladder**" on electrophoresis.

5. **Removal:** the dying cell flips **phosphatidylserine** to the outer leaflet (detected by annexin V binding). Thrombospondin and C1q also act as "eat-me" signals. Macrophages engulf the bodies without inflammation (they release anti-inflammatory TGF-β and IL-10).

Necrosis vs apoptosis

Feature	Necrosis	Apoptosis
Cell size	**Enlarged (swelling)**	**Reduced (shrinkage)**
Nucleus	Pyknosis → karyorrhexis → karyolysis	**Fragmentation into nucleosome-sized pieces** (DNA ladder)
Plasma membrane	**Disrupted**	**Intact**, with altered structure (PS exposure)
Cellular contents	Enzymatic digestion; may leak	**Intact, packaged in apoptotic bodies**
Inflammation	**Frequent**	**None**
Energy (ATP)	Not required	**Required**
Extent	Contiguous groups of cells	**Single cells** or small clusters
Role	Always pathological	Often physiological; may be pathological

Other forms of cell death

- **Necroptosis:** programmed necrosis via **RIPK1/RIPK3/MLKL**, caspase-independent (e.g. some viral infections).
- **Pyroptosis:** inflammasome → **caspase-1** → IL-1β + gasdermin pores (infection).
- **Ferroptosis:** iron-dependent lipid peroxidation.
- **Autophagy:** "self-eating" during nutrient deprivation (LC3, Beclin-1).

Clinical relevance

Disorder	Apoptosis	Examples
Too little apoptosis	↓	**Cancer**: **Bcl-2 overexpression from t(14;18) in follicular lymphoma**; **p53 mutation** (>50% of cancers). **Autoimmunity**: Fas mutation → autoimmune lymphoproliferative syndrome (ALPS)

Disorder	Apoptosis	Examples
Too much apoptosis	↑	Neurodegenerative diseases (Alzheimer's, Parkinson's, ALS), ischaemic injury, **$CD4^+$ T-cell loss in HIV**, viral hepatitis, GVHD

Therapeutic use: venetoclax (BH3 mimetic that inhibits Bcl-2) for CLL and AML.

> 🎤 **Viva corner**
>
> **Q:** What is the guardian of the genome and how does it work? **A: p53** (TP53 on chromosome 17p13). When DNA is damaged, it accumulates and stops the cell at **G_1/S** by inducing **p21**, which inhibits CDK. It then either activates DNA repair (GADD45) or, if repair fails, triggers **apoptosis** (by inducing BAX, PUMA and NOXA) or senescence.

Chapter 36 · The Cell Cycle and its Regulation

Phases

Phase	Events	Duration (typical)
G_1 (gap 1)	Cell growth; protein and RNA synthesis. **Restriction point (R)**: commitment to divide	Variable (hours to years)
S (synthesis)	**DNA replication** (2n → 4n), centrosome duplication	~6–8 h
G_2 (gap 2)	Preparation for mitosis; DNA damage check	~3–4 h
M (mitosis)	Prophase, prometaphase, metaphase, anaphase, telophase + cytokinesis	~1 h
G_0	Quiescent (resting) state outside the cycle	—

Cells classified by proliferative capacity

Type	Description	Examples
Labile (continuously dividing)	Constantly renewing; stem cells	**Surface epithelia** (skin, GI mucosa, oral cavity, cervix, urinary tract), **haemopoietic cells**
Stable (quiescent)	In G_0; can re-enter the cycle on stimulation	**Liver, kidney, pancreas**, fibroblasts, smooth muscle, endothelium
Permanent (non-dividing)	Have left the cycle; injury → scar	**Neurons, cardiac muscle**, (skeletal muscle has limited regeneration via satellite cells)

Regulators

- **Cyclins** (D, E, A, B) bind and activate **cyclin-dependent kinases (CDKs)**:

Cyclin–CDK complex	Transition / phase
Cyclin D–CDK4/6	G_1 progression
Cyclin E–CDK2	G_1/S

Cyclin–CDK complex	Transition / phase
Cyclin A–CDK2	S
Cyclin B–CDK1	G_2/M (MPF, maturation-promoting factor)

- **The RB "switch":**
  - Hypophosphorylated **RB** binds **E2F** and blocks the cycle.
  - Cyclin D–CDK4/6 phosphorylates RB, which releases E2F.
  - E2F transcribes the genes for S phase.
- **CDK inhibitors (CKIs):**
  - **INK4 family:** p16^INK4a, p15, p18, p19. These inhibit CDK4/6.
  - **CIP/KIP family: p21** (induced by p53), **p27**, p57. These inhibit all CDKs.
- **Checkpoints:**

Checkpoint	Controlled by	Checks
G_1/S	p53, RB	DNA damage before replication
Intra-S		Replication stress
G_2/M	ATM/ATR → Chk1/Chk2 → CDC25 inhibition	DNA damage before mitosis
Spindle assembly (metaphase/anaphase)	MAD2, BUB1	Proper chromosome attachment

- **Telomeres and telomerase:** telomeres shorten with each division, leading to senescence (the Hayflick limit). Cancer cells and stem cells **reactivate telomerase**.

Relevance

- **Oncogenesis:** mutations in cyclin D1 (mantle cell lymphoma, t(11;14)), CDK4 amplification, loss of p16 (melanoma, pancreatic cancer), RB loss (retinoblastoma), p53 loss.
- **HPV:** E6 degrades p53 and E7 binds RB, which explains cervical cancer.
- **Drug targets:** palbociclib and ribociclib (CDK4/6 inhibitors in breast cancer); cell-cycle-specific chemotherapy (Chapter 30).
- **Regeneration and repair:** healing by *regeneration* (labile/stable cells with intact stroma) vs *scar formation* (permanent cells or a damaged ECM framework).
- **Healing of wounds:**
  - *Primary intention:* clean, closely apposed edges.
  - *Secondary intention:* large defect, more granulation tissue, **wound contraction by myofibroblasts**.
  - **Granulation tissue** = new capillaries (angiogenesis via VEGF) + fibroblasts + inflammatory cells.
  - **Remodelling:** collagen type III is replaced by type I. Scar strength reaches ~70–80% of normal by 3 months.

- **Factors that impair healing:** infection (the most important), diabetes, malnutrition (vitamin C, zinc, protein), steroids, poor perfusion, foreign bodies, mechanical stress, radiation.
- **Complications: keloid** (excess collagen extending *beyond* the wound margin; **common in people of African descent**), hypertrophic scar (confined to the margin), dehiscence, contracture, Marjolin ulcer (SCC in a chronic scar).

Chapter 37 · Neoplasia

Neoplasia ("new growth") is the abnormal, autonomous, purposeless proliferation of cells that continues after the stimulus has stopped, due to **heritable genetic alterations** (Willis' definition, updated). A **tumour** is a neoplastic mass. **Oncology** is the study of tumours.

Nomenclature

Tissue	Benign	Malignant
Epithelium: glandular	**Adenoma**, cystadenoma	**Adenocarcinoma**
Epithelium: surface (squamous, transitional)	**Papilloma**	**Squamous cell / urothelial carcinoma**
Mesenchyme	Fibroma, lipoma, leiomyoma (fibroids; common in Black women), chondroma, osteoma, haemangioma	**Sarcomas**: fibrosarcoma, liposarcoma, leiomyosarcoma, chondrosarcoma, **osteosarcoma**, angiosarcoma, **Kaposi sarcoma (HHV-8)**
Haemopoietic	—	Leukaemias, lymphomas (**Burkitt**), myeloma
Melanocytes	Naevus	**Melanoma** (acral lentiginous type common in Africans)
Germ cells	Mature cystic teratoma (dermoid)	Seminoma, choriocarcinoma, immature teratoma
Embryonic (blastomas)	—	Nephroblastoma (Wilms), neuroblastoma, retinoblastoma, hepatoblastoma

Misnomers: lympho*ma*, melano*ma*, seminoma and hepatoma (old term) are all malignant. **Hamartoma** is a disorganised overgrowth of tissue native to the site (pulmonary hamartoma). **Choristoma (heterotopia)** is normal tissue in an abnormal place (pancreatic rest).

Benign vs malignant

Feature	Benign	Malignant
Differentiation	Well differentiated	Ranges from well to poorly differentiated; **anaplasia** (lack of differentiation)
Rate of growth	Slow	Usually rapid; mitoses frequent and atypical

Feature	Benign	Malignant
Local invasion	Expansile, **capsulated**, well demarcated	**Infiltrative**, non-capsulated, invades
Metastasis	**Absent**	**Present: the single most reliable feature of malignancy**
Recurrence after excision	Rare	Common
Effect on host	Pressure, hormone secretion	Cachexia, death

Anaplasia features: pleomorphism, abnormal nuclear morphology (hyperchromasia, ↑ N:C ratio up to 1:1, prominent nucleoli), atypical mitoses (tripolar), tumour giant cells, **loss of polarity**.

Routes of spread

Route	Typical tumours
Direct invasion	All malignant tumours
Lymphatic	**Carcinomas** (sentinel lymph node; Virchow's node = left supraclavicular)
Haematogenous	**Sarcomas**, but also renal cell carcinoma (renal vein), HCC (hepatic vein), follicular thyroid carcinoma, choriocarcinoma. **Liver and lungs are the commonest sites**
Transcoelomic (seeding of body cavities)	Ovarian carcinoma (peritoneum), gastric carcinoma (**Krukenberg tumour** of the ovaries), pseudomyxoma peritonei
Implantation / iatrogenic	Needle tracts, surgical wounds

Bone metastases:

- **Lytic:** breast, lung, kidney, thyroid, myeloma.
- **Sclerotic: prostate**, some breast cancers.
- Mnemonic for the primaries that go to bone: "**BLT with a Kosher Pickle**" = Breast, Lung, Thyroid, Kidney, Prostate.

Epidemiology (relevant to Nigeria and Africa)

- **Commonest cancers in Nigeria:**
  - Women: **breast** and **cervix**.
  - Men: **prostate**, liver (HCC from **HBV** and **aflatoxin**), colorectal.
  - Children: **Burkitt lymphoma** (EBV + malaria), retinoblastoma, Wilms, leukaemia.
  - **Kaposi sarcoma** (HIV/HHV-8).
  - **Bladder SCC** (schistosomiasis, northern Nigeria).

- **Precancerous conditions:**
  - Cirrhosis
  - Ulcerative colitis
  - Barrett's oesophagus
  - Leukoplakia
  - Actinic keratosis
  - Chronic scars and sinuses (Marjolin ulcer)
  - Familial adenomatous polyposis
  - Xeroderma pigmentosum
  - Albinism (skin SCC, common in Nigerian people with albinism)

Molecular basis: the hallmarks of cancer (Hanahan & Weinberg, 2000, 2011, 2022)

1. **Self-sufficiency in growth signals (oncogenes):** RAS (point mutations; ~30% of cancers), **MYC** (Burkitt t(8;14)), **ERBB2/HER2** amplification (breast), **BCR-ABL** (CML, t(9;22) Philadelphia chromosome), EGFR, BRAF V600E, KIT, cyclin D.
2. **Insensitivity to growth inhibition (tumour suppressor genes): RB** (Knudson's **two-hit hypothesis**: retinoblastoma), **TP53** (Li-Fraumeni syndrome), APC (FAP), **BRCA1/2** (breast and ovarian), NF1, WT1, VHL, PTEN, CDKN2A (p16).
3. **Evasion of apoptosis:** Bcl-2, p53.
4. **Limitless replicative potential:** telomerase.
5. **Sustained angiogenesis:** VEGF, HIF-1α.
6. **Invasion and metastasis:**
   - Loss of **E-cadherin**.
   - Degradation of the basement membrane by **matrix metalloproteinases (MMPs)**.
   - Attachment to ECM (laminin, fibronectin), migration, intravasation, **EMT** (epithelial–mesenchymal transition).
   - "Seed and soil" theory (Paget).
7. **Reprogramming of energy metabolism:** the **Warburg effect** (aerobic glycolysis). This is the basis of FDG-PET imaging.
8. **Evasion of immune destruction:** PD-L1, CTLA-4.
9. **Enabling characteristics: genomic instability** (DNA repair defects: HNPCC/Lynch syndrome, mismatch repair; xeroderma pigmentosum, nucleotide excision repair; BRCA, homologous recombination) and **tumour-promoting inflammation**.

Carcinogens

Type	Agent	Cancer
Chemical: direct-acting	Alkylating agents (cyclophosphamide)	Leukaemia

Type	Agent	Cancer
Chemical: indirect-acting (procarcinogens)	Polycyclic aromatic hydrocarbons (tobacco smoke)	Lung, oral, bladder cancer
	Aflatoxin B1 (*Aspergillus flavus* on poorly stored groundnuts, maize and grains)	**HCC (TP53 codon 249 mutation)**
	Aromatic amines (β-naphthylamine; dyes, rubber)	**Bladder**
	Nitrosamines (smoked and preserved foods)	Gastric, oesophageal
	Asbestos	Mesothelioma, lung
	Vinyl chloride	Hepatic angiosarcoma
	Arsenic	Skin, lung, angiosarcoma
	Benzene	AML
Radiation	UV-B (**pyrimidine (thymine) dimers**)	Skin: BCC, SCC, melanoma
	Ionising radiation	Leukaemia, thyroid, breast, lung
Microbial	**HPV 16, 18** (E6/E7)	Cervical, anal, oropharyngeal
	EBV	**Burkitt lymphoma**, **nasopharyngeal carcinoma**, Hodgkin lymphoma
	HBV, HCV	**HCC**
	HHV-8	**Kaposi sarcoma**, primary effusion lymphoma
	HTLV-1	Adult T-cell leukaemia/lymphoma
	Helicobacter pylori	Gastric adenocarcinoma, MALT lymphoma
	Schistosoma haematobium	**SCC of the bladder**
	Clonorchis sinensis / Opisthorchis	Cholangiocarcinoma

Multistep carcinogenesis:

1. **Initiation:** irreversible DNA mutation.
2. **Promotion:** reversible proliferation stimulated by promoters (phorbol esters, hormones).
3. **Progression:** further mutations.

Example: **adenoma–carcinoma sequence** in colon (APC → KRAS → SMAD4 → TP53).

Host defence: tumour immunity

- **Tumour antigens:** neoantigens from mutations, overexpressed proteins (HER2), oncofetal antigens (AFP, CEA), viral antigens.
- **Effectors:** $CD8^+$ CTLs, NK cells, macrophages.
- **Immune evasion:** ↓ MHC-I, PD-L1 expression, Tregs, TGF-β.

Clinical features of tumours

- **Local effects:** pressure, obstruction, ulceration, bleeding.
- **Hormone production:** insulinoma, phaeochromocytoma.
- **Cancer cachexia:** TNF, IL-6, proteolysis-inducing factor.
- **Paraneoplastic syndromes:**

Syndrome	Associated tumour
Hypercalcaemia (PTHrP)	Squamous cell carcinoma of lung, breast, renal cancer
Cushing syndrome (ectopic ACTH)	**Small cell lung carcinoma**
SIADH	Small cell lung carcinoma
Polycythaemia (EPO)	RCC, HCC, cerebellar haemangioblastoma
Lambert–Eaton syndrome	Small cell lung carcinoma
Trousseau's sign (migratory thrombophlebitis)	Pancreatic adenocarcinoma
Acanthosis nigricans	Gastric adenocarcinoma
Dermatomyositis	Various
Hypertrophic pulmonary osteoarthropathy (clubbing)	Lung cancer
Non-bacterial thrombotic endocarditis	Various

Grading and staging

- **Grading:** the degree of **differentiation**, from histology (G1 well → G3/4 poorly differentiated or anaplastic). Examples: Gleason/ISUP for prostate, Nottingham for breast.
- **Staging:** the **extent of spread**, using **TNM** (AJCC/UICC 8th ed.): T = primary tumour size and extent; N = regional nodes; M = distant metastasis. Others: FIGO (gynaecological), Ann Arbor/Lugano (lymphoma), Dukes (historical, colorectal).
- **Staging has greater prognostic value than grading.**

Laboratory diagnosis

- **Histopathology:** biopsy (incisional, excisional, core), frozen section (intra-operative).
- **Cytology: FNAC**, exfoliative cytology (**Pap smear**; Papanicolaou).
- **Immunohistochemistry** to identify the tumour type:

Marker	Identifies
Cytokeratin	Carcinoma
Vimentin	Sarcoma
S-100, HMB-45, Melan-A	Melanoma
CD45 (LCA)	Lymphoma
Desmin	Muscle
GFAP	Glial cells
ER/PR/HER2	Breast (prognostic and predictive)
Ki-67	Proliferation

- **Molecular diagnostics:** FISH, PCR, NGS (BCR-ABL, EGFR, KRAS, BRCA).
- **Tumour markers** (mainly for monitoring, not screening):

Marker	Tumour
AFP	HCC, yolk sac tumour
β-hCG	Choriocarcinoma, molar pregnancy, germ cell tumours
PSA	Prostate
CEA	Colorectal
CA-125	Ovary
CA 19-9	Pancreas
CA 15-3	Breast
Calcitonin	Medullary thyroid carcinoma
Thyroglobulin	Differentiated thyroid carcinoma
Paraprotein	Myeloma
LDH	Lymphoma, germ cell tumours

🎤 Viva corner

Q: What is the single most reliable feature that distinguishes a malignant tumour? **A:** Metastasis. (Basal cell carcinoma and gliomas rarely metastasise, but are still malignant because they invade.)

Q: Explain Knudson's two-hit hypothesis. **A:** Both alleles of a tumour suppressor gene must be inactivated before cancer develops. In *familial* retinoblastoma, the first hit is inherited in the germline, so a single somatic hit is enough: tumours are early, bilateral and multifocal. In *sporadic* cases, both hits are somatic, so the tumour is later and unilateral.

Chapter 38 · Inflammation

Inflammation is a protective response of vascularised tissue to infection or tissue injury. It brings cells and molecules of host defence from the circulation to where they are needed in order to eliminate the offending agent, clear necrotic tissue and start repair. The suffix "**-itis**" denotes it. It is beneficial but can cause harm (anaphylaxis, RA, atherosclerosis).

Cardinal signs:

- **Rubor** (redness), **calor** (heat), **tumor** (swelling), **dolor** (pain): Celsus.
- **Functio laesa** (loss of function): Virchow.

Acute inflammation

Stimuli: infections (via PAMPs on TLRs), tissue necrosis (via DAMPs: urate, ATP, HMGB1; NLRP3 inflammasome), foreign bodies, immune reactions.

1. Vascular changes

- **Transient vasoconstriction** (seconds).
- **Vasodilation** (histamine, NO, PGs) → ↑ blood flow → heat and redness.
- ↑ **Vascular permeability** → protein-rich **exudate** → swelling. Mechanisms:
  - **Endothelial cell contraction** in **post-capillary venules** (histamine, bradykinin, leukotrienes). This is the *immediate transient response* (15–30 min) and the most common mechanism.
  - Endothelial injury (burns, toxins): *immediate sustained* or *delayed prolonged* (sunburn).
  - Leucocyte-mediated injury.
  - Transcytosis (VEGF).
  - Leakage from new vessels (angiogenesis).
- **Stasis:** ↑ viscosity → leucocytes marginate.
- **Lymphatics:** increased flow drains the oedema. They can become inflamed themselves (**lymphangitis**, lymphadenitis).

Fluid	Protein	Specific gravity	Cells	Cause
Exudate	**High (>30 g/L)**	**>1.020**	Many	**Inflammation** (↑ permeability)
Transudate	Low (<30 g/L)	<1.012	Few	↑ hydrostatic or ↓ oncotic pressure (HF, nephrotic syndrome, cirrhosis)

(**Light's criteria** for pleural fluid: protein ratio >0.5, LDH ratio >0.6, or LDH >2/3 of the upper limit of normal → exudate.)

2. Cellular events (neutrophils dominate at 6–24 h; monocytes/macrophages at 24–48 h)

Step	Molecules
Margination and rolling	**Selectins**: E-selectin (endothelium), P-selectin (endothelium and platelets, from Weibel–Palade bodies), L-selectin (leucocytes) bind sialyl-Lewis X
Firm adhesion	**Integrins** on leucocytes (**LFA-1**/Mac-1 bind **ICAM-1**; VLA-4 binds VCAM-1), activated by chemokines. Endothelial ICAM/VCAM are induced by TNF and IL-1
Transmigration (diapedesis)	**PECAM-1 (CD31)**, mainly in post-capillary venules. Collagenases break down the basement membrane
Chemotaxis	Along a chemical gradient. **Exogenous:** bacterial N-formyl-methionine peptides. **Endogenous: C5a, LTB_4, IL-8 (CXCL8)**, kallikrein
Recognition and phagocytosis	**Opsonins** (IgG Fc, **C3b**, collectins/MBL). Steps: engulfment (pseudopods) → phagosome → phagolysosome
Killing	**O_2-dependent: NADPH oxidase (phagocyte oxidase)** "respiratory burst" → $O_2^{\bullet -}$ → H_2O_2 → **myeloperoxidase (MPO)** + Cl^- → **HOCl** (bleach): the most potent bactericidal system. **NO** (iNOS). **O_2-independent:** lysozyme, lactoferrin, major basic protein (eosinophils), defensins, BPI. **NETs** (neutrophil extracellular traps)

Defects in leucocyte function:

Disorder	Defect	Result
LAD-1	CD18 (β_2 integrin) defect	Delayed umbilical cord separation, recurrent infections with **no pus**
LAD-2	Fucosyl transferase (sialyl-Lewis X)	Infections, intellectual disability
Chédiak–Higashi	LYST gene: defective phagolysosome fusion	Giant granules, partial albinism, neuropathy
Chronic granulomatous disease (CGD)	NADPH oxidase deficiency (X-linked)	Infections with **catalase-positive** organisms (*S. aureus*, *Aspergillus*, *Serratia*, *Nocardia*, *Burkholderia*). Diagnose with **NBT test** (negative) or **DHR flow cytometry**
MPO deficiency		Mostly asymptomatic; *Candida* infection
Acquired	Diabetes, steroids, chemotherapy, malnutrition	

3. Outcomes of acute inflammation

1. **Complete resolution** (mild injury, short duration, regenerating tissue): lipoxins, resolvins and protectins help switch inflammation off.
2. **Healing by fibrosis (scarring):** extensive destruction, or permanent cells.
3. **Abscess formation:** pyogenic organisms (S. *aureus*).
4. **Progression to chronic inflammation.**

Morphological patterns

Pattern	Description	Example
Serous	Watery, protein-poor fluid	Skin blister (burns), effusions (pleural, peritoneal)
Fibrinous	Large vascular leak → **fibrin** deposition	**Fibrinous pericarditis** ("bread and butter"), lobar pneumonia; organisation → adhesions
Suppurative (purulent)	**Pus** (neutrophils, necrotic debris, oedema); **abscess**	Staphylococcal abscess, appendicitis, empyema
Ulceration	Local loss of surface epithelium	Peptic ulcer, aphthous ulcer, diabetic ulcer
Pseudomembranous	Necrotic epithelium + fibrin + exudate forming a membrane	***C. difficile* colitis**, **diphtheria**
Haemorrhagic / catarrhal		Plague, influenza (haemorrhagic); common cold (catarrhal)

Systemic effects (the acute phase response):

- **Fever:** IL-1, TNF, IL-6 → **PGE_2** in the hypothalamus.
- **Acute phase proteins** (IL-6 stimulates the liver):
  - **CRP**
  - Fibrinogen → ↑ **ESR** (rouleaux)
  - Serum amyloid A
  - Ferritin
  - **Hepcidin** → anaemia of chronic disease
  - ↓ Albumin and transferrin (negative acute phase proteins)
- **Leucocytosis:**
  - Neutrophilia: bacterial infection (**left shift**).
  - Lymphocytosis: viral infection.
  - Eosinophilia: parasites, allergy.
  - Leucopenia: typhoid, some viruses.
- Tachycardia, rigors, anorexia, somnolence.
- **Sepsis / septic shock:** high cytokine levels → DIC, hypotension, metabolic disturbance.

Chronic inflammation

Chronic inflammation lasts weeks to months. **Active inflammation, tissue destruction and attempts at repair (fibrosis, angiogenesis) happen at the same time.** The infiltrate is mainly **mononuclear**: **macrophages (the dominant cell)**, lymphocytes and plasma cells. Eosinophils are prominent in parasitic infection and allergy.

Causes:

- Persistent infections: TB, syphilis, leprosy, fungi, viruses (HBV), parasites (schistosomiasis).
- Prolonged exposure to toxic agents:
  - Exogenous: **silica** (silicosis), asbestos.
  - Endogenous: lipids (atherosclerosis).
- Autoimmunity: RA, SLE, IBD.
- Unresolved acute inflammation (chronic osteomyelitis, chronic abscess).

Macrophages:

- Blood monocytes → tissue macrophages: Kupffer cells, alveolar macrophages, microglia, osteoclasts, histiocytes.
- **M1 (classically activated):** IFN-γ, microbicidal.
- **M2 (alternatively activated):** IL-4/IL-13, repair and fibrosis (TGF-β, PDGF).

Granulomatous inflammation

A **granuloma** is a distinctive form of chronic inflammation. It is a microscopic aggregate of **activated macrophages transformed into epithelioid cells**, surrounded by a collar of mononuclear leucocytes (lymphocytes and occasional plasma cells). Older granulomas have a rim of fibroblasts. Epithelioid cells often fuse to form **multinucleated giant cells**:

- **Langhans type:** nuclei arranged in a peripheral horseshoe (TB).
- **Foreign-body type:** nuclei scattered.

Mechanism: a Th1 (type IV hypersensitivity) response. Macrophages present antigen; $CD4^+$ Th1 cells release **IFN-γ**, which activates macrophages into epithelioid cells. TNF maintains the granuloma (**so anti-TNF therapy reactivates TB**).

Type	Examples
Immune (caseating)	**Tuberculosis** (caseous centre)

Type	Examples
Immune (non-caseating)	**Sarcoidosis**, **leprosy** (tuberculoid), **Crohn's disease**, cat-scratch disease (stellate, necrotising), **schistosomiasis** (around eggs), **syphilis** (gumma), fungal infections (histoplasmosis), brucellosis
Foreign-body	Sutures, talc (IV drug users), silica, beryllium (berylliosis), splinters

Repair (tissue regeneration and fibrosis): see Chapter 36

🎤 Viva corner

Q: What is the main cell type of acute inflammation, and of chronic inflammation? **A:** Neutrophils in acute inflammation. Macrophages, along with lymphocytes and plasma cells, in chronic inflammation.

Q: What is an abscess? **A:** A localised collection of pus (neutrophils, necrotic tissue and bacteria) within a cavity formed by tissue destruction. It is surrounded by a pyogenic membrane (granulation tissue) and later a fibrous wall.

Chapter 39 · Chemical Mediators of Inflammation

General principles

Mediators are either **cell-derived** (preformed in granules, or synthesised *de novo*) or **plasma protein-derived** (circulating as inactive precursors and activated by proteolysis, mainly made in the liver). Most act on specific receptors. They are short-lived, and one mediator can trigger the release of others (amplification).

Cell-derived mediators

Mediator	Source	Main actions
Histamine (preformed)	**Mast cells** (the richest source), basophils, platelets	**Arteriolar dilation, ↑ venular permeability (immediate transient phase)**, bronchoconstriction (H_1). Released by: IgE cross-linking, C3a/C5a (**anaphylatoxins**), physical injury, cytokines, neuropeptides
Serotonin (5-HT) (preformed)	Platelets, enterochromaffin cells	Vasoconstriction; ↑ permeability (rodents)
Arachidonic acid metabolites (eicosanoids) (synthesised)	Most cells, via **PLA_2**	See the next table
Platelet-activating factor (PAF)	Leucocytes, endothelium	Platelet aggregation, vasoconstriction, bronchoconstriction; at low dose ↑ permeability (100–10,000× histamine) and vasodilation; leucocyte adhesion
Reactive oxygen species	Neutrophils, macrophages	Microbial killing; tissue damage
Nitric oxide (NO)	Endothelium (eNOS), macrophages (**iNOS**, induced by IFN-γ and TNF)	**Vasodilation**, ↓ platelet and leucocyte adhesion, microbicidal (with ROS: peroxynitrite)
Cytokines	Macrophages, lymphocytes, endothelium, mast cells	See the cytokines table below
Chemokines (~40)		**Chemotaxis**, integrin activation. **CXC (α): IL-8 → neutrophils. CC (β):** MCP-1 (monocytes), eotaxin (eosinophils), RANTES. C (γ): lymphotactin. **CX3C:** fractalkine

Mediator	Source	Main actions
Lysosomal enzymes	Neutrophils, monocytes	Tissue damage (elastase, collagenase); counteracted by **α_1-antitrypsin** (deficiency → **emphysema**)
Neuropeptides	Sensory nerves	**Substance P**: pain, ↑ permeability

Arachidonic acid (AA) pathways:

Pathway	Product	Action
COX-1/2	**PGI_2** (prostacyclin; endothelium)	**Vasodilation, inhibits platelet aggregation**
	TXA_2 (platelets)	**Vasoconstriction, platelet aggregation**
	PGE_2, PGD_2 (mast cells)	Vasodilation, ↑ permeability (potentiates histamine and bradykinin), **pain (hyperalgesia), fever**
	$PGF_2\alpha$	Uterine and bronchial smooth muscle contraction
5-Lipoxygenase	**LTB_4** (neutrophils)	**Potent chemotaxis** and neutrophil activation
	LTC_4, LTD_4, LTE_4 (cysteinyl leukotrienes), formerly "slow-reacting substance of anaphylaxis"	**Intense bronchospasm** (asthma), vasoconstriction, ↑ venular permeability (more potent than histamine)
12-Lipoxygenase / transcellular	**Lipoxins (LXA_4, LXB_4)**	**Anti-inflammatory**: inhibit neutrophil recruitment; promote resolution
Omega-3 derived	Resolvins, protectins, maresins	Pro-resolution

Drug relevance: steroids inhibit PLA_2 (and COX-2 expression); NSAIDs inhibit COX; zileuton inhibits 5-LOX; montelukast blocks cysteinyl LT receptors.

Key cytokines:

Cytokine	Actions
TNF and **IL-1**	**Endothelial activation** (↑ adhesion molecules, ↑ IL-6, procoagulant), **systemic acute phase response (fever, cachexia, sleep)**, leucocyte activation, fibroblast proliferation. **TNF in septic shock** (hypotension, DIC, insulin resistance). IL-1 is produced through the inflammasome (caspase-1)
IL-6	**Hepatic acute phase proteins (CRP)**, fever, hepcidin

Cytokine	Actions
IL-17	Neutrophil recruitment (Th17)
IFN-γ	Macrophage activation (granulomas)
IL-12	Th1 differentiation
IL-10, TGF-β	**Anti-inflammatory**
TGF-β, PDGF, FGF, VEGF	Fibrosis, angiogenesis

Plasma protein-derived mediators

1. The complement system

Pathways:

Pathway	Trigger
Classical	**C1** binds antigen–antibody complexes (IgM, IgG1/3)
Alternative	Microbial surfaces (LPS), spontaneous C3 hydrolysis; factor B, factor D, properdin
Lectin	**Mannose-binding lectin** binds microbial carbohydrates

All three converge on **C3 convertase** → C3a + **C3b** → **C5 convertase** → C5a + C5b → **C5b–C9 membrane attack complex (MAC)**.

Functions:

- **Inflammation: C3a, C5a (and C4a) = anaphylatoxins** → mast cell degranulation (histamine). **C5a is chemotactic** for neutrophils and activates the 5-LOX pathway.
- **Opsonisation and phagocytosis: C3b** (and iC3b).
- **Cell lysis: MAC (C5b–9)**.
- Immune complex clearance (C3b on red cells).

Regulators:

Regulator	Action	Deficiency causes
C1-inhibitor	Inhibits C1	**Hereditary angio-oedema** (↑ bradykinin; ACE inhibitors worsen it)
DAF (CD55), CD59	Protect cells from complement	**Paroxysmal nocturnal haemoglobinuria** (PIG-A mutation; GPI anchor defect)
Factor H, factor I		Factor H deficiency → **atypical HUS**

Deficiencies:

Deficiency	Result
C2/C4/C1q	**SLE-like** disease

Deficiency	Result
C3	Severe pyogenic infections
C5–C9 (MAC)	Recurrent ***Neisseria*** infections

2. The kinin system

Factor **XIIa (Hageman factor)** converts prekallikrein → **kallikrein**. Kallikrein cleaves **HMW kininogen** → **bradykinin**. Bradykinin:

- ↑ vascular permeability
- causes arteriolar dilation
- contracts bronchial smooth muscle
- causes **pain**

It is rapidly inactivated by **kininase / ACE**. Kallikrein is also chemotactic and converts C5 → C5a and plasminogen → plasmin.

3. Coagulation and fibrinolytic systems

- **Thrombin** acts on **PARs** (protease-activated receptors) → endothelial activation, adhesion molecules, chemokines.
- Fibrinopeptides ↑ permeability and are chemotactic.
- **Plasmin** activates C3.
- **Fibrin degradation products** ↑ permeability.

Hageman factor (XII) is the link that activates four systems: kinin, clotting, fibrinolytic and complement.

Summary by function

Inflammatory event	Principal mediators
Vasodilation	**Histamine, prostaglandins (PGI_2, PGE_2), NO**
↑ Vascular permeability	**Histamine, serotonin, C3a/C5a, bradykinin, $LTC_4/D_4/E_4$, PAF, substance P**
Chemotaxis, leucocyte recruitment and activation	**C5a, LTB_4, IL-8 (chemokines), TNF, IL-1, bacterial products**
Fever	**IL-1, TNF, IL-6, PGE_2**
Pain	**PGE_2, bradykinin, substance P**
Tissue damage	Lysosomal enzymes, ROS, NO

★ Exam pearls

Bradykinin and PGE_2 cause pain. IL-1, TNF and IL-6 (via PGE_2) cause fever. C5a, LTB_4 and IL-8 recruit neutrophils. Examiners favour this "function → mediator" table.

Chapter 40 · Haemodynamic Disorders, Thrombosis and Shock

A. Oedema and effusions

> **Oedema** is an abnormal accumulation of fluid in the **interstitial tissue spaces**. Fluid in body cavities is an **effusion**: hydrothorax, hydropericardium, **ascites** (hydroperitoneum). Severe generalised oedema is **anasarca**.

Starling forces: net filtration = $K[(Pc - Pi) - \sigma(\pi c - \pi i)]$

Mechanism	Causes
↑ **Hydrostatic pressure**	**Congestive heart failure**, constrictive pericarditis, cirrhosis (portal hypertension, giving ascites), venous obstruction (DVT), fluid overload, arteriolar dilation (heat)
↓ **Plasma oncotic pressure (hypoalbuminaemia)**	**Nephrotic syndrome**, liver cirrhosis, **protein-energy malnutrition (kwashiorkor)**, protein-losing enteropathy
Lymphatic obstruction (lymphoedema)	**Filariasis** (*Wuchereria bancrofti*: **elephantiasis**), malignancy, post-surgery or radiation (mastectomy with axillary node dissection), *peau d'orange* in breast cancer
Na^+ and water retention	Renal failure, RAAS activation (heart failure), corticosteroids
↑ **Vascular permeability**	Inflammation (exudate), burns, anaphylaxis

Morphology:

- **Pitting oedema:** subcutaneous; dependent in cardiac failure.
- **Periorbital oedema:** renal disease.
- **Pulmonary oedema:**
  - Left ventricular failure: heavy, wet lungs; "heart failure cells" (haemosiderin-laden macrophages) in chronic cases.
  - ARDS (non-cardiogenic).
- **Cerebral oedema:** vasogenic, cytotoxic or interstitial. Risk of herniation (**coning**).

B. Hyperaemia and congestion

	Hyperaemia (active)	Congestion (passive)
Mechanism	**Arteriolar dilation** → ↑ inflow	↓ **Venous outflow**
Colour	**Red (erythema)**, oxygenated blood	**Blue-red (cyanosis)**, deoxygenated blood
Examples	Inflammation, exercise, blushing	**Heart failure**, venous obstruction

Chronic venous congestion (CVC):

Organ	Cause	Appearance
Liver	**Right heart failure**	"**Nutmeg liver**": congested red-brown **centrilobular** areas (zone 3, hypoxic necrosis) alternating with pale fatty periportal areas. Severe cases → cardiac sclerosis / cirrhosis
Lung	**Left heart failure**, mitral stenosis	**Brown induration**: thickened fibrotic septa, **haemosiderin-laden macrophages ("heart failure cells")** in the alveoli
Spleen	Portal hypertension, RHF	Congestive splenomegaly; **Gamna–Gandy bodies** (siderofibrotic nodules)

C. Haemorrhage and haemostatic disorders

> **Haemorrhage** is the escape of blood from a ruptured vessel.

Type	Size / description
Petechiae	1–2 mm; ↓ platelets, platelet dysfunction, vasculitis
Purpura	3–10 mm
Ecchymoses (bruises)	1–2 cm; colour changes Hb (red-blue) → bilirubin (blue-green) → haemosiderin (golden-brown)
Haematoma	A localised mass of blood
Haemothorax, haemopericardium, haemoperitoneum, **haemarthrosis** (haemophilia)	Cavity bleeds

Clinical significance depends on the volume and rate of loss (>20% → hypovolaemic shock) and on the site (intracranial).

Classification of bleeding disorders:

Category	Causes
Vascular disorders (normal coagulation tests)	Scurvy, Ehlers–Danlos, senile purpura, steroids, Henoch–Schönlein purpura (IgA vasculitis), hereditary haemorrhagic telangiectasia, meningococcal septicaemia, dengue, Lassa fever (Nigeria)
Platelet disorders (mucocutaneous bleeding, petechiae)	**Thrombocytopenia**: ITP, aplastic anaemia, leukaemia, drugs, DIC, TTP, HUS, hypersplenism. **Platelet dysfunction**: aspirin, uraemia, Bernard–Soulier (GPIb), Glanzmann (GPIIb/IIIa), **von Willebrand disease** (the commonest inherited bleeding disorder)
Coagulation disorders (deep tissue bleeding, haemarthrosis)	**Haemophilia A (VIII), B (IX)** (X-linked), vitamin K deficiency (newborn, malabsorption, warfarin), liver disease, **DIC**

Laboratory screening:

Test	What it assesses
Platelet count, blood film	Platelet number and morphology
PFA-100 / bleeding time	Platelet function
PT	Extrinsic pathway (VII) + common pathway; INR
aPTT	Intrinsic pathway (XII, XI, IX, VIII) + common pathway
Thrombin time, fibrinogen, D-dimer	Fibrin formation and breakdown
Mixing studies	Factor deficiency vs inhibitor

DIC (disseminated intravascular coagulation):

- **Mechanism:** widespread activation of coagulation → microthrombi → **consumption** of platelets and factors → bleeding, plus secondary fibrinolysis.
- **Causes:**
  - Obstetric: **abruptio placentae**, amniotic fluid embolism, retained dead fetus, eclampsia.
  - **Sepsis** (Gram-negative bacteria, meningococcus).
  - Malignancy: **APL (M3)**, mucin-secreting adenocarcinomas.
  - Massive trauma and burns.
  - Snake bite (*Echis ocellatus*, carpet viper: common in northern Nigeria).
  - Viral haemorrhagic fevers (Lassa, Ebola).
- **Laboratory:** ↓ platelets, ↑ **PT/aPTT**, ↓ **fibrinogen**, ↑ **D-dimer / FDPs**, **schistocytes** on the film.
- **Management:** treat the cause. Replace with FFP, cryoprecipitate and platelets if bleeding.

D. Thrombosis

> **Thrombosis** is the formation of a solid mass (**thrombus**) from the constituents of blood **within the vascular system of a living person**. A clot outside the vessel or after death is not a thrombus.

Virchow's triad:

1. **Endothelial injury** (the dominant factor in the heart and arteries): atherosclerosis, MI, vasculitis, hypertension, smoking, toxins, valvulitis.
2. **Abnormal blood flow:**
   - **Stasis:** venous, AF, ventricular aneurysm, immobility.
   - **Turbulence:** arterial, aneurysms, bifurcations.
3. **Hypercoagulability (thrombophilia):**

Category	Causes
Primary (genetic)	**Factor V Leiden** (resistance to activated protein C; the commonest in Caucasians, rare in Africans), **prothrombin G20210A**, deficiencies of **antithrombin, protein C and protein S**, hyperhomocysteinaemia
Secondary (acquired)	**Prolonged immobilisation**, surgery, fractures, **pregnancy and the puerperium**, **oral contraceptives / HRT**, **malignancy (Trousseau)**, **antiphospholipid syndrome** (lupus anticoagulant, anticardiolipin, anti-β_2GP1), **HIT**, nephrotic syndrome (loss of antithrombin), smoking, obesity, **sickle cell disease**, myeloproliferative neoplasms, PNH

Morphology:

- **Lines of Zahn:** alternating pale layers (platelets and fibrin) and dark layers (RBCs). These indicate a thrombus formed in **flowing blood**, before death.
- **Arterial / cardiac thrombi** are pale ("white") and grow retrograde from the point of attachment. Mural thrombi form in the ventricles after MI and in the atria in AF. **Vegetations** on valves (infective endocarditis, NBTE, Libman–Sacks).
- **Venous thrombi (phlebothrombosis)** are red, rich in RBCs, and grow **in the direction of flow (propagate toward the heart). ~90% are in the deep veins of the legs.**

Post-mortem clot vs thrombus:

Feature	Post-mortem clot	Thrombus
Consistency	Gelatinous, rubbery, moist	**Firm, dry, friable**
Attachment	**Not attached** to the vessel wall	**Attached**
Shape	Takes the shape of the vessel	
Appearance	**"Chicken fat"** supernatant over a dark red **"currant jelly"** portion	**Lines of Zahn**

Fates of a thrombus:

1. **Propagation:** accumulates more platelets and fibrin.
2. **Embolisation:** dislodges.
3. **Dissolution:** fibrinolysis (recent thrombi).
4. **Organisation and recanalisation:** ingrowth of endothelium, smooth muscle and fibroblasts, with new channels.

Clinical: a DVT is often asymptomatic, or presents with a swollen, painful calf (Homans' sign is unreliable). Diagnose with **Wells score → D-dimer → Doppler ultrasound**.

E. Embolism

> An **embolus** is a detached intravascular solid, liquid or gaseous mass carried by the blood to a site distant from its origin. **~99% are thromboemboli.**

Type	Details
Pulmonary thromboembolism (PE)	**>95% from deep leg veins** (above the knee). Size determines the effect: **saddle embolus** at the bifurcation of the pulmonary artery → sudden death (acute right heart failure, cardiovascular collapse). Medium vessels → **pulmonary haemorrhage**, infarction only if the bronchial circulation is compromised (LVF). Multiple small emboli → pulmonary hypertension. **Paradoxical embolism** through a patent foramen ovale or ASD → systemic infarcts
Systemic thromboembolism	**~80% from intracardiac mural thrombi** (2/3 from LV infarcts, 1/4 from **AF** in a dilated left atrium), aortic aneurysms, ulcerated atheroma, valve vegetations. Sites: **lower limbs (75%)**, brain (10%), intestines, kidneys, spleen
Fat embolism	**Long-bone (femur) fractures**, soft tissue trauma, burns, liposuction. **Fat embolism syndrome** (24–72 h): **respiratory insufficiency (hypoxaemia), neurological symptoms (confusion), petechial rash** (axillae, conjunctiva), anaemia, thrombocytopenia. **Bone marrow embolism** also occurs in sickle cell crisis
Air (gas) embolism	Iatrogenic (IV lines, neck surgery), obstetric, chest trauma; >100 mL causes symptoms. **Decompression sickness ("the bends", caisson disease)**: N_2 bubbles in divers ascending rapidly → joint pain, "**chokes**" (lung), CNS damage. Chronic → **avascular necrosis** of femoral heads. Treat with hyperbaric O_2

Type	Details
Amniotic fluid embolism	Rare but high mortality (~20–60%). During labour or immediately postpartum → **sudden dyspnoea, cyanosis, shock, seizures, DIC**. Histology: **fetal squamous cells, lanugo hair, fat, mucin** in the maternal pulmonary microvasculature
Others	Tumour emboli, septic emboli (infective endocarditis), atheromatous (cholesterol) emboli, foreign body (talc, catheter), parasites (schistosome eggs)

F. Infarction

> An **infarct** is an area of **ischaemic necrosis** caused by occlusion of the arterial supply or venous drainage. ~99% are due to thrombotic or embolic arterial occlusion.

	Red (haemorrhagic) infarcts	White (anaemic) infarcts
Settings	**Venous occlusion** (ovarian torsion, testicular torsion); **loose tissues** (lung); **dual blood supply** (**lung, small intestine**, liver); previously congested tissue; reperfusion	**Arterial occlusion** in **solid organs with end-arterial circulation**: **heart, spleen, kidney**

Other features of infarcts:

- **Shape:** often **wedge-shaped**, with the occluded vessel at the apex and the organ surface at the base.
- **Septic infarcts** (infected emboli) → abscess.
- **Histology:** coagulative necrosis (except **brain = liquefactive**). Inflammation begins at the margins within hours. Repair by scar.

Factors that influence infarct development:

1. **Nature of the vascular supply:** dual supply (lung, liver, hand/forearm) resists infarction; end arteries (kidney, spleen) do not.
2. **Rate of occlusion:** slow occlusion allows collaterals to develop.
3. **Vulnerability to hypoxia:**
    - Neurons: 3–4 min.
    - Myocardium: 20–30 min.
    - Fibroblasts: very resistant.
4. **O_2 content of blood:** anaemia, cyanosis and CHF make infarction more likely.

Myocardial infarction timeline (Robbins):

Time	Change
0–½ h	No light-microscopic change
4–12 h	Early coagulative necrosis, wavy fibres, oedema
12–24 h	Contraction band necrosis, **neutrophil infiltration begins**
1–3 days	Neutrophilic infiltrate (**pericarditis** risk)
3–7 days	Macrophages; **maximum risk of rupture** (free wall, septum, papillary muscle)
7–10 days	Granulation tissue
>2 months	Dense collagen scar (risk of **ventricular aneurysm**, Dressler syndrome)

G. Shock

Shock is a state of **systemic hypoperfusion due to reduced cardiac output or reduced effective circulating blood volume**, causing **inadequate tissue perfusion and cellular hypoxia**. Initially reversible, it progresses to irreversible injury and multiorgan failure.

Type	Mechanism	Examples
Hypovolaemic	↓ Blood or plasma volume	**Haemorrhage** (trauma, PPH, GI bleed), fluid loss (**severe diarrhoea/cholera**, vomiting, **burns**), DKA
Cardiogenic	Myocardial pump failure	**MI**, arrhythmias, myocarditis, valve rupture
Obstructive (sometimes grouped with cardiogenic)	Extrinsic compression or obstruction	**Cardiac tamponade, tension pneumothorax, massive PE**
Septic (distributive)	**Peripheral vasodilation + pooling**; endothelial activation and injury; leucocyte-induced damage; DIC; metabolic disturbance. Triggered by microbial products (**LPS → TLR4**) → **TNF, IL-1, IL-6**, NO	**Gram-negative and Gram-positive bacteria, fungi**. The commonest cause of ICU death. **Sepsis-3 (2016):** life-threatening organ dysfunction (SOFA ≥2) from a dysregulated host response. **Septic shock:** vasopressors needed for MAP ≥65 and lactate >2 mmol/L despite fluids
Anaphylactic	IgE-mediated systemic vasodilation, ↑ permeability	Drugs (penicillin), insect stings, foods (peanuts)
Neurogenic	Loss of vascular (sympathetic) tone	Spinal cord injury, anaesthetic accidents

Type	Mechanism	Examples
Endocrine		**Addisonian crisis**, myxoedema

Stages:

1. **Non-progressive (compensated):** baroreceptor reflexes, catecholamines (tachycardia, peripheral vasoconstriction → **cool, pale skin**; septic shock initially gives warm, flushed skin), RAAS, ADH. Vital organ perfusion is maintained.
2. **Progressive:** tissue hypoperfusion, **anaerobic glycolysis → lactic acidosis**, arteriolar dilation → pooling, endothelial anoxic injury → **DIC**. Confusion and oliguria.
3. **Irreversible:** severe cellular and tissue injury. Lysosomal enzyme leakage, ↓ myocardial contractility (NO), ischaemic bowel → bacteraemia, **acute tubular necrosis**. Death follows even if haemodynamics are restored.

Morphology (shock organs):

- **Brain:** ischaemic encephalopathy.
- **Heart:** subendocardial haemorrhage and necrosis, **contraction band necrosis**.
- **Kidneys: acute tubular necrosis (ATN)**.
- **Lungs:** relatively resistant to hypoxia but develop **diffuse alveolar damage (ARDS, "shock lung")** in sepsis.
- **Adrenals:** lipid depletion. **Waterhouse–Friderichsen syndrome**: bilateral adrenal haemorrhage in **meningococcal septicaemia**.
- **GIT:** patchy mucosal haemorrhage (haemorrhagic enteropathy).
- **Liver:** fatty change, **central haemorrhagic necrosis**.

Clinical: hypotension (SBP <90 mmHg), tachycardia, tachypnoea, cold, clammy, cyanotic skin (except warm early septic shock), oliguria (<0.5 mL/kg/h), confusion, **raised lactate**.

Principles of management (ABC):

- **Hypovolaemic:** volume resuscitation, blood products, stop the bleeding.
- **Septic:** antibiotics within 1 h, fluids 30 mL/kg, **noradrenaline**, source control.
- **Cardiogenic:** inotropes, revascularisation.
- **Anaphylactic:** IM adrenaline.

🎤 Viva corner

Q: Name the components of Virchow's triad, and say which dominates in arterial and in venous thrombosis. **A:** Endothelial injury, abnormal blood flow (stasis or turbulence) and hypercoagulability. Endothelial injury dominates in arterial and cardiac thrombosis. Stasis and hypercoagulability dominate in venous thrombosis.

Q: Why are infarcts of the lung often haemorrhagic? **A:** The lung has a dual blood supply (pulmonary and bronchial arteries) and loose, spongy tissue. Blood from the bronchial circulation flows into the necrotic zone.

Chapter 41 · Environmental and Nutritional Pathology

A. Environmental pathology

Tobacco

Smoking is the most important preventable cause of death worldwide. Tobacco smoke contains >7,000 chemicals, ~70 of them carcinogenic (polycyclic aromatic hydrocarbons, nitrosamines, benzopyrene).

- **Cancers:** lung (squamous and small cell, centrally placed), larynx, oral cavity, oesophagus, bladder, kidney, pancreas, cervix, AML.
- **Cardiovascular:** atherosclerosis, MI, PAD (Buerger's disease), aortic aneurysm.
- **Respiratory: COPD** (emphysema, centrilobular), chronic bronchitis.
- **Pregnancy:** low birth weight, preterm birth, sudden infant death.
- **Passive smoking.**

Alcohol (ethanol)

Metabolism: alcohol dehydrogenase → **acetaldehyde** → (ALDH) → acetate. Microsomal MEOS (CYP2E1) is induced in chronic use. ↑ **NADH/NAD⁺ ratio** → ↑ fatty acid synthesis (steatosis), lactic acidosis, hypoglycaemia.

Acute effects	Chronic effects
CNS depression	**Liver:** fatty liver → **alcoholic hepatitis (Mallory–Denk bodies, neutrophils)** → **cirrhosis (micronodular)** → HCC
	GI: gastritis, ulcers, **varices**, **acute and chronic pancreatitis**
	Nervous system: Wernicke encephalopathy (thiamine deficiency: ophthalmoplegia, ataxia, confusion) and **Korsakoff psychosis**, peripheral neuropathy, cerebellar degeneration
	CVS: **dilated cardiomyopathy**, hypertension
	Fetal alcohol syndrome: microcephaly, smooth philtrum, thin upper lip, growth retardation
	Cancers: oral, oesophageal, liver, breast
	Macrocytosis, ↑ GGT, ↑ MCV

Drug abuse

- Cocaine: **MI and arrhythmias** (vasospasm), nasal septum perforation.
- Heroin: pulmonary oedema, **talc granulomas**, infection (HIV, HBV, HCV, endocarditis of the **right-sided tricuspid valve**), FSGS nephropathy.

- Cannabis (Indian hemp): psychosis.
- **Tramadol and codeine abuse** is a growing problem in Nigeria.

Adverse drug reactions and iatrogenic disease

See Book 1, Chapter 3.

Physical injury

Type	Key points
Mechanical trauma	**Abrasion**, **contusion** (bruise), **laceration** (tear; tissue bridges), **incised wound** (clean edges, longer than deep), **puncture/stab** (deeper than long), gunshot wounds (entry and exit features)
Thermal: burns	Depth: **superficial (1st degree, epidermis)**, partial thickness (2nd degree, blisters), **full thickness (3rd degree)**, painless. **Rule of nines** (adults) / Lund–Browder (children) for % TBSA. Complications: **hypovolaemic shock** (Parkland formula: 4 mL × kg × %TBSA over 24 h, half in the first 8 h), **sepsis** (*Pseudomonas*), inhalation injury (**CO and cyanide poisoning**), hypermetabolic state, **Curling ulcer** (stomach/duodenum), contractures, **Marjolin ulcer**
Hyperthermia	Heat cramps, heat exhaustion, **heat stroke** (>40 °C core with CNS dysfunction: rhabdomyolysis, DIC). Malignant hyperthermia
Hypothermia	Frostbite, trench foot (immersion)
Electrical injury	Burns at the entry and exit points; VF (AC current) and respiratory paralysis. Lightning: **Lichtenberg figures**
Ionising radiation	Measured in Gray (Gy) and Sievert (Sv). Damages DNA (directly and via hydroxyl free radicals). Most sensitive tissues: **gonads, haemopoietic and lymphoid tissue, GI epithelium** (rapidly dividing). Acute radiation syndromes: haematopoietic (2–10 Gy), GI (10–20 Gy), CNS (>50 Gy). Late effects: **fibrosis**, vascular damage (endarteritis obliterans), **cancer** (leukaemia, thyroid, breast), cataract, teratogenesis

Environmental pollution

- **Outdoor air:** ozone, SO_2, particulates (PM2.5), NO_2. Harmattan dust.
- **Indoor air: biomass fuel smoke** (firewood, kerosene) is common in Nigerian kitchens → COPD, pneumonia in children. **Carbon monoxide** from generators ("*I better pass my neighbour*") in closed rooms is a common cause of death in Nigeria:
  - CO binds Hb with ~240× the affinity of O_2 → **carboxyhaemoglobin** + left shift of the O_2 curve → tissue hypoxia.

  - Signs: **cherry-red** skin (at autopsy), headache, confusion, coma.
  - **Pulse oximetry is falsely normal.**
  - Treatment: **100% O_2**; hyperbaric O_2.

- **Heavy metals:**

Metal	Sources	Effects
Lead	Paint, batteries, mining (**Zamfara lead poisoning epidemic, 2010**: >400 child deaths), petrol (historical)	Inhibits **ALA dehydratase and ferrochelatase → microcytic anaemia with basophilic stippling**. **Encephalopathy** in children, **wrist/foot drop** in adults, **lead lines** on gums (Burton) and at metaphyses, nephropathy, abdominal colic. Treat with chelation: **succimer (DMSA), EDTA, dimercaprol**
Mercury	Minamata disease; artisanal gold mining	Neurotoxicity, nephrotoxicity
Arsenic	Groundwater	**Skin hyperkeratosis and pigmentation**, cancers (skin, lung, bladder)
Cadmium	Batteries	Itai-itai disease (osteomalacia), nephrotoxicity, emphysema

- **Occupational lung disease (pneumoconioses):**
  - Coal workers' pneumoconiosis: progressive massive fibrosis, Caplan syndrome.
  - **Silicosis:** upper lobes, eggshell calcification of hilar nodes, ↑ TB risk.
  - **Asbestosis:** lower lobes, **pleural plaques**, **ferruginous (asbestos) bodies**, **mesothelioma**, lung carcinoma (synergy with smoking).
  - Berylliosis: granulomas.
- **Agricultural hazards:** organophosphates, **aflatoxins**.
- **Crude oil pollution** in the Niger Delta: hydrocarbon exposure, benzene, polluted water.

B. Nutritional pathology

Protein-energy malnutrition (PEM)

Common in children under 5 in northern Nigeria.

Feature	Marasmus	Kwashiorkor
Deficiency	Severe deficiency of **total energy (calories)**	Protein deficiency **relative to** calories (classically after weaning: "the disease of the deposed child", Cicely Williams 1933). Also oxidative stress and aflatoxin theories

Feature	Marasmus	Kwashiorkor
Weight	**<60% of expected; "skin and bones"**	60–80% of expected (oedema masks wasting)
Oedema	**Absent**	**Present (pitting; hypoalbuminaemia)**: the defining feature
Muscle and fat	Severe wasting of both	Muscle wasting, but subcutaneous fat relatively preserved
Serum albumin	Normal or slightly ↓	**Markedly ↓**
Liver	Normal	**Enlarged fatty liver** (↓ apolipoprotein synthesis)
Skin and hair	Wrinkled, loose skin; "old man face"	**"Flaky paint" dermatosis** (alternating hyper- and hypopigmentation, desquamation); **hair is sparse, pale and reddish ("flag sign")**
Mood	Alert, hungry	**Apathetic, irritable, anorexic**
Other	Growth retardation, immune deficiency, infections	Anaemia, infection, diarrhoea, **micronutrient deficiencies**, hypothermia, hypoglycaemia
Prognosis	Better	**Worse**

WHO definition of severe acute malnutrition (SAM): weight-for-height < −3 SD, or **MUAC <115 mm**, or bilateral pitting oedema.

Management (WHO 10 steps):

1. Treat or prevent hypoglycaemia.
2. Hypothermia.
3. Dehydration (with **ReSoMal**).
4. Electrolytes.
5. Infection (antibiotics routinely).
6. Micronutrients (no iron initially).
7. Cautious feeding with **F-75**.
8. Catch-up growth with **F-100 / RUTF**.
9. Sensory stimulation.
10. Preparation for follow-up.

Watch for **refeeding syndrome** (hypophosphataemia).

Vitamin deficiencies

Vitamin	Function	Deficiency	Notes / toxicity
A (retinol)	Vision (**rhodopsin**), epithelial differentiation, immunity	**Night blindness** (earliest), **xerophthalmia**, **Bitot's spots**, keratomalacia → blindness; squamous metaplasia; ↑ infections (measles mortality)	Supplementation in Nigeria from 6–59 months. Toxicity: raised ICP (pseudotumour cerebri), hepatotoxicity, **teratogenic** (isotretinoin)
D	Ca^{2+} and PO_4^{3-} absorption, bone mineralisation	**Rickets** (children: craniotabes, **rachitic rosary**, Harrison's sulcus, bow legs, widened wrists, delayed fontanelle closure) and **osteomalacia** (adults: bone pain, Looser zones, proximal myopathy). Calcium-deficiency rickets is reported in northern Nigeria	Toxicity: hypercalcaemia, metastatic calcification
E (tocopherol)	Antioxidant	Haemolytic anaemia (premature infants), spinocerebellar degeneration, neuropathy	
K	γ-carboxylation (II, VII, IX, X, C, S)	**Haemorrhagic disease of the newborn** (give IM vitamin K at birth), malabsorption, warfarin	
B_1 (thiamine)	TPP cofactor (pyruvate dehydrogenase, α-KG dehydrogenase, transketolase)	**Beriberi**: dry (peripheral neuropathy), **wet (high-output cardiac failure)**, infantile. **Wernicke–Korsakoff** (alcoholics)	Give thiamine **before** glucose in alcoholics
B_2 (riboflavin)	FAD, FMN	Angular cheilitis, glossitis, corneal vascularisation	
B_3 (niacin)	NAD, NADP	**Pellagra: the 3 Ds (dermatitis** in sun-exposed areas: **Casal's necklace**, **diarrhoea, dementia**) → death. Seen in **maize-dependent diets**, **Hartnup disease**, carcinoid syndrome, isoniazid	
B_6 (pyridoxine)	PLP (transamination, ALA synthase)	Peripheral neuropathy (**isoniazid**), sideroblastic anaemia, seizures	

Vitamin	Function	Deficiency	Notes / toxicity
B_{12}, folate	DNA synthesis	**Megaloblastic anaemia** (see Chapter 48)	
C (ascorbic acid)	**Hydroxylation of proline and lysine in collagen**, antioxidant, iron absorption	**Scurvy: perifollicular haemorrhages, corkscrew hairs**, **swollen bleeding gums**, poor wound healing, subperiosteal haemorrhage (children), anaemia	

Mineral deficiencies

Mineral	Deficiency
Iron	Anaemia (see Book 3)
Iodine	**Goitre**, hypothyroidism, **cretinism** (endemic in some Nigerian regions before salt iodisation)
Zinc	Acrodermatitis enteropathica, growth retardation, poor wound healing, hypogonadism, diarrhoea
Selenium	Keshan cardiomyopathy
Copper	Anaemia, neutropenia, neuropathy
Fluoride	Dental caries; excess → fluorosis

Obesity

- **Definition: BMI ≥30 kg/m²**; overweight 25–29.9.
- **Regulation:** leptin (adipocytes → ↓ appetite), ghrelin (stomach → ↑ appetite), insulin, PYY, GLP-1.
- **Complications:**
  - **Type 2 diabetes** and **metabolic syndrome**
  - **Hypertension**, dyslipidaemia, IHD
  - **NAFLD**
  - Gallstones
  - **Obstructive sleep apnoea** (Pickwickian syndrome)
  - Osteoarthritis
  - **Cancers:** endometrial, breast, colon, kidney, oesophageal adenocarcinoma
  - PCOS

Diet and cancer

- **Increased risk:** aflatoxin; nitrosamines; low fibre and high red meat (colon cancer); obesity.
- **Possibly protective:** fibre, fruit and vegetables.

🎤 Viva corner

Q: Differentiate marasmus from kwashiorkor. **A:** Kwashiorkor has oedema, hypoalbuminaemia, fatty liver, flaky-paint dermatosis, hair changes and apathy. Marasmus has severe wasting of muscle and fat without oedema, with the child alert and hungry.

Q: Name the three Ds of pellagra. **A:** Dermatitis, diarrhoea and dementia. Death is sometimes given as the fourth.

Histopathology Self-Assessment

Section A: Single best answer MCQs

1. The commonest type of necrosis in a myocardial infarct is: A. Liquefactive · B. **Coagulative** · C. Caseous · D. Fat · E. Fibrinoid

> **B.** In coagulative necrosis the tissue architecture is preserved and the dead cells remain as "ghosts". This pattern is typical of hypoxic death in solid organs other than the brain.

2. Which feature is characteristic of apoptosis rather than necrosis? A. Cell swelling · B. Inflammation · C. **Internucleosomal DNA cleavage** · D. Membrane rupture · E. Always pathological

> **C.** Caspase-activated DNase cuts DNA between nucleosomes, which gives the "DNA ladder" on electrophoresis.

3. Squamous cell carcinoma of the urinary bladder in northern Nigeria is most strongly associated with: A. Aniline dyes · B. Smoking · C. ***Schistosoma haematobium*** · D. HPV · E. Cyclophosphamide

> **C.** Chronic irritation by schistosome eggs causes squamous metaplasia, which progresses to squamous cell carcinoma.

4. The most potent mediator of neutrophil chemotaxis among these is: A. Histamine · B. Bradykinin · C. **LTB_4** · D. PGI_2 · E. Serotonin

> **C.** LTB_4 is a potent neutrophil chemoattractant. C5a and IL-8 are the others to remember.

5. Lines of Zahn are characteristic of: A. Post-mortem clot · B. **Thrombus formed in flowing blood** · C. Infarct · D. Fat embolus · E. Granuloma

> **B.** Lines of Zahn are alternating pale layers of platelets and fibrin with dark layers of red cells.

6. Bilateral adrenal haemorrhage in septicaemia is called: A. Sheehan syndrome · B. Addison disease · C. **Waterhouse–Friderichsen syndrome** · D. Conn syndrome · E. Nelson syndrome

> **C.** It is classically caused by *Neisseria meningitidis*.

7. Which tumour suppressor is the target of HPV E7? A. p53 · B. **RB** · C. APC · D. BRCA1 · E. PTEN

> **B.** E7 binds RB. E6 degrades p53.

8. Metastatic calcification occurs in: A. Atheroma · B. Caseous TB · C. **Hyperparathyroidism** · D. Fat necrosis · E. Dead parasites

C. Metastatic calcification is driven by hypercalcaemia and deposits calcium in normal tissue. The other options are dystrophic calcification.

9. Kwashiorkor differs from marasmus by the presence of: A. Growth retardation · B. Muscle wasting · C. **Oedema** · D. Infection · E. Anaemia

C. Bilateral pitting oedema is the defining feature of kwashiorkor.

10. The type of giant cell in tuberculosis is: A. Foreign body · B. Touton · C. **Langhans** · D. Reed–Sternberg · E. Aschoff

C. Langhans giant cells have their nuclei arranged in a horseshoe around the periphery. (Touton cells occur in xanthomas; Reed–Sternberg cells in Hodgkin lymphoma; Aschoff bodies in rheumatic fever.)

11. Fat embolism syndrome is most commonly associated with: A. Burns · B. **Fracture of long bones** · C. Pancreatitis · D. Liposuction · E. Diabetes

B. It classically follows a fracture of a long bone such as the femur.

12. The "nutmeg liver" is caused by: A. Left heart failure · B. **Right heart failure** · C. Hepatitis B · D. Schistosomiasis · E. Alcohol

B. Chronic venous congestion from right heart failure causes congested, haemorrhagic centrilobular areas that give the cut surface its nutmeg pattern.

13. Aflatoxin B1 causes HCC through a mutation in: A. RAS · B. **TP53 codon 249** · C. MYC · D. APC · E. BCL2

B. Aflatoxin produces a characteristic mutation at codon 249 of TP53.

14. The most reliable criterion of malignancy is: A. Pleomorphism · B. Mitoses · C. **Metastasis** · D. Size · E. Necrosis

C. Metastasis is unequivocal evidence of malignancy.

15. Complement components acting as anaphylatoxins are: A. C3b and C4b · B. **C3a and C5a** · C. C5b–9 · D. C1q · E. Factor B

B. C3a and C5a trigger mast cell degranulation and histamine release.

Section B: Essay and short-answer questions (with outlines)

1. Define inflammation. Describe the vascular and cellular events in acute inflammation.

1. Definition.
2. Vascular events: vasodilation; increased permeability (with its mechanisms); stasis.
3. Cellular events: margination, rolling (selectins), adhesion (integrins), transmigration (PECAM-1), chemotaxis (C5a, LTB_4, IL-8), phagocytosis (opsonins), killing (respiratory burst, MPO–HOCl).
4. Outcomes.
5. Systemic effects.
6. Draw labelled diagrams.

2. Write short notes on: (a) granuloma; (b) apoptosis; (c) dystrophic calcification; (d) keloid; (e) Virchow's triad.

3. Classify shock. Discuss the pathogenesis and morphological changes of septic shock.

4. Discuss the molecular basis of carcinogenesis. Answer with the hallmarks of cancer, oncogenes, tumour suppressors, multistep carcinogenesis, and carcinogens with Nigerian examples.

5. Define oedema. Discuss its pathogenesis with relevant examples.

Section C: Viva voce bank (Histopathology)

#	Question	Model answer
1	Define metaplasia and give two examples.	Reversible change of one adult cell type to another. Examples: squamous metaplasia of the bronchus in smokers; Barrett's oesophagus.
2	What is the difference between hypertrophy and hyperplasia?	Hypertrophy is an increase in cell size; hyperplasia is an increase in cell number.
3	List the nuclear changes of necrosis.	Pyknosis, karyorrhexis, karyolysis.
4	What is caseous necrosis?	Cheesy necrosis surrounded by granulomatous inflammation, typical of TB.
5	What are Councilman bodies?	Apoptotic hepatocytes seen in viral hepatitis and yellow fever.
6	What is a hamartoma?	A tumour-like malformation made of tissue native to the site, arranged in a disorganised way.
7	What is anaplasia?	Lack of differentiation. It is a hallmark of malignancy.

#	Question	Model answer
8	Name three viruses associated with cancer.	EBV (Burkitt lymphoma, nasopharyngeal carcinoma), HPV (cervical cancer), HBV (HCC), HHV-8 (Kaposi sarcoma), HTLV-1 (adult T-cell leukaemia/lymphoma).
9	What is a paraneoplastic syndrome? Example?	Effects of a tumour that are not caused by local or metastatic spread, e.g. ectopic ACTH from small cell lung carcinoma.
10	Grading vs staging?	Grading measures differentiation; staging measures extent of spread (TNM). Staging is more prognostic.
11	What is pus?	Neutrophils (living and dead), necrotic debris, bacteria and exudate.
12	Exudate vs transudate?	Exudate is rich in protein (>30 g/L, SG >1.020) and inflammatory. Transudate is low in protein and caused by pressure changes.
13	What is an embolus? Commonest type?	A detached intravascular mass carried to a distant site. Thromboembolus.
14	Red vs white infarcts?	Red: venous occlusion, dual circulation, loose tissue (lung, bowel). White: end-arterial solid organs (heart, spleen, kidney).
15	What is chronic granulomatous disease?	NADPH oxidase deficiency, causing infections with catalase-positive organisms. NBT test is negative.
16	Name the mediators of fever.	IL-1, TNF, IL-6, PGE_2.
17	What are heart failure cells?	Haemosiderin-laden alveolar macrophages in chronic pulmonary congestion.
18	What is DIC and its lab findings?	Consumptive coagulopathy: ↓ platelets, ↑ PT/aPTT, ↓ fibrinogen, ↑ D-dimer, schistocytes.
19	Causes of granulomatous inflammation in Nigeria?	TB, leprosy, schistosomiasis, fungi (histoplasmosis), sarcoidosis, foreign bodies.
20	What is a keloid?	Excess collagen deposition that extends beyond the original wound margin. Common in Africans.

Book 3 — Haematology

Haemopoiesis and the haemopoietic microenvironment → erythropoiesis → granulocytes, monocytes and benign white cell disorders → megakaryocytopoiesis → lymphocyte ontogeny (B and T cells) → iron metabolism and iron deficiency anaemia → vitamin B_{12} and folate

Chapter 42 · Haemopoiesis and the Haemopoietic Inductive Microenvironment (HIM)

Haemopoiesis (haematopoiesis) is the formation of blood cells. All blood cells come from a common **pluripotent haemopoietic stem cell (HSC)**. HSCs can **self-renew** and **differentiate** into every lineage.

Sites of haemopoiesis through life

Stage	Site
Embryo (week 3 to 2 months)	**Yolk sac** (mesoblastic phase: primitive nucleated erythroblasts with embryonic Hb: Gower 1 and 2, Portland). Also the aorta–gonad–mesonephros (AGM) region, which is the source of definitive HSCs
Fetus (6 weeks to 7 months)	**Liver** (the main site) and **spleen** (hepatic phase). **HbF** predominates
Fetus (from ~5 months onwards)	**Bone marrow** (medullary phase)
Infants	Marrow of practically **all bones**
Adults	Marrow of the **axial skeleton**: **vertebrae, ribs, sternum, skull, sacrum, pelvis**, and the **proximal ends of the femur and humerus**

Extramedullary haemopoiesis means haemopoiesis returns to the liver and spleen. It occurs in myelofibrosis, **thalassaemia major** and severe haemolytic anaemia, and causes hepatosplenomegaly and bossing of the skull.

Marrow examination sites:

- Adults: **posterior iliac crest** for aspirate and trephine (the sternum for aspirate only).
- Infants: **tibia**.

The haemopoietic hierarchy

The **haemopoietic stem cell** ($CD34^+$, $CD38^-$; ~1 in 20 million nucleated marrow cells) divides asymmetrically. It gives rise to two lines:

1. **Common lymphoid progenitor**, which produces B cells, T cells, NK cells and lymphoid dendritic cells.
2. **Common myeloid progenitor (CFU-GEMM)**, which produces:
    - **MEP** (megakaryocyte–erythroid progenitor) → **BFU-E** → **CFU-E** → **erythroid cells**; and **CFU-Meg** → **megakaryocytes** → **platelets**.

- **GMP** (granulocyte–macrophage progenitor, CFU-GM) → **CFU-G** → **neutrophils**; **CFU-M** → **monocytes/macrophages**.
- **CFU-Eo** → **eosinophils** and **CFU-Bas** → **basophils / mast cells**.

As cells mature they lose their ability to self-renew and become committed to one lineage, which is when they gain specific morphology. The marrow produces about **10^{13} cells per day**.

Growth factors (haemopoietic cytokines)

Factor	Main source	Target and role
Stem cell factor (SCF, c-KIT ligand), FLT3 ligand	Stromal cells	Early stem cells and progenitors
IL-3 ("multi-CSF")	T cells	Multipotent progenitors
GM-CSF	T cells, endothelium, fibroblasts	Granulocytes, monocytes
G-CSF	Monocytes, endothelium	**Neutrophil production and release** (filgrastim is used clinically)
M-CSF		Monocytes
Erythropoietin (EPO)	**Peritubular interstitial cells of the kidney (~90%)**, liver (~10%)	**Erythroid progenitors (CFU-E)**. Driven by hypoxia via **HIF**
Thrombopoietin (TPO)	**Liver**	**Megakaryocytes**
IL-5		**Eosinophils**
IL-7		Lymphoid (B and T) development
IL-6, IL-11		Megakaryocytes and various

Clinical use: recombinant EPO (in CKD), G-CSF (neutropenia, stem cell mobilisation), TPO-receptor agonists (romiplostim, eltrombopag: ITP).

The haemopoietic inductive microenvironment (HIM)

> The **HIM** (also called the **stem cell niche**) is the specialised local environment in the bone marrow that supports the survival, self-renewal, proliferation, differentiation and release of haemopoietic stem cells. HSCs cannot survive or develop without it.

Components:

1. **Stromal cells:**
   - Mesenchymal stromal cells (CXCL12-abundant reticular cells)
   - Fibroblasts and **reticular cells**
   - **Macrophages**
   - **Adipocytes** (fat cells: these increase with age, giving "yellow marrow")

    - **Osteoblasts** (the **endosteal niche**: quiescent HSCs)
    - **Endothelial cells** of the sinusoids (the **perivascular / vascular niche**: proliferating HSCs)
2. **Extracellular matrix (ECM):** fibronectin, laminin, collagen, **proteoglycans and glycosaminoglycans** (heparan sulfate). These bind and present growth factors and anchor progenitors.
3. **Soluble and membrane-bound growth factors:** SCF, **CXCL12 (SDF-1)**, angiopoietin-1, thrombopoietin, Notch ligands, Wnt.
4. **Adhesion molecules:**
    - **VLA-4 on HSCs binds VCAM-1 on stroma.**
    - **CXCR4 on HSCs binds CXCL12** (retention).
    - Disrupting these releases HSCs into the blood. **G-CSF and plerixafor** (a CXCR4 antagonist) are used this way to **mobilise stem cells** for transplantation.
5. **Neural and vascular input:** sympathetic nerves regulate the circadian release of HSCs. The sinusoidal structure allows mature cells to exit.

Functions of the HIM:

- Homing of HSCs (important for engraftment after a bone marrow transplant).
- Maintaining quiescence and self-renewal.
- Directing lineage commitment through local cytokines. **Erythroblastic islands** are clusters of erythroblasts around a central macrophage that supplies iron and phagocytoses extruded nuclei.
- Controlled release of mature cells.

Clinical relevance:

- **Aplastic anaemia:** stem cell failure or immune destruction.
- **Myelofibrosis:** fibrosis of the niche → extramedullary haemopoiesis, teardrop cells.
- **Radiation and chemotherapy** damage stroma and HSCs.
- **Marrow infiltration** (leukaemia, metastases) → **leucoerythroblastic picture**.
- **Stem cell transplantation** depends on donor HSCs homing to and engrafting in the recipient's niche.

Development, maturation and function of blood cells: overview

Cell	Maturation sequence	Lifespan	Main function	Normal adult count
Red cells	Proerythroblast → basophilic → polychromatic → orthochromatic erythroblast → **reticulocyte** → RBC	**120 days**	O_2/CO_2 transport (Hb)	Male 4.5–6.5; female 3.9–5.6 × 10^{12}/L

Cell	Maturation sequence	Lifespan	Main function	Normal adult count
Neutrophils	Myeloblast → promyelocyte → myelocyte → metamyelocyte → band → segmented	6–10 h in blood; 1–4 days in tissues	**Phagocytosis** of bacteria	2.0–7.5 × 10^9/L (**benign ethnic neutropenia** in many Africans: 1.0–2.0 normal)
Eosinophils	Similar granulocytic series	Days to weeks in tissue	**Parasites**, allergy (MBP, ECP)	0.04–0.4 × 10^9/L
Basophils			Histamine, heparin; allergy	<0.1 × 10^9/L
Monocytes	Monoblast → promonocyte → monocyte → **tissue macrophage**	Blood 20–40 h; tissue months	Phagocytosis, antigen presentation	0.2–0.8 × 10^9/L
Lymphocytes	Lymphoid progenitor → B (marrow) / T (thymus) / NK	Weeks to years (memory cells)	Adaptive immunity	1.5–3.5 × 10^9/L
Platelets	Megakaryoblast → megakaryocyte (endomitosis) → platelets	**7–10 days**	Haemostasis	150–400 × 10^9/L

Chapter 43 · Erythropoiesis

Stages

Stage	Features
Proerythroblast (pronormoblast)	Large cell; deeply basophilic cytoplasm; nucleoli present
Basophilic (early) erythroblast	Smaller; no nucleoli; intense basophilia (ribosomes)
Polychromatic (intermediate) erythroblast	Hb accumulates, giving grey-blue cytoplasm; **last stage capable of division**
Orthochromatic (late) erythroblast	Pink cytoplasm; small, pyknotic nucleus that is **extruded**
Reticulocyte	Anucleate; residual RNA (stain with supravital **new methylene blue / brilliant cresyl blue**). Spends 1–2 days in the marrow and ~1 day in the blood
Mature red cell	Biconcave disc, 7–8 µm; no nucleus or mitochondria

- Each pronormoblast produces about **16 red cells**. The whole process takes about 7 days.
- **Ineffective erythropoiesis** (intramedullary death of erythroblasts) occurs in **thalassaemia, megaloblastic anaemia, sideroblastic anaemia** and myelodysplasia.

Regulation: erythropoietin

Hypoxia (anaemia, high altitude, lung disease, cyanotic heart disease) stabilises **HIF-1α/HIF-2α** (normally degraded by **VHL** under normoxia). This increases EPO transcription in the renal peritubular cells. EPO binds receptors on CFU-E (via JAK2–STAT5) → prevents apoptosis → more red cells.

Cause of high EPO / polycythaemia	Cause of low EPO
Appropriate: hypoxia, smoking, high altitude	**Renal failure (anaemia of CKD)**
Inappropriate: RCC, HCC, cerebellar haemangioblastoma, uterine fibroids, EPO abuse (athletes)	Anaemia of chronic disease (relative)
Polycythaemia vera: JAK2 V617F, **EPO low**	

Nutritional requirements for erythropoiesis: iron, vitamin B_{12}, folate, protein, vitamin C, B_6 (ALA synthase), copper, cobalt, thyroxine, androgens (explain the higher Hb in men).

Haemoglobin

- Structure: **4 globin chains + 4 haem** (protoporphyrin IX + Fe^{2+}).
- **Haem synthesis:** starts in the mitochondria with **succinyl-CoA + glycine** (ALA synthase; needs B_6). Ends with insertion of Fe^{2+} (**ferrochelatase**). See Book 4, Chapter 57.

Haemoglobin	Chains	Normal adult proportion
HbA	$\alpha_2\beta_2$	~97%
HbA_2	$\alpha_2\delta_2$	1.5–3.5% (↑ **in β-thalassaemia trait**)
HbF	$\alpha_2\gamma_2$	<1% (↑ in β-thalassaemia, HPFH, sickle cell disease on hydroxyurea)

- **Haemoglobinopathies** are common in Nigeria:
  - **HbS carrier frequency ~20–30%.** Nigeria has the **world's highest burden of sickle cell disease** (~150,000 births per year).
  - **HbC** is found in the southwest.
- **The O_2 dissociation curve** is sigmoid (cooperative binding).

Shift	Causes	Effect
Right shift (↓ affinity, more O_2 released to tissues)	↑ **CO_2**, ↑ **H^+ (Bohr effect)**, ↑ **temperature**, ↑ **2,3-BPG**, HbS	Better tissue delivery
Left shift	↓ CO_2, alkalosis, ↓ temperature, ↓ 2,3-BPG, **HbF, CO**, methaemoglobin	Tighter O_2 binding

Red cell metabolism

- **Embden–Meyerhof (glycolytic) pathway:** generates ATP. **Pyruvate kinase deficiency** → haemolysis. The **Rapoport–Luebering shunt** produces **2,3-BPG**.
- **Hexose monophosphate shunt (pentose phosphate pathway):** generates **NADPH** (via **G6PD**) → reduced glutathione → protects the cell from oxidant damage. **G6PD deficiency** → oxidant haemolysis with **Heinz bodies and bite cells**.
- **Methaemoglobin reductase (NADH-cytochrome b5 reductase):** keeps iron in the Fe^{2+} state. **Methaemoglobinaemia** (from dapsone, nitrites, prilocaine) → cyanosis with "chocolate" blood. Treat with **methylene blue** (avoid in G6PD deficiency).

Red cell destruction

- **~90% extravascular:** in macrophages of the spleen, liver and marrow.
  - Haem → biliverdin → **unconjugated bilirubin** (see Chapter 57).
  - Iron is recycled.
  - Globin goes to the amino acid pool.
- **~10% intravascular:**

- Free Hb binds **haptoglobin** (which falls in haemolysis) → haemoglobinaemia, **haemoglobinuria**, **haemosiderinuria**.

★ Exam pearls

- The **reticulocyte count** (normal 0.5–2.5%, or 50–100 × 10^9/L) is the best quick index of marrow erythroid activity.
  - ↑ in haemolysis and after bleeding.
  - ↓ in aplasia, nutritional deficiency and CKD.
- **Polychromasia** on a Leishman/Giemsa-stained film corresponds to reticulocytes.

Chapter 44 · Granulopoiesis, Monocytopoiesis and Benign White Cell Disorders

(The outline's item "Myelopoiesis (B and T cell)" is split here: **myelopoiesis**, the granulocyte and monocyte lineages, is covered in this chapter; **B- and T-cell development** is covered in Chapter 46.)

Granulopoiesis

Myeloblast → promyelocyte → myelocyte → metamyelocyte → band (stab) form → segmented neutrophil

Stage	Key features
Myeloblast	Large; fine chromatin; nucleoli; few or no granules. **Mitotic**
Promyelocyte	**Primary (azurophilic) granules**: MPO, elastase, defensins, BPI. Mitotic (**APL** arrests at this stage: Auer rods, t(15;17))
Myelocyte	**Secondary (specific) granules**: lactoferrin, lysozyme, collagenase, alkaline phosphatase. **Last stage that divides**
Metamyelocyte	Kidney-shaped nucleus; does not divide
Band form	Horseshoe-shaped nucleus
Neutrophil	**2–5 nuclear lobes**; tertiary granules (gelatinase)

Neutrophil kinetics:

- Marrow transit takes about **10–14 days**. The marrow holds a large **storage pool**, about 10× the blood pool.
- In blood, neutrophils are split between a **circulating pool** and a **marginating pool** (adherent to vessel walls). These are roughly equal in size.
- Neutrophils survive **6–10 h in blood**, then migrate into tissues.

Demargination (from exercise, adrenaline or steroids) shifts cells into the circulating pool and raises the count quickly.

Monocytopoiesis: monoblast → promonocyte → monocyte. Monocytes circulate for 1–3 days, then enter tissues as **macrophages** (the **mononuclear phagocyte system**: Kupffer cells, alveolar macrophages, microglia, osteoclasts, Langerhans cells (dendritic)).

Eosinophils: bilobed nucleus; orange-red granules (major basic protein, eosinophil cationic protein, peroxidase). IL-5 drives their production. **Basophils:** dark purple granules (histamine, heparin).

Benign (reactive) disorders of white cells

Disorder	Definition	Causes
Neutrophil leucocytosis (neutrophilia)	>7.5 × 10^9/L	**Bacterial infection** (with **left shift**, **toxic granulation**, **Döhle bodies**, vacuolation), inflammation and tissue necrosis (MI, burns, surgery), **corticosteroids**, pregnancy, acute haemorrhage or haemolysis, malignancy, myeloproliferative neoplasms, exercise, sickle cell crisis
Leukaemoid reaction	Very high WBC (>50 × 10^9/L) with immature cells, mimicking leukaemia	Severe infection (TB, pertussis), burns, haemolysis. **Distinguished from CML** by: ↑ **neutrophil alkaline phosphatase (NAP) score** (low in CML), absence of the Philadelphia chromosome (BCR-ABL negative), toxic granulation, no basophilia
Leucoerythroblastic reaction	Immature myeloid cells + nucleated red cells in the blood	**Marrow infiltration** (metastatic carcinoma, myelofibrosis, leukaemia, TB, myeloma), severe haemolysis or haemorrhage
Neutropenia	<1.5 × 10^9/L (mild 1.0–1.5; moderate 0.5–1.0; **severe <0.5**: high infection risk)	**Benign ethnic neutropenia** (Duffy-null phenotype; common in people of African descent, without infection risk); viral infection (HIV, influenza), **typhoid, overwhelming sepsis**; **drugs (carbimazole, clozapine, cotrimoxazole, chemotherapy, chloramphenicol)**; B_{12}/folate deficiency; aplastic anaemia; leukaemia; **hypersplenism**; autoimmune (SLE, Felty's syndrome); congenital (Kostmann, cyclical neutropenia)

Disorder	Definition	Causes
Eosinophilia	>0.4 × 10^9/L	**Parasites (tissue helminths: hookworm, *Ascaris* larval migration, *Strongyloides*, filariasis, schistosomiasis, onchocerciasis)**, allergy (asthma, eczema, drugs), skin disease, Hodgkin lymphoma, eosinophilic leukaemia, Addison's disease, Churg–Strauss (EGPA). **Mnemonic NAACP:** Neoplasm, Allergy/Asthma, Addison's, Collagen vascular disease, Parasites
Basophilia		**Myeloproliferative neoplasms (CML)**, hypothyroidism, ulcerative colitis
Monocytosis	>0.8 × 10^9/L	**Chronic bacterial infection (TB, brucellosis, SBE)**, protozoa (malaria, leishmaniasis), recovery from neutropenia, Hodgkin lymphoma, CMML, sarcoidosis
Lymphocytosis	>3.5 × 10^9/L	**Viral infections (EBV: atypical lymphocytes in infectious mononucleosis)**, CMV, **pertussis** (very high), TB, brucellosis, toxoplasmosis, CLL. Normal in children
Lymphopenia		**HIV**, steroids, chemotherapy, SLE, uraemia, lymphoma

Morphological abnormalities of neutrophils

Abnormality	Description / cause
Hypersegmented neutrophils (≥6 lobes, or >5% with 5 lobes)	**Megaloblastic anaemia (B_{12}/folate deficiency)**
Pelger–Huët anomaly	Bilobed "pince-nez" nucleus; inherited (lamin B receptor) or acquired (MDS)
Toxic granulation, Döhle bodies, vacuolation	Severe infection
May–Hegglin anomaly	Döhle-like inclusions + giant platelets
Chédiak–Higashi	Giant granules
Alder–Reilly anomaly	Dense granules (mucopolysaccharidoses)

Qualitative (functional) defects: see Chapter 38 (CGD, LAD, MPO deficiency).

Infectious mononucleosis (EBV):

- Features: fever, sore throat, lymphadenopathy, splenomegaly.
- Blood: **atypical (reactive) lymphocytes ($CD8^+$ T cells)**.
- Serology: **heterophile antibody (Monospot / Paul–Bunnell) positive**; EBV VCA IgM.
- **Ampicillin causes a rash.**

Chapter 45 · Megakaryocytopoiesis and Platelets

Development

HSC → CFU-Meg → megakaryoblast → promegakaryocyte → mature megakaryocyte → platelets

- **Endomitosis:** the nucleus replicates DNA without the cell dividing. This gives **polyploid** (8N–64N) giant cells, 50–100 µm across, the **largest cells in the marrow**, with a multilobed nucleus.
- The cytoplasm develops granules and a **demarcation membrane system**. Megakaryocytes sit next to sinusoids and extend **proplatelets** into them. Blood flow sheds platelets.
- Each megakaryocyte produces **1,000–5,000 platelets**. Maturation takes about 10 days.
- **Thrombopoietin (TPO)** from the liver acts on the **c-MPL receptor**.
  - Free TPO levels are **inversely related** to the total platelet and megakaryocyte mass, because platelets bind and remove TPO.
  - IL-6 increases TPO, which explains reactive thrombocytosis in inflammation.
  - IL-11 and SCF also contribute.

Platelet structure and function

- **Size and number:** 2–3 µm anucleate discs; 150–400 × 10^9/L.
- **Lifespan:** 7–10 days.
- **Distribution:** about **one-third pooled in the spleen** (up to 90% in splenomegaly, causing hypersplenism).

Component	Contents / role
Glycoproteins	**GPIb–IX–V** (binds **vWF** → **adhesion**; deficient in Bernard–Soulier). **GPIIb/IIIa** (αIIbβ3; binds fibrinogen → **aggregation**; deficient in Glanzmann). GPIa/IIa and GPVI (collagen)
α-granules	Fibrinogen, vWF, factor V, **PF4**, PDGF, TGF-β, P-selectin, thrombospondin
Dense (δ) granules	**ADP**, ATP, **Ca^{2+}**, **serotonin**
Open canalicular system, dense tubular system (Ca^{2+} storage, TXA_2 synthesis)	
Phospholipid surface (PS exposure)	Platelet factor 3 activity; assembly of the tenase and prothrombinase complexes

Functions:

1. **Adhesion:** to subendothelial collagen via vWF and GPIb.
2. **Shape change and activation.**

3. **Release reaction:** ADP, TXA_2.
4. **Aggregation:** GPIIb/IIIa + fibrinogen.
5. **Procoagulant activity.**
6. **Clot retraction** (thrombosthenin/actomyosin).
7. **Vessel repair:** PDGF.

Disorders

Disorder	Causes
Thrombocytopenia	↓ **Production:** marrow failure (aplastic anaemia, leukaemia, infiltration), megaloblastic anaemia, drugs (chemotherapy, alcohol), viral infection (HIV, dengue). ↑ **Destruction: ITP** (immune thrombocytopenia: anti-GPIIb/IIIa autoantibodies; children post-viral, self-limiting; adults chronic), drugs (heparin/HIT, quinine), **DIC**, **TTP** (ADAMTS13 deficiency: pentad of fever, MAHA, thrombocytopenia, renal impairment, neurological signs), **HUS** (*E. coli* O157:H7), **malaria**, Lassa fever, neonatal alloimmune thrombocytopenia, **HELLP**. **Sequestration:** hypersplenism. **Dilution:** massive transfusion. **Pseudothrombocytopenia:** EDTA clumping
Thrombocytosis	**Reactive (secondary; commonest):** iron deficiency, infection, inflammation, malignancy, post-splenectomy, haemorrhage. **Primary: essential thrombocythaemia** (JAK2, CALR, MPL mutations), other MPNs
Qualitative defects	**Inherited:** Bernard–Soulier (giant platelets), Glanzmann thrombasthenia, storage pool disease, **von Willebrand disease** (the commonest inherited bleeding disorder; a vWF defect). **Acquired: aspirin, NSAIDs, clopidogrel, uraemia**, MPNs, paraproteinaemia

★ Exam pearls

- **Bleeding pattern:**
  - **Platelet or vessel defect:** immediate, mucocutaneous bleeding (petechiae, epistaxis, menorrhagia).
  - **Coagulation factor defect:** delayed, deep bleeding (haemarthroses, muscle haematomas).
- Spontaneous bleeding is unusual until platelets fall below **20×10^9/L**. Serious spontaneous bleeding occurs below **10×10^9/L**.

Chapter 46 · Lymphocyte Ontogeny: B-Cell and T-Cell Development

Lymphocyte ontogeny describes how lymphocytes develop from HSCs into mature, antigen-specific, self-tolerant B and T cells. It happens in **primary (central) lymphoid organs**: the **bone marrow** (B cells, NK cells) and the **thymus** (T cells). Mature cells then circulate through **secondary (peripheral) lymphoid organs**, where they meet antigen: **lymph nodes, spleen** (white pulp), **MALT** (tonsils, Peyer's patches, appendix).

B-cell development (bone marrow; antigen-independent)

Development needs **IL-7** and stromal contact (CXCL12, SCF).

Stage	Immunoglobulin gene status	Markers
Common lymphoid progenitor	Germline	CD34
Pro-B cell	**Heavy chain D–J then V–DJ recombination** (enzymes: **RAG-1/RAG-2**, TdT)	CD19, CD10, CD34, TdT
Pre-B cell	μ heavy chain expressed with surrogate light chain (VpreB/λ5) = **pre-BCR** (checkpoint; signals through **BTK**). Then **light chain V–J recombination** (κ first, then λ)	Cytoplasmic μ, CD19, CD10
Immature B cell	Surface **IgM**. **Central tolerance:** self-reactive cells undergo **receptor editing**, **clonal deletion** (apoptosis) or anergy	sIgM, CD19, CD20
Mature naïve B cell	Surface **IgM + IgD** (alternative splicing). Leaves the marrow for secondary lymphoid organs	CD19, CD20, CD21, CD22

Antigen-dependent phase (in secondary lymphoid organs):

1. Antigen binding plus **T-cell help (CD40L–CD40, IL-4, IL-21)** → activation.
2. **Germinal centre reaction** (follicle):
   - **Somatic hypermutation** (affinity maturation, via **AID**).
   - **Class switch recombination** (IgM → IgG, IgA, IgE; also AID).
   - Selection by follicular dendritic cells.
3. Differentiation into **plasma cells** (antibody-secreting; $CD138^+$, $CD38^+$; lose CD20) or **memory B cells**.

T-independent antigens (polysaccharides) give IgM-dominant responses without memory (see Chapter 32).

T-cell development (thymus)

T-cell precursors leave the marrow and **home to the thymic cortex**.

Stage	Site	Events
Double negative (DN; $CD4^-CD8^-$)	Subcapsular cortex	**TCR β-chain rearrangement** (RAG). **Pre-TCR** (β + pre-Tα) → **β-selection** → proliferation. A minority become **γδ T cells**
Double positive (DP; $CD4^+CD8^+$)	Cortex	**TCR α-chain rearrangement** → αβ TCR
Positive selection	**Cortex** (on cortical thymic epithelial cells)	DP cells whose TCR binds **self-MHC** with modest affinity **survive** (**MHC restriction**). About 90% die by "neglect". Cells that bind **MHC class II become $CD4^+$**; those that bind **MHC class I become $CD8^+$** (single positive)
Negative selection	**Medulla / corticomedullary junction** (medullary thymic epithelial cells, dendritic cells)	Cells that bind **self-peptide–MHC too strongly** are **deleted (apoptosis)**: **central tolerance**. **AIRE** (autoimmune regulator) drives expression of tissue-restricted self-antigens. AIRE mutation → **APECED / autoimmune polyendocrine syndrome type 1**. Some high-affinity cells become **Tregs ($CD4^+CD25^+FOXP3^+$)**
Mature naïve single-positive T cells	Exit to the periphery	$CD4^+$ helper or $CD8^+$ cytotoxic. **Only ~2–5% of thymocytes survive**

Peripheral T-cell differentiation (after antigen presentation by dendritic cells):

$CD4^+$ subset	Driving cytokine	Products	Role
Th1	IL-12	IFN-γ	Intracellular pathogens, macrophage activation (TB, granulomas)
Th2	IL-4	IL-4, IL-5, IL-13	**Helminths**, allergy, IgE
Th17	IL-6 + TGF-β, IL-23	IL-17, IL-22	Extracellular bacteria, fungi, autoimmunity
Tfh		IL-21	B-cell help in germinal centres

CD4⁺ subset	Driving cytokine	Products	Role
Treg		IL-10, TGF-β	Suppression

CD8⁺ CTLs kill virus-infected and tumour cells (perforin/granzyme, Fas-L).

NK cells:

- Innate lymphoid cells (CD56⁺, CD16⁺; no TCR).
- They kill cells that **lack MHC class I** ("missing self").
- They also mediate ADCC.

Ontogeny of the thymus and clinical correlations

- The thymus develops from the **third pharyngeal pouch** (with the inferior parathyroids). It is largest relative to body size at birth and **involutes after puberty**.
- **Primary immunodeficiencies:**

Disorder	Defect	Result
DiGeorge syndrome (22q11.2 deletion)	Thymic aplasia	↓ T cells, **hypocalcaemia** (absent parathyroids), cardiac defects, facial anomalies (**CATCH-22**)
SCID	Common γ chain (X-linked), **ADA deficiency**, RAG deficiency	Absent T cells ± B cells
X-linked (Bruton) agammaglobulinaemia	**BTK** mutation	Arrest at the pre-B stage; no B cells or immunoglobulins; recurrent bacterial infections after 6 months
Hyper-IgM syndrome	**CD40L defect**	No class switching
Common variable immunodeficiency		Late-onset hypogammaglobulinaemia

- **HIV** infects CD4⁺ T cells (via the CD4 + CCR5/CXCR4 receptors).
- **Lymphoid malignancies** arise at specific maturation stages:

Malignancy	Stage of origin
B-ALL	Pro-/pre-B (TdT⁺, CD10⁺)
Burkitt lymphoma	Germinal centre B cell, t(8;14) MYC
Follicular lymphoma	Germinal centre, t(14;18) BCL2
CLL	Mature B cell (CD5⁺, CD23⁺)
Myeloma	Plasma cell
T-ALL	Thymocyte (mediastinal mass)
ATLL	HTLV-1 CD4⁺ T cell

Lab assessment of lymphocytes:

- **Flow cytometry immunophenotyping:** CD markers, κ/λ light chain restriction (clonality).
- **CD4 count** in HIV.
- **TCR and IgH gene rearrangement studies** (clonality).

Chapter 47 · Iron Metabolism and Iron Deficiency Anaemia

Body iron distribution (~3–4 g in adults)

Compartment	Amount
Haemoglobin	**65–70%** (2.5 g)
Storage (ferritin, haemosiderin): liver, marrow macrophages, spleen	~25–30% (men ~1 g; women ~0.3 g)
Myoglobin	~4%
Enzymes (cytochromes, catalase)	<1%
Transport (transferrin-bound)	**~0.1%** (3–4 mg) but the **most dynamic**

Iron balance

- **Daily losses:**
  - **~1 mg** in men and post-menopausal women (desquamation of gut and skin cells).
  - **~2 mg** in menstruating women.
  - **Pregnancy** needs ~1,000 mg in total.
- **Diet** supplies 10–20 mg/day, of which only **~10% (1–2 mg) is absorbed**. **There is no physiological excretion mechanism**, so **iron balance is controlled entirely at the level of absorption**.

Absorption (duodenum and upper jejunum)

Step	Mechanism
Haem iron (meat, fish: 20–30% absorbed, better than non-haem)	Absorbed intact (haem carrier protein HCP1) → haem oxygenase releases Fe^{2+}
Non-haem iron (Fe^{3+} in cereals, vegetables)	**Ferric reductase (duodenal cytochrome b, Dcytb)** reduces Fe^{3+} → **Fe^{2+}** → **DMT1** (divalent metal transporter 1) on the apical membrane
Inside the enterocyte	Stored as ferritin (lost when the cell is shed) **or** exported
Basolateral export	**Ferroportin** (the only known cellular iron exporter) → Fe^{2+} is oxidised to Fe^{3+} by **hephaestin** (in plasma, caeruloplasmin) → binds **transferrin**

Factors that affect absorption:

Enhance	Inhibit
Vitamin C (ascorbic acid), gastric acid (HCl), haem iron, meat ("meat factor"), **iron deficiency**, ↑ erythropoiesis, pregnancy, hypoxia	**Phytates** (cereals), **tannins (tea)**, phosphates, **calcium**, antacids, **PPIs/H_2 blockers**, achlorhydria, tetracyclines, iron overload, **inflammation (hepcidin)**

Regulation: hepcidin (the master regulator)

Hepcidin is a 25-amino-acid peptide made by the **liver**. It **binds ferroportin and causes it to be internalised and degraded**. This blocks:

- iron export from enterocytes (↓ absorption)
- iron release from macrophages (iron is trapped in stores)

↑ Hepcidin (↓ iron availability)	↓ Hepcidin (↑ iron availability)
Iron overload; **inflammation (IL-6) → anaemia of chronic disease**	**Iron deficiency**; ↑ **erythropoietic activity (erythroferrone from erythroblasts)**; **hypoxia**; **hereditary haemochromatosis** (HFE, hemojuvelin, TfR2, hepcidin mutations) → iron overload

Cellular iron regulation: the **IRE/IRP system**. In iron deficiency, iron regulatory proteins bind iron-responsive elements, which ↑ **transferrin receptor (TfR1)** translation and ↓ ferritin translation.

Transport and storage

- **Transferrin:** a β-globulin made in the liver. It carries 2 Fe^{3+} and delivers them to cells through **TfR1** (receptor-mediated endocytosis). It is normally **~30% saturated** (20–45%).
- **Ferritin:**
  - A water-soluble apoferritin shell holding up to ~4,500 Fe atoms.
  - **Serum ferritin reflects body iron stores** (1 µg/L ≈ 8–10 mg stored iron).
  - It is also an **acute phase reactant**, so it rises in inflammation, liver disease and malignancy.
- **Haemosiderin:** insoluble, partially degraded ferritin. It is **Perls' Prussian blue positive** in marrow macrophages.

Iron deficiency anaemia (IDA)

> **Iron deficiency anaemia** is anaemia caused by an insufficient iron supply for Hb synthesis. It is the **commonest cause of anaemia worldwide**, and among **children and pregnant women in Nigeria**.
>
> **Anaemia (WHO):** Hb <130 g/L in men; <120 g/L in non-pregnant women; **<110 g/L in pregnancy and in children 6–59 months**.

Causes:

Category	Examples
Chronic blood loss (the commonest cause in adults)	**Menorrhagia**. **GI bleeding:** peptic ulcer, NSAIDs, **colorectal cancer** (always investigate men and post-menopausal women), oesophageal varices, haemorrhoids, angiodysplasia. **Hookworm infestation** (*Necator americanus*, *Ancylostoma duodenale*): ~0.03–0.2 mL blood loss per worm per day. **Schistosomiasis** (*S. haematobium* haematuria; *S. mansoni*). Haematuria, epistaxis, frequent blood donation
Increased demand	**Pregnancy, infancy (especially prematurity), adolescence**, menstruation, EPO therapy
Poor intake	Vegetarian diets, poverty, elderly, prolonged exclusive milk feeding in infants (cow's milk)
Malabsorption	**Coeliac disease**, gastrectomy, *H. pylori*, achlorhydria, atrophic gastritis, bariatric surgery

Stages:

1. **Iron depletion (negative iron balance):** ↓ ferritin; Hb normal.
2. **Iron-deficient erythropoiesis:** ↓ serum iron, ↑ TIBC, ↓ transferrin saturation, ↑ free erythrocyte protoporphyrin / zinc protoporphyrin, ↑ soluble transferrin receptor; Hb still normal.
3. **Iron deficiency anaemia:** ↓ Hb, microcytic hypochromic.

Clinical features:

- **General anaemia symptoms:** fatigue, pallor, dyspnoea, palpitations, high-output cardiac failure.
- **Epithelial changes:**
  - **Koilonychia** (spoon nails)
  - **Angular cheilitis (stomatitis)**
  - **Glossitis** (smooth tongue)
  - Brittle hair
  - **Plummer–Vinson (Paterson–Kelly) syndrome:** post-cricoid **oesophageal web** + dysphagia + IDA (pre-malignant)
- **Pica:** craving for non-food items (clay/**geophagia**, ice/pagophagia).
- **Children:** impaired cognitive development, behavioural problems.
- **Pregnancy:** preterm delivery, low birth weight.

Laboratory findings:

Test	IDA	Anaemia of chronic disease (ACD)	β-thalassaemia trait	Sideroblastic anaemia
MCV, MCH	↓↓	Normal or ↓	↓↓ **(disproportionately low for Hb)**	↓ (congenital), ↑ (acquired MDS)

Test	IDA	Anaemia of chronic disease (ACD)	β-thalassaemia trait	Sideroblastic anaemia
Serum ferritin	**↓ (<15 µg/L diagnostic; <30 likely)**	**Normal or ↑**	Normal	↑
Serum iron	↓	↓	Normal	↑
TIBC / transferrin	↑	↓	Normal	Normal or ↓
Transferrin saturation	↓ (<16%)	↓ or normal	Normal	↑
Marrow iron stores	**Absent**	**Present (↑ in macrophages)**	Present	Present + **ring sideroblasts**
Soluble TfR	↑	Normal	↑	
Hb electrophoresis	Normal	Normal	↑ **HbA_2 (>3.5%)**	Normal
RDW	↑ (anisocytosis)	Normal	Normal	↑

Blood film in IDA:

- **Microcytic, hypochromic** red cells.
- **Anisopoikilocytosis.**
- **Pencil (elongated) cells.**
- Target cells (few).
- Often **reactive thrombocytosis**.

Mentzer index = MCV ÷ RBC count. <13 suggests thalassaemia trait; >13 suggests IDA.

Management:

1. **Find and treat the cause.**
   - Deworm (**albendazole**).
   - Treat menorrhagia.
   - **Upper and lower GI endoscopy** for men and post-menopausal women with IDA (BSG guidelines).
   - Coeliac serology.
2. **Oral iron:**
   - Preparations: **ferrous sulfate 200 mg** (65 mg elemental iron), ferrous fumarate, ferrous gluconate.
   - Dose: **once daily or on alternate days** (this improves absorption by avoiding the hepcidin surge). Take on an empty stomach with vitamin C.
   - Side effects: **nausea, epigastric pain, constipation or diarrhoea, black stools**.
   - **Expected response: reticulocytosis at 7–10 days**; Hb rises by **~10–20 g/L every 2–3 weeks**.
   - **Continue for 3 months after Hb normalises** to replenish stores.

3. **Parenteral iron:**
    - Preparations: **ferric carboxymaltose**, iron sucrose, iron dextran (anaphylaxis risk), ferric derisomaltose.
    - Indications: intolerance or non-adherence, malabsorption, ongoing losses, CKD on EPO, **late pregnancy** (rapid repletion), IBD.
    - **It does not raise Hb faster than oral iron**, but it **replaces stores faster**.
4. **Blood transfusion:** only for severe or symptomatic anaemia, cardiac compromise, or late pregnancy with severe anaemia.

Prevention in Nigeria:

- Iron and folic acid supplementation in pregnancy.
- Food fortification.
- Delayed cord clamping.
- Deworming of school children.
- Malaria control.

Causes of failure to respond to oral iron:

- Non-adherence (the commonest).
- Wrong diagnosis (thalassaemia, ACD).
- Continuing blood loss.
- Malabsorption.
- Concurrent B_{12}/folate deficiency.

> 🎤 **Viva corner**
>
> **Q:** How do you distinguish iron deficiency from anaemia of chronic disease? **A:** - In IDA, ferritin is low, TIBC is high, and marrow iron is absent. - In ACD, ferritin is normal or high (it is an acute phase reactant), TIBC is low, and marrow macrophage iron is present but trapped by high hepcidin. - If the patient has inflammation, a ferritin below 100 µg/L may still mean iron deficiency.

Chapter 48 · Vitamin B_{12} and Folate Metabolism; Megaloblastic Anaemia

Vitamin B_{12} (cobalamin)

Feature	Detail
Sources	**Only foods of animal origin** (meat, liver, fish, eggs, dairy). It is synthesised by microorganisms. **Strict vegans are at risk**
Daily requirement	1–3 µg (2.4 µg)
Body stores	**2–5 mg**, mainly in the **liver**. They **last 2–4 years**, so deficiency develops slowly
Absorption	1. Freed from food protein by **pepsin and gastric acid**. 2. Binds **haptocorrin (R-binder)** from saliva. 3. In the duodenum, **pancreatic proteases** digest haptocorrin. 4. B_{12} binds **intrinsic factor (IF)**, which is secreted by **gastric parietal cells**. 5. The **IF–B_{12} complex** is absorbed in the **terminal ileum** through the **cubilin–amnionless receptor (cubam)**. 6. ~1% is absorbed passively at high doses, which allows oral therapy
Transport	**Transcobalamin II (holoTC)**: the active, delivered fraction. Transcobalamin I (haptocorrin) carries most of the plasma B_{12}, but it is inactive
Enterohepatic circulation	Yes

Biochemical functions (two reactions):

1. **Methionine synthase:** homocysteine → **methionine** (needs **methyl-cobalamin**). This also converts **5-methyl-THF → THF**.
   - When B_{12} is lacking, folate is **trapped as methyl-THF** (the "**methyl-folate trap**") → ↓ THF → ↓ 5,10-methylene-THF → ↓ **thymidylate (dTMP)** synthesis → **impaired DNA synthesis** → **megaloblastic** change.
   - **Homocysteine rises.**
2. **Methylmalonyl-CoA mutase: methylmalonyl-CoA → succinyl-CoA** (needs **adenosyl-cobalamin**).
   - When B_{12} is lacking, **methylmalonic acid (MMA) rises**. Abnormal fatty acids then accumulate in myelin, and ↓ S-adenosylmethionine (SAM) impairs methylation. Both contribute to **neurological damage (subacute combined degeneration)**.
   - **This reaction does not need folate**, which explains why giving folate alone does not correct the neuropathy.

Causes of B_{12} deficiency:

Category	Causes
Inadequate intake	**Veganism**, poverty
Gastric causes (↓ IF)	**Pernicious anaemia** (the commonest cause in temperate countries; autoimmune gastritis with **antibodies against parietal cells (~90%, non-specific) and against intrinsic factor (~50%, specific)**; atrophic body gastritis; ↑ risk of gastric cancer). **Total or partial gastrectomy**, bariatric surgery. Congenital IF deficiency
Food-cobalamin malabsorption	Atrophic gastritis, **long-term PPIs**, **metformin**, elderly
Intestinal causes	**Ileal resection / Crohn's disease** (terminal ileum), **tropical sprue**, blind loop / **bacterial overgrowth**, ***Diphyllobothrium latum*** (fish tapeworm), Imerslund–Gräsbeck syndrome (cubam defect), pancreatic insufficiency
Other	**Nitrous oxide** (inactivates cobalamin), TC II deficiency, HIV

Folate (folic acid; pteroylglutamic acid)

Feature	Detail
Sources	**Green leafy vegetables**, liver, legumes, fruits, fortified foods. **Destroyed by prolonged cooking**
Daily requirement	~200–400 µg. **Pregnancy: 600 µg**
Body stores	**10–15 mg**. They **last only ~4 months**, so deficiency develops quickly
Absorption	**Duodenum and jejunum**. Dietary polyglutamates are converted to monoglutamates (by **folate conjugase**; inhibited by phenytoin and alcohol), then absorbed via the **proton-coupled folate transporter** and converted to **5-methyl-THF**
Function	**One-carbon (methyl, formyl) transfer**: **thymidylate synthesis** (5,10-methylene-THF), purine synthesis, amino acid interconversions (serine–glycine; homocysteine → methionine)

Causes of folate deficiency:

Category	Causes
Inadequate intake	Poor diet, poverty, elderly, **alcoholism**, overcooked food, goat's milk in infants, PEM
Malabsorption	**Coeliac disease**, **tropical sprue**, Crohn's disease, jejunal resection

Category	Causes
Increased requirement	**Pregnancy and lactation**, prematurity. **Chronic haemolytic anaemia, especially sickle cell disease** (so routine folic acid is given in SCD), thalassaemia. Malignancy, inflammatory disease (TB, RA), exfoliative skin disease (psoriasis), haemodialysis
Drugs	**Antifolates:** methotrexate, trimethoprim, **pyrimethamine** (DHFR inhibitors). **Anticonvulsants:** phenytoin, valproate, phenobarbital. Sulfasalazine, alcohol
Excess loss	Dialysis, liver disease

Megaloblastic anaemia

> **Megaloblastic anaemias** are a group of anaemias in which erythroblasts in the marrow show a characteristic abnormality: **nuclear maturation lags behind cytoplasmic maturation** ("**nuclear–cytoplasmic asynchrony**"). This is due to **defective DNA synthesis**, while RNA and protein synthesis continue normally. The usual causes are **B_{12} or folate deficiency**, or drugs that interfere with DNA synthesis (methotrexate, hydroxyurea, zidovudine, azathioprine).

Pathophysiology:

- Cells grow larger but cannot divide properly.
- **Ineffective erythropoiesis** (intramedullary haemolysis) causes:
  - ↑ unconjugated bilirubin (mild jaundice, "**lemon-yellow**" tint from pallor + jaundice).
  - ↑↑ **LDH**.
  - ↓ haptoglobin.
- **Pancytopenia** is possible, because all rapidly dividing cells are affected (including epithelia: glossitis, infertility, abnormal cervical cytology).

Clinical features:

- **Anaemia:** insidious onset. Patients may be quite well adapted even with very low Hb.
- **Mild jaundice.**
- **Glossitis** (beefy red tongue), angular cheilitis.
- Weight loss, diarrhoea.
- Mild splenomegaly.
- Skin hyperpigmentation (in Africans), sterility.
- **Neurological (B_{12} only):**
  - **Subacute combined degeneration of the spinal cord: dorsal columns** (↓ vibration and proprioception, positive Romberg, sensory ataxia) + **lateral corticospinal tracts** (spastic weakness, **extensor plantars**, with *absent ankle reflexes* if neuropathy coexists).
  - Peripheral neuropathy (symmetrical paraesthesiae).

    - Optic atrophy.
    - **Dementia and psychiatric disturbance** ("megaloblastic madness").
    - Neurological damage can occur **without anaemia** and may be irreversible.

- **Folate deficiency in pregnancy → neural tube defects**: give **folic acid 400 µg daily preconceptionally**, or **5 mg** for high risk (previous NTD, antiepileptics, diabetes, obesity, sickle cell disease).

Laboratory findings:

Test	Finding
FBC	↑ **MCV (>100 fL; often >110–120)**, ↓ Hb; ↓ **WBC and platelets** possible (pancytopenia); ↓ reticulocytes (inappropriately low)
Blood film	**Oval macrocytes (macro-ovalocytes)**, anisopoikilocytosis, **hypersegmented neutrophils (≥6 lobes) (an early and sensitive sign)**, Howell–Jolly bodies, basophilic stippling, occasional megaloblasts
Bone marrow (not usually required)	**Hypercellular**, with **megaloblasts** (large erythroblasts with open, sieve-like ("lacy") chromatin and well-haemoglobinised cytoplasm), **giant metamyelocytes** and band forms; hypersegmented megakaryocytes
Biochemistry	↑ **LDH (very high)**, ↑ unconjugated bilirubin, ↓ haptoglobin
Serum B_{12}	↓ (<200 ng/L or <148 pmol/L) in B_{12} deficiency
Serum and red cell folate	↓ in folate deficiency (red cell folate reflects tissue stores)
Methylmalonic acid (MMA)	↑ **in B_{12} deficiency only** (normal in folate deficiency)
Homocysteine	↑ **in both**
HoloTC (active B_{12})	↓ (early marker)
Pernicious anaemia work-up	**Anti-intrinsic factor antibodies** (specific), anti-parietal cell antibodies, ↑ gastrin, ↓ pepsinogen. **Schilling test** (historical: radioactive B_{12} absorption with and without IF)

Other causes of macrocytosis (non-megaloblastic):

- **Alcohol** (the commonest cause of macrocytosis)
- **Liver disease**
- **Hypothyroidism**
- **Reticulocytosis** (haemolysis, bleeding)
- **Myelodysplastic syndrome**
- Aplastic anaemia
- Pregnancy and the newborn

- Drugs: **hydroxyurea, zidovudine, methotrexate**
- Myeloma

Management:

1. **B_{12} deficiency:**
   - **Hydroxocobalamin 1 mg IM.**
     - **Without neurological involvement:** 3 times a week for 2 weeks.
     - **With neurological involvement:** on alternate days until there is no further improvement.
   - **Maintenance: 1 mg every 2–3 months for life** in pernicious anaemia.
   - **High-dose oral B_{12} (1–2 mg/day)** is effective even without IF (passive absorption).
2. **Folate deficiency: folic acid 5 mg daily orally for 4 months**, and continue if the cause persists (e.g. SCD).
3. ⚠ **Never give folic acid alone to a patient with unexplained megaloblastic anaemia before ruling out B_{12} deficiency.** It may correct the anaemia but **precipitate or worsen subacute combined degeneration**. If in doubt, give both.
4. **Monitor potassium: hypokalaemia** occurs in the first 48 h of treatment (K^+ moves into new cells). Watch for this in severe anaemia.
5. **Response:**
   - Patient feels better within 24–48 h.
   - **Reticulocyte peak at 5–7 days.**
   - Hb rises by ~10 g/L per week.
   - Hypersegmented neutrophils disappear within ~14 days.
6. **Transfusion:** only for severe anaemia with cardiac failure. Transfuse slowly (packed cells + diuretic), because of the **risk of circulatory overload**.

★ Exam pearls

- **MMA distinguishes B_{12} from folate deficiency.** MMA is ↑ only in B_{12} deficiency. Homocysteine is ↑ in both.
- **Stores:** B_{12} lasts years; folate lasts months.
- **Absorption site:** B_{12} in the terminal ileum; folate in the jejunum.
- **Neuropathy** occurs only in B_{12} deficiency.

Haematology summary: approach to anaemia by MCV

Microcytic (MCV <80 fL)	Normocytic (80–100)	Macrocytic (>100)
TAILS: Thalassaemia, **A**naemia of chronic disease (late), **I**ron deficiency, **L**ead poisoning, **S**ideroblastic anaemia (congenital)	**Acute blood loss**, **anaemia of chronic disease**, **CKD**, haemolysis (some), aplastic anaemia, marrow infiltration, mixed deficiency, **sickle cell anaemia**, endocrine (hypothyroidism, Addison's), pregnancy	**Megaloblastic** (B_{12}, folate, drugs); **non-megaloblastic** (alcohol, liver disease, hypothyroidism, reticulocytosis, MDS, myeloma, aplastic anaemia)

Haematology Self-Assessment

Section A: Single best answer MCQs

1. The main site of haemopoiesis in the fetus at 4 months is the: A. Yolk sac · B. **Liver** · C. Bone marrow · D. Thymus · E. Lymph nodes

B. The hepatic phase runs from about 6 weeks to 7 months of gestation.

2. Which of the following enhances non-haem iron absorption? A. Tea · B. Phytates · C. **Ascorbic acid** · D. Calcium · E. Omeprazole

C. Vitamin C reduces Fe^{3+} to Fe^{2+} and chelates it, which keeps it soluble.

3. Hepcidin reduces iron availability by: A. Binding transferrin · B. Inhibiting DMT1 directly · C. **Degrading ferroportin** · D. Increasing ferritin synthesis · E. Inhibiting hephaestin

C. Hepcidin binds ferroportin, which is then internalised and degraded.

4. A 40-year-old man has Hb 85 g/L, MCV 68 fL and ferritin 6 µg/L. The most important next step is: A. Blood transfusion · B. Bone marrow biopsy · C. **Endoscopic evaluation of the GI tract** · D. Haemoglobin electrophoresis · E. Start IV iron

C. In an adult man, iron deficiency is caused by GI blood loss until proven otherwise. Look for occult malignancy. In Nigeria, check stool for hookworm as well.

5. Which test is raised in B_{12} deficiency but normal in folate deficiency? A. Homocysteine · B. LDH · C. **Methylmalonic acid** · D. MCV · E. Bilirubin

C. Methylmalonyl-CoA mutase needs B_{12} but not folate, so only B_{12} deficiency raises methylmalonic acid.

6. Hypersegmented neutrophils are characteristic of: A. Iron deficiency · B. **Megaloblastic anaemia** · C. Pelger–Huët anomaly · D. Bacterial infection · E. CML

B. Hypersegmented neutrophils are an early sign of megaloblastic anaemia.

7. Positive selection of T cells takes place in the: A. Bone marrow · B. **Thymic cortex** · C. Thymic medulla · D. Lymph node · E. Spleen

B. Positive selection takes place in the thymic cortex. Negative selection takes place mainly in the medulla.

8. Bruton's agammaglobulinaemia arrests B-cell development at the: A. Stem cell stage · B. **Pre-B cell stage** · C. Plasma cell stage · D. Memory B cell stage · E. Germinal centre

B. Bruton's tyrosine kinase (BTK) is needed for pre-BCR signalling, so development stops at the pre-B cell.

9. Platelet adhesion to collagen is mediated mainly by: A. GPIIb/IIIa + fibrinogen · B. **GPIb + von Willebrand factor** · C. ADP · D. TXA_2 · E. P-selectin

B. GPIb binding vWF mediates adhesion. GPIIb/IIIa binding fibrinogen mediates aggregation.

10. The main stimulus to erythropoietin production is: A. Anaemia per se · B. **Renal tissue hypoxia** · C. Iron · D. Androgens · E. IL-3

B. Hypoxia stabilises HIF, which drives EPO production in the kidney.

11. A leukaemoid reaction differs from CML in having: A. Basophilia · B. Philadelphia chromosome · C. **High NAP score** · D. Splenomegaly · E. Myelocytes

C. The neutrophil alkaline phosphatase (NAP) score is high in a leukaemoid reaction and low in CML.

12. Which of these does NOT cause macrocytosis? A. Alcohol · B. Hypothyroidism · C. Hydroxyurea · D. **β-thalassaemia trait** · E. Liver disease

D. β-thalassaemia trait causes microcytosis, not macrocytosis.

Section B: Short answers and essays

1. Describe the haemopoietic inductive microenvironment and its clinical importance.

Cover: - Definition. - Components: stroma, ECM, growth factors, adhesion molecules (VLA-4/VCAM-1, CXCR4/CXCL12). - The endosteal and vascular niches. - Functions. - Clinical relevance: aplastic anaemia, myelofibrosis, stem cell mobilisation, transplantation.

2. Discuss iron metabolism. Outline the investigation and management of iron deficiency anaemia in a pregnant Nigerian woman.

3. Write short notes on: (a) erythropoietin; (b) subacute combined degeneration of the cord; (c) causes of neutrophilia; (d) megakaryocytopoiesis; (e) thymic selection.

4. Compare and contrast vitamin B_{12} and folate metabolism. A tabular answer is acceptable: sources, stores, absorption site, functions, causes of deficiency, tests and treatment.

Section C: Viva voce bank (Haematology)

#	Question	Model answer
1	Sites of haemopoiesis in an adult?	Axial skeleton: vertebrae, ribs, sternum, skull, pelvis, and the proximal femur and humerus.
2	What is extramedullary haemopoiesis?	Blood formation outside the marrow (liver, spleen), e.g. in myelofibrosis or thalassaemia.
3	Name the haemopoietic growth factors.	SCF, IL-3, GM-CSF, G-CSF, M-CSF, EPO, TPO, IL-5, IL-7.
4	What is the lifespan of a red cell, neutrophil and platelet?	120 days; 6–10 hours in blood; 7–10 days.
5	What is the reticulocyte and how is it stained?	An immature, anucleate RBC containing residual RNA. Supravital stain: new methylene blue.
6	Where is iron absorbed?	Duodenum and upper jejunum.
7	What is the best test for iron stores?	Serum ferritin. The gold standard is Perls' stain of the bone marrow.
8	Name three causes of IDA in Nigeria.	Hookworm infestation, menorrhagia, poor diet, pregnancy, schistosomiasis, GI bleeding.
9	Clinical signs of IDA?	Pallor, koilonychia, angular cheilitis, glossitis, pica.
10	Where is B_{12} absorbed and what does it need?	Terminal ileum; it needs intrinsic factor from gastric parietal cells.
11	What is the methyl-folate trap?	Without B_{12}, methionine synthase cannot demethylate 5-methyl-THF, so folate is trapped in an unusable form.
12	Why not give folate alone in B_{12} deficiency?	It can correct the anaemia but allow the neuropathy (subacute combined degeneration) to progress.
13	What is pernicious anaemia?	Autoimmune gastritis causing intrinsic factor deficiency and B_{12} malabsorption.
14	What are the stages of T-cell development?	DN → DP → positive selection (cortex) → negative selection (medulla) → SP CD4 or CD8.
15	What is AIRE?	The autoimmune regulator gene. It drives thymic expression of tissue antigens for negative selection. Mutation causes APS-1.

#	Question	Model answer
16	What is endomitosis?	DNA replication without cell division. It produces polyploid megakaryocytes.
17	What is benign ethnic neutropenia?	A lower normal neutrophil count in people of African descent (Duffy-null), with no increased infection risk.
18	Causes of eosinophilia?	Parasites (helminths), allergy, drugs, Hodgkin lymphoma, Addison's disease, vasculitis.
19	What is the expected response to oral iron?	Reticulocytosis at 7–10 days; Hb rises by ~20 g/L every 3 weeks.
20	What is hypersplenism?	Splenomegaly with cytopenias and a compensatory marrow, corrected by splenectomy.

Book 4 — Chemical Pathology

Laboratory medicine principles → potassium → acid–base → renal function → calcium, phosphate and metabolic bone disease → diabetes mellitus → lipoproteins (introduction)

Chapter 49 · Principles of Laboratory Medicine

This chapter brings together seven related topics from the outline:

- Requests for laboratory investigation
- Specimen collection
- Sources of laboratory error and their consequences
- The role of the physician in obtaining a valid result
- Point-of-care testing (POCT)
- Reference intervals and medical decision limits
- Decision-making using laboratory results

> **Chemical pathology (clinical chemistry / clinical biochemistry)** applies biochemical methods to diagnosing, screening, prognosing and monitoring disease. **Over 70% of clinical decisions depend on laboratory results.**

49.1 Why request a laboratory test?

Purpose	Example
Diagnosis (confirm or exclude)	Troponin in chest pain
Screening (asymptomatic population)	Neonatal TSH (congenital hypothyroidism), lipid profile, HbA1c, sickle cell screening
Prognosis	Creatinine in AKI, AFP in germ cell tumours
Monitoring (disease progress or treatment)	HbA1c in DM, INR on warfarin, drug levels (TDM), tumour markers (CEA, PSA)
Management (guide treatment)	K^+ before giving insulin in DKA
Medicolegal	Blood alcohol, drugs of abuse
Research	

The request form (the physician's responsibility)

A complete request form must include:

1. **Patient identification:** full name, hospital number, age/date of birth, sex, ward or clinic.
2. **Clinical details:** a provisional diagnosis and **relevant clinical information** (drugs, e.g. "on digoxin"; "pregnant"; "on IV fluids"), plus the specific question.
3. **Tests requested** (clearly written).
4. **Specimen type**, **date and time of collection** (critical for drug levels, cortisol and the glucose tolerance test).

5. **Requesting clinician's name, signature and contact** (bleep or phone), for communicating critical results.
6. **Urgency** (routine or emergency).
7. **Biohazard label** for known high-risk infections (HIV, HBV, Lassa fever).

Good requesting practice:

- Request tests that will **change management**. Ask: "Will the result alter what I do?"
- **Avoid**: "routine" batteries without indication, repeating tests too soon (HbA1c needs 3 months), and **"defensive" or "fishing" tests**.
- More tests increase the chance of a false-positive "abnormal" result. With 20 independent tests and 95% reference intervals, **the probability that all are "normal" in a healthy person is only $0.95^{20} \approx 36\%$**.

49.2 Specimen collection: types, containers and principles

Specimen	Container / additive (colour codes vary by manufacturer; common BD/Vacutainer and international convention)	Used for
Serum (clotted blood)	**Red** (plain) or **gold/yellow** (SST: clot activator + gel separator)	**Most chemistry**: U&E, LFT, lipids, Ca, hormones, proteins, drug levels
Plasma: lithium heparin	**Green**	Electrolytes, urgent chemistry, ammonia (on ice)
Plasma: fluoride oxalate	**Grey**	**Glucose** (fluoride inhibits **enolase** → stops glycolysis), **lactate**, alcohol
Whole blood: EDTA (K_2/K_3)	**Purple/lavender**	FBC, **HbA1c**, blood film, Hb electrophoresis, PTH, ACTH (on ice), red cell folate, blood grouping (pink)
Plasma: sodium citrate (1:9)	**Light blue**	**Coagulation** (PT, aPTT): the tube must be **filled exactly**
Sodium citrate (1:4)	Black	ESR (Westergren)
Arterial blood gas	Heparinised syringe, **no air bubbles**, analyse within 15–30 min or **on ice**	pH, pO_2, pCO_2, HCO_3^-, lactate
Urine	Random/spot (protein:creatinine or albumin:creatinine ratio), **early morning** (concentrated; pregnancy test), **24-hour urine** (with preservative, e.g. **HCl** for catecholamines/VMA and calcium; **discard the first void**, collect all others including the next morning's first void), midstream (MSU) for culture	

Specimen	Container / additive (colour codes vary by manufacturer; common BD/Vacutainer and international convention)	Used for
CSF	Sterile tubes. **Fluoride tube for glucose** (with a paired blood glucose). Protein, microscopy; xanthochromia (protected from light)	Meningitis, SAH
Others	Faeces (occult blood, elastase, calprotectin), sweat (chloride: cystic fibrosis), fluids (pleural, ascitic; with paired serum for Light's criteria), stones, amniotic fluid, saliva, hair, tissues	

Order of draw (CLSI GP41) to avoid additive carry-over:

1. Blood cultures
2. **Citrate (blue)**
3. Serum (red/gold)
4. **Heparin (green)**
5. **EDTA (purple)**
6. **Fluoride oxalate (grey)**

Mnemonic: "**B**ecause **B**lue **R**ed **G**reen **P**urple **G**rey". Drawing EDTA (potassium EDTA) before a serum tube gives **spurious hyperkalaemia and hypocalcaemia**: EDTA adds K^+ and chelates Ca^{2+}.

Principles of good venepuncture:

- Identify the patient positively (two identifiers). **Label tubes at the bedside**.
- Apply the **tourniquet for <1 minute**. Prolonged stasis → haemoconcentration (↑ protein, Ca^{2+}, lipids, cells) and ↑ K^+.
- Avoid **fist clenching** (raises K^+).
- **Do not sample from an arm with an IV infusion running** (dilution, contamination: high glucose or Na^+).
- Use a suitable needle size (≥21G) to avoid **haemolysis**. Do not force blood through a needle.
- Note the **timing**:

Test / situation	Timing requirement
Cortisol	9 am peak (circadian rhythm)
Trough and peak drug levels	Digoxin ≥6 h post-dose; lithium 12 h post-dose; gentamicin per protocol
Fasting tests (glucose, triglycerides)	Fast 8–12 h
Iron	Morning

Test / situation	Timing requirement
Prolactin	Avoid stress
Post-prandial tests	Per protocol

- **Posture:** standing increases protein-bound analytes by ~10% compared with lying down (protein, Ca^{2+}, lipids, enzymes). Measure renin and aldosterone in a standard posture.
- Transport promptly at the correct temperature:
  - **On ice:** ammonia, ABG (if delayed), ACTH, PTH (some), lactate.
  - **Protect from light:** bilirubin (neonatal), porphyrins, vitamin A.
- **Centrifuge within 1–2 hours.**
- **Safety:**
  - Universal precautions; dispose of sharps safely; **no re-capping**.
  - Post-exposure prophylaxis for HIV and HBV.
  - **Lassa fever** (endemic in Nigeria): special biosafety precautions.

49.3 Sources of laboratory error and their consequences

Errors are divided into three phases:

Phase	Proportion of errors	Examples
Pre-analytical (before analysis; **the commonest: ~60–70%**)	Mostly the **clinician's or phlebotomist's** responsibility	**Patient misidentification / wrong labelling** (most dangerous: wrong blood in tube → transfusion reactions), **wrong container**, **haemolysis** (↑ K^+, LDH, AST, phosphate, Mg^{2+}; interferes with bilirubin assays), **delayed separation** (glycolysis → ↓ glucose by ~5–7%/hour; K^+ leaks from cells → **spurious hyperkalaemia**; especially in a cold room), **"drip arm" sampling**, prolonged tourniquet, **EDTA contamination** (↑ K^+, ↓ Ca^{2+}, ↓ ALP, ↓ Mg^{2+}), insufficient sample or wrong blood:anticoagulant ratio, wrong timing, **non-fasting** samples, incomplete 24-h urine, lipaemia, **drugs** (biotin interferes with immunoassays: TFTs, troponin), patient factors (exercise, posture, diet, stress)

Phase	Proportion of errors	Examples
Analytical (~10–15%)	Laboratory	Instrument malfunction, reagent deterioration, calibration error, **interference** (haemolysis, icterus, lipaemia: "HIL"; heterophile antibodies; **hook effect** with very high analyte concentrations, e.g. prolactin or hCG), random and systematic error. Controlled by **quality control (internal QC, Westgard rules, Levey–Jennings charts)** and **external quality assessment (EQA)**; accreditation (ISO 15189)
Post-analytical (~15–20%)	Lab + clinician	Transcription errors, wrong units, delayed reporting, **failure to read, act on or follow up results**, misinterpretation, results filed without review, not communicating critical values

Error types:

Type	Characteristics	Cause	Detection
Random error	Affects **precision** (reproducibility); unpredictable scatter	Pipetting variation, electronic noise	**Coefficient of variation (CV%)** = SD/mean × 100
Systematic error (bias)	Affects **accuracy**; consistent shift in one direction	Calibration error, reagent deterioration	Shifts and trends on QC charts

- **Accuracy:** closeness to the true value.
- **Precision:** closeness of repeated measurements to each other.

Consequences of laboratory error:

- Misdiagnosis and inappropriate treatment. Examples: giving insulin to "hyperkalaemia" that was really haemolysis; missing true hypoglycaemia.
- Patient harm or death (transfusion of incompatible blood).
- Delayed treatment.
- Unnecessary repeat tests and further investigations, increasing cost.
- Anxiety.
- Medicolegal liability.
- Loss of confidence in the laboratory.

49.4 The role of the physician in getting a valid result

The physician controls most of the **pre-analytical** and **post-analytical** phases. **The total testing process is a loop**, and the clinician's responsibilities are spread around it:

1. **Request appropriately.** Choose the right test, at the right time, for the right reason.
2. **Prepare the patient.** Fasting, stopping interfering drugs, posture, timing relative to doses, avoiding exercise, and informed consent (HIV, genetic tests).
3. **Complete the request form fully and legibly**, with clinical details and drug history.
4. **Collect the specimen correctly** (or supervise collection). Right container, order of draw, technique, volume, labelling at the bedside.
5. **Ensure prompt, correct transport and storage** (on ice, protected from light, within the time limit).
6. **Communicate with the laboratory.** Discuss unusual requests, urgent samples and dynamic function tests. Give a contact for critical values.
7. **Interpret results in the clinical context:**
   - Think "Is this result plausible?" (**delta check** against previous results).
   - Consider pre-analytical artefacts before acting. Example: K^+ of 7.5 in a well patient with a normal ECG and a haemolysed sample → **repeat before treating**.
   - Use reference intervals appropriate for age, sex and pregnancy.
8. **Act on and document results.** Inform the patient. Follow up abnormal results.
9. **Audit and feedback** to the laboratory. Participate in quality improvement.

49.5 Point-of-care testing (POCT)

> **POCT (near-patient testing, bedside testing)** is any analytical test done **outside the central laboratory, close to the patient**, usually by non-laboratory staff (doctors, nurses, patients), where the result can lead to an immediate change in management.

Examples:

- **Glucose meters (glucometers)**
- **Urine dipsticks**
- **Blood gas analysers** (ICU, theatre)
- **Malaria RDTs** (HRP-2 / pLDH)
- **HIV rapid tests**
- **Pregnancy tests (hCG)**
- HbA1c analysers
- Hb meters (HemoCue)
- Cholesterol
- Troponin
- INR (CoaguChek)
- Bilirubinometers (transcutaneous)

- **Sickle cell POCT** (HemoTypeSC, Sickle SCAN)
- CRP
- Ketones
- Lactate
- Pulse oximetry

Advantages	Disadvantages
Rapid turnaround time (TAT) → faster clinical decisions (A&E, ICU, theatre, rural settings)	**Higher cost per test**
Small sample volume (capillary blood)	**Variable quality and accuracy**; often less precise
Convenient; patient empowerment (home glucose and INR monitoring)	**Operator-dependent errors** (untrained staff)
Useful in **remote and resource-limited settings** (Nigerian primary health centres)	**Poor documentation** and record keeping; results not in the patient record
Fewer pre-analytical transport errors	Inadequate quality control (QC/EQA); limited test menu; interferences (haematocrit affects glucometers)
Better patient satisfaction	Risk of over-reliance; no clinical scientist oversight

Governance of POCT (ISO 22870; national guidelines):

- A POCT committee with laboratory oversight.
- Choose devices carefully and evaluate them.
- **Train and certify operators.**
- SOPs, internal QC and EQA participation.
- Connect devices to the LIS.
- Maintain devices.
- Confirm critical results in the laboratory (glucometers are inaccurate at extremes; **DKA and HHS need laboratory glucose**).

49.6 Reference intervals and medical decision limits

> A **reference interval** (formerly "normal range") is the range of values found in a defined, healthy **reference population**. By convention it is the **central 95%** (mean ± 2 SD for Gaussian data, or the 2.5th–97.5th percentiles for non-Gaussian data).
>
> **Consequence: 5% of healthy people (1 in 20) will fall outside the reference interval for any given test.**

Establishing reference intervals (IFCC/CLSI EP28-A3c):

- Select ≥**120 healthy reference individuals** (a priori or a posteriori), with defined inclusion and exclusion criteria.
- **Partition** by age, sex, ethnicity, pregnancy and physiological state.

- Standardise the pre-analytical and analytical conditions.
- Use statistical analysis (non-parametric percentiles).
- A laboratory may **transfer and verify** an interval with 20 samples.
- **Locally derived intervals are important.** Nigerian/African populations differ, e.g. **lower neutrophil counts**, **higher CK** and creatinine, and higher IgG.

Factors affecting reference intervals:

Factor	Examples
Age	**ALP high in children** (bone growth) and in the elderly; creatinine; neonatal bilirubin
Sex	Hb, creatinine, urate, ferritin (higher in men)
Pregnancy	↓ urea, creatinine and albumin; ↑ ALP (placental), lipids, T_4 (TBG), fibrinogen
Ethnicity	CK, neutrophils
Diet, posture, time of day (circadian), exercise, altitude, menstrual cycle	

Medical (clinical) decision limits:

> A **decision limit (cut-off, action limit)** is a value of a test, set **on the basis of clinical outcome data** (risk), at which a specific medical action is recommended. It is **not** based on the distribution in healthy people.

Test	Decision limit
Fasting plasma glucose	**≥7.0 mmol/L (126 mg/dL) = diabetes**; 5.6–6.9 = impaired fasting glucose
HbA1c	**≥48 mmol/mol (6.5%) = diabetes**
LDL-cholesterol targets	<1.4–1.8 mmol/L in very high CV risk (ESC 2019)
Troponin	>99th percentile URL of a healthy population = myocardial injury
Potassium	**>6.5 mmol/L = emergency treatment**
Neonatal bilirubin	Phototherapy / exchange transfusion thresholds by age in hours
PSA, TSH	Thresholds for action

Other important values:

- **Critical (panic or alert) values:** results so abnormal they indicate an immediately life-threatening situation, and **must be telephoned at once** to the clinician. Examples:
  - K^+ <2.5 or >6.5 mmol/L
  - Na^+ <120 or >160

- Glucose <2.5
- Ca^{2+} <1.5 or >3.5
- pH <7.2

- **Therapeutic ranges** for drugs.

Reference change value (RCV, critical difference): is a change between two serial results significant?

RCV = 2^½ × Z × √(CVa² + CVi²)

where CVa = analytical variation, CVi = within-individual biological variation, and Z = 1.96 (95%). Example: for creatinine, a rise of >~15% between results is significant.

49.7 Decision-making using laboratory results

Test performance characteristics:

	Disease present	Disease absent
Test positive	**True positive (TP)**	**False positive (FP)**
Test negative	**False negative (FN)**	**True negative (TN)**

Measure	Formula	Meaning
Sensitivity	TP / (TP + FN)	Positivity in disease: the ability to **detect** disease. **High sensitivity → good for ruling OUT ("SnNout")** and for **screening**
Specificity	TN / (TN + FP)	Negativity in health. **High specificity → good for ruling IN ("SpPin")** and **confirmation**
Positive predictive value (PPV)	TP / (TP + FP)	Probability of disease given a positive test. **Depends on prevalence**
Negative predictive value (NPV)	TN / (TN + FN)	Probability of no disease given a negative test
Likelihood ratio (+)	Sensitivity / (1 − specificity)	>10 strongly rules in
Likelihood ratio (−)	(1 − sensitivity) / specificity	<0.1 strongly rules out
Efficiency (accuracy)	(TP + TN) / total	

Key concepts:

- **Effect of prevalence:**
  - The same test has a **lower PPV in a low-prevalence population** (screening) and a **higher PPV in a high-prevalence population** (symptomatic patients).
  - This is why HIV and other screening algorithms use **serial testing** (a sensitive test first, then a specific confirmatory test).

- **Choosing a cut-off** involves a trade-off between sensitivity and specificity. A **ROC curve** (receiver operating characteristic) plots sensitivity against (1 − specificity). The **area under the curve (AUC)** measures overall diagnostic accuracy: 1.0 is perfect; 0.5 is useless.
- **Bayesian reasoning:** pre-test probability (clinical judgement + prevalence) × likelihood ratio → post-test probability (use the **Fagan nomogram**).
- **Interpreting an abnormal result.** Ask:
  1. Is it a **real result** (pre-analytical error)?
  2. Is it **outside the reference interval** but clinically insignificant (the 5% of healthy people)?
  3. Is the change from the previous result significant (**RCV**)?
  4. Does it fit the clinical picture?
  5. Does it cross a **decision limit**?
- **Repeat** tests that are unexpected before taking a major action.
- **Evidence-based laboratory medicine:** test selection should be based on diagnostic accuracy studies (STARD) and outcome impact.

★ Exam pearls

- Most laboratory errors are **pre-analytical**. Haemolysis and delayed separation cause spurious hyperkalaemia.
- **Fluoride oxalate** for glucose; **EDTA** for HbA1c and FBC; **citrate** for clotting.
- Sensitivity rules out, specificity rules in. PPV falls when prevalence falls.

Chapter 50 · Potassium Homeostasis and its Disorders

Physiology

- **Total body K^+ is ~3,500 mmol: 98% intracellular** (~150 mmol/L) and only **2% extracellular** (~4 mmol/L). Plasma K^+ is therefore a poor guide to total body K^+.
- **Reference interval: 3.5–5.0 (5.3) mmol/L**. Serum is ~0.1–0.4 mmol/L higher than plasma, because K^+ is released during clotting.
- The **ICF:ECF gradient**, maintained by the **Na^+/K^+-ATPase**, determines the **resting membrane potential**. This is why both hypokalaemia and hyperkalaemia cause **cardiac arrhythmias and neuromuscular dysfunction**.
- **Intake:** 60–100 mmol/day. **Output:** ~90% renal, ~10% faecal.

Regulation:

1. **Internal balance (shifts between ICF and ECF; rapid):**

Moves K^+ into cells (↓ plasma K^+)	Moves K^+ out of cells (↑ plasma K^+)
Insulin (activates Na^+/K^+-ATPase)	Insulin deficiency
β_2-agonists (adrenaline, salbutamol)	β-blockers
Alkalosis (H^+ leaves cells in exchange for K^+)	**Acidosis** (especially mineral acidosis)
Aldosterone	Hyperosmolality (solvent drag)
Cell anabolism (B_{12} therapy, refeeding)	**Cell lysis**: tumour lysis, rhabdomyolysis, haemolysis, burns, crush injury
	Exercise; digoxin toxicity; suxamethonium

2. **External balance (kidney; hours to days):**
    - K^+ is freely filtered; ~90% is reabsorbed in the PCT and loop.
    - **Secretion by principal cells of the collecting duct** determines excretion. It is stimulated by:
        - **aldosterone** (↑ ENaC and ROMK)
        - **high distal Na^+ delivery and flow** (diuretics)
        - **hyperkalaemia**
        - **alkalosis**
        - non-reabsorbable anions (bicarbonate, penicillins)

Hypokalaemia (K^+ < 3.5 mmol/L)

Causes:

Mechanism	Causes
Redistribution into cells	Insulin (DKA treatment), β_2-agonists (salbutamol), alkalosis, **refeeding syndrome**, hypokalaemic periodic paralysis (thyrotoxic: Asian men), B_{12} treatment, hypothermia
GI losses (urinary K^+ <20 mmol/day)	**Diarrhoea** (cholera, infectious diarrhoea: common in Nigeria), **vomiting / nasogastric aspiration** (mostly *renal* loss via secondary hyperaldosteronism + metabolic alkalosis), laxative abuse, villous adenoma, fistulae, ureterosigmoidostomy
Renal losses (urinary K^+ >20 mmol/day)	**Diuretics (loop and thiazide: the commonest cause overall). Mineralocorticoid excess: primary hyperaldosteronism (Conn's)**, secondary hyperaldosteronism (renal artery stenosis, CCF, cirrhosis), **Cushing's syndrome / ectopic ACTH**, **liquorice** (inhibits 11β-HSD2), congenital adrenal hyperplasia (11β-hydroxylase). **Renal tubular disorders: Bartter** (loop-like) and **Gitelman** (thiazide-like; hypomagnesaemia, hypocalciuria) syndromes, **Liddle syndrome** (ENaC gain of function; hypertension), **RTA types 1 and 2. Hypomagnesaemia** (Mg^{2+} normally inhibits ROMK: correct Mg first). **Drugs:** amphotericin B, aminoglycosides, high-dose penicillins, steroids. Osmotic diuresis (DKA)
Inadequate intake	Rare alone: starvation, anorexia, alcoholism

Clinical features:

- **Neuromuscular:** muscle weakness, cramps, hyporeflexia, paralysis, **respiratory failure**, **paralytic ileus**, constipation, rhabdomyolysis (severe).
- **Cardiac:** arrhythmias (atrial and ventricular ectopics, VT, VF, torsades). **Increased digoxin toxicity.**
- **ECG (in order):**
  - **Flattened / inverted T waves**
  - **Prominent U waves**
  - ST depression
  - Prolonged QT (really QU)
  - **"No pot, no T, but a long PT and a U"**
- **Renal: nephrogenic diabetes insipidus** (polyuria), metabolic alkalosis (↑ H^+ secretion, ↑ NH_4^+ production), hypokalaemic nephropathy.
- Glucose intolerance (↓ insulin release).
- Encephalopathy in cirrhosis (↑ ammonia production).

Investigations:

- Repeat U&E. Check **Mg^{2+}**, glucose, **ABG/HCO_3^-** (alkalosis or acidosis).
- **Spot urine K^+** (or **TTKG**) to separate renal from extrarenal loss.

- If renal loss with hypertension: renin and aldosterone (aldosterone:renin ratio for Conn's), cortisol.
- Drug history, ECG.

Management:

Severity	Treatment
Mild (3.0–3.5)	**Oral KCl** (Slow-K, Sando-K: e.g. 40–80 mmol/day in divided doses). Dietary sources: **banana, plantain, orange, coconut water**, tomatoes, beans
Severe (<2.5) or symptomatic, or ECG changes	**IV KCl**: **max concentration 40 mmol/L via peripheral line** (higher via central line), **max rate 10–20 mmol/h** (up to 40 mmol/h in ICU with cardiac monitoring). Use **saline, not dextrose** (dextrose → insulin → lowers K^+ further). **Never give KCl as a bolus (fatal cardiac arrest)**

Also correct **hypomagnesaemia**, treat the cause, consider **K^+-sparing diuretics** (amiloride, spironolactone) if loss continues, and monitor.

Hyperkalaemia (K^+ > 5.5 mmol/L; severe ≥6.5)

First exclude pseudohyperkalaemia (spurious):

- **Haemolysis** (difficult venepuncture, small needle, vigorous shaking)
- **Delayed separation** / sample stored in the cold
- **EDTA contamination**
- Prolonged tourniquet or **fist clenching**
- **Drip arm** (K^+-containing fluid)
- **Thrombocytosis / leucocytosis** (K^+ released during clotting: compare a plasma (heparin) sample)
- Familial pseudohyperkalaemia

True hyperkalaemia causes:

Mechanism	Causes
↓ **Renal excretion (the most common)**	**Acute kidney injury** and **CKD** (GFR <15–20). **Hypoaldosteronism: Addison's disease**, hyporeninaemic hypoaldosteronism (type 4 RTA: diabetes), congenital adrenal hyperplasia (21-hydroxylase). **Drugs: ACE inhibitors, ARBs**, **K^+-sparing diuretics** (spironolactone, amiloride), **NSAIDs**, trimethoprim, ciclosporin/tacrolimus, heparin
Redistribution out of cells	**Acidosis**, **insulin deficiency (DKA)**, **tissue damage** (rhabdomyolysis, crush injury, **burns**, **tumour lysis syndrome**, haemolysis, massive transfusion of stored blood), **suxamethonium**, **digoxin toxicity**, β-blockers, hyperkalaemic periodic paralysis, hyperosmolality

Mechanism	Causes
Excess intake	IV K^+ (iatrogenic), K^+ supplements, salt substitutes (with ↓ renal function), traditional herbal preparations, **stored blood**

Clinical features:

- Often **asymptomatic until cardiac arrest**.
- Muscle weakness, paraesthesiae, flaccid paralysis.
- **Arrhythmias:** bradycardia, heart block, **VF, asystole**.

ECG changes (progressive):

1. **Tall, peaked ("tented") T waves** (earliest)
2. Prolonged PR interval
3. **Flattened or absent P waves**
4. **Widened QRS**
5. **"Sine wave" pattern** (QRS merges with T)
6. **VF or asystole**

Emergency management (K^+ ≥6.5 mmol/L, or ECG changes; UK Renal Association 2023 / Resuscitation Council):

Step	Treatment	Mechanism / onset
1. Protect the heart (stabilise the myocardium)	**10% calcium gluconate 30 mL IV** (or 10 mL of 10% calcium chloride) over 5–10 min. Repeat until the ECG improves	Raises the threshold potential. Onset 1–3 min; lasts 30–60 min. **Does not lower K^+**. Caution with digoxin
2. Shift K^+ into cells	**Insulin–glucose: 10 units soluble (Actrapid) insulin + 25 g glucose** (e.g. 50 mL of 50% or 250 mL of 10%) IV over 15 min	↓ K^+ by 0.6–1.0 mmol/L in 15–30 min. Lasts 4–6 h. **Monitor glucose for 6 h** (hypoglycaemia)
	Salbutamol 10–20 mg nebulised (adjunct)	β_2-agonist; ↓ K^+ by 0.5–1 mmol/L in 30 min. Tachycardia
	Sodium bicarbonate	Only with severe metabolic acidosis
3. Remove K^+ from the body	**Potassium binders**: **sodium zirconium cyclosilicate (Lokelma®)**, **patiromer**, calcium resonium (calcium polystyrene sulfonate)	GI K^+ exchange (hours)
	Loop diuretics	If the kidneys respond
	Haemodialysis	**Definitive** for refractory hyperkalaemia, AKI/CKD, or anuria

Step	Treatment	Mechanism / onset
4. Prevent recurrence	Stop offending drugs (ACEI, spironolactone, NSAIDs, K^+ supplements); low-K^+ diet; treat the cause; monitor	

🎤 Viva corner

Q: A patient's K^+ is 7.2 mmol/L but they are well, the ECG is normal and the sample is haemolysed. What do you do? **A:** Suspect pseudohyperkalaemia. Repeat the sample urgently: free-flowing, no prolonged tourniquet, correct tube and order, separated promptly. Before you treat, check the clinical context: renal function, drugs and the ECG.

Chapter 51 · Acid–Base Balance and Disorders

Physiology

- **Normal arterial pH: 7.35–7.45**, which is an [H^+] of 35–45 nmol/L. The compatible range for life is about 6.8–7.8.
- **Daily acid production:**
  - **Volatile acid (CO_2):** ~15,000–20,000 mmol/day, excreted by the **lungs**.
  - **Non-volatile (fixed) acid: ~50–100 mmol H^+/day** (sulfuric acid from sulfur-containing amino acids, phosphoric acid, organic acids), excreted by the **kidneys**.

Henderson–Hasselbalch equation (the bicarbonate buffer system):

pH = 6.1 + log [HCO_3^-] / (0.03 × pCO_2 in mmHg), or with pCO_2 in kPa: pH = 6.1 + log [HCO_3^-] / (0.23 × pCO_2).

So **pH ∝ HCO_3^- (metabolic, renal) / pCO_2 (respiratory)**.

Three lines of defence:

1. **Buffers** (seconds):
   - **Bicarbonate/carbonic acid** (the most important ECF buffer, because it is an *open system*: CO_2 is removed by the lungs).
   - **Haemoglobin** (the most important RBC buffer).
   - Proteins, phosphate (intracellular and urinary).
   - Bone.
2. **Respiratory compensation** (minutes to hours): changes in ventilation alter pCO_2 (chemoreceptors).
3. **Renal compensation** (hours to days, 3–5 days to be complete):
   - **Reclaims filtered HCO_3^-** (PCT, ~85%; carbonic anhydrase).
   - **Generates new HCO_3^-** by excreting H^+ as **titratable acid** ($H_2PO_4^-$) and **ammonium (NH_4^+)**. NH_4^+ is the main adaptive response to acidosis (glutamine metabolism).

Normal arterial blood gas values:

Parameter	Normal
pH	7.35–7.45
pCO_2	**4.7–6.0 kPa (35–45 mmHg)**
HCO_3^-	**22–26 mmol/L**
Base excess	**−2 to +2 mmol/L**
pO_2 (on air)	10.6–13.3 kPa (80–100 mmHg)
Lactate	<2 mmol/L

Parameter	Normal
Anion gap	**$(Na^+ + K^+) - (Cl^- + HCO_3^-) = 10–20$ mmol/L**; or without K^+: 8–16 (assays vary). Unmeasured anions are mostly **albumin** (correct: add ~2.5 per 10 g/L fall in albumin), phosphate, sulfate

Systematic interpretation of ABG ("step-by-step")

1. **Assess oxygenation** (pO_2 against FiO_2; P/F ratio).
2. **pH:** acidaemia (<7.35) or alkalaemia (>7.45)?
3. **Primary disorder:**
   - Is pCO_2 moving in the direction that explains the pH? → **respiratory**.
   - Is HCO_3^- moving in the direction that explains the pH? → **metabolic**.
4. **Compensation:** is it appropriate? (Compensation **never overcorrects** the pH.)
5. **Anion gap** (if metabolic acidosis); then the **delta ratio** to look for mixed disorders.
6. **Clinical correlation.**

Disorder	pH	Primary change	Compensation	Expected compensation
Metabolic acidosis	↓	↓ **HCO_3^-**	↓ pCO_2 (hyperventilation: **Kussmaul respiration**)	**Winter's formula:** expected pCO_2 (mmHg) = **1.5 × HCO_3^- + 8 ± 2**
Metabolic alkalosis	↑	↑ **HCO_3^-**	↑ pCO_2 (hypoventilation; limited by hypoxia)	pCO_2 ↑ by ~0.7 mmHg per 1 mmol/L ↑ HCO_3^-
Respiratory acidosis	↓	↑ **pCO_2**	↑ HCO_3^-	Acute: HCO_3^- ↑ 1 per 10 mmHg ↑ pCO_2. Chronic: ↑ 3.5 per 10
Respiratory alkalosis	↑	↓ **pCO_2**	↓ HCO_3^-	Acute: ↓ 2 per 10 mmHg. Chronic: ↓ 5 per 10

Metabolic acidosis

High anion gap (gain of acid with an unmeasured anion) — mnemonic MUDPILES or GOLDMARK	Normal anion gap (hyperchloraemic) — loss of HCO_3^- or failure of H^+ excretion
Methanol (formate)	**GI HCO_3^- loss: diarrhoea** (the commonest), pancreatic or biliary fistula, **ureterosigmoidostomy** / ileal conduit

High anion gap (gain of acid with an unmeasured anion) — mnemonic MUDPILES or GOLDMARK	Normal anion gap (hyperchloraemic) — loss of HCO_3^- or failure of H^+ excretion
Uraemia (CKD, AKI: sulfate, phosphate)	**Renal tubular acidosis (RTA)**: type 1 (distal; urine pH >5.5, stones, ↓ K^+), type 2 (proximal; Fanconi, ↓ K^+), type 4 (hypoaldosteronism; ↑ K^+)
Diabetic (and alcoholic and starvation) **ketoacidosis** (β-hydroxybutyrate, acetoacetate)	**Carbonic anhydrase inhibitors** (acetazolamide)
Paraldehyde, **P**henformin/**metformin** (lactate)	**Excess saline infusion (hyperchloraemic)**
Iron, **I**soniazid (seizure → lactate)	Addison's disease
Lactic acidosis: **type A** (tissue hypoxia: **shock, sepsis, severe anaemia, CO poisoning, cardiac arrest**) and **type B** (no hypoxia: **metformin**, liver failure, malignancy, **thiamine deficiency**, **severe malaria**)	Early renal failure; NH_4Cl administration
Ethylene glycol (glycolate, **oxalate crystals**)	Hyperalimentation (TPN)
Salicylates (late)	

Urinary anion gap ($Na^+ + K^+ - Cl^-$) in a normal-AG acidosis:

- **Negative** (↑ NH_4^+ excretion) → GI loss.
- **Positive** → RTA (impaired NH_4^+ excretion).

Osmolal gap = measured osmolality − calculated osmolality [$2Na^+$ + glucose + urea]. A gap **>10** suggests **toxic alcohols** (methanol, ethylene glycol), ethanol or mannitol.

Clinical features of metabolic acidosis:

- **Kussmaul breathing** (deep, sighing)
- Confusion, coma
- ↓ myocardial contractility, arrhythmias
- Peripheral vasodilation → hypotension
- **Hyperkalaemia**
- Bone demineralisation (chronic)

Management:

- **Treat the cause:**
  - DKA: insulin + fluids.
  - Sepsis: fluids + antibiotics.
  - Toxic alcohols: **fomepizole** or ethanol, plus dialysis.
- **IV sodium bicarbonate** only in severe acidosis (pH <7.1) with specific indications: hyperkalaemia, TCA overdose, salicylate poisoning, RTA, or bicarbonate loss. Risks: hypernatraemia, fluid overload, ↓ ionised Ca^{2+}, paradoxical intracellular acidosis.

Metabolic alkalosis

Category	Causes
Saline (chloride)-responsive (urine Cl^- <20 mmol/L)	**Vomiting / NG suction** (loss of HCl; e.g. **pyloric stenosis**: hypochloraemic hypokalaemic metabolic alkalosis with **paradoxical aciduria**), **diuretics** (after the event), contraction alkalosis, post-hypercapnia, villous adenoma, congenital chloride diarrhoea
Saline-resistant (urine Cl^- >20)	**Mineralocorticoid excess** (Conn's, Cushing's, liquorice, Bartter, Gitelman, Liddle), current diuretic use, **severe hypokalaemia**
Alkali administration	Antacids, **milk-alkali syndrome**, massive transfusion (citrate → HCO_3^-), bicarbonate therapy

Maintenance factors (why the kidney fails to excrete the excess HCO_3^-):

- **Volume depletion** (↑ Na^+ reabsorption)
- **Chloride depletion**
- **Hypokalaemia**
- Hyperaldosteronism

Clinical features:

- **Hypoventilation**
- Confusion
- **Tetany and paraesthesiae** (↓ ionised Ca^{2+}, because alkalosis increases protein binding)
- Arrhythmias
- **Hypokalaemia**

Treatment:

- **0.9% saline + KCl** for chloride-responsive alkalosis.
- Spironolactone for mineralocorticoid excess.
- Acetazolamide in some cases.
- Treat the cause.

Respiratory acidosis (↑ pCO_2; hypoventilation)

Category	Causes
CNS depression	**Opioids**, benzodiazepines, anaesthesia, head injury, stroke, encephalitis
Neuromuscular	**Guillain–Barré syndrome**, myasthenia gravis, **organophosphate poisoning**, polio, **tetanus** (Nigeria), **snake bite (neurotoxic: cobra)**, high spinal cord injury, motor neurone disease, hypokalaemia
Chest wall	Flail chest, kyphoscoliosis, obesity hypoventilation

Category	Causes
Airway / lung disease	**COPD** (chronic), **severe acute asthma** (late, ominous sign), pneumonia, pulmonary oedema, ARDS, airway obstruction (foreign body), pneumothorax
Iatrogenic	Inadequate mechanical ventilation

Clinical features:

- **CO_2 narcosis:** headache, confusion, drowsiness, **flapping tremor (asterixis)**.
- Bounding pulse, **warm peripheries**, papilloedema (↑ ICP from cerebral vasodilation).

Treatment:

- Treat the cause.
- **Controlled O_2 (24–28% Venturi, target SpO_2 88–92%) in COPD** (risk of CO_2 retention).
- Non-invasive ventilation (BiPAP) or intubation.
- Naloxone for opioid toxicity.

Respiratory alkalosis (↓ pCO_2; hyperventilation)

Causes:

- **Anxiety / panic attack (hyperventilation syndrome)**
- Pain, fever
- **Hypoxia** (high altitude, pneumonia, **pulmonary embolism**, asthma (early), pulmonary oedema)
- **Pregnancy** (progesterone; normal pCO_2 ~4.0 kPa)
- **Salicylate poisoning (early)**
- **Sepsis** (early)
- **Liver failure**
- CNS disorders (stroke, meningitis, head injury)
- Over-ventilation on a ventilator

Features:

- **Paraesthesiae (perioral and digital)**
- **Carpopedal spasm (tetany)**
- Lightheadedness
- **Chvostek and Trousseau signs**

All of these come from ↓ ionised Ca^{2+}.

Treatment:

- Treat the cause.
- Reassure the patient in anxiety hyperventilation. (Rebreathing into a paper bag is no longer recommended; exclude hypoxia first.)

Mixed disorders

Mixed disorder	Example
Salicylate poisoning	**Respiratory alkalosis + high-AG metabolic acidosis**
Septic shock	Metabolic (lactic) acidosis + respiratory alkalosis
Cardiac arrest	Metabolic (lactic) acidosis + **respiratory acidosis** (the most severe acidaemia)
Vomiting + diuretics in COPD	Metabolic alkalosis + respiratory acidosis
DKA with vomiting	High-AG acidosis + metabolic alkalosis. Clue: **delta ratio** ($\Delta AG/\Delta HCO_3^-$) **>2**

Delta ratio:

Value	Interpretation
<0.4	Hyperchloraemic normal-AG acidosis
1–2	Pure high-AG acidosis
>2	High-AG acidosis + metabolic alkalosis, or pre-existing compensated respiratory acidosis

★ Exam pearls

Worked example. pH 7.25, pCO_2 3.2 kPa (24 mmHg), HCO_3^- 10 mmol/L, Na^+ 138, Cl^- 100.

1. **Acidaemia.**
2. Low HCO_3^- → **metabolic acidosis**.
3. **Winter's formula:** 1.5 × 10 + 8 = 23 ± 2. Measured pCO_2 is 24 → **appropriate respiratory compensation**.
4. **Anion gap:** 138 − (100 + 10) = 28 → **high-AG metabolic acidosis** (e.g. DKA, lactic acidosis, severe malaria).
5. **Delta ratio:** (28 − 12) / (24 − 10) ≈ 1.1 → **pure high-AG acidosis**.

Chapter 52 · Renal Function Tests

Functions of the kidney

- **Excretory:** urea, creatinine, urate, drugs.
- **Homeostatic:** water, electrolytes, acid–base.
- **Endocrine:**
  - **EPO**
  - **1α-hydroxylation of vitamin D**
  - **Renin**
  - Prostaglandins
- Gluconeogenesis.
- Degradation of peptide hormones (insulin).

Glomerular function tests

Test	Details / limitations
Serum creatinine	**Reference:** men 60–110 µmol/L; women 45–90 µmol/L (≈0.6–1.2 mg/dL). Formed from **creatine phosphate in muscle** at a constant rate in proportion to **muscle mass**. Freely filtered, **not reabsorbed, slightly secreted (~10–20%). Limitations:** (1) **Insensitive: GFR must fall by ~50% before creatinine rises above the reference range** (the hyperbolic relationship); (2) depends on **muscle mass** (low in the elderly, women, amputees and malnourished patients → falsely "normal"; high in muscular people and **Black individuals**); diet (cooked meat); (3) **tubular secretion is inhibited by trimethoprim and cimetidine** (↑ creatinine without a fall in GFR); (4) assay interference (**Jaffe method**: ketones, cefoxitin, bilirubin); (5) not in steady state during AKI
Serum urea	**2.5–7.8 mmol/L**. Made in the **liver (urea cycle)** from protein and amino acid breakdown. Filtered, then **~40–50% passively reabsorbed** (more when flow is low). **Less reliable than creatinine. ↑ with:** high-protein diet, **GI bleeding** (protein meal), **dehydration / pre-renal states** (↑ reabsorption), catabolic states (sepsis, burns, steroids, tetracyclines), heart failure. **↓ with:** liver failure, malnutrition, pregnancy, overhydration. **Urea:creatinine ratio** is ↑ in pre-renal uraemia and GI bleeding

Test	Details / limitations
Creatinine clearance	**CrCl = (U_cr × V) / P_cr** (24-h urine). Normal ~**90–140 mL/min**. **Overestimates GFR** by ~10–20% (because of tubular secretion). **Collection errors** are common
Estimated GFR (eGFR)	**CKD-EPI 2021 equation** (now **race-free**: the earlier race coefficient was removed by NKF-ASN). Uses creatinine, age and sex (± cystatin C). Older: **MDRD** (4-variable). **Cockcroft–Gault** CrCl = [(140 − age) × weight (kg) × (0.85 if female)] / (0.815 × serum creatinine µmol/L) or /(72 × Cr mg/dL). Still used for **drug dosing**. eGFR is **not valid** in AKI (non-steady state), at extremes of body habitus, in pregnancy, amputees, or children (use **Schwartz** in children)
Cystatin C	A low-molecular-weight protein (13 kDa) made by **all nucleated cells at a constant rate**. Freely filtered, reabsorbed and **catabolised** in the PCT. **Independent of muscle mass**, so it is a better GFR marker in the elderly, children and cirrhosis. Affected by thyroid status and steroids
Measured GFR (gold standard)	Clearance of an **ideal marker**: freely filtered, not reabsorbed or secreted, not metabolised, physiologically inert. **Inulin** (the classic gold standard), **^{51}Cr-EDTA, ^{99m}Tc-DTPA, iohexol**
Proteinuria	**Dipstick** (detects mainly albumin, ≥300 mg/L; false positives in alkaline urine). **Urine albumin:creatinine ratio (ACR)** (preferred for diabetics): normal <3 mg/mmol; **moderately increased (microalbuminuria) 3–30 mg/mmol** (early diabetic nephropathy); severely increased >30. **Protein:creatinine ratio (PCR)**: normal <15 mg/mmol; nephrotic >300 mg/mmol (≈>3 g/24 h). **24-h urine protein:** normal <150 mg/day; **nephrotic syndrome >3.5 g/day** (adults). Glomerular (albumin) vs tubular (β_2-microglobulin, retinol-binding protein, α_1-microglobulin) vs overflow (**Bence Jones protein**: free light chains in myeloma; not detected by dipstick)
Urinalysis and microscopy	**Red cell casts** (glomerulonephritis), **dysmorphic RBCs**, white cell casts (pyelonephritis, interstitial nephritis), **granular / "muddy brown" casts (ATN)**, hyaline casts (normal, concentrated urine), crystals

Tubular function tests

Function	Test
Concentrating ability	**Urine osmolality / SG**. Early morning urine osmolality >600 mOsm/kg excludes a concentrating defect. **Water deprivation test** (8 h) followed by **desmopressin (DDAVP)**: distinguishes **cranial DI** (urine concentrates after DDAVP) from **nephrogenic DI** (no response) and primary polydipsia
Acidification	Urine pH after an **ammonium chloride load** (or furosemide + fludrocortisone test). In distal RTA (type 1), urine pH **cannot fall below 5.5**
Proximal tubule (Fanconi syndrome)	Glycosuria with normal blood glucose, generalised aminoaciduria, phosphaturia, bicarbonaturia, uricosuria, tubular proteinuria
Fractional excretion of sodium (FENa)	FENa = (U_Na × P_Cr) / (P_Na × U_Cr) × 100. **<1% = pre-renal AKI** (avid Na^+ retention); **>2% = intrinsic AKI (ATN)**. Unreliable with diuretics (use **FEurea <35%** for pre-renal)

Pre-renal vs renal (ATN) AKI:

Index	Pre-renal	ATN
Urine Na^+ (mmol/L)	<20	>40
FENa	<1%	>2%
Urine osmolality (mOsm/kg)	>500	<350 (isosthenuria)
Urine:plasma urea ratio	>8	<3
Urine:plasma creatinine ratio	>40	<20
Serum urea:creatinine	High	Normal
Sediment	Bland (hyaline casts)	Muddy brown granular casts

Acute kidney injury (KDIGO 2012)

Definition — any one of:

- Creatinine **rise ≥26.5 µmol/L (0.3 mg/dL) within 48 h**
- Creatinine **≥1.5× baseline within 7 days**
- **Urine output <0.5 mL/kg/h for 6 h**

Stages:

Stage	Creatinine	Urine output
1	1.5–1.9× baseline	<0.5 mL/kg/h for 6–12 h
2	2.0–2.9× baseline	<0.5 mL/kg/h for ≥12 h

Stage	Creatinine	Urine output
3	≥3× baseline, or ≥354 µmol/L, or RRT	<0.3 mL/kg/h for ≥24 h, or anuria ≥12 h

Causes in Nigeria:

- **Pre-renal: gastroenteritis**, sepsis, haemorrhage (obstetric).
- **Renal:**
  - **Severe malaria (*P. falciparum*: "blackwater fever")**
  - **Nephrotoxins:** herbal remedies, NSAIDs, aminoglycosides
  - Snake envenomation
  - **Lassa fever**
  - Glomerulonephritis
  - Sepsis
  - Haemolysis (G6PD deficiency)
- **Post-renal:** BPH, prostate cancer, cervical cancer, stones, schistosomiasis.

Chronic kidney disease (KDIGO 2024)

Definition: abnormalities of kidney **structure or function** (eGFR <60 and/or markers of damage such as ACR ≥3 mg/mmol) present for **>3 months**.

Staging by GFR (G1–G5) and albuminuria (A1–A3):

Stage	eGFR (mL/min/1.73 m²)
G1	≥90
G2	60–89
G3a	45–59
G3b	30–44
G4	15–29
G5 (kidney failure)	**<15**

Commonest causes in Nigeria:

- **Chronic glomerulonephritis**
- **Hypertension**
- **Diabetic nephropathy**
- HIV-associated nephropathy (HIVAN; **APOL1** risk variants in people of African descent)
- Sickle cell nephropathy
- Obstructive uropathy

Biochemical abnormalities in CKD:

- ↑ urea, creatinine

- ↑ **K^+**
- **Metabolic acidosis**
- ↓ **Ca^{2+}**, ↑ **PO_4^{3-}**, ↑ **PTH (secondary → tertiary hyperparathyroidism)**
- ↓ 1,25-$(OH)_2D$
- **CKD–mineral and bone disorder (CKD-MBD; renal osteodystrophy):** osteitis fibrosa cystica, adynamic bone disease, osteomalacia, vascular calcification
- **Normocytic anaemia (↓ EPO)**
- ↑ urate
- Dyslipidaemia

Chapter 53 · Calcium, Phosphate and Magnesium Metabolism; Metabolic Bone Disease

(The hormonal regulation of calcium and the drugs that act on bone are covered in Book 1, Chapter 24.)

Calcium

Body calcium: ~1 kg (25 mol); **99% is in bone** as hydroxyapatite.

Plasma calcium: reference **2.20–2.60 mmol/L** (8.8–10.4 mg/dL). It exists in three forms:

- **~45–50% ionised (free; biologically active**: 1.1–1.3 mmol/L)
- ~40–45% **protein-bound** (mainly albumin)
- ~5–10% complexed (citrate, phosphate)

Adjusted (corrected) calcium = measured total Ca^{2+} + **0.02 × (40 − albumin in g/L)** mmol/L.

- Example: Ca 2.0, albumin 25 → 2.0 + 0.02 × 15 = 2.3 mmol/L (normal).
- Correction is **unreliable** in critical illness. Measure **ionised Ca^{2+}** (blood gas analyser) instead.

Factors affecting ionised Ca^{2+}:

- **Alkalosis ↑ protein binding → ↓ ionised Ca^{2+}** → tetany (e.g. hyperventilation).
- **Acidosis ↓ binding → ↑ ionised Ca^{2+}.**
- **Citrate** (massive transfusion) chelates Ca^{2+}.

Functions:

- Bone mineralisation
- Neuromuscular excitability
- Muscle contraction (excitation–contraction coupling)
- Coagulation (factor IV)
- Second messenger
- Hormone secretion
- Cardiac action potential (plateau)

Phosphate

- **Reference:** 0.8–1.5 mmol/L. ~85% is in bone. The **major intracellular anion**.
- **Functions:**
  - **ATP**
  - **2,3-BPG**
  - Nucleic acids
  - Phospholipids
  - Buffer

    - Mineralisation

- **Regulated by:** PTH (phosphaturic), FGF-23 (phosphaturic), calcitriol (↑ absorption), kidney.

Hyperphosphataemia:

- **Renal failure (the commonest)**
- Hypoparathyroidism
- **Tumour lysis**, rhabdomyolysis, haemolysis (spurious if the sample is haemolysed)
- Vitamin D toxicity
- Acromegaly
- Phosphate enemas

Hypophosphataemia:

- **Refeeding syndrome** (malnutrition, alcoholism)
- **DKA treatment**
- Hyperparathyroidism
- Vitamin D deficiency
- Respiratory alkalosis
- Antacids (Al^{3+})
- Fanconi syndrome
- **X-linked hypophosphataemic rickets** (↑ FGF-23; PHEX mutation)
- Tumour-induced osteomalacia

Severe hypophosphataemia (<0.3 mmol/L) → muscle weakness, **respiratory failure**, rhabdomyolysis, haemolysis, heart failure, confusion.

Magnesium

- **Reference:** 0.7–1.0 mmol/L.
- **Functions:** cofactor for ATP-dependent enzymes; Na^+/K^+-ATPase; **needed for PTH secretion and action**.

Hypomagnesaemia causes:

- Diarrhoea, malabsorption
- **Alcoholism**
- **Diuretics**
- **PPIs**
- Aminoglycosides, amphotericin B, cisplatin
- Refeeding
- DKA
- Gitelman syndrome

Hypomagnesaemia consequences:

- **Refractory hypocalcaemia and hypokalaemia** (correct Mg^{2+} first)

- **Arrhythmias (torsades)**
- Tetany, seizures

Hypermagnesaemia:

- Causes: renal failure, **$MgSO_4$ therapy in pre-eclampsia**.
- Signs of toxicity, by level:

Mg^{2+} (mmol/L)	Effect
>4	**Loss of patellar reflexes**
>5	Respiratory depression
>7	Cardiac arrest

- **Antidote: 10% calcium gluconate**.

Hypercalcaemia

Causes — ~90% are primary hyperparathyroidism or malignancy:

PTH status	Causes
PTH-dependent (PTH ↑ or inappropriately normal)	**Primary hyperparathyroidism** (the commonest cause in outpatients; parathyroid **adenoma ~85%**, hyperplasia ~15% (MEN 1, 2A), carcinoma <1%). **Tertiary hyperparathyroidism** (CKD, after prolonged secondary). **Familial hypocalciuric hypercalcaemia** (FHH; CaSR mutation; **low urine Ca:Cr clearance ratio <0.01**; benign). Lithium
PTH-independent (PTH suppressed)	**Malignancy** (the commonest in hospital inpatients): **humoral hypercalcaemia via PTHrP** (squamous cell carcinomas of lung, head and neck; renal; breast); **local osteolytic metastases** (breast, myeloma: IL-6, RANKL); **calcitriol production** (lymphoma). **Granulomatous disease** (↑ 1α-hydroxylase in macrophages): **sarcoidosis, TB**, histoplasmosis. **Vitamin D intoxication**. **Thiazides**. **Milk-alkali syndrome**. Thyrotoxicosis. Immobilisation (Paget's disease, adolescents). Addison's disease. Vitamin A toxicity

Clinical features ("bones, stones, abdominal groans, psychic moans, thrones (polyuria)"):

- Bone pain, **osteitis fibrosa cystica** (subperiosteal resorption of the phalanges; "**brown tumours**"; "**salt-and-pepper**" skull)
- **Renal stones**, nephrocalcinosis, **polyuria / polydipsia** (nephrogenic DI), AKI
- Constipation, nausea, anorexia, **peptic ulcer**, **acute pancreatitis**
- Depression, fatigue, confusion, coma
- **Short QT interval**, arrhythmias, hypertension

Investigations:

1. Repeat **adjusted Ca^{2+}**.
2. **PTH (key test).**
3. Phosphate, ALP, U&E, vitamin D.
4. 24-h urinary calcium (FHH).
5. PTHrP.
6. Myeloma screen: serum and urine protein electrophoresis, free light chains.
7. CXR, ACE (sarcoidosis), imaging.

Typical patterns:

Condition	PTH	PO_4^{3-}	Other
Primary hyperparathyroidism	↑ or inappropriately normal	↓	↑ urinary Ca^{2+}
Malignancy	↓	Variable	↑ PTHrP

Management: see Book 1, Chapter 24 (fluids, bisphosphonates). **Parathyroidectomy** for symptomatic primary hyperparathyroidism.

Hypocalcaemia

Causes:

Category	Causes
↓ **PTH (hypoparathyroidism)**	**Post-surgical** (thyroidectomy/parathyroidectomy: the commonest cause), autoimmune, **DiGeorge**, infiltration (haemochromatosis, Wilson's), **hypomagnesaemia** (↓ PTH secretion and action), irradiation
PTH resistance	**Pseudohypoparathyroidism** (Albright's hereditary osteodystrophy: short 4th/5th metacarpals, short stature; ↑ PTH; GNAS)
Vitamin D deficiency / resistance	↓ sunlight, diet, malabsorption, **CKD** (↓ 1α-hydroxylation), liver disease, anticonvulsants (enzyme induction), vitamin D-dependent rickets types 1 and 2
Other	**Acute pancreatitis** (saponification), **hyperphosphataemia** (renal failure, rhabdomyolysis, tumour lysis), **hungry bone syndrome** (post-parathyroidectomy), **massive transfusion (citrate)**, sepsis, bisphosphonates/denosumab, **EDTA contamination** (spurious), **hypoalbuminaemia** (spurious total Ca^{2+}, with normal ionised Ca^{2+}), neonatal hypocalcaemia

Clinical features (neuromuscular irritability):

- **Paraesthesiae** (perioral, fingers)
- **Tetany: carpopedal spasm**, laryngospasm
- **Chvostek's sign:** tapping the facial nerve anterior to the ear → twitching of the facial muscles
- **Trousseau's sign:** BP cuff above systolic for 3 min → carpal spasm. More specific than Chvostek's
- **Seizures**
- **Prolonged QT**, arrhythmias, heart failure
- **Chronic:** cataracts, basal ganglia calcification, dental hypoplasia, depression

Treatment:

- **Acute / severe: 10% calcium gluconate 10–20 mL IV over 10 min**, then an infusion. Check and correct Mg^{2+}.
- **Chronic:** oral calcium + **alfacalcidol or calcitriol** (in hypoparathyroidism and CKD), or colecalciferol (in vitamin D deficiency).

Metabolic bone disease

Disease	Defect	Ca^{2+}	PO_4^{3-}	ALP	PTH	Key features
Osteoporosis	↓ **Bone mass (matrix + mineral in normal proportion)**; microarchitectural deterioration → fragility fractures (vertebrae, **hip**, wrist/Colles)	**Normal**	**Normal**	**Normal**	Normal	**Diagnosis: DXA T-score ≤ −2.5** (WHO); osteopenia −1 to −2.5. Risk factors: age, **post-menopausal oestrogen loss**, steroids, smoking, alcohol, low BMI, immobility, hyperthyroidism, hypogonadism. **FRAX** tool. Treat with bisphosphonates, etc. Biochemistry is normal (useful to exclude other causes)

Disease	Defect	Ca^{2+}	PO_4^{3-}	ALP	PTH	Key features
Osteomalacia (adults) / rickets (children)	**Defective mineralisation** of osteoid (↑ unmineralised osteoid). Usually **vitamin D deficiency** (also Ca^{2+} deficiency or phosphate deficiency)	↓ or N	↓	↑	↑ **(secondary)**	Bone pain and tenderness, **proximal myopathy (waddling gait)**, **Looser's zones (pseudofractures)**. ↓ 25-OH vitamin D (<25 nmol/L). Rickets: bowing, rachitic rosary, widened metaphyses (cupping, fraying on X-ray), delayed walking, craniotabes. **Calcium-deficiency rickets is common in Nigerian children** (low dietary Ca^{2+})
Primary hyperparathyroidism	Excess PTH → ↑ resorption	↑	↓	↑ or N	↑	Osteitis fibrosa cystica
Secondary hyperparathyroidism (CKD)	Physiological response to ↓ Ca^{2+} / ↓ vitamin D / ↑ PO_4^{3-}	↓ or N	↑ **(CKD)**	↑	↑↑	CKD-MBD
Tertiary hyperparathyroidism	Autonomous PTH after long-standing secondary	↑	↑	↑	↑↑	Post-transplant; parathyroidectomy

Disease	Defect	Ca^{2+}	PO_4^{3-}	ALP	PTH	Key features
Paget's disease of bone (osteitis deformans)	**Disordered remodelling**: excess osteoclastic resorption followed by chaotic formation → **mosaic pattern** of woven bone	**Normal**	**Normal**	↑↑↑ **(very high)**	Normal	Elderly (rare in Africans). Bone pain, deformity (bowed tibia, enlarged skull: **"hat no longer fits"**), deafness, nerve compression, **high-output cardiac failure**, **osteosarcoma** (<1%). ↑ urinary hydroxyproline / NTX. Treat with bisphosphonates (zoledronate)
Renal osteodystrophy (CKD-MBD)	Combination	↓	↑	↑	↑↑	Osteitis fibrosa, adynamic bone, osteomalacia, vascular calcification
Bone metastases / myeloma	Lytic	↑	N/↑	↑ **(metastases); usually NORMAL in myeloma** (pure lytic, no osteoblastic activity)	↓	

Markers of bone turnover:

Type	Markers
Formation	**Bone-specific ALP**, **P1NP** (procollagen type 1 N-propeptide), osteocalcin
Resorption	**CTX** (C-telopeptide of type 1 collagen), NTX, urinary deoxypyridinoline, hydroxyproline

ALP isoenzymes: liver, **bone**, placenta, intestine. Raised ALP with raised GGT suggests a liver source; with normal GGT, a bone source.

🎤 Viva corner

Q: A 65-year-old woman has a fracture after minor trauma. Calcium, phosphate and ALP are normal. Diagnosis? **A:** Osteoporosis. Biochemistry is normal. Confirm with DXA (T-score ≤ −2.5).

Q: Low Ca^{2+}, low PO_4^{3-} and high ALP? **A:** Osteomalacia (vitamin D deficiency) with secondary hyperparathyroidism.

Chapter 54 · Diabetes Mellitus

Diabetes mellitus (DM) is a group of metabolic disorders characterised by **chronic hyperglycaemia** resulting from defects in **insulin secretion, insulin action, or both**. It causes disturbed carbohydrate, fat and protein metabolism and long-term **microvascular** and **macrovascular** complications (WHO; ADA 2024).

Glucose homeostasis

Hormone	Effect on glucose	Actions
Insulin (β cells of the islets; from proinsulin → insulin + **C-peptide** in equimolar amounts)	**The only hormone that lowers blood glucose**	↑ glucose uptake (**GLUT4** in muscle and fat), ↑ glycogenesis, ↑ lipogenesis, ↑ protein synthesis, ↓ gluconeogenesis, ↓ glycogenolysis, ↓ lipolysis, ↓ ketogenesis, K^+ moves into cells
Glucagon (α cells)	Raises glucose	Counter-regulatory: ↑ glycogenolysis, gluconeogenesis, ketogenesis
Adrenaline	Raises glucose	Counter-regulatory
Cortisol	Raises glucose	Counter-regulatory
Growth hormone	Raises glucose	Counter-regulatory
Incretins: GLP-1, GIP (gut)	Lower glucose	↑ glucose-dependent insulin secretion. The "incretin effect": oral glucose provokes more insulin than IV glucose

Glucose-stimulated insulin secretion:

1. Glucose enters the β cell via **GLUT2**.
2. **Glucokinase** (the glucose sensor) phosphorylates it.
3. Metabolism produces ↑ **ATP**.
4. ATP **closes K_ATP channels** (the target of sulfonylureas).
5. The cell depolarises.
6. Voltage-gated **Ca^{2+} channels open** → Ca^{2+} influx.
7. Insulin granules are exocytosed.

Classification (WHO 2019; ADA 2024)

Type	Features
Type 1 DM (~5–10%)	**Autoimmune destruction of β cells → absolute insulin deficiency**. Autoantibodies: **GAD65**, IA-2, insulin (IAA), ZnT8, ICA. HLA-DR3/DR4. Usually children and young adults; lean; **prone to ketoacidosis**; **low C-peptide**. LADA (latent autoimmune diabetes in adults)
Type 2 DM (~90%)	**Insulin resistance + relative insulin deficiency** (progressive β-cell failure). Obesity (central), physical inactivity, family history, ethnicity. **Prevalence in Nigeria is ~3–6% and rising** (IDF). Insidious onset; often detected with complications. **HHS** rather than DKA (although "ketosis-prone T2DM" is common in Africans)
Gestational DM (GDM)	Hyperglycaemia first recognised in pregnancy (24–28 weeks). **75 g OGTT** with IADPSG/WHO 2013 criteria: **fasting ≥5.1, 1 h ≥10.0, 2 h ≥8.5 mmol/L** (any one). Risks: macrosomia, shoulder dystocia, neonatal hypoglycaemia, later T2DM
Other specific types	**MODY** (maturity-onset diabetes of the young; monogenic, autosomal dominant: HNF1A, glucokinase, HNF4A). **Pancreatic disease:** chronic pancreatitis, **fibrocalculous pancreatic diabetes** (tropical, historically "malnutrition-related"), cystic fibrosis, haemochromatosis, pancreatectomy, carcinoma. **Endocrinopathies: Cushing's**, acromegaly, phaeochromocytoma, glucagonoma, thyrotoxicosis. **Drugs: glucocorticoids**, thiazides, **antipsychotics**, protease inhibitors, tacrolimus. Infections (congenital rubella), genetic syndromes

Diagnostic criteria (WHO / ADA)

Diabetes is diagnosed by **any one** of the following. In an asymptomatic person, confirm with a repeat test.

Test	Diabetes	Pre-diabetes / intermediate hyperglycaemia
Fasting plasma glucose (≥8 h fast)	**≥7.0 mmol/L (126 mg/dL)**	**Impaired fasting glucose (IFG):** 6.1–6.9 (WHO) / 5.6–6.9 (ADA)
2-h plasma glucose in 75 g OGTT	**≥11.1 mmol/L (200 mg/dL)**	**Impaired glucose tolerance (IGT):** 7.8–11.0
HbA1c (standardised, DCCT/IFCC aligned)	**≥48 mmol/mol (6.5%)**	42–47 mmol/mol (6.0–6.4%) [ADA 39–47; 5.7–6.4%]

Test	Diabetes	Pre-diabetes / intermediate hyperglycaemia
Random plasma glucose with **classic symptoms** (polyuria, polydipsia, weight loss) or hyperglycaemic crisis	**≥11.1 mmol/L (200 mg/dL)**	

Conversion: mg/dL ÷ 18 = mmol/L.

The oral glucose tolerance test (OGTT):

1. **Preparation:** unrestricted carbohydrate diet (≥150 g/day) for 3 days; overnight fast (8–14 h); no smoking; seated during the test.
2. **Fasting sample.**
3. **75 g anhydrous glucose** (82.5 g monohydrate) in 250–300 mL water, drunk over 5 min. Children: 1.75 g/kg, max 75 g.
4. **Sample at 2 h** (and 1 h in pregnancy).
5. Collect samples in **fluoride oxalate tubes**.

HbA1c (glycated haemoglobin):

- **What it is:** non-enzymatic glycation of the **N-terminal valine of the β-chain**.
- **What it reflects:** mean glycaemia over the previous **8–12 weeks** (red cell lifespan).
- **Uses:** monitoring and diagnosis.
- **Treatment target:** usually **≤53 mmol/mol (7%)**; individualise.
- **Unreliable when red cell turnover is altered:**

HbA1c falsely low	HbA1c falsely high
Haemolysis, **sickle cell disease** and other haemoglobinopathies (method-dependent), recent transfusion, **pregnancy**, blood loss, EPO therapy, CKD	**Iron deficiency anaemia**, B_{12}/folate deficiency, asplenia

In these situations use **fructosamine** (2–3 weeks) or glucose.

Other tests in DM:

- **Urine glucose:** the renal threshold is ~10 mmol/L. **Renal glycosuria** occurs in pregnancy and Fanconi syndrome.
- **Ketones:** urine acetoacetate by nitroprusside dipstick; **blood β-hydroxybutyrate** is preferred.
- **C-peptide:** endogenous insulin production. Low in T1DM. Distinguishes **factitious insulin administration** (high insulin, low C-peptide) from **insulinoma or sulfonylurea use** (both high).
- **Autoantibodies:** GAD.
- **Urine ACR, eGFR, lipids.**

Acute metabolic complications

Feature	Diabetic ketoacidosis (DKA)	Hyperosmolar hyperglycaemic state (HHS)
Typical patient	**T1DM** (any age; often the first presentation), also ketosis-prone T2DM	**Elderly T2DM**
Pathogenesis	**Absolute insulin deficiency** + ↑ counter-regulatory hormones → **lipolysis → FFA → hepatic ketogenesis** (β-oxidation; glucagon ↑ CPT-1) → **ketoacidosis**; gluconeogenesis + glycogenolysis → hyperglycaemia → osmotic diuresis → dehydration, electrolyte loss	**Relative insulin deficiency**: enough insulin to **suppress lipolysis and ketogenesis**, but not hyperglycaemia → profound dehydration and hyperosmolality
Precipitants	**Infection** (the commonest; malaria, pneumonia, UTI, sepsis), **missed insulin** (cost, availability in Nigeria), new diagnosis, MI, stroke, pancreatitis, drugs	Infection, dehydration, stroke, MI, steroids, thiazides
Onset	**Hours to 1–2 days**	**Days to weeks**
Diagnostic criteria	**Ketonaemia ≥3.0 mmol/L** (or urine ketones ≥2+), **glucose >11 mmol/L** (or known DM; euglycaemic DKA with **SGLT2 inhibitors**), **venous pH <7.3 and/or HCO_3^- <15 mmol/L** (JBDS 2023)	**Glucose ≥30 mmol/L**, **osmolality ≥320 mOsm/kg**, pH >7.3, HCO_3^- >15, ketones <3, marked hypovolaemia
Clinical	Polyuria, polydipsia, vomiting, abdominal pain, **Kussmaul respiration, ketotic (fruity/acetone) breath**, dehydration, drowsiness, coma	Severe dehydration, confusion, coma, focal neurology, seizures, **thrombosis (VTE, stroke)**
K^+	**Total body K^+ depleted**, but serum K^+ may be normal or high initially (acidosis + insulin deficiency shift K^+ out of cells)	Depleted
Mortality	<1–5%	**10–20% (higher)**

DKA management (JBDS 2023 / ADA):

Step	Treatment
1. Fluids	**0.9% saline**: 1 L over 1 h, then 1 L over 2 h, 2 h, 4 h, 4 h, 6 h (adjust for age, cardiac and renal status). Average deficit ~5–7 L (100 mL/kg). **Children: careful, slower rehydration to avoid cerebral oedema**

Step	Treatment
2. Insulin	**Fixed-rate IV insulin infusion (FRIII) 0.1 unit/kg/h** of soluble insulin. **Continue long-acting basal insulin.** Add **10% glucose** when blood glucose falls <14 mmol/L (so insulin can continue until ketoacidosis resolves)
3. Potassium	**Insulin drives K^+ into cells → dangerous hypokalaemia.** Add **KCl 40 mmol/L to fluids if K^+ 3.5–5.5**. If K^+ >5.5, give none. If K^+ <3.5, seek senior help: delay or slow insulin and replace K^+ urgently
4. Monitor	Hourly capillary glucose and ketones, 2-hourly (or at 1, 2, 6 h) venous gas (pH, HCO_3^-, K^+). **Resolution:** ketones <0.6 mmol/L and pH >7.3. Then switch to subcutaneous insulin (overlap 30–60 min)
5. Treat the precipitant	Antibiotics, antimalarials. Prevent VTE. Insert an NG tube if vomiting or drowsy. Avoid bicarbonate (unless pH <6.9 in some guidelines). Phosphate only if severely low

Complications of DKA treatment:

- **Hypokalaemia**
- **Hypoglycaemia**
- **Cerebral oedema** (children)
- Hyperchloraemic acidosis
- ARDS
- Aspiration

HHS management:

- **Fluids are the mainstay.** Use 0.9% saline, and aim to lower **Na^+ by no more than 10 mmol/L in 24 h** and **osmolality by 3–8 mOsm/kg/h**.
- **Low-dose insulin (0.05 unit/kg/h)** only once glucose stops falling with fluids alone, or if ketones are significant.
- **Prophylactic LMWH.**

Calculated serum osmolality = **$2Na^+$ + glucose + urea** (all in mmol/L). Normal is 275–295 mOsm/kg.

Corrected Na^+ (for hyperglycaemia):

- Add 2.4 mmol/L per 5.6 mmol/L (100 mg/dL) of glucose above 5.6 (Katz/Hillier), or
- 1.6–2.4 per 100 mg/dL.

Hypoglycaemia (plasma glucose <4.0 mmol/L in treated diabetes; **<3.0 is clinically significant**):

Aspect	Details
Causes in DM	Insulin, **sulfonylureas** (glibenclamide: long-acting; avoid in the elderly and in CKD), missed meals, exercise, alcohol, renal failure

Aspect	Details
Other causes	**Insulinoma**, liver failure, Addison's, hypopituitarism, sepsis, **severe malaria and quinine therapy**, neonates, alcohol, non-islet cell tumours (IGF-2), factitious
Whipple's triad (to confirm a hypoglycaemic disorder)	Symptoms of hypoglycaemia + low plasma glucose + relief when glucose is raised
Symptoms	**Autonomic** (sweating, tremor, palpitations, hunger) and **neuroglycopenic** (confusion, behavioural change, seizures, coma)
Treatment (conscious)	**15–20 g fast-acting carbohydrate orally** (glucose tablets, juice)
Treatment (unconscious)	**IV glucose:** 10% dextrose 150–200 mL (or 20% 75–100 mL); avoid 50% (extravasation injury). **Or glucagon 1 mg IM** (not effective in liver failure or alcohol-related hypoglycaemia). Children: **2 mL/kg of 10% dextrose**
Follow-up	Give a long-acting carbohydrate afterwards. Sulfonylurea-induced hypoglycaemia may recur (admit; consider octreotide)

Chronic complications

Type	Complications
Microvascular (specific to diabetes; related to hyperglycaemia)	**Retinopathy** (background → pre-proliferative → **proliferative** (neovascularisation; VEGF) → maculopathy; the leading cause of blindness in working-age adults). **Nephropathy:** glomerular hyperfiltration → **microalbuminuria (the earliest clinical sign)** → overt proteinuria → declining GFR → ESRD. Histology: **Kimmelstiel–Wilson nodules** (nodular glomerulosclerosis), diffuse glomerulosclerosis, hyaline arteriolosclerosis (afferent and efferent). **Neuropathy:** symmetrical sensory ("glove and stocking"), autonomic (gastroparesis, postural hypotension, erectile dysfunction), mononeuropathy, amyotrophy
Macrovascular (accelerated atherosclerosis)	**Coronary artery disease** (the commonest cause of death), stroke, **peripheral arterial disease**
Diabetic foot	Neuropathy + ischaemia + infection → ulcers, **gangrene, amputation**. A major problem in Nigeria
Other	Cataracts, infections (TB, UTI, candida, mucormycosis), periodontal disease, skin (necrobiosis lipoidica)

Mechanisms of hyperglycaemic damage:

1. **Advanced glycation end products (AGEs)** → RAGE → inflammation and cross-linking.

2. **Polyol pathway:** aldose reductase converts glucose to **sorbitol** → osmotic damage and ↓ NADPH (lens, nerves).
3. **Protein kinase C activation** (DAG) → VEGF, TGF-β.
4. **Hexosamine pathway.**
5. **Oxidative stress** (the unifying mechanism; Brownlee).

Biochemical monitoring of DM:

- **HbA1c** every 3–6 months.
- Home glucose monitoring or CGM.
- **Annual urine ACR and eGFR.**
- **Lipids.**
- **Annual retinal screening and foot examination.**
- BP.

Chapter 55 · Lipoprotein Metabolism and Disorders; Metabolic Syndrome

Lipoprotein structure

Lipoproteins are spherical particles:

- **Hydrophobic core:** triglycerides (TG) and cholesteryl esters.
- **Amphipathic surface:** phospholipids, free cholesterol and **apolipoproteins**.

Lipoprotein	Density	Main lipid	Key apolipoproteins	Origin / function
Chylomicrons	Lowest (<0.95 g/mL; largest)	**Dietary TG (~85%)**	**B-48**, C-II, E, A-I	Intestine. Carry **exogenous** dietary lipid to tissues. Cause the **creamy layer** on standing plasma
VLDL	0.95–1.006	**Endogenous TG**	**B-100**, C-II, E	Liver. Carries endogenous TG to tissues
IDL	1.006–1.019	TG, cholesterol	B-100, E	VLDL remnant
LDL	1.019–1.063	**Cholesterol esters (the main cholesterol carrier; ~70% of plasma cholesterol)**	**B-100** (one per particle)	From VLDL/IDL. Delivers cholesterol to peripheral tissues via the **LDL receptor**. **The most atherogenic** ("bad cholesterol"). Small dense LDL is especially atherogenic
HDL	1.063–1.21 (smallest, densest)	Phospholipids, cholesterol	**A-I**, A-II, C, E	Liver and intestine. **Reverse cholesterol transport** from the periphery to the liver. **Anti-atherogenic** ("good cholesterol")
Lp(a)		Like LDL + apo(a)	B-100 + apo(a) (homologous to plasminogen)	Genetic; independent CVD risk factor

Key apolipoproteins and enzymes:

Protein	Function
Apo C-II	**Activator of lipoprotein lipase (LPL)** on capillary endothelium (adipose and muscle). LPL hydrolyses TG in chylomicrons and VLDL → free fatty acids
Apo B-100	**Ligand for the LDL receptor**
Apo E	Ligand for remnant receptors (LRP) and the LDL receptor. Clears chylomicron remnants and IDL. **E2/E2** genotype → dysbetalipoproteinaemia; **E4** → Alzheimer's risk

> **⚠ Caution**
>
> The rest of this chapter (lipoprotein pathways and disorders, metabolic syndrome) and the chapters on the thyroid axis, haem and bilirubin metabolism, and the porphyrias are **not included in this edition**. Study them from Marshall & Bangert *Clinical Chemistry* (chapters on lipids, thyroid, liver and porphyrias) and your lecture notes.

Chemical Pathology Self-Assessment

Section A: Single best answer MCQs

1. The commonest source of laboratory error is the: A. Analytical phase · B. **Pre-analytical phase** · C. Post-analytical phase · D. Calibration · E. Reagents

B. About 60–70% of laboratory errors happen before the sample is analysed: misidentification, haemolysis, the wrong tube, or delayed separation.

2. Blood glucose is best collected into a tube containing: A. EDTA · B. Lithium heparin · C. **Fluoride oxalate** · D. Citrate · E. Plain

C. Fluoride inhibits enolase, which stops the red cells using up glucose (glycolysis) while the sample waits.

3. A reference interval covers what proportion of healthy individuals? A. 68% · B. 90% · C. **95%** · D. 99% · E. 100%

C. It is the central 95%. This means 1 in 20 healthy people will have a result outside the interval.

4. As disease prevalence falls, the positive predictive value of a test: A. Rises · B. **Falls** · C. Stays the same · D. Equals sensitivity · E. Equals specificity

B. PPV depends on prevalence. In a low-prevalence population, a larger share of positive results are false positives.

5. The earliest ECG sign of hyperkalaemia is: A. U waves · B. **Tall peaked T waves** · C. Wide QRS · D. Sine wave · E. ST depression

B. Tall, peaked T waves appear first. Wide QRS and a sine-wave pattern come later and are more dangerous.

6. The first drug given in hyperkalaemia with ECG changes is: A. Insulin–glucose · B. Salbutamol · C. **Calcium gluconate** · D. Bicarbonate · E. Resonium

C. Calcium gluconate protects the myocardium within minutes but does not lower K^+. Insulin–glucose and salbutamol then shift K^+ into cells.

7. pH 7.28, pCO_2 8.5 kPa, HCO_3^- 30 mmol/L in a patient with COPD. The disorder is: A. Metabolic acidosis · B. **Respiratory acidosis (partially compensated)** · C. Metabolic alkalosis · D. Respiratory alkalosis · E. Normal

B. The low pH with a high pCO_2 makes the primary problem respiratory acidosis. The raised HCO_3^- shows the kidneys are partly compensating.

8. Which causes a normal anion gap metabolic acidosis? A. DKA · B. Lactic acidosis · C. **Diarrhoea** · D. Methanol · E. Uraemia

C. Diarrhoea loses bicarbonate, and chloride rises to take its place, so the anion gap stays normal. The other options all add unmeasured acids and raise the gap.

9. A rise in serum creatinine above the reference range occurs only after GFR has fallen by about: A. 10% · B. 25% · C. **50%** · D. 75% · E. 90%

C. The relationship between creatinine and GFR is curved (hyperbolic), so creatinine stays in the normal range until about half of GFR is lost.

10. HbA1c is falsely low in: A. Iron deficiency · B. **Haemolytic anaemia** · C. Asplenia · D. B_{12} deficiency · E. Alcoholism

B. In haemolysis, red cells live for a shorter time, so they have less time to become glycated.

11. Low calcium, low phosphate and raised ALP suggest: A. Osteoporosis · B. Primary hyperparathyroidism · C. **Osteomalacia** · D. Paget's disease · E. Myeloma

C. This is the pattern of vitamin D deficiency. In osteoporosis, by contrast, the biochemistry is normal.

12. The insulin regimen in adult DKA (JBDS) is: A. 1 unit/kg bolus · B. **Fixed-rate IV insulin 0.1 unit/kg/h** · C. Subcutaneous sliding scale · D. 10 units IM hourly · E. No insulin until K^+ is normal

B. Give a fixed-rate infusion of 0.1 unit/kg/h. If K^+ is below 3.5 mmol/L, replace potassium urgently before starting insulin.

Section B: Short-answer and essay questions

1. **Discuss the role of the physician in obtaining a valid laboratory result.** Structure your answer around the pre-analytical, analytical and post-analytical phases of the total testing loop (see 49.4).
2. **Write short notes on:**
    - a. point-of-care testing;
    - b. reference intervals versus decision limits;
    - c. sensitivity and specificity.
3. **Classify the causes of hyperkalaemia. Outline the emergency management.**
4. **Describe the approach to interpreting arterial blood gases, using a worked example.**

5. **Discuss the diagnosis and management of diabetic ketoacidosis.**

Section C: Viva voce bank (Chemical Pathology)

#	Question	Model answer
1	What colour tube for glucose, FBC, clotting, U&E?	Grey (fluoride oxalate); purple (EDTA); light blue (citrate); gold or red (serum) or green (heparin).
2	What causes spurious hyperkalaemia?	Haemolysis, delayed separation, EDTA contamination, a prolonged tourniquet or fist clenching, a drip arm, or a very high platelet or white cell count.
3	Define sensitivity and specificity.	Sensitivity = TP/(TP+FN); specificity = TN/(TN+FP).
4	What is a critical value?	A result that is life-threatening and must be telephoned to the clinician at once, e.g. K^+ >6.5 mmol/L.
5	Advantages of POCT?	Rapid results, small samples, and useful in remote settings and emergencies.
6	What is the anion gap and its normal value?	$(Na^+ + K^+) - (Cl^- + HCO_3^-)$, normally about 10–20 mmol/L. It reflects unmeasured anions, mostly albumin.
7	Causes of high anion gap acidosis?	MUDPILES: methanol, uraemia, DKA, paraldehyde/metformin, iron/isoniazid, lactate, ethylene glycol, salicylates.
8	What is Winter's formula?	Expected pCO_2 (mmHg) = 1.5 × HCO_3^- + 8 ± 2.
9	What is the ideal GFR marker?	One that is freely filtered, not reabsorbed, secreted or metabolised. Inulin is the classic example.
10	What does FENa <1% suggest?	Pre-renal AKI.
11	How do you correct calcium for albumin?	Adjusted Ca^{2+} = measured Ca^{2+} + 0.02 × (40 − albumin in g/L).
12	Two commonest causes of hypercalcaemia?	Primary hyperparathyroidism and malignancy.
13	Signs of hypocalcaemia?	Chvostek's and Trousseau's signs, tetany, paraesthesiae, prolonged QT.

#	Question	Model answer
14	WHO diagnostic criteria for diabetes?	FPG ≥7.0 mmol/L, 2-h OGTT ≥11.1, HbA1c ≥48 mmol/mol (6.5%), or random glucose ≥11.1 with symptoms.
15	Why add dextrose during DKA treatment?	So the insulin infusion can continue until the ketones have cleared, without the glucose falling too low.

Book 5 — Medical Microbiology

Part A: Virology · Part B: Parasitology (Protozoology) · Part C: Parasitology (Helminthology)

Part A — Virology

Chapter 56 · Introduction to Virology

Viruses are the smallest (20–300 nm) infectious agents. They are **obligate intracellular parasites** with a **single type of nucleic acid (DNA or RNA)**, no cellular structure, no metabolic machinery of their own and no ribosomes. They therefore cannot reproduce outside a living host cell.

Structure

Component	Description
Genome (core)	DNA or RNA; single- or double-stranded; linear or circular; segmented or not; **+sense** (mRNA-like) or **−sense**
Capsid	A protein coat made of subunits (**capsomeres**). Symmetry: **icosahedral**, **helical**, or complex (poxviruses)
Nucleocapsid	Capsid + genome
Envelope	A lipid bilayer from host membrane, with viral **glycoprotein spikes** (e.g. influenza haemagglutinin). **Enveloped viruses are fragile** (sensitive to heat, drying, detergents, alcohol, bile), so they usually spread by close contact, droplets or body fluids. **Naked (non-enveloped) viruses are tough** and resist the environment, acid and detergents, so they spread by the faecal–oral route and survive on surfaces

Classification (Baltimore groups, simplified)

Group	Example families
I. dsDNA	Herpesviruses, Poxviruses, Adenoviruses, Papillomaviruses, Polyomaviruses
II. ssDNA	Parvoviruses
III. dsRNA	Reoviruses (Rotavirus)
IV. +ssRNA	Picornaviruses, Flaviviruses, Togaviruses, Coronaviruses, Caliciviruses
V. −ssRNA	Orthomyxoviruses, Paramyxoviruses, Rhabdoviruses, Filoviruses, Arenaviruses, Bunyaviruses
VI. +ssRNA with reverse transcriptase (retroviruses)	**HIV, HTLV**
VII. dsDNA with reverse transcriptase	**Hepatitis B**

Useful rules:

- **All DNA viruses are double-stranded except Parvovirus.**
- **All RNA viruses are single-stranded except Reovirus (Rotavirus).**
- **DNA viruses replicate in the nucleus except Poxvirus** (which carries its own polymerase and replicates in the cytoplasm).
- **RNA viruses replicate in the cytoplasm except Orthomyxovirus (influenza) and Retroviruses.**

Viral replication cycle

1. **Attachment (adsorption):** viral proteins bind host receptors (HIV gp120 → **CD4 + CCR5/CXCR4**; rabies → ACh receptor; EBV → CD21; SARS-CoV-2 → ACE2; rhinovirus → ICAM-1).
2. **Penetration:** endocytosis or membrane fusion.
3. **Uncoating:** the genome is released.
4. **Synthesis:** replication of the genome and expression of viral proteins. (+)RNA acts directly as mRNA; (−)RNA needs its own **RNA-dependent RNA polymerase** carried in the virion; retroviruses use **reverse transcriptase** then integrate (**provirus**).
5. **Assembly (maturation).**
6. **Release: lysis** (naked viruses) or **budding** (enveloped viruses; the cell may survive).

The **eclipse period** is the time when no infectious virus can be detected inside the cell.

Effects on the host cell

- **Cytolytic (cytopathic effect, CPE):** cell death (polioviruses). Inclusion bodies may form: **Negri bodies** (rabies, cytoplasmic), **Cowdry type A** (herpes, CMV "owl's eye", intranuclear), **Guarnieri bodies** (smallpox).
- **Latent:** the virus persists silently and can reactivate (herpesviruses).
- **Transformation:** immortalisation → cancer (oncogenic viruses, Chapter 37).
- **Persistent / chronic** (HBV, HCV).
- Syncytium (giant cell) formation (RSV, measles, HIV).

Pathogenesis

Entry → primary replication → spread (**local, lymphatic, viraemia (primary and secondary), or neural**) → target organ → shedding. Outcomes: acute self-limiting, chronic, latent, slow (SSPE, prions), or transforming.

Chapter 57 · Laboratory Diagnosis of Viral Infections

Approach	Methods	Notes
1. Direct detection	**Electron microscopy** (rabies, pox, rota). **Antigen detection:** ELISA, immunofluorescence, lateral-flow **rapid tests** (HBsAg, HIV, SARS-CoV-2, rota, RSV, dengue NS1). **Histology/cytology:** inclusion bodies (Negri, Cowdry), Tzanck smear (herpes)	Rapid
2. Nucleic acid amplification (molecular)	**PCR, RT-PCR, real-time quantitative PCR (viral load)**, GeneXpert, sequencing, LAMP	**Most sensitive and specific.** Essential for **HIV viral load**, HCV RNA, HBV DNA, CMV, and **Lassa and Ebola** diagnosis. Genotyping and resistance testing
3. Virus isolation (culture)	**Cell lines** (CPE, haemadsorption), embryonated eggs (influenza), animals. The historical "gold standard"	Slow (days to weeks), needs expertise and biosafety; largely replaced by molecular methods
4. Serology (antibody detection)	**ELISA**, immunofluorescence, neutralisation, complement fixation, haemagglutination inhibition, Western blot	**IgM = recent/acute** infection. **IgG = past infection or immunity.** A **four-fold rise** in titre between acute and convalescent sera (2–3 weeks apart) confirms recent infection. Used for diagnosis and immune status (rubella)

Specimens: depend on the illness — respiratory (nasopharyngeal swab or aspirate), stool (enteroviruses, rota), blood/serum (arboviruses, HIV, hepatitis), CSF (encephalitis), vesicle fluid (herpes), saliva and brain (rabies). Transport viruses in **viral transport medium**, cold.

Biosafety: Lassa fever and Ebola are handled at BSL-4 (or high-containment); liaise with reference laboratories (in Nigeria, **NCDC** and the Irrua and Lagos reference centres).

Chapter 58 · Antiviral Chemotherapy

Antivirals are **mostly virustatic** (they suppress replication) and act at specific steps. Resistance develops readily, so **combinations** are used (HIV, HCV).

Target / class	Drugs	Uses	Notes
Entry / uncoating	Enfuvirtide (fusion), **maraviroc** (CCR5), amantadine (influenza A M2; now resistant), docosanol	HIV; herpes labialis	
DNA polymerase inhibitors (herpes)	**Aciclovir** (prodrug; activated by viral thymidine kinase → chain terminator), valaciclovir, famciclovir; **ganciclovir/valganciclovir** (CMV), foscarnet, cidofovir	HSV, VZV; CMV (ganciclovir)	Aciclovir: safe; nephrotoxic if rapid IV. Ganciclovir: myelosuppression. Foscarnet: nephrotoxicity, ↓ Ca^{2+}
Reverse transcriptase inhibitors	**NRTIs:** zidovudine, lamivudine, **tenofovir**, abacavir, emtricitabine. **NNRTIs:** efavirenz, nevirapine, rilpivirine, doravirine	HIV, HBV (tenofovir, lamivudine)	NRTIs: lactic acidosis, lipodystrophy; abacavir (**HLA-B*57:01**); zidovudine (anaemia). NNRTIs: rash, hepatotoxicity, CYP interactions
Integrase inhibitors (INSTIs)	**Dolutegravir**, raltegravir, bictegravir	**HIV first-line (dolutegravir-based)**	Well tolerated; weight gain
Protease inhibitors	Lopinavir, atazanavir, darunavir (+ **ritonavir** booster); for HCV: boceprevir	HIV; HCV	GI upset, dyslipidaemia, CYP3A4
Neuraminidase inhibitors	**Oseltamivir** (oral), zanamivir	**Influenza A and B** (within 48 h)	
Direct-acting antivirals (HCV)	**Sofosbuvir** (NS5B), ledipasvir/velpatasvir (NS5A), glecaprevir (NS3/4A protease)	**Hepatitis C (cure >95%)**	Oral, 8–12 weeks
Nucleotide analogues (HBV)	**Tenofovir, entecavir**	Chronic hepatitis B	Long-term suppression
Interferon-α (pegylated)		HBV, HCV (historical)	Flu-like, cytopenias, depression

Target / class	Drugs	Uses	Notes
Broad-spectrum	**Ribavirin** (Lassa fever, RSV, HCV with others), **remdesivir** (RdRp; Ebola, COVID-19), favipiravir, molnupiravir, nirmatrelvir/ritonavir (Paxlovid)		Ribavirin: haemolytic anaemia, teratogenic

HIV treatment principle: combination antiretroviral therapy (ART) with ≥3 drugs. Nigeria (NASCP/WHO): first line **tenofovir + lamivudine + dolutegravir (TLD)** for all, **started regardless of CD4 count ("treat all")**. Monitor **viral load** (aim undetectable). **PMTCT** (prevention of mother-to-child transmission), **PrEP** and **PEP**.

Chapter 59 · Picornaviruses, Coxsackieviruses and Rhinoviruses

Picornaviridae ("pico-RNA": small RNA): **naked, icosahedral, +ssRNA** viruses. They include the **enteroviruses** (poliovirus, coxsackie A and B, echovirus, enterovirus 68–71, hepatitis A) and the **rhinoviruses**.

Poliovirus

- Three serotypes (1, 2, 3). Spread by the **faecal–oral** route.
- Replicates in the gut, then spreads via blood to the **anterior horn cells of the spinal cord**.
- **Outcomes:** ~95% asymptomatic; abortive (minor febrile illness); non-paralytic (aseptic meningitis); **paralytic poliomyelitis (~0.1–1%)** = asymmetrical **flaccid paralysis**, lower motor neurone, no sensory loss; bulbar polio.
- **Vaccines:**
  - **OPV (Sabin):** live attenuated, oral, gives mucosal (IgA) and herd immunity; rarely **vaccine-associated paralytic polio (VAPP)** and **circulating vaccine-derived poliovirus (cVDPV2)** — now addressed with **nOPV2**.
  - **IPV (Salk):** killed, injectable, safe in immunocompromised patients; no mucosal immunity.
- **Nigeria was declared free of wild poliovirus in 2020** (Africa certified wild-polio-free), a major public health achievement, though cVDPV2 outbreaks still occur. **AFP surveillance** continues.

Coxsackieviruses and echoviruses

Virus	Diseases
Coxsackie A	**Herpangina** (vesicles on the soft palate), **hand-foot-and-mouth disease** (A16, EV71), aseptic meningitis, conjunctivitis
Coxsackie B	**Pleurodynia (Bornholm disease, "devil's grip")**, **myocarditis and pericarditis**, dilated cardiomyopathy, aseptic meningitis, (link to type 1 diabetes)
Echovirus / enteroviruses	Aseptic meningitis (the commonest cause), febrile rash, neonatal sepsis, EV-D68 (flaccid paralysis)

Rhinoviruses

- Over 100 serotypes; bind **ICAM-1**; grow best at **33 °C** (nasal passages).
- **The commonest cause of the common cold.** Also trigger asthma and COPD exacerbations.

- No vaccine (too many serotypes).

Hepatitis A (an enterovirus/Hepatovirus)

- **Naked +ssRNA; faecal–oral** spread (contaminated food and water; common in Nigeria). **No chronic carrier state.**
- Incubation 2–6 weeks. Often anicteric in children; more severe in adults.
- Diagnosis: **anti-HAV IgM** (acute).
- Self-limiting; rare fulminant hepatitis. **Vaccine (inactivated)** available.

Chapter 60 · Hepatitis Viruses

Feature	HAV	HBV	HCV	HDV	HEV
Family / genome	Picornavirus; **naked RNA**	**Hepadnavirus; enveloped, partial dsDNA** (reverse transcriptase)	Flavivirus; **enveloped RNA**	Deltavirus; **defective RNA** (needs HBsAg)	Hepevirus; **naked RNA**
Transmission	**Faecal–oral**	**Blood, sexual, vertical (perinatal)**	**Blood** (transfusion, IVDU, needlestick)	Blood (with HBV)	**Faecal–oral** (water)
Incubation	2–6 wk	1–6 months	2 wk–6 months	—	2–9 wk
Chronicity	**No**	**Yes** (~90% if perinatal, ~5% if adult)	**Yes (~75–85%)**	Yes	**No** (chronic in immunosuppressed)
HCC / cirrhosis	No	**Yes**	**Yes**	Yes (accelerates)	No
Prevention	Vaccine, hygiene	**Vaccine**, HBIG	No vaccine; DAAs cure	**HBV vaccine**	Sanitation; vaccine (China)
Special note	Children mild	**Most common cause of HCC in Nigeria** (with aflatoxin)	"Silent"	Co-infection vs superinfection	**High mortality (~20%) in pregnancy**

Hepatitis B serology (crucial)

Marker	Meaning
HBsAg	**Surface antigen: current infection** (acute or chronic)
Anti-HBs	**Immunity** (recovery or **vaccination**)
HBeAg	**High infectivity / active replication**
Anti-HBe	Lower infectivity (seroconversion)
IgM anti-HBc	**Acute (recent) infection**; the only marker in the "window period"
IgG anti-HBc	Past or chronic infection (not from vaccination)
HBV DNA	Viral load; guides treatment and monitors therapy

Pattern	Interpretation
HBsAg +, IgM anti-HBc +	Acute hepatitis B
HBsAg + (>6 months), IgG anti-HBc +	Chronic hepatitis B
Anti-HBs + only	**Vaccinated**
Anti-HBs + and anti-HBc +	Natural immunity (past infection, recovered)
IgM anti-HBc + alone	Window period

Prevention in Nigeria: HepB birth dose + pentavalent vaccine; screen blood donors and antenatal women; **tenofovir in pregnancy** for highly viraemic mothers (PMTCT); HBIG + vaccine for exposed neonates and needlestick injuries.

Chapter 61 · Herpesviridae

Herpesviruses are large, **enveloped, dsDNA** viruses whose hallmark is **latency with reactivation**. "Herpes" means to creep.

Virus	Primary disease	Latency site	Reactivation / notes
HSV-1	Gingivostomatitis, **herpes labialis (cold sores)**, keratitis, **encephalitis (temporal lobe)**, herpetic whitlow	**Trigeminal ganglion**	Stress, sunlight. Tzanck smear: multinucleated giant cells. Treat with **aciclovir**
HSV-2	**Genital herpes**, neonatal herpes	**Sacral ganglia**	STI; risk to the neonate at vaginal delivery
VZV (HHV-3)	**Chickenpox (varicella)**	**Dorsal root / cranial ganglia**	**Shingles (herpes zoster)** in a dermatome; postherpetic neuralgia; Ramsay Hunt syndrome. Vaccines (varicella; Shingrix)
EBV (HHV-4)	**Infectious mononucleosis** (fever, pharyngitis, lymphadenopathy, splenomegaly, atypical lymphocytes)	B lymphocytes (CD21)	**Burkitt lymphoma** (jaw, Africa, with malaria), **nasopharyngeal carcinoma**, Hodgkin lymphoma, hairy leukoplakia, PTLD. **Heterophile (Monospot) positive**. Ampicillin rash
CMV (HHV-5)	Usually asymptomatic; mononucleosis (Monospot-negative)	Monocytes, others	**Congenital CMV** (commonest congenital infection: deafness, periventricular calcification, "blueberry muffin"); **immunocompromised** (retinitis, colitis, pneumonitis in HIV/transplant). **Owl's-eye inclusions**. Treat with **ganciclovir**
HHV-6, HHV-7	**Roseola infantum** (exanthem subitum, sixth disease)		High fever then rash; febrile seizures

Virus	Primary disease	Latency site	Reactivation / notes
HHV-8	**Kaposi sarcoma** (HIV/AIDS), primary effusion lymphoma, multicentric Castleman disease		Common in Africa

Chapter 62 · Rhabdoviridae and Rabies Virus

Rhabdoviridae are **bullet-shaped, enveloped, −ssRNA** viruses. The genus *Lyssavirus* contains **rabies virus**.

Rabies

- **Reservoir:** dogs (the main source in Nigeria and Africa), bats, foxes, jackals.
- **Transmission:** the **bite or saliva** of an infected animal (rarely aerosol or transplant).
- **Pathogenesis:** the virus binds **nicotinic acetylcholine receptors** at the neuromuscular junction → travels by **retrograde axonal transport** to the CNS → replicates → spreads centrifugally to the salivary glands and other organs.
- **Incubation:** 1–3 months (depends on the distance of the bite from the brain; shortest for head and neck bites).
- **Clinical (almost always fatal once symptomatic):**
  - Prodrome: fever, **paraesthesiae at the bite site**.
  - **Furious rabies (~80%):** hydrophobia, aerophobia, hyperexcitability, hypersalivation, pharyngeal spasms, autonomic instability.
  - **Paralytic (dumb) rabies (~20%):** ascending flaccid paralysis.
  - Coma, death from respiratory failure.
- **Diagnosis:** clinical; **Negri bodies** (eosinophilic cytoplasmic inclusions in hippocampal and Purkinje neurons); direct fluorescent antibody on brain, skin biopsy (nape of neck), saliva RT-PCR, corneal impression.
- **Post-exposure prophylaxis (PEP), the key message — it works if given before symptoms:**
  1. **Immediate, thorough wound washing** with soap and running water for ≥15 minutes (plus a virucidal such as povidone-iodine). This is critical.
  2. **Rabies vaccine** (cell-culture vaccine) on days 0, 3, 7 and 14–28 (Essen regimen), or the WHO-abridged intradermal schedule.
  3. **Rabies immunoglobulin (RIG)** infiltrated around the wound for category III exposures (in previously unimmunised people).
  4. Do **not** suture the wound immediately. Give tetanus prophylaxis and antibiotics.
- **Pre-exposure prophylaxis** for high-risk groups (vets, lab workers).
- **Prevention:** dog vaccination and control. Rabies is a neglected disease in Nigeria.

Chapter 63 · Prions

A **prion** (proteinaceous infectious particle) is an **infectious agent made only of protein, with no nucleic acid**. Disease arises when the normal cellular prion protein (**PrP^{C}, α-helical) misfolds into an abnormal, protease-resistant, β-sheet-rich form (PrP^{Sc})**, which then recruits more normal protein to misfold (a self-propagating chain reaction).

- **Properties:** extraordinarily **resistant** to heat, formalin, UV, radiation and standard sterilisation. Destroyed only by autoclaving at 134 °C for prolonged periods, or by 1 N NaOH / hypochlorite.
- **Pathology: spongiform (vacuolar) change** in grey matter, neuronal loss, astrogliosis and amyloid plaques, **without inflammation**. These are the **transmissible spongiform encephalopathies**.
- **Human prion diseases:**

Disease	Features
Creutzfeldt–Jakob disease (CJD) — sporadic (most), familial, iatrogenic	Rapidly progressive **dementia, myoclonus**, ataxia; death in months. EEG periodic complexes; ↑ 14-3-3 protein in CSF; RT-QuIC
Variant CJD (vCJD)	From **bovine spongiform encephalopathy ("mad cow disease")**; younger patients, psychiatric onset
Kuru	"Laughing death"; among the Fore people of Papua New Guinea, from **ritual cannibalism**; cerebellar ataxia
GSS, fatal familial insomnia	Inherited (PRNP mutations)

- **Animal:** scrapie (sheep), BSE (cattle).
- **No treatment; uniformly fatal.** Prevention centres on not recycling animal protein and on decontaminating surgical instruments.

Chapter 64 · Myxoviruses: Orthomyxoviruses and Paramyxoviruses

Orthomyxoviruses (Influenza)

> **Influenza viruses** are **enveloped, segmented (8 segments), −ssRNA** viruses. Types **A** (animals and humans; pandemics), **B** (humans; epidemics) and **C** (mild).

- **Surface glycoproteins:**
  - **Haemagglutinin (HA, 18 types):** attachment to **sialic acid** and entry. The main target of neutralising antibody.
  - **Neuraminidase (NA, 11 types):** releases new virions. The target of **oseltamivir**.
- **Antigenic variation (the reason for seasonal epidemics and pandemics):**

	Antigenic drift	Antigenic shift
Mechanism	**Point mutations** in HA/NA	**Reassortment** of genome segments between strains (e.g. human + avian in a pig)
Magnitude	Minor	**Major (new HA/NA)**
Result	Seasonal **epidemics**	**Pandemics** (no population immunity)
Examples	Yearly strains	H1N1 (1918, 2009), H2N2, H3N2; avian H5N1, H7N9

- **Replicates in the nucleus** (unusual for an RNA virus).
- **Complications:** primary viral and **secondary bacterial pneumonia** (*S. pneumoniae*, *S. aureus*), myositis, Reye syndrome (with aspirin).
- **Prevention:** annual **inactivated vaccine** (updated by WHO; trivalent/quadrivalent), live attenuated nasal vaccine. **Treatment:** oseltamivir within 48 h.

Paramyxoviruses

Enveloped, non-segmented, −ssRNA; cause **syncytia (giant cells)**. Includes measles, mumps, parainfluenza, RSV and the zoonotic Nipah/Hendra viruses. (Measles, mumps and rubella follow in Chapters 65–67.)

Virus	Disease	Notes
Parainfluenza	**Croup** (laryngotracheobronchitis; “barking cough”, stridor), bronchiolitis	
Respiratory syncytial virus (RSV)	**Bronchiolitis and pneumonia in infants** (the commonest cause)	Syncytia. **Palivizumab** (monoclonal) and nirsevimab for prophylaxis in high-risk infants; new maternal vaccine
Human metapneumovirus	Bronchiolitis	
Nipah, Hendra	Encephalitis (zoonotic)	

Chapter 65 · Measles (Rubeola)

- **Paramyxovirus** (genus *Morbillivirus*); **enveloped −ssRNA**; one serotype.
- **Transmission:** respiratory droplets and airborne; **extremely contagious** (R_0 12–18). Still a major cause of childhood death in Nigeria.
- **Pathogenesis:** replicates in the respiratory epithelium and lymphoid tissue → viraemia. Causes profound (but transient) **immunosuppression** ("immune amnesia").
- **Clinical (the "3 Cs" prodrome):**
  - Prodrome: high fever, **Cough, Coryza, Conjunctivitis**, plus **Koplik spots** (pathognomonic: white spots on the buccal mucosa).
  - **Rash:** maculopapular, starts behind the ears and on the face, spreads downwards (cephalocaudal), becomes confluent, then desquamates.
- **Complications (common in malnourished and vitamin A-deficient children):**
  - **Otitis media** (the commonest)
  - **Pneumonia** (the commonest cause of death; giant cell pneumonia)
  - **Diarrhoea** → dehydration
  - **Croup**
  - **Acute encephalitis / ADEM**
  - **SSPE (subacute sclerosing panencephalitis):** a late, fatal, slow degenerative disease years later
  - **Keratitis → blindness** (with vitamin A deficiency)
  - Worsens malnutrition
- **Diagnosis:** clinical; **measles-specific IgM**.
- **Treatment:** supportive; **vitamin A** (reduces mortality and eye complications); treat bacterial complications.
- **Prevention: live attenuated vaccine** (MCV1 at 9 months and MCV2 at 15 months in Nigeria; MMR elsewhere). Measles outbreaks follow dips in coverage.

Chapter 66 · Mumps

- **Paramyxovirus** (*Rubulavirus*); enveloped −ssRNA.
- **Transmission:** respiratory droplets and saliva.
- **Clinical:**
  - Prodrome, then **painful bilateral parotitis** (parotid swelling) — the classic feature.
  - **Complications: orchitis** (in post-pubertal males; may cause infertility; usually unilateral), **oophoritis**, **aseptic meningitis** and encephalitis, **pancreatitis**, **sensorineural deafness**, myocarditis.
- **Diagnosis:** clinical; mumps IgM; raised **serum amylase** (parotid source).
- **Prevention: live attenuated vaccine (MMR).**

Chapter 67 · Rubella (German Measles)

- **Togavirus** (*Rubivirus*); **enveloped +ssRNA** (note: it is a togavirus, not a paramyxovirus).
- **Transmission:** respiratory droplets; **transplacental** (the important route).
- **Postnatal rubella:** mild — low fever, a fine maculopapular rash (face → body), and **tender posterior auricular and suboccipital lymphadenopathy**. Polyarthralgia in adults.
- **Congenital rubella syndrome (CRS)** — the reason rubella matters. Maternal infection in the **first trimester** is teratogenic (highest risk <12 weeks):

Classic triad	Other features
Sensorineural deafness (the commonest)	"Blueberry muffin" rash, microcephaly
Cataracts (and glaucoma, "salt-and-pepper" retinopathy)	Hepatosplenomegaly, thrombocytopenia
Congenital heart disease (**patent ductus arteriosus**, pulmonary stenosis)	Intellectual disability, growth restriction, diabetes

- **Diagnosis:** rubella **IgM** (acute); IgG for immune status in antenatal screening.
- **Prevention: live attenuated vaccine (MMR). Do not give in pregnancy** (live vaccine); vaccinate women of childbearing age beforehand. Check rubella immunity in antenatal care.

Virology Self-Assessment

Section A: MCQs

1. Which virus is DNA and replicates in the cytoplasm? A. Herpes · B. **Poxvirus** · C. Adenovirus · D. HBV · E. HPV

B. Poxviruses carry their own DNA polymerase, so they replicate in the cytoplasm. Other DNA viruses replicate in the nucleus.

2. The marker of hepatitis B immunity after vaccination is: A. HBsAg · B. **Anti-HBs** · C. HBeAg · D. IgM anti-HBc · E. HBV DNA

B. Vaccination produces anti-HBs alone. Anti-HBc is absent, because the vaccine contains only surface antigen.

3. Negri bodies are seen in: A. Measles · B. Herpes · C. **Rabies** · D. CMV · E. Smallpox

C. Negri bodies are cytoplasmic inclusions in neurons, characteristic of rabies.

4. The single most important step in rabies post-exposure care is: A. Antibiotics · B. **Immediate wound washing** · C. Suturing · D. Steroids · E. Observation

B. Wash the wound thoroughly with soap and running water for at least 15 minutes, then give vaccine and immunoglobulin.

5. Antigenic shift in influenza is caused by: A. Point mutation · B. **Genome segment reassortment** · C. Recombination · D. Latency · E. Translation error

B. Reassortment of segments produces a new HA or NA, so the population has no immunity. This causes pandemics.

6. Koplik spots are pathognomonic of: A. Rubella · B. **Measles** · C. Mumps · D. Chickenpox · E. Roseola

B. Koplik spots are white spots on the buccal mucosa, seen in the measles prodrome.

7. Congenital rubella classically causes: A. Limb defects · B. **Cataracts, deafness and PDA** · C. Neural tube defects · D. Phocomelia · E. Microcephaly alone

B. The classic triad is sensorineural deafness, cataracts and congenital heart disease (PDA).

8. First-line antiretroviral therapy in Nigeria is based on: A. Zidovudine · B. Efavirenz · C. **Dolutegravir (TLD)** · D. Nevirapine · E. Lopinavir

> **C.** Tenofovir + lamivudine + dolutegravir (TLD) is the recommended first-line regimen.

Section B: Essays

1. Classify viruses and describe the steps of viral replication.
2. Discuss the laboratory diagnosis of viral infections.
3. Describe the serological diagnosis and prevention of hepatitis B.
4. Write short notes on: rabies PEP; Negri bodies; prions; antigenic drift and shift; Koplik spots.

Section C: Viva voce bank (Virology)

#	Question	Model answer
1	What is a virus?	An obligate intracellular agent with one type of nucleic acid, a protein capsid, no cellular organelles and no independent metabolism.
2	Enveloped vs naked viruses?	Enveloped viruses are fragile and spread by close contact or fluids. Naked viruses are hardy and spread by the faecal–oral route.
3	Which DNA virus is single-stranded?	Parvovirus.
4	Which RNA virus is double-stranded?	Reovirus (including rotavirus).
5	Most sensitive diagnostic method?	Nucleic acid amplification (PCR).
6	Mechanism of aciclovir?	A prodrug activated by viral thymidine kinase, then a chain terminator of viral DNA polymerase.
7	HBsAg positive for more than 6 months means?	Chronic hepatitis B.
8	Commonest cause of the common cold?	Rhinovirus.
9	Commonest cause of bronchiolitis in infants?	Respiratory syncytial virus.
10	What is SSPE?	Subacute sclerosing panencephalitis, a late fatal complication of measles.

#	Question	Model answer
11	Why is rabies almost always fatal?	Once it reaches the CNS and symptoms begin, there is no effective treatment; prophylaxis must be given before then.
12	What are prions resistant to?	Heat, formalin, UV and standard sterilisation.
13	EBV-associated tumours?	Burkitt lymphoma, nasopharyngeal carcinoma, Hodgkin lymphoma.
14	Owl's-eye inclusions?	Cytomegalovirus.
15	Why give vitamin A in measles?	It reduces mortality and the risk of blindness.

Part B — Parasitology: Protozoology

Chapter 68 · Introduction to Parasitology

A **parasite** is an organism that lives on or in a host and obtains nourishment at the host's expense. **Medical parasitology** studies the parasites of humans: **protozoa** (single-celled), **helminths** (worms) and **arthropods** (vectors and ectoparasites). Parasitic diseases are a huge burden in Nigeria and tropical Africa.

Key terms

Term	Meaning
Definitive host	Harbours the **adult / sexual** stage (for *Plasmodium*, the **mosquito**)
Intermediate host	Harbours the **larval / asexual** stage (for *Plasmodium*, **humans**)
Reservoir host	Maintains the parasite in nature; a source of human infection
Vector	Transmits the parasite (biological: part of the life cycle; mechanical: just carries it)
Paratenic host	A transport host in which no development occurs
Zoonosis	A disease transmissible from animals to humans
Obligate vs facultative	Must parasitise vs can live freely
Ectoparasite vs endoparasite	On the surface vs inside the body

Classification of medically important protozoa

Group	Locomotion	Examples
Sarcodina (amoebae)	Pseudopodia	*Entamoeba histolytica*, *Acanthamoeba*, *Naegleria*
Mastigophora (flagellates)	Flagella	*Giardia*, *Trichomonas*, *Leishmania*, *Trypanosoma*
Ciliophora (ciliates)	Cilia	*Balantidium coli*
Apicomplexa (sporozoa)	Gliding (non-motile adults)	*Plasmodium*, *Toxoplasma*, *Cryptosporidium*, *Cyclospora*, *Isospora/Cystoisospora*, *Babesia*

Laboratory diagnosis in parasitology:

- **Microscopy** (the mainstay): thick and thin blood films (malaria, trypanosomes, microfilariae), stool microscopy (ova, cysts, trophozoites; concentration techniques such as formol-ether), urine (schistosome eggs, microfilariae), tissue aspirates (leishmania).
- **Stains:** Giemsa/Leishman (blood), **modified Ziehl–Neelsen** (*Cryptosporidium*, *Cyclospora*, *Isospora*), trichrome, iron-haematoxylin.
- **Antigen detection / RDTs:** malaria (HRP-2, pLDH), *Entamoeba*, *Giardia*, *Cryptosporidium*.
- **Serology:** toxoplasmosis, amoebic liver abscess, cysticercosis.
- **Molecular (PCR)**, culture, xenodiagnosis.
- **Concentration and special techniques:** Kato-Katz (helminth egg counts), Knott's concentration (microfilariae), Baermann (*Strongyloides* larvae).

Chapter 69 · Apicomplexans I: Plasmodium (Malaria)

Malaria is caused by protozoa of the genus ***Plasmodium***, transmitted by the bite of the **female *Anopheles* mosquito**. It is the most important parasitic disease in the world and a leading cause of death in Nigerian children. **Nigeria carries ~27% of the global malaria burden** — the highest of any country (WHO).

The five species

Species	Features
P. falciparum	**The most dangerous**; the predominant species in Nigeria. Causes **severe and cerebral malaria**. No hypnozoites (no relapse). Infects red cells of all ages → high parasitaemia
P. vivax	**Hypnozoites → relapse.** Rare in West Africa because most people are **Duffy antigen-negative** (the Duffy antigen is the receptor)
P. ovale	Hypnozoites → relapse; mild; present in West Africa
P. malariae	**Quartan** fever (72 h); **chronic; nephrotic syndrome** (immune complex)
P. knowlesi	Zoonotic (macaques); Southeast Asia; can be severe

Life cycle

Human (intermediate host): asexual cycle (schizogony)

1. An infected mosquito injects **sporozoites** with its saliva.
2. **Exo-erythrocytic (liver) stage:** sporozoites invade **hepatocytes → schizonts** → rupture, releasing **merozoites**. In *P. vivax* and *P. ovale*, some become dormant **hypnozoites** (the cause of relapse).
3. **Erythrocytic stage:** merozoites invade red cells → **ring trophozoite → trophozoite → schizont → merozoites** → the red cell ruptures (causing **fever**) → the merozoites invade new cells. The synchronised rupture explains the periodicity of fever (**tertian** 48 h; **quartan** 72 h).
4. Some merozoites become **gametocytes** (male and female), which are taken up by a feeding mosquito.

Mosquito (definitive host): sexual cycle (sporogony)

5. In the mosquito gut, gametocytes form gametes → **zygote → ookinete → oocyst → sporozoites** migrate to the salivary glands.

Pathogenesis

- **Haemolysis** (intravascular and extravascular) → **anaemia** (the main cause of death in children), jaundice.
- ***P. falciparum*** causes **cytoadherence and sequestration**: infected red cells express **PfEMP-1** and stick to the endothelium of deep capillaries (via ICAM-1, CD36) and to uninfected cells (**rosetting**) → microvascular obstruction → **cerebral malaria**, hypoxia, lactic acidosis.
- **Splenomegaly** (clearance; hyperreactive malarial splenomegaly).
- **Hypoglycaemia** (consumption; quinine-induced insulin release).
- Cytokines (TNF) → fever.
- **Blackwater fever:** massive intravascular haemolysis → haemoglobinuria → AKI.

Protective host polymorphisms (balanced by malaria selection, common in Nigeria)

- **Sickle cell trait (HbAS)**
- HbC, HbE
- **Thalassaemias**
- **G6PD deficiency**
- **Duffy negativity** (against *P. vivax*)

Clinical features

- **Uncomplicated:** fever with chills and rigors, sweating, headache, myalgia, nausea, splenomegaly, anaemia. Classical paroxysms (cold → hot → sweating) are often absent.
- **Severe/complicated malaria (*P. falciparum*; WHO criteria): cerebral malaria** (coma, seizures), **severe anaemia** (Hb <5 g/dL children, <7 adults), **hypoglycaemia**, **metabolic acidosis / respiratory distress**, AKI, pulmonary oedema/ARDS, DIC and bleeding, shock ("algid malaria"), haemoglobinuria, jaundice, hyperparasitaemia (>10%), prostration.
- **In pregnancy:** severe anaemia, **placental sequestration**, low birth weight, abortion, maternal death.
- **Tropical splenomegaly** and **nephrotic syndrome** (*P. malariae*) are chronic sequelae.

Diagnosis

- **Thick film** (sensitive: detects parasites) and **thin film** (species identification and parasite count). Giemsa stain. **The gold standard.**
- **Rapid diagnostic tests (RDTs):** HRP-2 (*falciparum*) or pLDH antigen. Widely used in Nigeria. (HRP-2 deletions can cause false negatives.)
- PCR (research, mixed infections), serology (epidemiology).
- **"Test before you treat"** (WHO): confirm with RDT or microscopy.

Treatment and prevention

See Book 1, Chapter 28 (antimalarials), which has the full Nigerian treatment policy: **ACTs** for uncomplicated malaria, **IV artesunate** for severe malaria, **IPTp-SP** and **SMC** for prevention, and the **R21/RTS,S vaccines**. Prevention also includes **insecticide-treated nets (ITNs)**, indoor residual spraying and larval control.

🎤 Viva corner

Q: Why is *P. falciparum* the most dangerous species? **A:** It causes cytoadherence and sequestration (cerebral malaria), infects red cells of all ages (high parasitaemia), and causes severe anaemia, hypoglycaemia and acidosis.

Q: Why does *P. vivax* cause relapse but *P. falciparum* does not? **A:** *P. vivax* (and *P. ovale*) form dormant liver hypnozoites. Radical cure needs primaquine or tafenoquine, after G6PD testing.

Chapter 70 · Apicomplexans II: Babesia, Toxoplasma, and the Coccidia

(This chapter groups the remaining apicomplexans from the outline, including the **coccidia/sporozoa**.)

Babesia

- *Babesia microti/divergens*; transmitted by **Ixodes ticks** (and transfusion).
- Infects red cells → **intraerythrocytic "Maltese cross" tetrads**; no pigment (unlike malaria).
- Causes a **malaria-like illness with haemolysis**, severe in asplenic and immunocompromised patients.
- Treatment: atovaquone + azithromycin, or clindamycin + quinine.

Toxoplasma gondii

- **Definitive host: the cat.** Humans are intermediate hosts.
- **Transmission:** eating undercooked meat (tissue cysts/bradyzoites), **cat faeces (oocysts)**, or **transplacental**.
- **Immunocompetent:** usually asymptomatic, or mild mononucleosis-like illness with lymphadenopathy.
- **Immunocompromised (AIDS, CD4 <100): reactivation → encephalitis** with **ring-enhancing brain lesions** (the commonest cause in HIV), chorioretinitis.
- **Congenital toxoplasmosis** (first-trimester primary infection): the classic triad of **chorioretinitis, intracranial (diffuse) calcification and hydrocephalus**, plus seizures.
- **Diagnosis:** serology (IgM/IgG avidity), PCR, imaging.
- **Treatment: pyrimethamine + sulfadiazine + folinic acid**. Spiramycin in pregnancy. Cotrimoxazole prophylaxis in HIV.

Coccidia / Sporozoa (intestinal; cause diarrhoea, severe in HIV)

> The intestinal **coccidia** are apicomplexan parasites that infect the gut epithelium and cause watery diarrhoea. They are **opportunistic** and especially important in **HIV/AIDS**. They are acquired by the **faecal–oral route** (contaminated water and food) and most are identified on stool microscopy with a **modified Ziehl–Neelsen (acid-fast) stain**.

Parasite	Features	Diagnosis	Treatment
Cryptosporidium	**Profuse, watery, self-limiting diarrhoea** in the immunocompetent; **chronic, life-threatening** in AIDS (CD4 <100). Waterborne outbreaks (resists chlorine)	Small (4–6 µm) **acid-fast oocysts**; antigen tests	**Restore immunity (ART)**; nitazoxanide; supportive
Cyclospora cayetanensis	Prolonged relapsing watery diarrhoea; associated with imported fresh produce	Acid-fast oocysts (8–10 µm), **autofluorescent**	**Cotrimoxazole**
Cystoisospora (Isospora) belli	Diarrhoea in AIDS	**Large ellipsoidal acid-fast oocysts**; **Charcot–Leyden crystals, eosinophilia**	**Cotrimoxazole**

Chapter 71 · Pathogenic Flagellates

Flagellates move by whip-like flagella. The medically important ones are the **luminal/urogenital** flagellates (*Giardia*, *Trichomonas*) and the **blood and tissue** flagellates (*Leishmania*, *Trypanosoma*).

Giardia lamblia (intestinalis)

- **Transmission:** faecal–oral; cysts in contaminated water (travellers, children, poor sanitation).
- Trophozoite: **pear-shaped, bilaterally symmetrical, two nuclei ("owl's/monkey face"), four pairs of flagella, a sucking disc**. It lives in the duodenum.
- **Clinical: foul-smelling, fatty, non-bloody diarrhoea (steatorrhoea)**, bloating, flatulence, malabsorption, weight loss. No eosinophilia (luminal).
- **Diagnosis:** cysts/trophozoites in stool (examine 3 samples), **stool antigen (ELISA)**, "string test" (duodenal sampling).
- **Treatment: metronidazole or tinidazole**; nitazoxanide.

Trichomonas vaginalis

- **Transmission:** sexually (an STI). No cyst stage; only a trophozoite.
- Pear-shaped, with an **undulating membrane** and **jerky motility** on a wet mount.
- **Clinical:** women — **frothy, yellow-green, offensive vaginal discharge**, itch, dysuria, **"strawberry cervix"**. Men — usually asymptomatic (urethritis). Associated with preterm birth and HIV acquisition.
- **Diagnosis:** wet-mount microscopy (motile trophozoites), NAAT, culture. Vaginal pH >4.5.
- **Treatment: metronidazole** (treat the partner too).

Leishmania (sandfly-borne)

- **Vector: the female sandfly (*Phlebotomus*, *Lutzomyia*).** Reservoir: dogs, rodents.
- **Amastigotes** (**Leishman–Donovan bodies**) inside macrophages; promastigotes in the sandfly.

Form	Species	Features
Visceral (kala-azar)	*L. donovani*, *L. infantum*	**Fever, massive splenomegaly and hepatomegaly, pancytopenia, weight loss, darkening of the skin ("kala-azar" = black fever), high mortality untreated.** Post-kala-azar dermal leishmaniasis
Cutaneous (oriental sore)	*L. major*, *L. tropica*	Chronic skin ulcer that heals with scarring

Form	Species	Features
Mucocutaneous (espundia)	*L. braziliensis*	Destruction of the nasopharyngeal mucosa

- **Diagnosis:** amastigotes in splenic, bone marrow or lymph node aspirate; **rK39 serology**; culture (NNN medium); PCR.
- **Treatment: liposomal amphotericin B** (first line for visceral), miltefosine (oral), pentavalent antimonials (stibogluconate), paromomycin.

Trypanosoma (African and American)

Feature	African trypanosomiasis (sleeping sickness)	American trypanosomiasis (Chagas disease)
Species	***T. brucei gambiense*** (West/Central Africa, chronic) and ***T. b. rhodesiense*** (East Africa, acute)	***T. cruzi***
Vector	**Tsetse fly (*Glossina*)** — bite	**Reduviid ("kissing") bug** — faeces rubbed into the bite or mucosa
Found in	Sub-Saharan Africa (foci in Nigeria)	Latin America
Clinical	**Trypanosomal chancre** → haemolymphatic stage (fever, **Winterbottom's sign** = posterior cervical lymphadenopathy) → **CNS stage: daytime somnolence, sleep-cycle reversal, coma, death**	Acute: **Romaña's sign** (periorbital oedema), fever. **Chronic: cardiomyopathy, megacolon, megaoesophagus**
Diagnosis	**Trypomastigotes** in blood, lymph node aspirate, CSF; **CATT** serology	Blood film (acute); serology/PCR (chronic); xenodiagnosis
Treatment	**Fexinidazole** (oral, both stages); older: pentamidine/suramin (early), melarsoprol or NECT (late)	**Benznidazole, nifurtimox** (acute)

Chapter 72 · Entamoeba histolytica (Amoebiasis)

Entamoeba histolytica is a pathogenic amoeba that causes **amoebiasis** (amoebic dysentery and liver abscess). It is acquired by the **faecal–oral** route (ingestion of **cysts** in contaminated food or water) and is common where sanitation is poor. It must be distinguished from the non-pathogenic *E. dispar* (identical morphology) and *E. coli*.

Life cycle and forms

- **Cyst** (infective; **mature cyst has 4 nuclei**; resistant): ingested → excysts in the small intestine.
- **Trophozoite** (invasive; moves by pseudopodia; a single nucleus with a central karyosome): lives in the colon; can invade. **Trophozoites with ingested red cells (erythrophagocytosis) indicate the pathogenic *E. histolytica*.**
- Transmission by the **cyst** (trophozoites die in the stomach).

Clinical forms

Form	Features
Asymptomatic carrier	Passes cysts (~90%); a source of infection
Intestinal amoebiasis (amoebic dysentery)	Gradual onset; **bloody, mucoid diarrhoea** with lower abdominal pain and tenesmus; usually **afebrile** (contrast bacillary dysentery). **"Flask-shaped" ulcers** in the caecum and colon
Amoeboma	A granulomatous mass mimicking colon cancer
Amoebic liver abscess	The commonest **extraintestinal** form. **Right-lobe abscess** with **"anchovy sauce" (reddish-brown) pus**; fever, tender hepatomegaly, right upper quadrant pain, raised right hemidiaphragm. May rupture into the lung or pericardium
Other	Pulmonary, cerebral, cutaneous amoebiasis

Diagnosis

- **Stool microscopy:** trophozoites (with RBCs) in fresh dysenteric stool; cysts in formed stool. Examine ≥3 specimens.
- **Stool antigen (ELISA)** and **PCR** distinguish *E. histolytica* from *E. dispar*.
- **Serology** (positive in invasive disease, especially liver abscess).
- **Liver abscess:** ultrasound/CT; **aspiration is usually not needed** (serology + imaging + response to treatment). Aspirate shows trophozoites at the abscess wall, not in the pus.

Treatment

- **Invasive disease (dysentery or abscess): metronidazole or tinidazole**, **followed by a luminal agent** (**diloxanide furoate**, paromomycin, or iodoquinol) to clear intestinal cysts.
- **Asymptomatic cyst passers:** a **luminal agent** alone.
- Prevention: sanitation, safe water, hand hygiene.

Other amoebae (free-living):

- *Naegleria fowleri*: warm fresh water → nasal entry → **primary amoebic meningoencephalitis** (rapidly fatal). Treat with amphotericin B + miltefosine.
- *Acanthamoeba*: **keratitis** (contact lens wearers) and granulomatous encephalitis.

Chapter 73 · Ciliates: Balantidium coli

- *Balantidium coli* is the **only ciliate pathogenic to humans** and the **largest protozoan parasite**.
- **Reservoir: the pig**; transmission is faecal–oral (cysts).
- Trophozoite: large, oval, covered in **cilia**, with a **macronucleus (kidney-shaped)** and micronucleus, and a funnel-shaped **cytostome**.
- **Clinical:** often asymptomatic; may cause **balantidial dysentery** with colonic ulcers resembling amoebiasis.
- **Diagnosis:** trophozoites/cysts in stool.
- **Treatment: tetracycline** (first line), metronidazole.

Protozoology Self-Assessment

Section A: MCQs

1. The definitive host of *Plasmodium* is: A. Humans · B. **The female Anopheles mosquito** · C. The male mosquito · D. The tsetse fly · E. The pig

B. The sexual cycle takes place in the mosquito, which is therefore the definitive host. Humans are intermediate hosts.

2. Relapse in malaria is due to: A. Merozoites · B. **Hypnozoites** · C. Gametocytes · D. Sporozoites · E. Schizonts

B. Dormant liver hypnozoites of *P. vivax* and *P. ovale* cause relapse.

3. The classic triad of congenital toxoplasmosis is: A. Deafness, cataract, PDA · B. **Chorioretinitis, hydrocephalus, intracranial calcification** · C. Rash, fever, arthritis · D. Jaundice, anaemia, oedema · E. Seizures alone

B. This is the triad of congenital toxoplasmosis. (Congenital rubella gives deafness, cataract and PDA.)

4. "Anchovy sauce" pus is characteristic of: A. Pyogenic liver abscess · B. **Amoebic liver abscess** · C. Hydatid cyst · D. Hepatoma · E. Cholangitis

B. The reddish-brown pus of an amoebic liver abscess is described as anchovy sauce.

5. Foul-smelling fatty diarrhoea with no blood and no eosinophilia suggests: A. Amoebiasis · B. **Giardiasis** · C. Shigellosis · D. Ascariasis · E. Cholera

B. Giardia causes malabsorptive, fatty, non-bloody diarrhoea. It is a luminal parasite, so there is no eosinophilia.

6. The vector of African sleeping sickness is: A. Sandfly · B. Reduviid bug · C. **Tsetse fly** · D. Anopheles · E. Ixodes tick

C. The tsetse fly (*Glossina*) transmits *Trypanosoma brucei*.

7. The only ciliate pathogenic to humans is: A. Giardia · B. Balantidium coli · C. Entamoeba · D. Trichomonas · E. Isospora

B. *Balantidium coli*, whose reservoir is the pig.

8. Which stain is used for Cryptosporidium? A. Giemsa · B. Gram · C. **Modified Ziehl–Neelsen** · D. India ink · E. Silver

> **C.** The coccidian oocysts are acid-fast on a modified Ziehl–Neelsen stain.

Section B: Essays

1. Describe the life cycle of *Plasmodium falciparum* and the pathogenesis of severe malaria.
2. Discuss the laboratory diagnosis and management of amoebiasis.
3. Write short notes on: giardiasis, visceral leishmaniasis, toxoplasmosis in HIV, and the intestinal coccidia.

Section C: Viva voce bank (Protozoology)

#	Question	Model answer
1	Which *Plasmodium* species is commonest and most dangerous in Nigeria?	*P. falciparum*.
2	Why is *P. vivax* rare in West Africa?	Most West Africans are Duffy antigen-negative, and the Duffy antigen is the receptor for *P. vivax*.
3	What is the infective stage of malaria to humans?	The sporozoite, injected by the mosquito.
4	Gold standard for malaria diagnosis?	Giemsa-stained thick and thin blood films.
5	What protects against malaria?	Sickle cell trait, G6PD deficiency, thalassaemia, HbC, Duffy negativity.
6	Infective form of *Entamoeba histolytica*?	The mature (4-nucleated) cyst.
7	How do you distinguish *E. histolytica* from *E. dispar*?	Antigen detection or PCR; erythrophagocytosis by trophozoites favours *E. histolytica*.
8	Treatment of amoebic liver abscess?	Metronidazole, then a luminal agent such as diloxanide furoate.
9	Diagnostic feature of *Giardia* trophozoite?	Pear-shaped, two nuclei (“owl face”), four pairs of flagella, a sucking disc.
10	Leishman–Donovan bodies?	Amastigotes of *Leishmania* within macrophages.
11	Winterbottom’s sign?	Posterior cervical lymphadenopathy in African trypanosomiasis.

#	Question	Model answer
12	Which protozoa are important in HIV/AIDS?	*Toxoplasma*, *Cryptosporidium*, *Cystoisospora*, *Cyclospora*.

Part C — Parasitology: Helminthology

Chapter 74 · Introduction to Helminths

Helminths are multicellular parasitic worms. The medically important ones fall into two phyla: the **Nematodes** (roundworms; cylindrical, with a complete gut and separate sexes) and the **Platyhelminths** (flatworms), which include the **Cestodes** (tapeworms; segmented, hermaphrodite, no gut) and the **Trematodes** (flukes; leaf-shaped, mostly hermaphrodite except schistosomes).

Feature	Nematodes	Cestodes	Trematodes
Shape	Cylindrical	Segmented, ribbon-like	Leaf-shaped
Sexes	Separate	Hermaphrodite	Hermaphrodite (except schistosomes)
Gut	Complete	**Absent** (absorb through tegument)	Blind (incomplete)
Body cavity	Present	Absent	Absent
Examples	*Ascaris*, hookworm, *Enterobius*, *Trichuris*, *Strongyloides*, filariae	*Taenia*, *Echinococcus*, *Hymenolepis*	*Schistosoma*, *Fasciola*, *Clonorchis*, *Paragonimus*

- **Eosinophilia** is a hallmark of **tissue-invasive** helminths (hookworm migration, *Strongyloides*, *Ascaris*, filariae, schistosomes). Luminal worms cause little eosinophilia.
- Treatment of most intestinal nematodes: **albendazole/mebendazole**; drugs are summarised in Book 1, Chapter 29.

Chapter 75 · Intestinal Nematodes (Round, Hook, Pin and Whip Worms)

Ascaris lumbricoides (the large roundworm)

- **The largest and commonest intestinal nematode** (up to 30 cm). Faecal–oral; eggs in soil (a "soil-transmitted helminth").
- **Life cycle:** ingest eggs → larvae hatch in the gut → **penetrate the gut wall → liver → lungs** (Löffler's syndrome) → coughed up and swallowed → mature in the small intestine.
- **Clinical:** often asymptomatic; **Löffler's pneumonitis** (cough, wheeze, eosinophilia during migration); malnutrition and growth retardation in children; **intestinal obstruction** (a bolus of worms), **biliary and pancreatic obstruction**, appendicitis.
- **Diagnosis: eggs (bile-stained, mamillated) in stool**; adult worms passed per rectum, mouth or nose.
- **Treatment:** albendazole, mebendazole, pyrantel.

Hookworm (Necator americanus, Ancylostoma duodenale)

- A major cause of **iron deficiency anaemia** in the tropics, including Nigeria.
- **Life cycle: filariform larvae penetrate the skin (bare feet)** → blood → **lungs** → swallowed → attach to the small intestine with cutting plates/teeth and suck blood.
- **Clinical: "ground itch"** at entry; pulmonary migration; chronic intestinal blood loss → **microcytic hypochromic anaemia**, hypoproteinaemia (oedema), and in children cognitive and growth impairment. Each *Necator* worm takes ~0.03 mL blood/day; *Ancylostoma* more.
- **Diagnosis: eggs in stool** (oval, thin-shelled, segmented); Kato-Katz for egg counts.
- **Treatment:** albendazole/mebendazole + **iron replacement**. Mass deworming.

Enterobius vermicularis (pinworm / threadworm)

- The commonest helminth in temperate climates; common in children worldwide. Faecal–oral; **autoinfection**.
- The gravid female migrates to the **perianal skin at night** to lay eggs.
- **Clinical: nocturnal perianal itching (pruritus ani)**, disturbed sleep, irritability; rarely vulvovaginitis, appendicitis.
- **Diagnosis:** the **"Sellotape (cellophane tape) test"** in the early morning → eggs (D-shaped, flattened on one side); adult worms seen on the perianal skin.
- **Treatment: albendazole/mebendazole or pyrantel**, repeated after 2 weeks; **treat the whole household**; hygiene.

Trichuris trichiura (whipworm)

- Soil-transmitted; faecal–oral. The worm has a thin whip-like anterior and thick posterior.
- **Clinical:** usually asymptomatic; heavy infection → **chronic bloody diarrhoea, growth retardation and rectal prolapse** in children (part of "Trichuris dysentery syndrome").
- **Diagnosis: barrel-shaped eggs with bipolar plugs** in stool.
- **Treatment:** albendazole/mebendazole.

Strongyloides stercoralis (threadworm)

- **Capable of autoinfection**, so it can persist for decades.
- **Life cycle:** filariform larvae penetrate skin → lungs → gut; some larvae re-invade the gut wall or perianal skin (**autoinfection**).
- **Clinical: larva currens** (a rapidly moving itchy rash), abdominal pain, diarrhoea, eosinophilia.
- ⚠ **Hyperinfection / disseminated strongyloidiasis** in the **immunosuppressed** (steroids, HTLV-1): massive larval invasion → Gram-negative sepsis and meningitis, often fatal. **Screen before giving steroids or chemotherapy.**
- **Diagnosis: larvae (not eggs) in stool**; Baermann concentration; agar plate culture; serology.
- **Treatment: ivermectin** (drug of choice); albendazole.

Visceral/ocular larva migrans: *Toxocara canis/cati* (dog and cat roundworm); children with pica; eosinophilia, hepatomegaly, and ocular granuloma mimicking retinoblastoma. **Cutaneous larva migrans:** *Ancylostoma braziliense* (a serpiginous creeping eruption).

Chapter 76 · Tissue Nematodes: Filariae, Onchocerca and Dracunculus

Lymphatic filariasis (Wuchereria bancrofti, Brugia malayi)

- **Vector:** *Culex* (and other) mosquitoes.
- Adult worms live in the **lymphatics**; **microfilariae** circulate in the blood (with **nocturnal periodicity** for *W. bancrofti*).
- **Clinical:** acute lymphangitis and fever; chronic **lymphoedema → elephantiasis** (legs, scrotum → **hydrocele**); chyl-uria; **tropical pulmonary eosinophilia**.
- **Diagnosis: microfilariae on a night (midnight) blood film** (Giemsa; Knott's concentration); **antigen test (ICT card)**; ultrasound ("filarial dance sign").
- **Treatment: diethylcarbamazine (DEC)** ± albendazole; **ivermectin + albendazole** in areas co-endemic for onchocerciasis (DEC is avoided there). **Doxycycline** kills *Wolbachia* and sterilises the adults. Nigeria is part of the **Global Programme to Eliminate Lymphatic Filariasis** (mass drug administration).

Onchocerca volvulus (river blindness)

> **Onchocerciasis** is caused by the filarial nematode ***Onchocerca volvulus***, transmitted by the bite of the **blackfly (*Simulium*)**, which breeds in **fast-flowing rivers** — hence "river blindness". It is a major cause of blindness in Nigeria and West Africa.

- Adult worms live coiled in **subcutaneous nodules (onchocercomata)**; microfilariae migrate through the skin and eyes.
- **Clinical:**
  - **Severe itching (the commonest symptom)**, papular dermatitis.
  - **Skin changes:** "lizard/leopard skin" (depigmentation), "hanging groin", lichenification.
  - **Subcutaneous nodules.**
  - **Eye disease:** punctate then sclerosing keratitis, iridocyclitis, chorioretinitis, optic atrophy → **blindness** (dead microfilariae provoke inflammation).
- **Diagnosis: skin snip** (microfilariae emerge in saline; no blood needed — *Onchocerca* is not in blood); slit-lamp (microfilariae in the cornea/anterior chamber); nodule biopsy; the **Mazzotti test** (now rarely used); PCR; **DEC patch test**.
- **Treatment: ivermectin** (kills microfilariae; given 6–12-monthly; **community-directed treatment with ivermectin, CDTI**); **doxycycline** (anti-*Wolbachia*, kills adults); moxidectin. **DEC is contraindicated** (severe Mazzotti reaction and eye damage). The **Mectizan (ivermectin) Donation Programme** and **APOC/ESPEN** have hugely reduced the disease.

Loa loa (African eye worm)

- Vector: the **Chrysops (deer/mango) fly**; Central/West African rainforest.
- **Clinical: Calabar swellings** (transient subcutaneous angioedema), and the dramatic sign of an **adult worm crossing the conjunctiva** ("eye worm").
- ⚠ **Caution:** heavy *Loa loa* microfilaraemia + **ivermectin or DEC → fatal encephalopathy**. Check *Loa* load first.

Dracunculus medinensis (Guinea worm)

- **Transmission:** drinking water containing **copepods (Cyclops)** that carry larvae.
- **Life cycle:** larvae released in the stomach → mature; the gravid female (up to 1 m long) migrates to the **lower limb** → a skin blister/ulcer; on water contact it releases larvae.
- **Clinical:** a painful blister, then a cord-like ulcer with the emerging worm; secondary infection; arthritis.
- **Diagnosis:** clinical (visible worm).
- **Treatment: NO specific drug.** The classic management is **slow extraction of the worm by winding it on a stick**, a few centimetres each day, over weeks; wound care; analgesia; tetanus prophylaxis.
- **Prevention and eradication:** filter drinking water (cloth filters), treat water with the larvicide **temephos (Abate®)**, health education, case containment. **Nigeria was certified free of dracunculiasis (Guinea worm) in 2013** — one of its greatest public health successes, led by the Guinea Worm Eradication Programme.

Chapter 77 · Cestodes (Tapeworms)

Cestodes (tapeworms) are ribbon-like, segmented, **hermaphrodite** flatworms with **no gut** (they absorb nutrients through the tegument). The adult has a **scolex** (head, with suckers ± hooks for attachment), a neck, and a chain of segments (**proglottids**) called the strobila. Humans may be the definitive host (adult worm in the gut), the intermediate host (larval cysts in tissue), or both.

Tapeworm	Source	Human as	Disease
Taenia saginata (beef)	Undercooked **beef**	Definitive	Mild intestinal taeniasis; **motile proglottids** passed; **no cysticercosis** (no hooks, 4 suckers)
Taenia solium (pork)	Undercooked **pork** (larvae → adult worm); **ingested eggs → cysticercosis**	Definitive **and** intermediate	**Cysticercosis**, and especially **neurocysticercosis (a leading cause of acquired epilepsy worldwide)**: cysts in the brain, seizures; subcutaneous and ocular cysts. Scolex has a **rostellum with hooks**
Echinococcus granulosus	**Dog** faeces (eggs); sheep are intermediate	**Intermediate (accidental)**	**Hydatid disease (cyst)** in the **liver** (commonest) and lungs; anaphylaxis if it ruptures. "Eggshell"/daughter cysts; **hydatid sand**. Casoni test; serology. **Do not aspirate carelessly** → spill/anaphylaxis. Treat: albendazole + surgery/**PAIR**. Dogs are the definitive host
Hymenolepis nana (dwarf tapeworm)	Faecal–oral (eggs); **autoinfection**	Both	The commonest tapeworm; usually asymptomatic; no intermediate host needed
Diphyllobothrium latum (fish)	Raw/undercooked **freshwater fish**	Definitive	**Vitamin B_{12} deficiency (megaloblastic anaemia)** — the worm competes for B_{12}; the longest human tapeworm

- **Diagnosis:** eggs or proglottids in stool (*Taenia* eggs of *saginata* and *solium* are identical — examine proglottids or use PCR); imaging and serology for tissue forms (neurocysticercosis, hydatid).
- **Treatment: praziquantel** for intestinal tapeworms; **niclosamide**; **albendazole** for cysticercosis and hydatid (with steroids and anticonvulsants in neurocysticercosis). (See Book 1, Chapter 29.)

Chapter 78 · Trematodes (Flukes) — General and the Food-Borne Flukes

Trematodes (flukes) are leaf-shaped flatworms. Most are **hermaphrodite** (except the schistosomes, which have separate sexes). They have **two suckers** (oral and ventral). Their life cycles involve a **snail as the first intermediate host** and the release of swimming larvae (**cercariae**). Humans are infected by eating encysted larvae (**metacercariae**) on vegetation or in fish/crabs, or (for schistosomes) by cercariae penetrating the skin.

General fluke life cycle: egg → **miracidium** → **snail** (sporocyst → rediae → cercariae) → metacercaria (on plants/in fish) → definitive host.

Food-borne flukes (acquired by eating)

Fluke	Acquired from	Site in host	Disease
Fasciola hepatica (sheep **liver fluke**)	**Watercress** / aquatic plants (metacercariae)	Bile ducts	Liver and biliary disease, eosinophilia, biliary obstruction. **Treatment: triclabendazole** (praziquantel does not work)
Clonorchis sinensis / *Opisthorchis* (Chinese/Southeast Asian **liver flukes**)	Raw/undercooked **freshwater fish**	Bile ducts	Cholangitis, biliary obstruction, **cholangiocarcinoma**
Paragonimus westermani (**lung fluke**)	Undercooked **crabs/crayfish**	Lungs	Chronic cough, **haemoptysis** (mimics TB), eosinophilia
Fasciolopsis buski (giant **intestinal fluke**)	Aquatic plants (water chestnut)	Small intestine	Abdominal pain, diarrhoea, malabsorption
Heterophyes, *Metagonimus* (small **intestinal flukes**)	Freshwater fish	Intestine	Mild diarrhoea

- **Diagnosis:** eggs in stool (or sputum for *Paragonimus*); operculated eggs; serology.
- **Treatment: praziquantel** for all except ***Fasciola*** **(triclabendazole)**.

This chapter covers the outline items *liver flukes, lung flukes and intestinal flukes*. The **blood flukes (schistosomes)** are covered next because of their importance in Nigeria.

Chapter 79 · Blood Flukes: Schistosomiasis (Bilharzia)

Schistosomiasis (bilharzia) is caused by blood flukes of the genus ***Schistosoma***. Unlike other trematodes, schistosomes have **separate sexes**, and the slender male carries the female in a groove (the **gynaecophoric canal**). It is one of the most important parasitic diseases in Nigeria after malaria, affecting millions, especially children near fresh water.

Species

Species	Eggs	Veins inhabited	Disease
S. *haematobium*	**Terminal spine**	**Vesical (bladder) plexus**	**Urinary schistosomiasis**: haematuria, bladder fibrosis, **squamous cell carcinoma of the bladder**. **The main species in Nigeria**
S. *mansoni*	**Lateral spine**	**Inferior mesenteric (intestinal) veins**	**Intestinal and hepatosplenic** disease; periportal ("pipe-stem") fibrosis, portal hypertension. Also present in Nigeria
S. *japonicum*	Small, rudimentary spine	Superior mesenteric veins	Severe intestinal/hepatic disease (Asia)

Life cycle

1. Eggs are passed in **urine (*haematobium*)** or **stool (*mansoni*)** into fresh water.
2. Eggs hatch → **miracidia** → infect **freshwater snails** (*Bulinus* for *haematobium*; *Biomphalaria* for *mansoni*).
3. **Cercariae** (fork-tailed) emerge and **penetrate human skin** during water contact (swimming, farming, washing).
4. They become **schistosomula** → migrate via blood/lungs → mature in the liver → the adult pairs migrate to their final veins and lay eggs.
5. Eggs penetrate the bladder or bowel wall to be excreted (the **spine and egg enzymes** help), but **many eggs are trapped in tissues → granulomas (the cause of disease)**.

Clinical features

Phase	Features
Cercarial dermatitis ("swimmer's itch")	Itchy papular rash at penetration
Acute (Katayama fever), 4–8 weeks	Fever, urticaria, eosinophilia, hepatosplenomegaly (an immune-complex reaction to egg-laying; more with *mansoni/japonicum*)
Chronic S. *haematobium*	**Painless terminal haematuria** (classic in Nigerian schoolchildren), dysuria, bladder wall calcification and fibrosis, obstructive uropathy, hydronephrosis, **squamous cell carcinoma of the bladder**, and genital schistosomiasis (a risk factor for HIV)
Chronic S. *mansoni*	Colonic polyps, bloody diarrhoea, **hepatosplenic schistosomiasis** with **periportal ("pipe-stem"/Symmers') fibrosis → portal hypertension, varices, splenomegaly** (with *preserved* liver function early)
Other	Pulmonary hypertension, CNS schistosomiasis (transverse myelitis, seizures)

Diagnosis

- ***S. haematobium*:*** eggs (terminal spine) in urine** — collect a **midday (10 am–2 pm) terminal urine** specimen; filtration; **dipstick haematuria/proteinuria** for screening.
- ***S. mansoni*:*** eggs (lateral spine) in stool** (Kato-Katz); rectal snip.
- Serology and antigen tests (CCA/CAA in urine); PCR.
- Biopsy (bladder/rectum) shows eggs and granulomas; eosinophilia.

Treatment and control

- **Praziquantel** (40 mg/kg, single dose; active against adult worms) is the drug of choice for all species.
- **Control (WHO): mass drug administration (preventive chemotherapy) of praziquantel to school-age children**, snail control (molluscicides), safe water and sanitation, health education, avoiding contact with infested water. Nigeria runs national schistosomiasis control programmes.

★ Exam pearls

- **Terminal spine → terminal haematuria → bladder → *S. haematobium*. Lateral spine → *mansoni* → liver and bowel.**
- ***S. haematobium* is a recognised cause of squamous cell carcinoma of the bladder** (a WHO Group 1 carcinogen link).
- Praziquantel for everything except *Fasciola* (triclabendazole).

Helminthology Self-Assessment

Section A: MCQs

1. The commonest cause of iron deficiency anaemia from a helminth in the tropics is: A. Ascaris · B. **Hookworm** · C. Enterobius · D. Trichuris · E. Taenia

B. Hookworms attach to the small intestine and suck blood, causing chronic blood loss.

2. The "Sellotape test" is used to diagnose: A. Hookworm · B. Ascaris · C. **Enterobius vermicularis** · D. Strongyloides · E. Trichuris

C. Pinworm eggs are collected from the perianal skin with adhesive tape in the morning.

3. Which worm causes a fatal hyperinfection syndrome in immunosuppressed patients? A. Ascaris · B. **Strongyloides stercoralis** · C. Enterobius · D. Taenia saginata · E. Trichuris

B. Strongyloides can autoinfect and disseminate, causing fatal Gram-negative sepsis — screen before steroids.

4. River blindness is caused by: A. Loa loa · B. Wuchereria · C. **Onchocerca volvulus** · D. Brugia · E. Dracunculus

C. *Onchocerca volvulus*, transmitted by the *Simulium* blackfly.

5. DEC is contraindicated in onchocerciasis because it causes: A. Hepatitis · B. **A severe Mazzotti reaction with eye damage** · C. Marrow suppression · D. Renal failure · E. Cardiotoxicity

B. Rapid killing of microfilariae causes severe inflammation. Use ivermectin instead.

6. Terminal haematuria in a Nigerian schoolchild suggests: A. S. mansoni · B. **S. haematobium** · C. S. japonicum · D. Fasciola · E. Clonorchis

B. S. *haematobium* lives in the bladder venous plexus and its terminal-spined eggs cause haematuria.

7. The drug of choice for schistosomiasis is: A. Albendazole · B. Ivermectin · C. **Praziquantel** · D. Triclabendazole · E. DEC

C. Praziquantel treats all schistosome species. (Triclabendazole is for *Fasciola*.)

8. *Taenia solium* ingested eggs cause: A. Intestinal taeniasis · B. **Cysticercosis** · C. B_{12} deficiency · D. Hydatid disease · E. Anaemia

B. Eggs (not larvae) cause cysticercosis, including neurocysticercosis, a major cause of epilepsy.

9. Guinea worm (dracunculiasis) is treated by: A. Praziquantel · B. Albendazole · C. **Slow mechanical extraction (no specific drug)** · D. Ivermectin · E. DEC

C. There is no drug. The worm is wound out slowly over days. Nigeria eradicated it in 2013.

10. Megaloblastic anaemia can be caused by which tapeworm? A. Taenia saginata · B. Echinococcus · C. Hymenolepis · D. **Diphyllobothrium latum** · E. Taenia solium

D. The fish tapeworm competes for vitamin B_{12}.

Section B: Essays

1. Describe the life cycle, clinical features and control of urinary schistosomiasis in Nigeria.
2. Discuss the soil-transmitted helminths and their public health control (mass deworming).
3. Write short notes on: onchocerciasis, hookworm anaemia, neurocysticercosis, hydatid disease, and Guinea worm eradication.
4. Compare the three classes of helminths (nematodes, cestodes, trematodes).

Section C: Viva voce bank (Helminthology)

#	Question	Model answer
1	Classify helminths.	Nematodes (roundworms), cestodes (tapeworms) and trematodes (flukes).
2	Which helminths migrate through the lungs?	Ascaris, hookworm and Strongyloides (Löffler's syndrome).
3	How is hookworm acquired?	Filariform larvae penetrate the skin of bare feet.
4	Drug of choice for Strongyloides?	Ivermectin.
5	What transmits lymphatic filariasis?	*Culex* and other mosquitoes.
6	When do you take blood for *Wuchereria*?	At midnight (nocturnal periodicity).
7	How do you diagnose onchocerciasis?	Skin snip for microfilariae; slit-lamp examination; nodule biopsy.
8	Why avoid ivermectin in heavy Loa loa infection?	Risk of fatal encephalopathy from dying microfilariae.
9	Which tapeworm causes neurocysticercosis?	*Taenia solium* (from ingested eggs).

#	Question	Model answer
10	Definitive host of *Echinococcus*?	The dog. Humans are accidental intermediate hosts with hydatid cysts.
11	Which fluke needs triclabendazole?	*Fasciola hepatica*.
12	How do schistosomes differ from other flukes?	They have separate sexes and live in blood vessels.
13	Snail hosts of schistosomes?	*Bulinus* (*S. haematobium*), *Biomphalaria* (*S. mansoni*).
14	Cancer linked to *S. haematobium*?	Squamous cell carcinoma of the bladder.
15	Which Nigerian parasitic disease has been eradicated?	Dracunculiasis (Guinea worm), certified in 2013.

Appendices

Appendix A · Common Reference Intervals

⚠ Caution

Reference intervals vary between laboratories, methods and populations. **Always use your own laboratory's values.** Those below are typical adult values for orientation only.

Haematology

Test	Reference interval
Haemoglobin	Men 130–175 g/L; women 120–155 g/L
Haematocrit (PCV)	Men 0.40–0.52; women 0.36–0.47
MCV	80–100 fL
MCH	27–33 pg
White cell count	4.0–11.0 × 10^9/L
Neutrophils	2.0–7.5 × 10^9/L (lower in benign ethnic neutropenia)
Lymphocytes	1.5–3.5 × 10^9/L
Eosinophils	0.04–0.4 × 10^9/L
Platelets	150–400 × 10^9/L
Reticulocytes	0.5–2.5% (50–100 × 10^9/L)
ESR	<20 mm/h (age-dependent)
PT	11–14 s (INR 0.8–1.2)
aPTT	25–38 s
Fibrinogen	1.5–4.0 g/L
Ferritin	Men 30–400 µg/L; women 15–200 µg/L
Serum iron	10–30 µmol/L
TIBC	45–72 µmol/L
Transferrin saturation	20–45%
Vitamin B_{12}	160–900 ng/L
Serum folate	3–20 µg/L

Chemistry (serum/plasma)

Test	Reference interval
Sodium	135–145 mmol/L
Potassium	3.5–5.0 mmol/L
Chloride	95–105 mmol/L
Bicarbonate	22–28 mmol/L
Urea	2.5–7.8 mmol/L
Creatinine	60–110 µmol/L (M); 45–90 (F)
eGFR	>90 mL/min/1.73 m²
Glucose (fasting)	3.5–5.5 mmol/L
Calcium (adjusted)	2.20–2.60 mmol/L
Phosphate	0.8–1.5 mmol/L
Magnesium	0.7–1.0 mmol/L
Total protein	60–80 g/L
Albumin	35–50 g/L
Bilirubin (total)	3–17 µmol/L
ALT	10–40 U/L
AST	10–40 U/L
ALP	30–130 U/L (higher in children)
GGT	M <55, F <38 U/L
Amylase	25–125 U/L
Urate	M 0.2–0.43; F 0.14–0.36 mmol/L
Total cholesterol	<5.0 mmol/L (desirable)
LDL cholesterol	<3.0 mmol/L
HDL cholesterol	>1.0 (M), >1.2 (F) mmol/L
Triglycerides (fasting)	<1.7 mmol/L
CRP	<5 mg/L
HbA1c	<42 mmol/mol (<6.0%)
TSH	0.4–4.0 mU/L

Arterial blood gas

Test	Reference interval
pH	7.35–7.45
pCO_2	4.7–6.0 kPa (35–45 mmHg)
pO_2	10.6–13.3 kPa (80–100 mmHg)
HCO_3^-	22–26 mmol/L
Base excess	−2 to +2
Lactate	0.5–2.0 mmol/L

Appendix B · Key Poisons and their Antidotes

Poison / drug	Antidote
Paracetamol	**N-acetylcysteine**
Opioids	**Naloxone**
Benzodiazepines	**Flumazenil**
Warfarin	**Vitamin K**, PCC, FFP
Heparin	**Protamine sulfate**
Dabigatran	**Idarucizumab**
Factor Xa inhibitors	**Andexanet alfa**
Organophosphates	**Atropine + pralidoxime**
Beta-blockers	**Glucagon**, atropine, high-dose insulin
Digoxin	**Digoxin-specific antibody (Fab)**
Iron	**Desferrioxamine**
Lead / heavy metals	**DMSA (succimer), EDTA, dimercaprol (BAL)**
Copper (Wilson's)	**Penicillamine**
Methanol / ethylene glycol	**Fomepizole** or ethanol; dialysis
Cyanide	**Hydroxocobalamin**, sodium thiosulfate, dicobalt edetate
Carbon monoxide	**100% / hyperbaric oxygen**
Methotrexate	**Folinic acid (leucovorin)**
Methaemoglobinaemia	**Methylene blue**
Isoniazid overdose	**Pyridoxine (vitamin B_6)**
Local anaesthetic toxicity	**20% lipid emulsion (Intralipid)**
Magnesium toxicity	**Calcium gluconate**
Snake bite (carpet viper, Nigeria)	**Polyvalent antivenom**

Appendix C · High-Yield Treatment Regimens (Quick Reference)

- **Uncomplicated malaria:** artemether–lumefantrine (AL) × 3 days.
- **Severe malaria:** IV artesunate (2.4 mg/kg at 0, 12, 24 h, then daily).
- **Drug-susceptible TB:** 2 months HRZE, then 4 months HR (2HRZE/4HR) + pyridoxine.
- **MDR-TB:** BPaLM (bedaquiline, pretomanid, linezolid, moxifloxacin) × 6 months.
- **Leprosy (MB):** rifampicin + clofazimine + dapsone × 12 months.
- **HIV (Nigeria first line):** tenofovir + lamivudine + dolutegravir (TLD).
- ***H. pylori*:** PPI + clarithromycin + amoxicillin × 14 days (or bismuth quadruple therapy).
- **Hyperkalaemia:** calcium gluconate → insulin–glucose + salbutamol → binders/dialysis.
- **DKA:** fluids + fixed-rate insulin 0.1 u/kg/h + potassium replacement.
- **Anaphylaxis:** IM adrenaline 0.5 mg (1:1000), repeat after 5 min.
- **Status/severe hypocalcaemia:** 10% calcium gluconate 10–20 mL IV.
- **Schistosomiasis:** praziquantel 40 mg/kg single dose.
- **Onchocerciasis:** ivermectin (community-directed).

Appendix D · Useful Mnemonics

Topic	Mnemonic
CYP inducers	**"GPRS Cell Phone"** (Griseofulvin, Phenytoin, Rifampicin, St John's wort, Carbamazepine, Phenobarbital, chronic alcohol)
Cholinergic effects	**DUMBBELSS** (Diarrhoea, Urination, Miosis, Bradycardia, Bronchospasm, Emesis, Lacrimation, Salivation, Sweating)
Loop diuretic side effects	**OH DANG** (Ototoxicity, Hypokalaemia, Dehydration, Allergy, Nephritis, Gout)
Thiazide side effects	**hyper-GLUC** (Glucose, Lipids, Urate, Calcium)
ACE inhibitor effects	**CAPTOPRIL** (Cough, Angio-oedema, Proteinuria, Taste, Orthostatic hypotension, Pregnancy contraindication, Renal failure, Increased K^+, Leucopenia)
TB drug effects	INH–neuropathy/hepatitis; Rifampicin–red; Pyrazinamide–urate; Ethambutol–eyes; Streptomycin–ears
Amiodarone	Lungs, Liver, Thyroid (+ skin, eyes, nerves)
High anion gap acidosis	**MUDPILES**
Causes of eosinophilia	**NAACP** (Neoplasm, Allergy/Asthma, Addison's, Collagen vascular, Parasites)

Topic	Mnemonic
Pellagra	3 Ds: Dermatitis, Diarrhoea, Dementia
Hypercalcaemia	“Bones, stones, abdominal groans, psychic moans”
Bone metastases (primaries)	**BLT with a Kosher Pickle** (Breast, Lung, Thyroid, Kidney, Prostate)
Measles prodrome	3 Cs: Cough, Coryza, Conjunctivitis (+ Koplik spots)
Congenital rubella	Deafness, Cataract, PDA

Appendix E · Abbreviations

ACE angiotensin-converting enzyme · ACT artemisinin-based combination therapy · ADME absorption, distribution, metabolism, excretion · AKI acute kidney injury · ALP alkaline phosphatase · ART antiretroviral therapy · BBB blood–brain barrier · CKD chronic kidney disease · CNS central nervous system · COX cyclooxygenase · CYP cytochrome P450 · DKA diabetic ketoacidosis · DMARD disease-modifying antirheumatic drug · DOT directly observed therapy · eGFR estimated glomerular filtration rate · ENL erythema nodosum leprosum · EPO erythropoietin · G6PD glucose-6-phosphate dehydrogenase · GFR glomerular filtration rate · HBV/HCV hepatitis B/C virus · HIM haemopoietic inductive microenvironment · HSC haemopoietic stem cell · IDA iron deficiency anaemia · IF intrinsic factor · INR international normalised ratio · IPTp intermittent preventive treatment in pregnancy · MAC minimum alveolar concentration · MDT multidrug therapy · NSAID non-steroidal anti-inflammatory drug · OPV/IPV oral/inactivated polio vaccine · ORS oral rehydration solution · PCR polymerase chain reaction · POCT point-of-care testing · PPI proton pump inhibitor · PTH parathyroid hormone · RAAS renin–angiotensin–aldosterone system · RDT rapid diagnostic test · SAM severe acute malnutrition · TB tuberculosis · TDM therapeutic drug monitoring · TNM tumour–node–metastasis · WHO World Health Organization.

References and Further Reading

These notes summarise and synthesise material from standard, internationally accepted textbooks and current clinical guidelines. They are a study aid and **not a substitute for the primary texts**. Students should read the recommended textbooks in full and confirm all doses and protocols against a current formulary and local guidelines.

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A note to the student. These notes rearrange your departmental course outline into a logical, exam-oriented sequence and explain each topic at the depth expected of a 400-level MBBS student, with a Nigerian clinical context. Use them alongside the textbooks above, your lectures and your clinical postings. Confirm every drug dose and treatment protocol with a current formulary and the latest national guidelines before applying it to a patient. Good luck in your examinations.